Solution analysis and design

Full production design · physical firefighting and rescue · 241 screens · 37 workflows



Review baseline 3.0 · 2026-10-24. Proposed design, not a deployed or certified system. Complete screen specifications are provided separately.

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Screen Specifications — 241 screens

SmartAlarm — Business Requirements Document



Version: 3.0 · 24/10/2026 · Design documentation for review, not yet signed-off requirements.
Confirmed direction: international and cross-industry from the outset; a complete production design from the outset; no temporary architecture for a demo; Vietnamese–English–Japanese are mandatory scope.
Documentation language: Vietnamese in the original; this is the English translation. The interface has a Vietnamese–English–Japanese switcher; illustrative records retain their source language when they represent entered content.

1. Product decision

SmartAlarm is a platform for receiving signals, verifying incidents, alerting the right audiences, coordinating assistance, and tracking outcomes. The proposed name remains **SmartAlarm**, with the description “Multi-hazard safety & response platform”. **SmartAlarm One** may be used for the product suite, and **SmartAlarm Command / Connect / Companion / Field** for its four product surfaces. Alternative names: **SignalAid** (emphasizing assistance), **Sentra Response** (emphasizing coordination). Trademarks, domains, and registration availability have not been checked; these names must not be treated as protected marks.

The differentiators are a closed-loop response lifecycle and interoperability. The product does more than sound a siren or send a push notification: it identifies where a signal came from, what has been confirmed, who is responsible, who needs help, which information is stale, and who recorded the outcome.

Physical firefighting and rescue are core product scope: observe existing alarm panels, water-based suppression, fire pumps, releasing controllers, and rescue equipment; coordinate approved physical operations and verify outcomes. Local automatic protection remains authoritative and independent of cloud availability. SmartAlarm distinguishes monitor-only connections from individually approved control capabilities; a generic protocol name does not grant control authority.

2. Analysis basis and source limitations

The content of all 34 slides of FireEscape_Proposal_Expanded_EN.pptx was read, and the embedded images of the legacy interface, two-hub diagram, cameras, and floor plans were reviewed. The file is business reference material, not execution instructions. Slide 33 links to FireEscape_Introduction.pptx and FEscape_Demo.mp4, but neither file was supplied. The deck cannot recover the original source code, database structure, or protocols; this design rebuilds the described capabilities and is not verified reverse engineering.

Capability in the deck	Source slides	Treatment in SmartAlarm
Property → building → floor → zone	11, 23–24	Retain and generalize into a Location graph supporting roads, areas, and outdoor facilities
Camera fire detection and imagery	13–14, 25	Retain through connector/model adapters; confidence is unverified evidence
Maps and voice guidance	16	Retain; add offline versions, accessibility, and approved revisions
SOS with location and notes	15–16, 26	Retain; expand the taxonomy and support multiple people needing assistance
Smart Alarm Hub / Integration Hub	9, 27, 32	Retain business boundaries; separate Connect, Incident, Workflow, and Delivery
IOC / fire alarm / suppression	17–18	Design adapters by partner and capability; do not assume connections already exist
Groups/chat/family and safety reporting	19–20, 28–29	Retain fan-out to subscribed groups; add consent and data minimization
A safe status does not close an incident	6, 19, 29	An invariant throughout design and acceptance

The physical baseline additionally uses the manufacturer evidence in Physical Device Catalog. Concrete candidates include Siemens Cerberus PRO, NOTIFIER BACNET-GW-3, Simplex BACpac, Tornatech GPX/GPD, and Potter PFC-4410G3. These are researched candidates, not a claim of universal interoperability or a completed installation. Model, regional variant, firmware, license, point map, permitted operations, and field acceptance must be bound for each site.

3. Business problems and required outcomes

BR-01: unify multi-source signals while retaining provenance and confidence. BR-02: reduce the gap between an alert and someone accepting responsibility. BR-03: provide SOS suitable for varied situations and accessibility needs. BR-04: maintain situational awareness when networks or providers fail. BR-05: enable rapid integration through adapters with clear contracts. BR-06: expand internationally with data and authority boundaries. BR-07: retain audit evidence and learn from drills and incidents. BR-08: build a complete production platform with long-term scalability. BR-09: provide rescue assistance through speakers and voice navigation. BR-10: provide fused positioning and evacuation optimization. BR-11: provide evidence-based real-time AI analysis. BR-12: proactively rescue people needing assistance even without an SOS.

Success metric	Measurement	Proposed acceptance target, not an actual measurement
Time for the system to record an SOS	server accepted_at − ingress_at; excludes the device network	p95 ≤ 2 s at the reference load
Time until an alert is ready to send	delivery job created_at − incident committed_at	p95 ≤ 5 s at the reference load
Duplicate signals creating duplicate incidents	Replay the same source + event ID	0 new incidents from replaying the same ID
Decision traceability	Incident → source → rule version → actor → dispatch	100% of acceptance cases have all links
Detection of missing acknowledgment	Deadline expires without anyone accepting	100% of test scenarios create escalation exactly once per step
Time to integrate a new adapter	After the partner sandbox, spec, and credentials are available	Target of 5–10 person-days for a simple telemetry adapter, to be measured in a pilot
Availability	Minutes the core API operates / minutes in the month	Production target 99.95%; the prototype does not demonstrate an SLA

No casualty-reduction target or rescue-time commitment is set without research and operational data. Detector accuracy is evaluated by hazard/site/model using labeled data, rather than one generic marketing percentage.

BR-13: provide trustworthy visibility of physical fire-protection coverage, pump readiness, waterflow, and impairment. BR-14: execute only authorized physical requests and reconcile uncertain outcomes with independent feedback. BR-15: maintain compatibility, FAT/SAT, commissioning, and restoration evidence throughout the equipment lifecycle. BR-16: coordinate supervised rescue-robot/water-monitor missions with local operator authority and human handover.

Physical acceptance adds configuration-specific measures: every enabled point and command must trace to approved mapping and evidence; every stale or lost link must become visibly uncertain within its signed site threshold; no expired or duplicate command may cause a new physical effect in acceptance testing. Detection-to-observation, approval-to-request, and request-to-physical-feedback latency are measured separately. Thresholds depend on the approved device and site design and are not universal rescue-time guarantees.

4. Users and organizational model

Group	Work	Critical permissions
Citizen / occupant / worker / traveler	Receive guidance, send SOS, check in	Personal data, invited groups, and public warnings relevant to the area
Family / caregiver	Monitor shared status and assist	No automatic access to precise location or health profiles
Operator	Classify, verify, and coordinate incidents	Locations within the shift and tenant scope
Dispatcher / responder	Assign, accept, arrive, and hand over work	Assigned incidents; task-necessary data
Supervisor / incident commander	Approve warnings and commands; close incidents	Separation of author and approver according to policy
Site/device/integration admin	Configure assets and integrations	No automatic permission to view sensitive SOS content
Privacy officer / auditor	Handle data requests and audits	Controlled read access; audit history cannot be edited
Platform operator	Infrastructure health and recovery	No tenant payload access by default; time-limited, audited break-glass access
Partner / city IOC	Receive agreed information fields	Service identity, field-level scope, and application ACK

A tenant is a security and commercial boundary; an organization has multiple sites and users. A person may have membership in multiple tenants, but each operation must have an explicit tenant context. A response federation does not imply sharing the entire database. Personal data belongs to the selected regional cell; cross-region operational metrics are aggregated and deidentified.

Additional operating stakeholders are the fire-protection designer, OT owner, authorized commissioning/maintenance provider, local pump/releasing-system operator, robot/water-monitor pilot, and the authority having jurisdiction (AHJ), where applicable. The designer owns the physical cause-and-effect and hydraulic design; the OT owner approves network access; the local operator owns immediate physical control and restoration; the incident commander coordinates operational authorization. Software administrators cannot self-authorize physical actuation. Procurement must obtain vendor permission, support terms, regional approvals, and evidence for the exact assembly.

5. Scope of the complete solution

The domains are Access, Companion, Field, Command, Sites & Detection, Connect & Drone, Administration, Platform, Developer, Localization, Public Address, Care, Intelligence, and System states. The official catalog is in Screen-Catalog.csv; each ID has one mockup and a corresponding specification in Screen-Specifications.md. “All screens” means the entire baseline catalog, not a claim to cover undiscovered requirements of every country/vendor. Shared modals/states are recorded in UIUX and the X states; screen counts are not inflated by counting colors or viewports.

Business packages use the same core:

Package	Input	Coordination	Mandatory distinctions
Fire & evacuation	Cameras, smoke/heat controllers, manual reports	Verification, site guidance, SOS, accountability	Versioned floor plans; do not infer a safe route from a single camera
Disaster	Authorized CAP/feed, sensors, manual reports	Affected areas, shelters, check-in, broadcast	Retain issuer, validity, and Update/Cancel; do not replace source content with AI
Personal & care	Wearables, permitted cameras, SOS, timers	Check-in, caregiver, operator, responder	Distinguish suspected from verified falls; no medical diagnosis
Road safety	Telematics, cameras, user reports	Verify location/direction of travel, dispatch, perimeter	Speed/hard braking does not automatically confirm a crash; avoid requiring interaction while driving
General SOS	Fire, medical, threat, lost, welfare, assistance, custom	Playbooks by tenant and sensitivity	Provide an “unknown type” path; threat supports discreet mode and does not post to groups by default

Outside the scope of uncontrolled automation: controlling suppression without a compatible/authorized controller, autonomous drone flight without authority and validation, replacing certified fire alarm systems, automatically calling public authorities without an agreement, medical certification, AI independently choosing evacuation routes, comprehensive facial recognition, and real payments. All required integrations are part of the product design; simulated mockups are for review only. Real hardware requires vendor binding and on-site acceptance.

The physical domain covers WF-28–WF-33 and screens F01–F20, extending the baseline from 213 to 233 screens and from 27 to 33 workflows. The original workflows and screen requirements remain in force. Physical scope includes approved equipment onboarding, point/zone mapping, telemetry supervision, authorized actuation, physical outcome reconciliation, impairment/test/restoration, FAT/SAT evidence, and supervised unmanned missions.

Wet-pipe sprinklers and controlled-release pre-action/deluge systems must have distinct asset types and operating models. SmartAlarm does not assume an ordinary thermally operated sprinkler head is individually software-addressable. A supported releasing controller remains responsible for its approved local release sequence. Pump-start demand, pump running, release-output activation, valve position, and confirmed waterflow are different observations; none alone proves extinguishment. Retrofit monitoring is allowed even where remote control is prohibited. Real water-discharge and motion tests require the approved site test arrangement, qualified personnel, and a documented impairment/restoration plan.

6. Before–during–after journeys

Before: onboard the organization and region; import sites and floor plans; provision device identities; test connectors; configure playbooks and rosters; invite users; establish consent; conduct drills. During: receive event → assess → incident → alert/dispatch → citizen SOS/check-in → responder updates → partner synchronization. After: complete each task, confirm statuses with provenance, have the supervisor close or hand over the incident, report, and take improvement actions. Do not infer “everyone is safe” from the registered-person list: unregistered people, visitors, and disconnected devices are gaps that must be visible.

The physical journey adds an approved device survey and compatibility pack before onboarding; evidence-based monitoring and operator-approved requests during an incident; and independent outcome verification, manual handover, restoration, and signed records afterward. When feedback is lost, the operator sees UNKNOWN and the responsible site contact, not a fabricated success/failure. A cloud outage must not interrupt local detection, automatic pump start, or approved releasing logic. An impairment creates a visible coverage gap and compensating measures rather than silently hiding alarms.

7. Architecture selected according to confirmed requirements

Full production domain services + regional cells + edge adapters. A complete design from the outset, including HA, event streaming, durable workflows, database ownership, field-level access, AI governance, PA, positioning, and proactive assistance. A modular monolith or microservices only for a demo are no longer the baseline. Higher investment in SRE, contract testing, and

operations is accepted in exchange for long-term independent development boundaries. Trade-offs and topology are in Basic Design.

Physical integration uses site edge adapters and segmented OT connectivity. The cloud records intent, authorization, and evidence; local controllers enforce their approved interlocks and protection functions. Read and command capabilities are separate contracts. AI may summarize physical evidence but cannot invent supported actions, override local authority, or issue autonomous release/start/motion commands. Regional cells preserve equipment/site ownership and audit boundaries.

8. Complete development plan through quality gates

The full functional scope is retained. Workstreams organize execution and verification; they do not split the product into a feature-reduced demo/MVP.

Gate	Outcome	Completion criteria
G0 — Architecture & product baseline	BRD/FRD/Basic/Detail/UX/contracts	Stakeholder review, domain safety review, confirmed assumptions
G1 — Production foundations	IAM/tenant/regions/event/workflow/data/edge/security	Contract tests, tenant isolation, HA, observability
G2 — Full domain workflows	Fire/disaster/personal/road/SOS/care/voice/PA/AI	Complete flows, roles, vi/en/ja, and error states
G3 — Integration & resilience	Vendor adapters, localization, HIL, load/DR/security	Site acceptance, measured SLOs, and recovery evidence
G4 — Operational readiness	Training, support, runbooks, contracts, go-live controls	Independent UAT, approved operational ownership

Budget and staffing are not yet available to commit to a completion schedule. Estimates by workstream, load scale, and vendor are needed after baseline review. The team requires product/domain, architecture, backend/integration, web/mobile, UX/localization, QA, security/privacy, SRE, and site safety/hardware specialists.

Each gate explicitly includes physical evidence: G0 approves system boundaries and cause-and-effect; G1 validates OT separation and edge resilience; G2 completes WF-28–WF-33 and all physical roles/error states; G3 requires compatibility tests and FAT/SAT for each enabled assembly; G4 requires local operator training, maintenance ownership, manual fallback, and restoration procedures. Simulator success is insufficient for physical go-live.

9. Commercial model and operations

Proposed commercial units: tenant/region, site count, devices, protected users, connectors, and support tier; SMS/voice/video are budgeted usage. Do not silently drop SOS when quota is exhausted: reserve emergency capacity, alert operations, and record degraded status. The buyer must designate 24/7 recipients or an explicit service schedule; an app does not create a rescue team. Contracts must distinguish software processing time from human response time.

Physical delivery depends on licensed interfaces, gateways, supported firmware, hydraulic and electrical design, qualified installers, commissioning providers, and local approval. Quotations must distinguish software integration, equipment, installation, FAT/SAT, recurring maintenance, connectivity, and vendor support. The buyer must identify who can authorize start/release/motion, who is physically present, and who can restore an impaired system; software support is not a substitute for those roles.

10. Risks and open decisions

ID	Risk / decision	Baseline assumption	Confirmation owner
Q01	First country/region not selected	Demonstrate EU and APAC with synthetic data; region pinning	Product + legal
Q02	Authority to issue public warnings	Only configured and authorized sources/issuers	Operating partner

ID	Risk / decision	Baseline assumption	Confirmation owner
Q03	Authority to control suppression/drones	Read-only first; commands subject to capability and approval	Site engineer / pilot
Q04	Detectors and devices not selected	Adapter simulator; no accuracy commitment	Integration + domain lead
Q05	Retention and data of children/dependents	Design values are proposals; consent is versioned	Privacy owner
Q06	Assurance level and certification	No compliance/certification claims yet	Compliance + engineering
Q07	Indoor maps and positioning	User-entered/QR first; do not assume GPS knows the floor	Site owner
Q08	Roster coverage	Separate registered/estimated/unknown; do not infer total occupancy	Command owner

These decisions do not prevent preparation of documents and review mockups, but are gates before real deployment. Country-specific legal assessment has not been performed; the architecture provides configuration and audit points rather than guaranteeing compliance by itself.

ID	Physical risk / decision	Baseline treatment	Confirmation owner
Q09	Exact installed equipment/firmware and interface rights unknown	Monitor-only until signed capability and compatibility evidence	Vendor + site engineer
Q10	Physical command outcome unavailable or conflicting	UNKNOWN, reconcile, manual handover; no blind retry	Local operator + incident commander
Q11	Test or maintenance reduces protection	Authorized impairment, compensating coverage, verified restoration	Site safety owner
Q12	Robot/water-monitor interface and operating envelope unverified	Mission authorization and vendor-bound controls; no autonomous deployment claim	Equipment vendor + qualified operator

These gates apply separately to each country, site, model, firmware, and permitted operation. Research evidence does not establish certification of SmartAlarm or authorize a change to a protected system.

10A. Existing BMS integration is a core product capability

The BMS retains building-services operation and local protection authority. SmartAlarm adds cross-system observations, incident correlation, human warnings, evacuation support and rescue coordination. Replacing the BMS is not required, and vendor branding is not a plug-and-play guarantee. Deployment qualifies the exact product/build/interface/license/adapter/mapping/site combination before enabling a capability.

ID	Business outcome	Acceptance condition	Accountable owner
BR-17	Use existing BMS evidence to reduce situational blind spots and retain building-services investment	Critical points/alarms carry provenance, topology, quality and freshness; online gateways do not imply healthy points	Site owner + BMS integrator
BR-18	Scale across vendors through verifiable interoperability	11 versioned engineering profiles retain evidence; read/history/ACK/write are qualified separately, never inferred	Integration owner + security
BR-19	Coordinate permitted effects while respecting local authority	Demonstrate approvals, interlocks, ownership, feedback and handover; BMS ACK cannot close rescue	Incident commander + local authority
BR-20	Operate through outages, upgrades and retirement	Offline journals, visible gaps, single-writer fencing, drift quarantine, retention and retirement sign-off exist	Platform operations + site owner

Additional roles. The BMS integrator confirms product/build/interface and point dictionaries; the site safety engineer approves cause-and-effect and affected zones; mapping engineers prepare changes while independent approvers review consequential changes; security administrators manage principals/licenses/certificates; local authorities retain local priority; FAT/SAT witnesses sign capability-specific evidence. Small sites may assign several roles to one person, but high-risk actions still follow the approved separation-of-duties policy.

Required delivery scope. Inventory, qualification, bounded discovery/import, unit/semantic/topology mapping, quality/freshness, native alarms/ACK, correlation and anti-echo, minimal incident writeback, authorized effects, feedback/readback, offline/HA, upgrades, investigation and retirement are included. Read-only operation is a valid production configuration, not a demo stage. Undocumented or unlicensed APIs are not available capabilities.

Four new journeys. WF-34 onboards and qualifies BMS/point maps; WF-35 reconciles source alarms, acknowledgements and rescue without merging their states; WF-36 governs commands and physical evidence; WF-37 operates, recovers, upgrades and retires connections. The 40 FR-BMS entries in FRD implement these journeys. WF-01–33 identifiers remain unchanged.

Measured outcomes and acceptance. Establish site baselines for qualified critical-point coverage, stale data, source→edge→operator latency, duplicates/echoes, unresolved unknown outcomes, recovery time and people not yet reached. Define critical points, clock quality, sampling, percentiles and accountable signatories before an SLA commitment. Without benchmarks, record planning assumptions rather than a universal BMS latency promise. Passing reads does not authorize commands. Estimate adapter engineering, vendor licenses, gateway/OT networking, mapping volume/complexity, FAT/SAT, operations and version changes separately.

Operational admission gates. (1) Site/interface evidence; (2) approved mappings and fixtures; (3) read/health/alarm-reconciliation FAT; (4) site SAT and runbooks; (5) individual ACK/write/release privileges enabled only after separate approval and evidence. Loss of certificates, licensing, valid mappings or local authority revokes affected capabilities while retaining evidence under policy. Alarm intake, predetermined warnings and local protection do not depend on an LLM.

11. Business approval

V2 experience acceptance conditions

The v2 baseline retains every BR, WF, FR and all 233 screen IDs. It changes how decisions are presented, not operational authority. Emergency priority is **hazard → place → immediate action → uncertainty**. Floor plans, exit landmarks, assistance needs and equipment diagrams must support a decision. Decorative imagery or metric dashboards must not displace the action. Administration uses matrices, traces and comparisons when a map is inappropriate.

Business outcome to verify	Screens / acceptance criterion
Do not direct people from insufficient evidence	C35–C39, O35/O36: unknown heading removes relative arrows; route withdrawal removes the old route and speech together; reported-location entry and assistance remain available.
Do not create false rescue expectations	C08, R02–R04, O08/O09: assignment, acceptance, approach and arrival remain separate; no ETA without a confirmed source.
Do not turn receipts into outcomes	C07/C08/X01, P04, F11/F12: local saving, server receipt, human acknowledgment, playback and physical effect remain distinct.
Preserve discreet and accessible completion	C27 never starts sound or vibration automatically; emergency flows provide text labels, audio transcripts, keyboard access and reflow; one-handed and participant testing precede acceptance.

Three-second recognition is a research hypothesis in UX-Redesign-Audit, not a measured achievement. Acceptance records correctness, dangerous misunderstandings and recovery by role and language; average speed must not hide errors. Prototype v2 demonstrates illustrative data and local interactions only. It does not establish field safety, WCAG conformity or real rescue submission.

Reviewers: product owner, command operator, responder, site engineer, privacy/security, integration partner, and accessibility reviewer. Minimum review scenarios: fire with trapped people, updated/canceled flood warnings, unresponsive fall detection, unverified collision, discreet SOS, network loss, duplicate source submissions, access from another tenant, and delivery provider failure. Record results in the acceptance matrix; the document has not been signed off.

Physical reviewers additionally include the fire-protection designer, OT security owner, vendor/authorized installer, commissioning provider, and qualified mission operator. Review must cover local protection during cloud loss, stale/cross-zone telemetry, denied control, expired/duplicate requests, ACK without physical feedback, conflicting feedback, impairment expiry

without proven restoration, and mission link loss. Acceptance-Matrix.csv adds 16 cross-system physical cases and 20 physical screen cases; product execution remains unperformed.

Mandatory appendices within the same baseline

Rescue speakers and voice navigation adds WF-21–22. Positioning, optimization, AI, and care profiles adds WF-23–27. These two documents are part of the full design, not optional features for a later phase. Screen specifications provides FRs for each screen ID.

SmartAlarm — Functional Requirements Document



Version 3.0 · 24/10/2026 · Proposed design. WF IDs are cross-screen requirements; FR- denotes detailed interface specifications in Screen-Specifications.md. Together they form the complete FRD for the baseline.

Revision 3.0 is a complete design baseline, not a deployed or certified product. WF-01–WF-20 below, mandatory WF-21–WF-27 appendices, and WF-28–WF-33 physical workflows together with WF-34–WF-37 form 37 workflows. The screen catalog contains 241 screens, including F01–F20. Physical integration must use configuration-specific evidence from Physical Device Catalog; the functional requirements below do not assert unsupported vendor capabilities.

1. Terminology and state models

Signal/Event: an immutable observation from a source, with observed_at and received_at. **Incident:** an operational record collecting evidence, with severity and an owner. **SOS:** a request for assistance for a person/group, with its own state and an optional incident link. **Warning:** a message issued by version and audience. **Delivery:** one sending attempt through one channel. **Task:** an assignment for a unit. **Personal status:** a sourced statement/status, not an inference of absolute safety. **Command:** an instruction to a device, separate from an event.

Object	Valid states	Transition condition / actor
Signal	received → validated → correlated / quarantined / rejected	Gateway validates schema/source/tenant; invalid signals do not enter live workflows
Incident	new → triaging → active → contained → resolved → closed	Operator triage; supervisor closes after checklist completion; reopening requires a reason
Incident exception	new/triaging → dismissed; any incident not yet closed → linked	Dismissal requires a reason; linking does not delete IDs or evidence

Object	Valid states	Transition condition / actor
SOS	submitted → acknowledged → assigned → assisting → resolved	Server receipt differs from operator ACK; responders update only assigned tasks
SOS cancel	submitted/acknowledged/assigned → cancellation_requested → canceled or continued processing	After dispatch, an operator decision is required; citizens cannot delete the record
Person status	unknown, in_danger, help_requested, self_reported_safe, responder_confirmed_assisted	Append observations with timestamp/source; may return to danger; safe does not close the incident
Warning	draft → review → approved → sending → active → expired / canceled	New versions require reapproval; authors cannot approve their own work when policy requires separation
Task	offered → accepted → en_route → on_scene → completed	Offer deadline → expired → reassign; decline requires a reason; a task has one current owner
Command	requested → approved → dispatched → accepted → executing → succeeded / failed / unknown / expired	Device acceptance does not prove success; a timeout does not automatically mean physical failure

Physical asset: an installed, configuration-bound controller, pump, valve, releasing system, or rescue device. **Capability:** independently approved observe/control permission for an exact operation. **Physical feedback:** observed evidence distinct from transport/device acknowledgment. **Impairment:** an authorized reduction in protection requiring compensating measures and verified restoration. **Mission:** a supervised equipment operation with local authority and handover. UI label UNKNOWN maps to the existing machine state `unknown`; it is not a new terminal success/failure state. Impairment lifecycle: requested → approved → active → restoration_pending → verified_closed; expiry without restoration stays restoration_pending and escalates. Mission lifecycle: draft → review → approved → locally_accepted → executing → completed / aborted / unknown; completion requires outcome evidence.

2. Authorization and common controls

Every endpoint checks the principal, tenant membership, region, site scope, object ownership, and action permission on the server. Hiding frontend buttons is not a security control. Family members see only shared fields; operators see information needed for the incident; platform admins have no automatic data access. Warning approvers and command approvers have separate permissions. Break-glass access requires a reason, expiry, audit, and supervisor notification; it does not bypass tenant boundaries. Access to out-of-scope objects returns 404 or an error that does not disclose their existence.

Lists use cursor pagination, stable sorting (created_at,id), and a maximum of 100 records/page. All mutations require validation, idempotency when creating effects, and optimistic concurrency on updates. List screens provide search/filter for appropriate fields and loading/empty/error/denied states; forms provide required/invalid/conflict/unsaved states; details provide stale/not-found states. Prototype data is not real telemetry.

Physical control additionally checks the site/controller capability, vendor permission, OT access policy, current impairment/lockout, local authority, and operation-specific approval. The UI must show monitor-only, control eligible, inhibited, stale, and UNKNOWN distinctly. Engineering access does not imply incident command authority. Segment OT from public networks; allowlist supported endpoints and operations, use controlled maintenance access, and retain credential/change audit. Local automatic detection, pump starting, and releasing protection must not depend on cloud approval or connectivity.

3. Cross-cutting functional requirements

WF-01 — Onboarding and identity (BR-06)

An admin creates a tenant with an immutable home_region; selects locale/timezone, configures SSO/MFA, and invites memberships by site. Users select a workspace before entering the console. Recovery returns the same message for existing and nonexistent emails to reduce enumeration. Invitations are single-use, have a TTL, and enforce email/domain policy. When SSO is unavailable, a controlled break-glass account is available within the appropriate tenant; MFA is not automatically bypassed.

AC: expired invitations are rejected; revoked roles cannot use old tokens after the proposed 60 s revocation propagation limit; switching workspaces does not display the previous tenant's cache. Screens A01–A10, T02–T04.

WF-02 — Location and map registration (BR-01)

Site admins create a country/region/site/building/floor/zone/road-segment/geofence graph. Not every site must have floors. Device binding checks that the device belongs to the same tenant. Floor plans have revisions, scale, exits, accessibility, and an approver. Each incident stores the plan revision used. Locations carry method, accuracy, and age; GPS does not automatically infer a floor. Without positioning, users can enter a location manually and send an SOS.

AC: plans without a reviewer cannot be published; zones with active incidents cannot be hard-deleted; routes without freshness information show “not verified” and are not presented as certainly safe. Screens S01–S05, C03, C06.

WF-03 — Multi-source event ingestion (BR-01,05)

Adapters check credentials, tenant binding, schema, clock skew, and payload limits, then durably store the event before issuing a receipt. A duplicate key (tenant,source,event_id) returns the previous receipt. Late events are retained for audit, but live rules evaluate TTL by hazard. Unknown sources/permissions/units/locations are quarantined. Inputs support HTTPS/CloudEvents, MQTT adapters, CAP, camera metadata, wearables, telematics, and manual entry.

AC: replaying the same ID 100 times creates one event; two tenants using the same event ID do not collide; sources spoofing a tenant are blocked; stale events do not initiate commands. Screens O03,O05,I01–I06.

WF-04 — Verification and correlation (BR-01,02)

Versioned rules provide candidate incidents by time window, zone, hazard, and source. Operators review evidence and confidence, then create or link incidents; splitting preserves audit records and source links. Different hazards are not combined merely because they are nearby; cross-hazard relationships have a type (caused_by/nearby/possible_duplicate). Camera detection is labeled “possible fire”; fire panel policy may immediately initiate a response through a site-approved workflow without waiting for AI.

AC: merging does not lose SOS/tasks; splitting does not duplicate dispatch; rule changes do not rewrite history; dismissal requires a reason. Screens O03–O07,S11–S14.

WF-05 — Multi-category SOS (BR-03)

Users select a category or Other/Unknown; enter/update location, notes, and the number of people. The emergency flow does not require a complete profile. The server returns an SOS ID and received_at after commit. The client displays pending when the request is only stored locally. Resending uses client_request_id, TTL, and reconciled state. Threat/discreet requests default to no sound and no group fan-out; ordinary assistance uses rule-based severity rather than treating every SOS as P1.

AC: network loss does not show “received”; resending does not create a second SOS; users who deny GPS can still enter location manually; changing category does not lose the ID. Screens C05–C10,C26–C27,X01–X02.

WF-06 — Fall detection and check-in (BR-03)

An approved detector emits possible_fall; the app displays a proposed 30 s check-in with “I'm OK” and “Get help now”. Without a response, the server timer continues even if the app closes, creating an operator/caregiver request according to consent. An offline device produces unknown status, not a conclusion that the person fell. Check-in duration is configurable by service/group, with an accessible alternative; it is not automatically used as a diagnosis.

AC: escalation continues after the app is killed; a late “I'm OK” updates the record without silently withdrawing an ongoing task; false positives retain a review reason. Screens C24–C25,T07–T08.

WF-07 — Road incident (BR-03)

Reports/telematics provide position, heading if available, road segment, vehicle count, and injuries=unknown. Correlation distinguishes opposite traffic directions and bridge levels. Users can submit a minimal quick report and add details later. Dispatch targets configured units; without an agreement, the system does not automatically call a public emergency endpoint.

AC: missing injury data is accepted as unknown; one acceleration sample does not auto-confirm a collision; sensitive data goes only to the assigned team. Screens C26,O04,O08.

WF-08 — Disaster and CAP (BR-01,06)

Connectors allowlist issuers and validate XML/schema, identifiers, sent, status, scope, info/area, and expiry. Deduplicate sender+identifier; Update/Cancel references the previous version and must preserve lineage. Official warnings and local responses are separate objects. Expired warnings do not continue live broadcast; cancellation does not delete previously sent versions. Polygons use CAP's coordinate order and are explicitly converted to GeoJSON; lat/lon must not be implicitly swapped.

AC: Test/Exercise does not enter the Actual flow; unauthorized issuers are quarantined; Update replaces the active version while retaining the previous one; users outside the polygon are not selected because of coordinate errors. Screens O28–O29,C14–C17.

WF-09 — Warning and audience (BR-02,06)

Authors select the incident, area/site/group, severity, languages, content, and expiry. Audience preview is a snapshot of eligible recipients and exclusion reasons; an independent approver signs the version. Changing content/audience after approval requires reapproval. Rapid break-glass issuance requires special permission and audit; not every P1 must wait for approval if the site playbook permits preapproved templates. Machine translation is draft-only; safety content must be reviewed.

AC: approval of v3 does not permit sending v4; an audience ID targeting the wrong tenant is blocked; scheduled warnings recheck expiry and permissions at issuance. Screens O13–O15,T06.

WF-10 — Delivery and escalation (BR-02,04)

Notifications create individual delivery attempts by recipient/channel/version; queued/submitted/provider_accepted/delivered/read/human_ack/failed/unknown remain distinct states. Provider callbacks are authenticated and replay-protected. An ACK deadline triggers durable escalation with an owner and maximum steps. Channel fallback does not automatically send private content by SMS; SMS templates are minimal and use authenticated links when needed. Push notifications cannot guarantee bypassing the operating system's silent mode.

AC: provider acceptance does not increase the human ACK count; duplicate callbacks do not duplicate escalation; provider failures use bounded retries and alternate providers, and 429 respects retry_after. Screens O16–O17,T05.

WF-11 — Dispatch and response (BR-02)

Dispatchers filter units by capability, availability, site authorization, and latest location. ETA is only a timestamped estimate. Offers have a deadline; units accept atomically using a version. If nobody accepts, escalate; do not automatically mark dispatch as completed. Responders transition through en_route/on_scene, record observations and handover, and complete tasks; task completion does not close the incident.

AC: two dispatchers cannot assign two owners to one task; offline acceptance must remain pending until server confirmation; reassignment retains the previous responder's notes as observations. Screens O08–O10,R01–R12.

WF-12 — Personal safety and groups (BR-03)

Safety reports store status, person, incident scope, observed_at, reporter, and location precision. Fan-out reaches all eligible subscribed groups at report time, plus verified contacts; recipients are deduplicated. Each group receives status; detailed location requires specific consent. Revoked memberships/consent are rechecked before delivery. Group chat includes membership, abuse reporting, and individual blocking for social interaction; blocking cannot disable mandatory site warnings under the published policy.

AC: one contact in 3 groups does not receive 3 identical SMS messages; location consent off does not leak coordinates; a safe report does not close a building incident; unknown is not counted as safe. Screens C11–C23,O11–O12.

WF-13 — Shelter and guidance (BR-03,04)

Operators maintain shelter status, estimated capacity, timestamp, accessibility, and valid_until. Offline packs store versioned, expiring plans/guidance/contacts and do not guarantee live routes. Voice uses the same approved text with a transcript and pause/replay; autoplay must not expose discreet mode. Without a verified route, display only the plan and approved guidance rather than automatically drawing an optimized fire-avoidance route.

AC: a shelter past valid_until becomes unknown; missing capacity data is not shown as 0; offline plans display their age. Screens C03–C04,C16–C18,O26–O27.

WF-14 — Devices and maintenance (BR-01,04)

Enrollment tokens are single-use; device certificates support rotation/revocation. Heartbeat thresholds depend on device class; stale/disconnected is distinct from the area's safety status. Maintenance suppression has a start/end, reason, approver, and replacement coverage; blind spots are always visible. When maintenance ends, monitoring automatically resumes and is logged.

AC: revoked certificates cannot ingest; suppression does not continue after maintenance ends; firmware upgrades do not automatically run fleet-wide during an active incident. Screens S06–S10.

Physical maintenance additionally follows WF-31: ending a software suppression window does not prove that physical protection has been restored; required local checks and acknowledgment remain mandatory.

WF-15 — Physical command (BR-05)

Commands are limited to declared capabilities; parameter schema, zone, expiry, interlock, authorized approver, and local controller policy are all mandatory. The web interface cannot send arbitrary raw commands. Gateway receipt, device accepted, executing, and verified result are separate. An unknown timeout does not blindly retry a non-idempotent command. Cloud actuation does not replace certified local protection loops. Closed-head sprinklers typically activate thermally; the prototype does not assume software can open individual sprinkler heads.

AC: commands past TTL do not execute after reconnect; zone mismatches are rejected; acceptance without feedback remains unknown/awaiting confirmation; simulators have no live credentials. Screens O18–O19.

WF-30 refines this command workflow for installed suppression and rescue equipment. Wet-pipe monitoring, releasing-controller requests, pump requests, and mobile equipment actions remain separate capabilities; use WF-28 binding and WF-32 acceptance before enabling any control.

WF-16 — Drone (BR-05)

Fleet adapters primarily provide telemetry/media and health. A mission draft includes purpose, incident, route/geofence, altitude reference, pilot, clearance, weather check, and lost-link policy. The local pilot decides deployment according to the integrated system's capabilities. Approval does not mean takeoff has occurred. AI detectors do not automatically issue TAKEOFF. Abort/return commands require capability support and flight-stack confirmation; the UI does not promise universal cancellation.

AC: no approval without pilot/clearance; stale telemetry disables position-dependent commands; link loss shows unknown while local policy continues; mission evidence has hash/provenance. Screens I11–I15.

WF-17 — Connector lifecycle (BR-05)

Install manifest → configure secret reference → map → sandbox test → approve → activate → monitor → rotate/deprecate. Connectors run with least privilege, outbound allowlists, quota/backpressure, and a kill switch. DLQ replay requires a reason, a valid schema, and retention of the original ID; command replay is blocked by default. Cross-tenant exchange requires an agreement and payload allowlist.

AC: adapter failures do not block direct SOS; oversized payloads return 413; incompatible major schemas are quarantined; secrets do not appear in logs/exports. Screens I01–I10.

WF-18 — Internationalization and data (BR-06)

Use UTC for storage, IANA timezones for display/schedules, BCP47 locales, Unicode, and E.164 phone numbers according to country. The UI includes RTL layout tests, pluralization, and pseudo-localization. Do not hardcode one worldwide emergency number; the emergency contact registry is verified by country/site. Residency covers databases, object storage, backups, logs, support, and exports; global routing does not carry raw SOS data.

AC: schedules crossing DST do not send twice, using a schedule occurrence ID; long locales do not obscure CTAs; changing region is a reviewed migration, not merely editing a dropdown. Screens A01,A10,T02,T06.

WF-19 — Retention, audit, and privacy (BR-06,07)

Data classes have retention policies, legal holds, and deletion jobs with reports. Export/deletion verifies the requester, scope, and exceptions; held evidence is not deleted. Audit is append-only with actor, action, object/version, reason, time, correlation, and before/after digests. Do not embed tokens, raw secrets, or complete medical notes in audit records.

AC: precise-location access is logged; deletion does not remove held records; exports exclude other tenants; break-glass access automatically revokes at TTL expiry. Screens C29–C30,T09–T11.

WF-20 — Closure, analytics, exercise, and handover (BR-07)

Closure requires an open SOS/task check, disposition of people with unknown status, partner handover, a reason, and a supervisor. An “All clear” warning is a separate action. Drills require environment+mode isolation in the broker, credentials, and recipient directory; a banner alone is insufficient. Reports separate transport latency, operator latency, and response latency; samples with missing data explicitly state their denominator.

AC: safety check-ins do not close incidents; drills do not call live webhooks/actuators; shift handovers have incoming acknowledgment; reports are deidentified when the user lacks PII permission. Screens O20–O25,T16–T17.

WF-28 — Physical onboarding and point/zone mapping (BR-13,15)

Bind the installed regional model, serial number, firmware, gateway, license, vendor permission, and compatibility dossier before activating an adapter. Import and independently review point addresses, semantic states, units/scaling, site/zone associations, and cause-and-effect revisions. Distinguish wet-pipe, pre-action/deluge, pump, and mobile rescue assets. Unknown capability defaults to monitor-only; onboarding cannot create control authority.

FR-PHY-01: retain the exact equipment/interface identity and approved evidence revision. **FR-PHY-02:** validate point and zone mapping against physical labels and reject cross-tenant, duplicate, or unresolved mappings. **FR-PHY-03:** version the cause-and-effect and read/control capability contracts; changes require impact review and reacceptance where affected.

AC: an unbound firmware or license change suspends dependent control; a wrong-zone mapping cannot be activated; wet-pipe heads are not offered as individually controllable outputs. Screens F02–F06,F20.

WF-29 — Physical telemetry and supervision (BR-01,13)

Ingest manufacturer-supported panel, pump, valve, release, waterflow, power, and link states with provenance and quality. Preserve controller alarm priority and local protection when cloud/edge is unavailable. Separate demand, motor running, release output, valve position, waterflow, and confirmed field observations. Missing or stale values remain unknown; do not turn a link fault into a safe indication.

FR-PHY-04: store source/receive time, source address, unit, quality, mapping revision, and raw evidence reference for each observation. **FR-PHY-05:** supervise freshness/link health using approved device/site thresholds and expose coverage gaps and compensating measures. **FR-PHY-06:** correlate without suppressing local protective actions; deduplicate transport replay while retaining state transitions and restoration evidence.

AC: stale pump-running data is visibly uncertain; pressure alone does not confirm discharge; loss of cloud leaves local automatic protection available; duplicate telemetry does not create duplicate incidents. Screens F01,F07–F08.

WF-30 — Authorized actuation, feedback, and UNKNOWN reconciliation (BR-14)

Only a site-accepted, explicitly control-capable operation may create a physical request. Validate actor authority, incident/site/zone, approved parameter envelope, current physical readiness, local interlocks, command expiry, and the policy's independent approval requirements. Site/controller authority prevails. The system never assumes that BACnet/Modbus writes, a relay, or a network ACK authorizes or proves suppression. AI cannot originate autonomous release/start/motion actions.

FR-PHY-07: reject monitor-only or undocumented operations server-side and at the edge; preserve the approved request and command-policy version. **FR-PHY-08:** enforce TTL, unique request ID, duplicate-effect prevention, and bounded delivery; do not replay expired requests after reconnect or blindly retry non-idempotent actions. **FR-PHY-09:** distinguish transport receipt, device acceptance, execution, and physical feedback; evaluate success only against operation-specific approved evidence. **FR-PHY-10:** display UNKNOWN for missing/conflicting outcomes; assign reconciliation to a named operator and support local override and acknowledged manual handover without erasing history.

AC: ACK without required feedback cannot become succeeded; duplicate/expired requests produce no new effect; loss of feedback opens reconciliation; local override blocks conflicting remote control. Screens F09–F13; WF-15 remains applicable.

WF-31 — Impairment, test, and restoration (BR-13,15)

An authorized impairment identifies equipment, affected zones, time window, reason, owner, compensating protection, and the local test/lockout arrangement. Keep test and production credentials, routes, and outputs distinct; a UI banner is insufficient. Expiry prompts restoration verification and escalation rather than claiming protection has returned. Local safety ownership remains explicit during shift handover.

FR-PHY-11: require approved impairment and physical isolation/lockout evidence before relevant tests; block unauthorized production commands while preserving agreed local protection arrangements. **FR-PHY-12:** close an impairment only after required physical state checks, restoration evidence, and responsible-person acknowledgment; track overdue restoration and residual gaps.

AC: a simulated test cannot reach a live output; expiration without verification remains impaired/awaiting restoration; maintenance handover has an incoming owner. Screens F14,F18.

WF-32 — FAT/SAT and compatibility evidence (BR-15)

Maintain a configuration-specific dossier of manufacturer documents, interface permissions, regional listings/limitations, point maps, electrical/hydraulic dependencies, approved cause-and-effect, FAT, SAT, commissioning, calibration, and operator training. Manufacturer research and simulator results are not field acceptance. No whole-system certification is inferred from a component listing.

FR-PHY-13: track document revision, provenance, reviewer, expiry where applicable, and acceptance scope per model/firmware/site/capability. **FR-PHY-14:** gate production capability activation on signed required tests and approvals; equipment, firmware, mapping, or policy changes trigger impact assessment and invalidate affected acceptance until revalidated.

AC: missing SAT blocks production control even after simulator PASS; an unrelated site's certificate cannot satisfy the gate; changed firmware leaves unaffected read capabilities available only if their contract remains valid. Screens F15–F17.

WF-33 — Rescue robot and water-monitor mission (BR-16)

A mission binds the actual supported robot or water-monitor interface, purpose, incident, qualified operator, operating area, route/no-go zones, water/energy supply, telemetry freshness, and lost-link/local-emergency policy. Observe-only equipment remains observe-only. Approval is not movement or discharge. The local operator determines deployment and remains able to take over according to the vendor-supported design; no universal stop/return function is promised.

FR-PHY-15: require mission-specific authority, capability evidence, operating envelope, and readiness checks; separate movement, aiming, and water discharge permissions. **FR-PHY-16:** record telemetry and action feedback, enforce approved lost-link behavior locally, and provide acknowledged manual handover and outcome evidence; never infer rescue completion from command acceptance.

AC: missing vendor-supported control or operator blocks dispatch of physical actions; stale position blocks dependent aiming/motion; link loss shows UNKNOWN and the approved local policy; a handover retains responsibility and evidence. Screen F19, with F11–F13 for execution and reconciliation.

3A. Existing BMS integration — implementation contracts

This section is normative within design baseline3.0, replacing annex-only treatment. Preserve WF-01–33, FR-PHY and screen FRs; add WF-34–37 and FR-BMS-01–40. These are development and acceptance specifications, not evidence of implemented software.

WF-34 — BMS onboarding → discovery/import → mapping → qualification (BR-17,18)

Screens: BMS01, BMS02, BMS05, I01–I04, F03–F05, F15–F17. Related FR-BMS suffixes: 01–10, 31–32, 35–36, 39.

WF-35 — BMS alarm → native lifecycle → correlation → rescue → reconciliation (BR-17,18)

Screens: BMS03, BMS08, O03–O17, H08–H11, Q04–Q09. Related FR-BMS suffixes: 11–21, 30, 33, 38.

WF-36 — Preview → authorize → dispatch → observe → reconcile / hand over (BR-19)

Screens: BMS04, BMS07, F09–F13. Related FR-BMS suffixes: 22–27.

WF-37 — Operate → degrade/recover → upgrade → retire (BR-20)

Screens: BMS01, BMS05–BMS08, I05, G02–G10. Related FR-BMS suffixes: 28–29, 31–37, 40.

Mandatory common contract. Every mutation carries tenant/site scope, actor, authorization result, correlation/request IDs, expected revision and an idempotency key for side effects. Same key/same payload returns the prior result; same key/different payload conflicts. Backend authorization is independent of UI. Point identity includes tenant/site/connection/native namespace/native ID; display names are not keys. BMS observations cannot confirm people safe. BMS ACK is a separate capability and cannot close SmartAlarm incidents. No raw write-any-point API is exposed.

Error classification. 400 invalid schema/type/enum;403 missing scope/capability;409 revision/idempotency conflict;422 mapping/business precondition failure;429 budget exceeded;503 source/edge unavailable. Side-effect timeouts must not be described as retryable failures: return the intent ID and unknown outcome for readback. Retain native status in evidence rather than replacing it with an internal HTTP code.

FR-BMS-01 — Site BMS inventory

Item	Contract
Actor / permission	bms.connection.create
Inputs	tenant/site, product/build, profile ID, endpoint, evidence refs, secretRef
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Create a draft BmsSystem; validate endpoint allowlist, account scope and evidence; run planned read-only probes; record observed build/license/interface.
Outputs and failure	BMSConnection qualification revision; missing evidence remains unavailable, never infer compatibility from the vendor name.

Item	Contract
Acceptance	Activation requires build, interface and evidence.
Screens / traceability	BMS01, I01–I02, F17

FR-BMS-02 — Capability-specific access

Item	Contract
Actor / permission	IAM security administrator; capability-policy reviewer
Inputs	requested capabilities, role policy, vendor evidence, license, FAT/SAT refs
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Intersect implemented/vendor/licensed/authorized/tested capabilities; separate read/history/ACK/command/release permissions; re-evaluate on revocation or configuration changes.
Outputs and failure	Versioned CapabilitySnapshot with denial reasons; UI is only a projection and APIs reject operations outside the set.
Acceptance	Read, history, ACK and commands have separate server permissions.
Screens / traceability	I02, I07, F03

FR-BMS-03 — Bounded discovery

Item	Contract
Actor / permission	bms.discovery.run
Inputs	connection ID, approved roots/subnets/object classes, request budget, cancel token
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Create a discovery job, approve scope, browse with pagination and rate limits; never scan the entire network automatically; support cancellation and separate checkpointed partial results.
Outputs and failure	DiscoveryJob complete/partial/cancelled/failed; throttle below alarm intake priority.
Acceptance	Scan approved scope only; cancellable; alarm intake remains available.
Screens / traceability	I02, S08, BMS01

FR-BMS-04 — Point and object import

Item	Contract
Actor / permission	bms.mapping.edit
Inputs	CSV/API export, content hash, source namespace, encoding, expected revision
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.

Item	Contract
Processing	Parse into staging; validate duplicate native keys, type, unit and required fields; show row-level errors; commit an approved revision atomically, not live row-by-row.
Outputs and failure	ImportPreview and immutable ImportRevision; parsing failure does not create partial live points.
Acceptance	Preview row errors, duplicate IDs and encoding; commit a revision.
Screens / traceability	F04, S06

FR-BMS-05 — Unit and semantic mapping

Item	Contract
Actor / permission	bms.mapping.edit + independent qualification reviewer
Inputs	raw address/object ID, data type, scale/offset, unit, enum, endian, quality mapping
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Run golden fixtures for normal, zero, missing, boundary, NaN and unknown enums; verify unit conversion; retain raw values; publish only after reviewer approval.
Outputs and failure	PointMapping revision and test evidence; overflow/type mismatch is quarantined rather than silently coerced.
Acceptance	Golden fixtures verify scale, enums, types and byte order.
Screens / traceability	I03, F04

FR-BMS-06 — Equipment and zone topology

Item	Contract
Actor / permission	bms.mapping.edit + site engineer
Inputs	equipment ID, point ID, room, HVAC zone, fire compartment, route-edge IDs
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Create typed locatedIn/serves/observes/affects relationships; validate tenant/site and dependencies; allow multiple zones with explicit roles; do not collapse all graphs into one zone label.
Outputs and failure	TopologyRevision; reject dangling/cross-tenant edges; unmapped points cannot drive automated routing.
Acceptance	Distinguish room, HVAC zone, fire compartment and route edge.
Screens / traceability	BMS02, F05, S02–S04

FR-BMS-07 — Mapping publication and impact review

Item	Contract
Actor / permission	bms.mapping.validate; bms.qualification.submit; separate bms.connection.activate

Item	Contract
Inputs	candidate revision, expected current version, impact graph, fixtures, approval
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Show affected alarm rules, routes, PA zones and command permissions; require If-Match; publish one revision; retain mapping rollback and audit without reversing physical effects.
Outputs and failure	MappingPublished; conflicts return the current version and require review, never last-write-wins.
Acceptance	Show dependencies and approve revisions; rollback does not undo physical effects.
Screens / traceability	BMS05, F04, F17

FR-BMS-08 — Observation quality and freshness

Item	Contract
Actor / permission	connector runtime; bms.point.read
Inputs	source value/status, source time quality, receivedAt, sequence/epoch, freshness policy
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Normalize quality independently of value; calculate age only with trustworthy clocks; retain late observations in history without overwriting newer projections; emit freshness transitions.
Outputs and failure	Observation and current projection; unknown/bad/stale never becomes false/zero/normal.
Acceptance	Zero differs from missing; late data cannot overwrite newer observations.
Screens / traceability	F01, F07, S07

FR-BMS-09 — Subscriptions and polling

Item	Contract
Actor / permission	connector runtime; bms.lifecycle.change
Inputs	subscription ID/lease, supported COV/SSE/events, approved poll groups/budget
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Renew before lease expiry according to the vendor contract; detect silent gaps; reconcile before reporting healthy; polling fallback uses an approved profile and stable identity, never automatic protocol switching.
Outputs and failure	SubscriptionHealth and reconciliation job; rate-limit/quota errors use bounded jittered backoff, and history cannot starve alarm queues.
Acceptance	Renew/reconcile lost subscriptions; polling follows measured budgets.
Screens / traceability	I05, BMS06

FR-BMS-10 — History and investigation

Item	Contract
Actor / permission	qualified history reader / investigator
Inputs	point IDs, bounded UTC interval, source timezone, aggregation, cursor
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Query history within quotas; retain raw/interpolated/aggregate and gap flags; deduplicate by native identity; never feed historical values into live control.
Outputs and failure	HistoryPage, continuation and coverage; expose partial/backfill gaps and never interpolate them into safety evidence.
Acceptance	Preserve UTC, timezone, gaps and raw/interpolated distinctions.
Screens / traceability	F07, O21–O22

FR-BMS-11 — Native alarm lifecycle

Item	Contract
Actor / permission	connector runtime; bms.alarm.read
Inputs	native alarm key, occurrence/branch, active, ACK, confirmed, retained/suppressed, quality
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Persist immutable native transitions; keep independent state dimensions and unsupported=null; build a versioned projection; active=false does not imply acknowledgement or incident closure.
Outputs and failure	BmsAlarmProjection and alarm.transitioned; lost occurrence identity requires reconciliation, never text-only merging.
Acceptance	Active, ACK, confirmed and suppressed are independent; unsupported=null.
Screens / traceability	BMS03, O03, F07

FR-BMS-12 — Priority and severity mapping

Item	Contract
Actor / permission	site safety engineer + qualification reviewer
Inputs	native priority/severity range, class, source context, approved response matrix
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Map per source class; test limits and ordering; retain native severity; mapping changes require impact review of active escalations.
Outputs and failure	SeverityMapping revision; unmapped classes become needs-review with raw priority, never default-low.
Acceptance	Do not invert priority numbers globally; test per source.
Screens / traceability	S11–S13, O05

FR-BMS-13 — Correlation and provenance

Item	Contract
Actor / permission	correlation service + incident operator
Inputs	panel/native provenance, source path, occurrence, zone, timing uncertainty
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Retain canonical observations and all receipt paths; establish mirror lineage before confidence scoring; correlation carries evidence and can be reversed; a BMS mirror and direct panel are not independent evidence.
Outputs and failure	CorrelationDecision and lineage links; ambiguity remains separate/pending and never deletes the original alarm.
Acceptance	A BMS mirror and direct panel feed are not independent evidence.
Screens / traceability	O05–O07, BMS03

FR-BMS-14 — Controlled external ACK

Item	Contract
Actor / permission	bms.alarm.ack_native
Inputs	alarm ID, expected occurrence/version, native ACK token, actor, comment, idempotency key
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Check source ACK capability and permission; re-read changed versions; durably record intent before ACK to the exact occurrence; await receipt/native state; never acknowledge history.
Outputs and failure	AcknowledgementIntent pending/confirmed/failed/unknown; timeout never creates local confirmation and the incident remains open.
Acceptance	Target the correct occurrence/version; failed receipt remains pending.
Screens / traceability	BMS03, O03, F07

FR-BMS-15 — Observe external acknowledgement

Item	Contract
Actor / permission	connector runtime; bms.alarm.read
Inputs	native ACK transition, actor/time, occurrence, provenance
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Record source actor separately from SmartAlarm actor; link matching intents by correlation ID; update acknowledgement projection without changing rescue/closure checklists.
Outputs and failure	acknowledgement_changed; missing actor remains null/unknown, never attributed to the currently signed-in user.
Acceptance	Preserve source actor; ACK does not close the rescue incident.

Item	Contract
Screens / traceability	BMS03, O04

FR-BMS-16 — Publish incident summaries to BMS

Item	Contract
Actor / permission	approved integration writeback service + operator
Inputs	incident summary schema, permitted BMS namespace, correlation/causation IDs, minimal audience fields
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Build a minimal projection; remove unnecessary medical/identity data; record lineage and origin; reject echoes using source plus causation; never mirror raw SOS/chat.
Outputs and failure	WritebackReceipt and loop ledger; loops trip the writeback circuit breaker while alarm intake continues.
Acceptance	Approved namespace, minimal data and anti-echo; no medical records.
Screens / traceability	BMS08, I08–I10

FR-BMS-17 — Operator alarm grouping

Item	Contract
Actor / permission	incident operator; bms.alarm.read
Inputs	alarm flood group keys, severity, acknowledgement state, raw event references
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Group display by versioned rules and windows; expose counts, earliest/latest and drill-down; pin unacknowledged P1; never change or delete raw history.
Outputs and failure	GroupedAlarmView; operators can ungroup; actual suppression requires a separate maintenance workflow.
Acceptance	Retain raw events and grouping counts; unacknowledged P1 remains visible.
Screens / traceability	O03, BMS03

FR-BMS-18 — Maintenance and suppression

Item	Contract
Actor / permission	maintenance authority + site operator
Inputs	asset/zone scope, reason, permit, owner, start/end, alternative protection
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Approve impairment; mark affected rules/routes; adjust presentation under policy; require restoration checks at expiry rather than assuming healthy equipment.

Item	Contract
Outputs and failure	ImpairmentRecord active/restoration-pending/closed; overdue escalation and explicit ownership.
Acceptance	Require reason, owner and expiry; elapsed maintenance does not prove restoration.
Screens / traceability	F14, F18

FR-BMS-19 — BMS evidence for evacuation

Item	Contract
Actor / permission	route service + site safety engineer
Inputs	door/fan/lift observation, topology revision, freshness, capability and approved route policy
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Resolve affected route edges; use only qualified evidence; stale/conflicting data lowers confidence or atomically withdraws route and voice revisions; an open door does not prove a safe exit.
Outputs and failure	RouteConstraintEvidence and guidance revision; unknown routes offer assistance rather than fabricated directions.
Acceptance	Stale door/fan/lift evidence lowers confidence or withdraws guidance.
Screens / traceability	Q04–Q05, C35–C38

FR-BMS-20 — Proactive rescue

Item	Contract
Actor / permission	care triage operator + authorized responder
Inputs	functional needs with consent, identity presence evidence, zone exposure, observation ages
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Combine scoped presence with functional needs; occupancy counts only create area-check tasks; explain prioritization; authorized personnel review and assign teams.
Outputs and failure	ProactiveRescueCandidate and evidence; absence of SOS/presence does not establish safety or death.
Acceptance	Combine needs and fresh presence; occupancy counts are not identities.
Screens / traceability	H08–H11, O11–O12

FR-BMS-21 — Approved PA and app response

Item	Contract
Actor / permission	warning authority + PA operator
Inputs	qualified BMS alarm, approved cause/effect, message revision/locale, audience/PA zones

Item	Contract
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Deterministic rules create warning intents; validate actual/exercise and dedup before fan-out; do not wait for an LLM; send approved content and separate delivery/playback/human acknowledgement.
Outputs and failure	WarningIntent and per-channel receipts; replay/echo cannot rebroadcast and unreviewed translations are not automatically released.
Acceptance	Do not wait for an LLM; echo/history cannot repeat broadcasts.
Screens / traceability	O13–O17, P01–P09

FR-BMS-22 — Control impact preview

Item	Contract
Actor / permission	bms.command.preview
Inputs	target, desired effect, expected state/version, affected zones, local owner/priority, current policy
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Preview cause/effect, consequences and feedback prerequisites before submission; provide no raw-write editor; block non-allowlisted targets and stale states.
Outputs and failure	CommandPreview bound to payload hash/revisions; target/value changes require new preview and approval.
Acceptance	Show target, owner, priority, preconditions and affected zones.
Screens / traceability	F09–F10, BMS04

FR-BMS-23 — Approved commands with expiry

Item	Contract
Actor / permission	bms.command.submit + separate authorized physical-command approver
Inputs	CommandPreview hash, requester/approver, TTL, live state, interlocks
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Reuse the physical Command aggregate; separate duties by risk; revalidate at edge before dispatch; reject expired approval or changed payload; do not dispatch under local override.
Outputs and failure	Command intent/audit; TTL limits dispatch authority and does not promise retained BMS state clears after power loss.
Acceptance	Parameter changes invalidate approval; local override retains priority.
Screens / traceability	F10, BMS04

FR-BMS-24 — Priority lease and relinquishment

Item	Contract
Actor / permission	authorized physical-command approver + local authority
Inputs	approved priority slot, ownership proof, expected current slot value, release intent
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Use only site-approved priority slots; record lease/owner/current value; relinquish through the same command guards after readback; never clear another actor's slot; define a separate watchdog/manual cleanup plan.
Outputs and failure	OwnershipProjection and ReleaseIntent; unproven ownership requires local handover.
Acceptance	Use permitted slots only; never clear an unproven foreign slot.
Screens / traceability	BMS04, F13

FR-BMS-25 — Physical outcome reconciliation

Item	Contract
Actor / permission	command reconciler + authorized site observer
Inputs	transport receipt, controller execution status, independent feedback points, time bounds
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Track three evidence layers separately; compare expected effect with qualified feedback; HTTP200/accepted alone cannot mean completed; discrepancies create tasks and retain evidence.
Outputs and failure	PhysicalOutcome verified/contradictory/unknown under the existing aggregate; never fabricate completion.
Acceptance	Separate transport receipt, controller report and independent feedback.
Screens / traceability	F11–F12

FR-BMS-26 — Timeout and unknown outcomes

Item	Contract
Actor / permission	bms.command.reconcile
Inputs	command ID, last receipt/sequence, source availability, readback plan
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	On timeout mark outcome unknown; query status/readback first; retry requests only where the adapter proves safe semantics under an approved plan, never blanket-retry physical effects.
Outputs and failure	ReconciliationCase; uncertain outcomes require local escalation and retain dispatch fencing.
Acceptance	Do not resend unknown effects; query/read back first.

Item	Contract
Screens / traceability	F12, BMS07

FR-BMS-27 — Local override and handover

Item	Contract
Actor / permission	site operator + command authority
Inputs	local override status, open intents/leases, handover owner, cleanup evidence
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Revoke remote eligibility on override; hand over outstanding effects; reconnect reconciles without restoring old commands; local owner confirms release before renewed authority.
Outputs and failure	HandoverRecord and blocked capabilities; unknown override state is insufficient for writes.
Acceptance	Do not restore old commands on reconnect; assign cleanup ownership.
Screens / traceability	F13, BMS04, BMS07

FR-BMS-28 — Durable journal and offline operation

Item	Contract
Actor / permission	edge runtime + operations
Inputs	durable journal quota, sequence/checkpoint, offline policy, disk thresholds
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Commit incoming events before durable-intake acknowledgement; prioritize alarm/control evidence; throttle history first; never falsely acknowledge at disk limits; replay is explicitly marked.
Outputs and failure	JournalCheckpoint and gap/backpressure alarms; local protection remains independent.
Acceptance	Do not acknowledge transport ingestion before durable commit; this is separate from human/native alarm ACK. Local protection remains independent.
Screens / traceability	I05, G02, BMS07

FR-BMS-29 — HA and single writer

Item	Contract
Actor / permission	edge operations + security
Inputs	leader epoch, egress fence, lease health, standby credentials, failover approval
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.

Item	Contract
Processing	Only the active writer has authority at the actual egress; cached leases are insufficient; failover fences/revokes the old writer; disable automatic write failover when fencing cannot be proven.
Outputs and failure	WriterEpoch/Health evidence; split brain falls back to observe-only with escalation.
Acceptance	Fence at the actual dispatch point; standby cannot write.
Screens / traceability	G03–G04, BMS07

FR-BMS-30 — Alarm resynchronization after gaps

Item	Contract
Actor / permission	bms.reconciliation.run
Inputs	last cursor/source epoch, paginated active snapshot, transition gaps
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Read active snapshots under source contracts; ConditionRefresh/active lists do not replace historical replay; commit completion markers before evaluating missing entries; retain unverified alarms across gaps.
Outputs and failure	SnapshotReconciliation complete/partial/gapped; never clear from one missing page or failed snapshot.
Acceptance	Uncertain snapshot/cursor exposes gaps; one page cannot clear missing alarms.
Screens / traceability	BMS03, BMS07

FR-BMS-31 — Schema drift and upgrades

Item	Contract
Actor / permission	bms.mapping.edit + bms.lifecycle.change + qualification reviewer
Inputs	native schema diff, connector/mapping version, dependency impact, rollback artifact
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Renames retain stable keys; type/unit/enum/capability changes quarantine affected objects; shadow/test new adapters with fixtures; drain before cutover; version rollback cannot resurrect commands.
Outputs and failure	DriftDecision and rollout records; affected routes/commands remain invalidated until requalified.
Acceptance	Rename preserves identity; type/unit changes quarantine affected capabilities.
Screens / traceability	BMS05, G09–G10

FR-BMS-32 — License, certificate and account lifecycle

Item	Contract
Actor / permission	security administrator + site integrator

Item	Contract
Inputs	secretRef, certificate fingerprint/expiry, license entitlement, rotation/revoke plan
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Use a secret store; test new principals with least privilege; overlap certificates under policy; revocation makes capabilities unavailable; never log/export tokens or passwords.
Outputs and failure	Credential/LicenseEvidence revision; pre-expiry escalation and affected-capability shutdown.
Acceptance	Test rotation; revocation blocks privileges; never log secrets.
Screens / traceability	I05, I07, F17

FR-BMS-33 — Data- and zone-aware health

Item	Contract
Actor / permission	bms.health.read
Inputs	point freshness distribution, source latency, queue age, source clock quality, edge connectivity
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Aggregate by zone/source and criticality; healthy heartbeat cannot hide stale points; measure only trustworthy timestamps; expose gaps and bad quality; alarm against site budgets.
Outputs and failure	ConnectorHealth and affected zones; gateway-online never implies all-clear.
Acceptance	Online gateway does not hide stale points; alarm latency is measured separately.
Screens / traceability	I05, F01, BMS01

FR-BMS-34 — Capacity, backfill and retention

Item	Contract
Actor / permission	platform operations + site integrator
Inputs	point groups, intervals, batch limits, change/burst rate, retention/offline window
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Size samples separately from batched requests; benchmark actual sources; separate alarm/history queues; throttle backfill and per-BMS quotas; cloud autoscaling cannot increase BMS capacity.
Outputs and failure	ApprovedCapacityPlan and measured thresholds; excess load degrades visibly without silent alarm drops.
Acceptance	Per-BMS quotas; historian work cannot starve alarm queues.
Screens / traceability	BMS06, G02

FR-BMS-35 — Actual and exercise isolation

Item	Contract
Actor / permission	exercise controller + site commissioning authority
Inputs	mode actual/exercise/commissioning, isolated namespace, approved test plan
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Exercises use simulators and synthetic identities; block production ACK/writes; real commissioning needs permits and local personnel; mode transitions require audit and queue reconciliation.
Outputs and failure	ModeBoundSession; changing a UI flag cannot bypass server isolation.
Acceptance	Simulation never sends real ACK/writes; commissioning has a separate plan.
Screens / traceability	I04, F15–F16

FR-BMS-36 — Configuration-specific FAT/SAT

Item	Contract
Actor / permission	bms.qualification.submit + independent FAT/SAT witness
Inputs	exact build/license/profile/map versions, fixtures, expected outcomes, measured evidence
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Run tests per capability and exact configuration; retain raw evidence, observer, pass/fail and deviations; passing reads grants no ACK/write permission; require separate site SAT sign-off.
Outputs and failure	AcceptanceEvidence and capability admission; failed/untested capabilities remain disabled.
Acceptance	Sign off per capability; passing reads does not authorize commands.
Screens / traceability	F15–F17, BMS01

FR-BMS-37 — Support and forensic export

Item	Contract
Actor / permission	scoped support investigator
Inputs	incident/connection IDs, bounded time range, export purpose, redaction policy
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	Build timelines with source/revision/actor/quality; redact secrets and health details; authorize async export and download; expire and audit access; separate original evidence from presentation.
Outputs and failure	ForensicExport manifest/hash/expiry; reject scope violations, with no cross-tenant support bypass.
Acceptance	Timeline includes revisions/sources; redact secrets and health data.

Item	Contract
Screens / traceability	O21–O22, G07–G08

FR-BMS-38 — Evidence-based BMS AI assistance

Item	Contract
Actor / permission	AI analyst + incident commander
Inputs	scoped observations, provenance, quality/freshness, authorized tools, model/prompt version
Preconditions	Valid tenant/site; identified source and applicable profile/mapping revision; authorized read or action scope and qualified capability; actual/exercise mode distinguished. Missing qualification blocks source-side actions while evidence remains reviewable.
Processing	LLM produces analysis with evidence references and uncertainty; source text is data, not instructions; deterministic policy decides eligibility; AI has no raw-write/ACK tools.
Outputs and failure	AdvisoryAnalysis with revisions/citations; missing sources/LLM failure cannot block core alarm/routing.
Acceptance	LLM reads scoped, sourced evidence; it never controls directly.
Screens / traceability	Q06–Q09

FR-BMS-39 — Tenant/site isolation and residency

Item	Contract
Actor / permission	tenant/site administrator + regional security
Inputs	tenant/site/region IDs, native namespace, connection principal, allowed projections
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Partition registry, cache, journal, events and queries by composite scope; authorize cross-site/region projections; residency covers backups/exports; repeated native IDs cannot collide.
Outputs and failure	Scoped resource identities and audit; unauthorized scope returns denied/not-found under policy without leakage.
Acceptance	Repeated native IDs remain isolated; cross-site access uses authorized projections.
Screens / traceability	G01, T02–T04, BMS01

FR-BMS-40 — Controlled retirement and handover

Item	Contract
Actor / permission	bms.lifecycle.change + local owner + security administrator
Inputs	connection ID, active alarms/intents/leases, retention plan, handover acceptance
Preconditions	Valid tenant/site and configuration authority; candidate profiles/maps may be drafts. Qualification is required before activation, not before drafting. Any external operation uses only its current qualified capability.
Processing	Freeze new commands; drain/check outstanding effects; hand over local ownership; revoke credentials/subscriptions; verify no egress; retain evidence/tombstones under policy.

Item	Contract
Outputs and failure	RetirementRecord completes only after handover/revoke verification; unresolved effects block completion.
Acceptance	Review open alarms/leases, hand over, revoke, verify and retain audit.
Screens / traceability	BMS01, F13, G06

BMS release gate

Execute designed BMS-T01–20 business and BMS-P-01–28 protocol cases per profile/site; overlapping cases are not additive independent coverage. BMS01 displays separate read/history/ACK/command/release acceptance. Evidence records exact build, adapter/profile/mapping revisions, environment, redacted raw requests/responses, local observations, actors and time. Build/type/unit/authority changes invalidate related evidence and require reassessment. BMS research retains vendor sources; Detail Design specifies schemas/APIs, state machines and transactions. **Canonical API integration.** This baseline implements the typed `/bms/connections/{connection_id}/...` resource family in API-Contract.json, including discovery, mapping revisions, qualification/activation, native acknowledgements, command previews/submission and lifecycle transitions. Earlier `/bms-systems` and `/bms-release-intents` names in research are draft aliases, not additional production endpoints. BMS command submission delegates the existing physical Command executor and cannot bypass its policy.

4. NFRs and acceptance environment

ID	Proposed requirement	Verification
NFR-01	Production regional core 99.95% monthly	Synthetic + server metrics; include dependency failures affecting users
NFR-02	Ingest p95 ≤ 2 s; fan-out enqueue p95 ≤ 5 s	Load of 100 event/s steady, 1,000 event/s for 60 s; 10,000 recipients/incident; metadata payload ≤ 64 KB
NFR-03	Priority fairness	P1 reserved queues; tenant noisy-neighbor test; bulk telemetry must not block SOS
NFR-04	Multi-AZ target RPO 0 for committed records, RTO ≤ 15 minutes	Kill primary, verify durable commit and failover; commit only after testing
NFR-05	Regional disaster RPO ≤ 5 minutes, RTO ≤ 60 minutes	Restore/cold-region drill within residency policy; immutable backup restore
NFR-06	WCAG 2.2 AA target	Keyboard, screen reader, zoom, contrast, and error announcements; no certification claim yet
NFR-07	Web interactive p95 ≤ 3 s on the reference network	4G simulation, representative low-end device; map loads separate from core content
NFR-08	Offline durable outbox and stale visibility	Airplane mode, app kill, reconnect, clock skew, and reassignment conflict
NFR-09	Security isolation	Tenant/object permission tests, upload validation, credential revocation, audit coverage
NFR-10	Backward compatibility	API/event contract tests; additive minor changes; parallel major versions with sunset

Reference loads are engineering targets, not measured benchmarks. Media uploads use object storage separately from the critical event pipeline. NFRs apply to the product to be built; the static prototype does not demonstrate that these requirements are met.

ID	Physical requirement	Verification
NFR-11	Local automatic protection independent of cloud/edge availability	Witness cloud/link/adapter failure without inhibiting the approved local protection sequence
NFR-12	Configuration-specific telemetry/feedback timing and quality	Signed freshness, delivery, and feedback deadlines per device/site; measure separately; UNKNOWN on deadline violation
NFR-13	OT and command isolation	Prove monitor-only enforcement, denied writes, least privilege, maintenance access control, and no test-to-live path
NFR-14	Configuration-specific physical assurance	Complete provenance and signed required FAT/SAT; changes invalidate affected capability acceptance

These physical NFRs do not substitute software latency targets for physical response or hydraulic performance. Device/site timing and pass criteria must be approved before tests, and results must identify firmware, hardware, and operating conditions.

5. Overall acceptance gate

V2 interaction gate mapped to existing contracts

These criteria extend acceptance of the corresponding FRs without changing IDs or API enums.

Contract / screens	Condition and observable result
WF-05, FR-C06/C07/C08, FR-X01	Without connectivity, save a locally pending request; do not display server or team receipt. Reconcile using the original identifier when connectivity returns. Reported location remains editable before submission.
WF-11, FR-R02/R03/R04, FR-O08/O09	Selecting a team does not mean acceptance or arrival. Show the owner and source-backed state; an unsupported ETA is unconfirmed, not an illustrative time presented as a promise.
WF-22/24, FR-C35/C36/C37/C38/C39	Unknown heading removes turn arrows; unverified floor removes turn-by-turn guidance. Withdrawal cancels old route graphics and queued speech before a replacement can open. Only an approved replacement with the correct revision may guide.
WF-12, FR-C27	Entering or reloading discreet mode starts no sound, vibration or speech; quick exit and clearly labeled assistance remain available.
WF-21/30, FR-P04/P09/F11/F12	Receipt, actual playback, human ACK and physical effect have separate sources and timestamps. Missing independent feedback remains UNKNOWN; no automatic retry of a non-idempotent operation.

Cross-cutting gate: VI/EN/JA diagrams have equivalent text, icons have labels and color is never the sole meaning. Keyboard and focus remain usable; 320 CSS px reflow and text enlargement retain the action. Measure 48–56 px touch targets for emergency/field flows and 44 px for ordinary controls. Test assistive technology and real devices; screenshots or CSS inspection alone do not establish conformity.

Prototype v2 uses scene navigation and sample records. Navigation to C08 or a confirmation screen is not backend-receipt evidence. These are implementation gates and must not be marked PASS from prototype navigation alone.

Run complete Fire, Disaster, Fall, Road, discreet SOS, and generic assistance flows; test duplicates, late data, out-of-order data, expiry, network partitions, unauthorized tenants, quota saturation, provider failures, and restore. For physical/drone integrations, simulator PASS does not replace on-site acceptance. The detailed matrix is in Acceptance-Matrix.csv. Screen-level acceptance is in Screen-Specifications.md and linked to each mockup ID.

The physical gate additionally executes PHY-01–PHY-16 and AC-F01–AC-F20, checks every FR-PHY-01–FR-PHY-16 requirement, and requires the approved physical test arrangement and authorized site team. Acceptance-Matrix.csv retains the 298 baseline cases and adds 48 BMS cases (346 planned cases before any separately managed screen acceptance additions). Designed cases are not execution evidence. Supplier support, physical permits, local operator readiness, and signed FAT/SAT are dependencies for enabling production control; missing evidence leaves the relevant capability unavailable or monitor-only.

Mandatory appendices within the same baseline

Rescue speakers and voice navigation adds WF-21–22. Positioning, optimization, AI, and care profiles adds WF-23–27. These two documents are part of the full design, not optional features for a later phase. Screen specifications provides FRs for each screen ID.

SmartAlarm — Basic Design / Overall Production Architecture



Baseline 3.0 · 24/10/2026 · design, not deployed. Complete architecture for an international, cross-industry product; this is not a demo architecture. The user confirmed a complete build in one effort, with deep, long-term design. Workstreams/quality gates support implementation management and do not remove features from the baseline. The prototype is a UI review tool, not a backend product or evidence of SLO attainment.

1. Architectural principles and decisions

1. Independent domain services communicate through versioned APIs and events; no cross-database queries.
2. Tenant, region, authorization, provenance, and idempotency are invariants, not later additions.
3. Separate the control plane from regional data planes; a regional cell operates independently during control-plane disruption.
4. Each incident has one home region and one fenced writer; no multi-master modification of the same incident.
5. The critical path does not depend on generative AI, search, BI, or the global control plane.
6. At-least-once event delivery + consumer idempotency; no promise of exactly-once delivery across the Internet/providers/devices.
7. Edge retains independent local safety functions; cloud does not replace certified protective control loops.
8. AI provides evidence/suggestions; automation follows approved, bounded rules.
9. Three release locales: vi-VN, en, ja-JP; key-based design allows languages to be added without changing business schemas.
10. Full functionality does not mean enabling every capability for every tenant. Feature entitlements and safety policies determine access, with auditing.
11. Physical firefighting and rescue are part of the full design. Automatic local protection remains autonomous; SmartAlarm coordinates and requests only approved capabilities through approved controllers. No LLM participates in the actuation control loop.

12. Observation, authorized request, controller acceptance, and verified physical outcome are distinct. A monitoring-only gateway cannot be promoted to an actuator by an entitlement or writable protocol property.

2. System context

```
flowchart TB
  U[Companion · residents / employees / families] --> GW
  F[Field · response teams] --> GW
  C[Command · operations / administration] --> GW
  P[Developer / partner portal] --> GW
  GW[Regional API gateway + WAF + BFF] --> CORE[Regional domain services]
  E[Edge gateways / cameras / wearables / vehicles] --> ING[Connect ingestion]
  AUTH[Authority CAP feeds / IOC] --> ING
  ING --> CORE
  BMS_EXISTING[Existing BMS / native alarm authority] -->|qualified observation| SE
  SE -->|BMS normalized journal| ING
  SE -->|approved BMS request with local authority| BMS_EXISTING
  CORE --> N[Notification providers / PA gateway]
  CORE --> CMD[Command policy + approval]
  CMD --> DMZ[OT DMZ / authenticated conduit]
  DMZ --> SE[Site edge / ledger / fencing]
  SE --> VG[Approved vendor gateway / supervised interface]
  VG --> EDGE[Local fire / releasing / pump / voice controller or pilot gateway]
  EDGE --> PHY[Physical protection / rescue devices]
  LS[Local initiating inputs / approved manual controls] --> EDGE
  CORE --> DATA[Regional data + evidence + audit]
  G[Global control plane · tenant routing / catalog] -. signed config .-> GW
```

Channels: responsive web Command/Administration/Developer portal; native iOS/Android Companion and Field; optional kiosk/wallboard according to scope. Native mobile is preferred for background capabilities, secure storage, and notification integration; operating systems may restrict background delivery, so the app is not treated as the only channel.

The local controller/device sequence does not traverse the cloud or site edge. Heat-operated wet-pipe sprinkler heads are not individually activated by a generic software command; controlled-release systems require their own approved releasing controller and cause-and-effect design. Physical interfaces may provide observation only. The physical catalog, rescue/IoT catalog, and standards/deployment design provide dated official-source evidence, exact candidate restrictions, and procurement/acceptance gates. These researched candidates are not deployed integrations.

3. Service boundaries and ownership

Service	Responsibilities / owned data	Main APIs / emitted events	Permitted dependencies
Identity & membership	Tenant membership, roles, site scopes, consent identities	memberships, sessions; membership.revoked	OIDC provider; policy store
Tenant & policy	Region assignment, entitlements, configuration version	tenants, policies; policy.published	Global routing metadata
Location & mapping	Site graph, geofences, plan revisions, shelters	locations, plans; plan.published	Regional object store
Device registry	Identity/cert binding, capability, firmware, maintenance	devices; device.health.changed	PKI, connector registry
Integration runtime	Adapter execution, mapping, source receipt, DLQ	events ingest; signal.accepted	Device/policy caches, broker
BMS integration domain	Connections, immutable point maps, qualification, native alarm mirror, lifecycle policies; physical commands remain in Command domain	/bms/connections; com.smartalarm.bms.*.v1	Integration runtime, asset graph, policy, existing physical executor
Detection & correlation	Rule versions, candidate groupings, model metadata	candidates; incident.candidate	Signal stream, location snapshot

Service	Responsibilities / owned data	Main APIs / emitted events	Permitted dependencies
Incident	Incident aggregate, source references, state and version	incidents; incident.created/updated	Local policy cache, outbox
Assistance & accountability	SOS, person observations, check-in sessions	sos, safety-reports; sos.created	Incident references, consent
Workflow orchestration	Durable deadlines, escalation, human approvals	workflows; escalation.due	Service APIs with idempotency
Dispatch	Units, skills, shifts, offers, tasks, handover	tasks; task.assigned/completed	Incident projection, team roster
Warning & content	Warning version, translated content, audience approval	warnings; warning.approved	Approved content, audience snapshot
Delivery	Channel attempts, provider feedback, fallback, receipts	deliveries; delivery.status.changed	Provider adapters, recipient snapshot
Collaboration	Groups, memberships, restricted incident chat	conversations; message.created	IAM/consent, attachment service
Command & mission	Equipment commands, interlocks, approvals, mission state	commands, missions; command.result	Capability registry, edge adapter
Evidence/media	Upload session, hash, scan status, retention, stream auth	evidence; evidence.available	Object store, antivirus, media gateway
Audit & privacy	Audit ledger, legal hold, data requests, access logs	audits, privacy-requests	Events, policy, secure export
Analytics & reporting	Read models, metrics, deidentified warehouse	reports; report.ready	CDC/event sink, no source writes
Platform operations	SLOs, restore tests, region health, support access	operational endpoints	Metrics, traces, config metadata
Physical integration	Versioned site/controller/point bindings, quality observations, capability evidence and compatibility acceptance	physical-bindings, physical-points; physical.point.observed / physical.binding.accepted	Registry, approved vendor adapters, location revisions; no ownership of local protection logic
Physical readiness	Impairment records, test authorizations, restoration checks and acceptance evidence	impairments, physical-test-runs, compatibility-assessments; physical.impairment.changed / physical.test.completed	Physical binding references, maintenance roster, evidence service
Rescue equipment operations	Equipment mission scope, qualified operator assignment, payload/route constraints, handover and results	rescue-equipment-missions; rescue.equipment_mission.updated	Dispatch, equipment capabilities, pilot/vendor-console workflow; no inferred driving API

Each service owns its schema and migrations. Production uses a separate logical database per service; managed clusters may be shared according to workload, but credentials and schema boundaries are not shared. Incident and SOS are independent aggregates, linked with eventual consistency through events, without distributed DB transactions. Workflow sagas manage failures and compensation; they do not roll back real-world effects already sent.

Command & mission retains the authoritative physical-command aggregate and its approvals/ledger; Physical integration owns point semantics and transport bindings, not the command decision. Physical readiness owns impairment/test state; local controller state remains the source for actual equipment mode and interlocks. A mismatch produces an explicit conflict, not a cloud overwrite. Rescue equipment operations owns equipment missions linked to existing Dispatch tasks and Command flight missions where applicable, avoiding two independent assignment authorities. Site fire engineers/vendor integrators approve cause-and-effect; the incident commander does not gain reset/inhibit authority merely by owning the incident.

4. Regional cells and multi-region

A cell contains API ingress, multi-AZ compute, an event broker, workflow workers, transactional DB, cache, object store, search, and observability. Each tenant is pinned to home_region; large deployments have a dedicated cell or database. The global directory contains only opaque tenant IDs, region routing, service entitlements, and aggregate health. Credential secrets, camera media, personal coordinates, and SOS data do not enter global metadata.

Within a region: stateless active-active multi-AZ APIs; multi-AZ broker replication; PostgreSQL HA with synchronous durability against measured targets; workers scale by partition/queue lag. Across regions: single-writer incident ownership, asynchronous replication according to residency policy. During disaster recovery: fence the old writer (epoch/lease), promote an approved DR destination, increment the epoch, route clients, replay the outbox, and reconcile receipts. No automatic failover to an unauthorized country. RTO/RPO are design targets that require drills as evidence, not deductions from the existence of a replica.

If the global control plane fails: the cell uses signed cached policy with an appropriate TTL and continues receiving and processing for existing tenants; it does not onboard/migrate tenants or grant new permissions when verification is unavailable. The identity revocation feed has a bounded delay and an emergency local denylist; authentication outage policy fails closed for privileged actions, while existing restricted sessions follow approved policy.

Regional disaster recovery does not automatically transfer authority to operate site equipment. An approved edge/controller pairing binds tenant, site, home region, capability revision and command writer epoch; a new region must reconcile outstanding intents and pass the site's transfer policy before remote writes resume. Local automatic protection continues while cloud command authority is unavailable. Site-edge replacement restores its durable ledger and verified binding revisions; it must not replay uncertain or expired physical requests.

5. Data flow and integration layer

Edge → mTLS MQTT/HTTPS → adapter → canonical CloudEvent → durable ingest → broker → correlation → incident/SOS → workflow → warning/dispatch/command → delivery/edge → receipts → projections/UI. Websocket/SSE only updates views; reconnect uses a sequence cursor and GET snapshot, and socket loss does not mean incident loss. Large media uploads go directly through signed URLs and are scanned before becoming available; events carry only references/hashes.

The adapter framework has a manifest containing kind/version/vendor/protocol, supported schema, capabilities, auth modes, direction, rate, health, retry class, and safety class. A telemetry adapter cannot be promoted to command capability through unreviewed configuration. The contract test suite includes valid/invalid/duplicate/stale/out-of-order/revoked/cross-tenant/oversize/backpressure/reconnect. Connector certification is an internal process with test evidence, not certification by a standards authority.

Integration profiles: MQTT 5 for IoT transport; HTTPS CloudEvents for events; CAP 1.2 for public warning; SensorThings for observation models where supported by partners; ONVIF/RTSP through vendor-specific media gateways; BACnet/Modbus through OT-zone gateways, without exposing the OT bus directly to the Internet; MAVLink/vendor SDK through a drone gateway and local pilot. Selecting a protocol does not make all devices automatically compatible: semantics, units, status, auth, and capabilities must be mapped.

Physical acquisition normalizes point observations with raw address, source and receive time, quality, engineering unit/scaling, controller identity and mapping revision. Read and write conduits are separately authorized. For example, catalog candidates include monitoring-only Siemens US BACnet interfaces, supplementary NOTIFIER/BACpac status paths, model-specific Tornatech maps and a Potter local releasing panel: none is a universal suppression API. Robots or drone payloads without a verified public control interface use task assignment to a qualified vendor-console operator. The loss of a link is UNKNOWN/STALE, not proof that a pump is stopped or a zone is safe.

6. Data, consistency, and recovery

Transactional state uses PostgreSQL/PostGIS. A Kafka-compatible broker manages durable streams and consumer offsets. Temporal or an equivalent workflow engine manages human steps/timers; this choice requires benchmarking and license/deployment review, without binding every domain module to its SDK. Redis serves cache/rate limiting, not the source of truth for SOS/deadlines. An S3-compatible object store holds media and exports. A search engine supports investigative search; a separate analytics warehouse avoids querying OLTP directly during incident bursts.

The outbox is written in the same transaction as the aggregate; its publisher retries with a stable event ID. Consumer inbox uniqueness (consumer,event_id) + a mutation transaction; external side effects use an effect key persisted in advance/receipt reconciliation. Exactly-once side effects are unavailable with providers that do not support idempotency; show unknown and deduplicate by provider message ID when available. Entity projections retain source_version and discard stale updates; latest arrival is not treated as latest observation.

Physical integration persists immutable point observations and versioned controller/zone/capability bindings; projections distinguish alarm, supervisory, trouble, test and communication quality. Command & mission persists effect intent before sending and retains SENT/ACCEPTED/EXECUTING/COMPLETED_VERIFIED/UNKNOWN evidence separately. A lost ACK after send requires reconciliation, not automatic retry of a non-idempotent actuation. Physical readiness stores impairment/restoration and FAT/SAT evidence linked to exact hardware, firmware, license and configuration. Restore procedures reapply impairments, expired capabilities and revocations before enabling writes.

7. Security & trust boundaries

Internet → WAF/API gateway → service identity network; device network/OT → DMZ edge gateway; third-party adapter sandbox → canonical ingestion; authenticated user → object-level authorization; tenant → partner agreement; region → approved transfer. OIDC/OAuth2, PKCE for native/browser, short-lived access tokens, rotated refresh tokens, MFA for sensitive permissions. Workload identity and service-to-service mTLS; vault for secrets; KMS/envelope encryption, keys per region and optionally per tenant tier. RLS is defense in depth, not a replacement for service policy enforcement.

Threat-model focus: forged device alerts, tenant ID spoofing, broken object authorization, malicious callbacks/SSRF, command replay, compromised adapters, poisoned AI evidence, leaked location, notification flooding, account takeover, malicious file uploads, and support overreach. Specific controls are in Detail Design and the Acceptance Matrix. Partner prompts or metadata are data only; they cannot affect system rules/policies/code.

The OT boundary adds a site DMZ, approved conduits, per-capability write allowlists, isolated adapter execution and supervised interfaces where approved. No generic relay bypasses local fire, releasing, pump or voice-alarm controllers. Local manual controls and protective interlocks keep their documented authority; cloud denial blocks its own request without inhibiting local automatic protection. Assess forged point maps, stolen gateway credentials, reset/inhibit abuse, mixed test/actual traffic, stale feedback and malicious firmware/configuration. Use a site IEC 62443 risk assessment; protocol support or TLS alone is not a security-level certification.

8. Capacity model and scale triggers

Reference regional production profile: 100 tenants, 1,000 sites, 100,000 devices, 1,000,000 registered users; this is not a promise that one node handles the load. A 60 s heartbeat produces approximately 1,667 messages/s; do not send every heartbeat to the Incident service. Telemetry is routed/batched separately; the critical-event profile is 100/s steady, 1,000/s burst for 60 s. One warning to 100,000 recipients × 2 channels = 200,000 delivery jobs; if the provider actually allows 1,000/s, dispatch takes at least approximately 200 s, excluding latency, so broadcast delivery to everyone within 5 s is not promised. The 5 s NFR covers only job creation, not end-device delivery.

Storage example: 100 critical events/s × 2 KB × 86,400 ≈ 17.3 GB/day raw metadata; additional replicas/indexes/audit must be measured. Video: 1 Mbps × 10,000 continuously recording cameras ≈ 108 TB/day; the design prioritizes event clips and tenant-specific retention, without uploading continuous video to the cloud by default. Clips: 15 s × 2 Mbps ≈ 3.75 MB/clip, 100,000 clips/day ≈ 375 GB/day before replicas. The capacity sheet must be updated with actual site/vendor parameters.

Scale using CPU together with p95 latency, broker lag, workflow task latency, DB locks/IO, and provider quotas; use tenant+incident partition keys to preserve aggregate ordering and avoid a giant tenant occupying only one partition. Regional shards follow tenant/site groups; hot tenants have dedicated cells. Delivery uses fair scheduling by tenant and priority; emergency reserves are not unlimited capacity. Load shedding reduces noncritical telemetry, preserves SOS/command receipts, and emits degradation events when reserves are exhausted.

Physical capacity budgets are site-specific: record point count, update/event rate, polling minimum, gateway subscription limits, edge retention, critical-event reserve and controller command limits. Avoid flooding a panel during rediscovery or recovery. Physical command latency and feedback deadlines follow the approved capability/site test criteria; the cloud job-creation target does not promise pump start, waterflow, suppression effectiveness or acoustic coverage. Partial point failures remain visible by zone rather than being hidden behind aggregate gateway health.

9. Deployment & release

Production deploys containers on Kubernetes/equivalent managed orchestration, with IaC, immutable images, signed artifacts, SBOM, dependency scanning, secret scanning, and migration review. Network policy is deny by default. Dev/test/staging/production have separate accounts/networks/credentials/provider sandboxes; exercise mode also isolates topic + recipient + device namespaces.

Release follows expand → migrate → switch → contract; binary rollback does not roll back destructive DB migrations. Canary releases target test tenants/sites; minor schema changes are additive; major events are dual-published/consumed during a transition window, without silently reinterpreting fields. Consumer lag and DLQ are release gates. Mobile supports a minimum supported API version with a grace period and does not block cached emergency guidance during app updates.

Physical release gates are RFI/site survey → approved point/zone/cause-and-effect binding → FAT → installed SAT → signed handover. Pin manufacturer, regional model, firmware, gateway, license, point map and certificates; a firmware or mapping change reopens affected acceptance tests. Hardware tests require an authorized impairment/test/restoration plan; exercise mode in software alone cannot isolate physical outputs. Deploy remote write capabilities disabled until exact-configuration acceptance. Existing local protection remains operational during cloud rollout, and software rollback is not a promise to reverse a physical effect.

10. Observability and 24/7 operations

End-to-end tracing uses trace_id, correlation_id, causation_id, and event_id; metric-label cardinality is limited and excludes user/lat/lon. Monitor SLO burn-rate alerts, synthetic SOS with a genuinely separate test tenant, provider health, certificate expiry, clock skew, stale plans, uncovered shifts, unacknowledged incidents, and orphan workflows. Logs redact secrets/PII and follow the same residency routing.

Runbooks: ingestion lag, provider outage, regional loss, database failover, compromised device, adapter quarantine, widespread false positives, missed escalation, stale location, data leak, failed deletion, and lost drone link. Each runbook has an owner, detection, containment, recovery, verification, and communication, detailed in Detail Design. Restore drills are proposed at least quarterly and after topology changes; site exercises follow operational agreements.

Physical observability adds per-point age/quality, alarm/supervisory/trouble, controller mode, active impairment, overdue restoration, command-to-feedback discrepancy and unverified physical completion. Runbooks cover loss of cloud, loss of edge, partial gateway failure, wrong-zone mapping, power trouble, duplicate writes, lost feedback and voice coverage gaps. FMEA and failure tests must show independent local operation and honest UNKNOWN states. Inspection schedules and compensatory protection are set by the adopted code, manufacturer and site authority, not invented by cloud software.

11. UI information architecture

V2: presentation boundaries follow decisions

The illustrative v2 renderer separates `prototype-v2.js` (citizen, responder, command, care and public address), `enterprise-v2.js` (the remaining 124 screens) and `visuals-v2.js` (schematics). These are presentation modules, not control services. The browser remains outside the OT trust boundary; drawing a pump or valve grants no write capability. Every screen ID still maps to Screen-Registry.

Decision family	Primary composition / boundary
C35–C39, R05, O35/O36	Plans and landmarks bound to route revision, floor, position source and heading state; the valid route layer is separate from the surveyed base plan.
C05–C10, R01–R04, O08/O09	Assistance category → reported location → review → receipt chain; owner, ACK and arrival remain separate.
P01–P09, F01–F20	Speaker coverage / water-system mimic, capability dossier, authorization gate and readback; local priority is never inferred by the UI.
T04, D02/D03, L02/L03, Q06/Q07	Permission matrix, request/response, parallel translation and claim/evidence; neither cards nor maps are universal layouts.

Production requires a consistent presentation snapshot carrying source data, revision, observed time, quality and scope. A shared state selector must withdraw route overlay, text instructions and speech together when the snapshot becomes ineligible. Rendering or reloading does not refresh observation age. Missing heading cannot be inferred from screen orientation or a QR anchor that establishes position only.

The reviewer inspector keeps technical rules, FR/WF references and original fields outside the emergency flow. Progressive disclosure must not hide uncertainty that changes a decision. Equivalent text, discreet sound policy and focus management are shared capabilities rather than properties of one diagram. Current modules are prototypes; they do not replace production snapshots, validated routing or the production audio controller.

Companion prioritizes one task at a time: awareness → action selection → receipt → tracking. Field prioritizes glove-friendly targets, offline evidence, and assignment ownership. Command provides an incident workspace with map/floor, queue, people, timeline, dispatch, and delivery; dashboards do not replace workflows. Administration separates site ops / integration / tenant / platform to prevent excessive permissions. The Developer portal provides schemas, API clients, conformance tests, and version lifecycle. Global operations does not display tenant PII.

Shell design: sidebar with region/site scope; header with incident context, language, and freshness; dangerous actions clearly state target+effect+expiry. Red denotes critical/urgent action, not every function. White/cool ivory for workspaces, graphite for the command canvas, teal for states supported by good evidence; amber for unknown/stale. Dark mode does not alter severity semantics.

Physical operations adds F01–F20 to the coordinated 233-screen design set and WF-28–WF-33 to the 33-workflow baseline: point/zone onboarding, supervision, authorized actuation and UNKNOWN reconciliation, impairment/test/restoration, FAT/SAT compatibility, and rescue-equipment missions. The UI shows capability provenance, target/zone revision, controller mode, feedback quality and local authority before an action. It must not label ACCEPTED as physical success or show an unsupported robot/drone command simply because a mission exists. FR-PHY-01–FR-PHY-16 define the related functional requirements.

12. Key ADRs

ADR	Decision	Rationale / trade-off
001	Domain services from the start	Supports the complete development direction; accepts higher SRE/contract-testing costs than a modular monolith
002	Regional single-writer incident	Prevents split-brain dispatch/command; federation uses read-only shared copies

ADR	Decision	Rationale / trade-off
003	Broker + outbox/inbox	Recoverable at-least-once processing; no distributed transaction with providers
004	Durable workflow engine	Timers and human steps survive restarts; requires operating an additional stateful platform
005	Native mobile with two role experiences	Background/offline support and ergonomics; codebase may be shared without mixing permissions
006	CAP at the boundary, CloudEvents internally	CAP for public advisories; internal events retain extensible schemas and provenance
007	AI outside safety decision authority	Confidence is insufficient for conclusions; humans/site-approved automation retain decision authority
008	First-class vi/en/ja internationalization	Versioned translations; no string concatenation; alert publication checks locale completeness
009	Local automatic protection remains independent	Cloud/edge outages cannot interrupt its sequence; integration is supplementary unless specifically assessed/approved
010	Exact-configuration physical capability admission	Protocol writes and component certificates do not establish permitted actuation; FAT/SAT binds the approved assembly
011	UNKNOWN physical outcomes require reconciliation	Prevents duplicate discharge/motion and false success; accepts manual verification when feedback is unavailable
012	Impairment and test state are operational aggregates	Prevents test/actual confusion and silent loss of protection; restoration requires site evidence

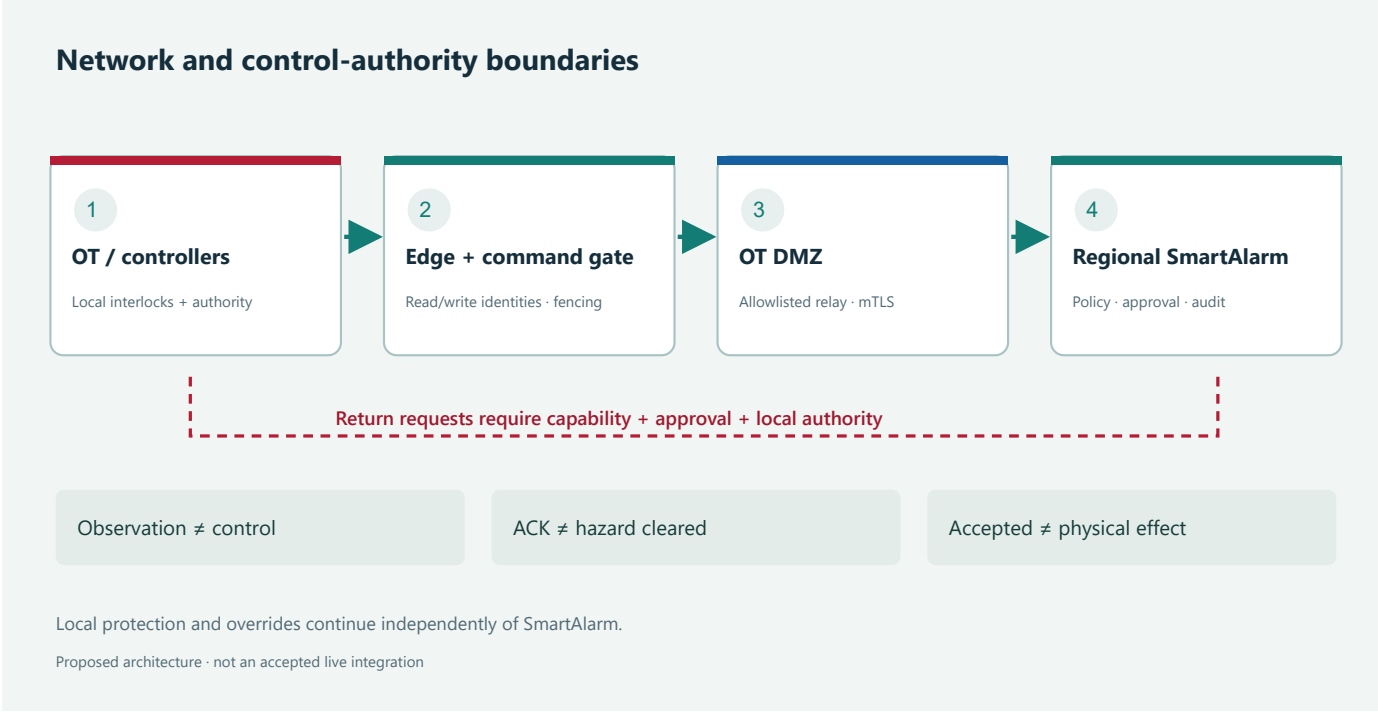
13. Standards sources and application

Consulted on 23/10/2026. These are design references, not claims of product certification.

- OASIS CAP 1.2: all-hazard warning exchange; preserve identifiers, updates, cancellations, areas, and expiration.
- CloudEvents: shared event envelope; SmartAlarm versions business-data schemas separately.
- MQTT 5: IoT pub/sub; transport QoS does not replace business deduplication.
- OGC SensorThings: sensor interoperability profiles supported per adapter.
- MAVLink Command Protocol: ACK may mean only accepted and does not prove physical completion; flight-stack capabilities determine behavior.
- WCAG 2.2: AA target for web; keyboard, contrast, focus, target sizing, and accessible errors.
- Material Design and Apple HIG Accessibility: native UI references; pages requiring JavaScript have not been read in full and are not used to claim detailed compliance.

14. BMS integration domain in the production architecture

This section directly incorporates WF-34–37 and FR-BMS-01–40 into the existing architecture. BMS integration is a managed domain rather than a generic BACnet/Modbus gateway. The APIs, tables and runtimes below are required development scope; the 11 researched vendor profiles remain uncommissioned designs. BMS retains ownership of plant control and native alarm state. SmartAlarm owns incident correlation, rescue tasks, messaging and human-safety evidence. A normal BMS state never automatically closes a rescue incident.



14.1 Component boundaries and ownership

Component	Implementation responsibility	Output / placement
BMS Connection Registry	Own site, profile, vendor build, entitlement, endpoint/secret references and revision; no passwords in manifests.	Regional control plane; BMS01/BMS02
Capability Admission	Intersect implemented adapter, vendor support, license, site authority and FAT/SAT; issue signed deployment projection.	Regional policy + edge cache; BMS04
Discovery & Mapping	Budgeted import/browse; map native ID to point/asset/space/zone; diff, unit validation and immutable publish.	Regional worker + site probe; BMS02
Site Integration Edge	Own vendor sessions, COV/subscriptions, polling, journal/outbox, cursor and writer epoch. Separate read/write workers and accounts.	OT management segment; outbound through DMZ
Normalization & Evidence	Normalize type/unit/quality/time, preserve raw hash; quarantine mapping errors; emit canonical events.	Site edge; regional validation repeats
Alarm Mirror & Correlation	Maintain native occurrences/branches and snapshot generations; link sources to an incident without discarding provenance.	Regional domain; BMS03
Command Orchestrator	Preview/approval digest; separate native ACK authority; physical commands delegate to the existing physical executor. No raw-write bypass.	Regional + fenced edge; F09–F13/BMS04
Connector Operations	SLOs, certificates, drift, drain, upgrade, rollback and local handback.	Regional operations; BMS05–BMS08

14.2 Data flow and network boundaries

Observation path: BMS/native API or qualified gateway → site adapter → validation/normalization → durable journal → outbound mTLS through OT DMZ → regional ingest → event bus → alarm/asset/incident read models → apps, command and rescue. Transport acknowledgment follows journal commit; business receipt follows inbox/projection commit. Never expose BACnet/IP, Modbus or OPC DA/DCOM directly to the Internet. OPC DA requires a managed Windows-zone bridge; it is not relabeled OPC UA.

Mutation path: authorized actor → impact preview → digest-bound approval → regional command ledger/outbox → edge rechecks epoch, revision, expiry and local mode/interlocks → admitted adapter → controller → feedback. Each action has one authority owner. Request acceptance, controller execution and physical feedback remain separate milestones. The existing

Physical Command aggregate is the authority for physical actions; the BMS resource is a façade linked through physical_command_id.

Egress allowlists approved hosts/IPs, ports and endpoint certificates; public APIs accept endpoint_ref rather than arbitrary URLs. Edge retrieves secrets using site-scoped workload identity and short TTL; rotation uses an approved overlap window and read test. A read account must not inherit administrator command authority. Inbound BMS webhooks terminate at an authenticated, replay-protected, quota-limited DMZ reverse proxy and never call the control service directly.

14.3 Adapter families and selection gates

Each deployment pins an interface contract and adapter digest; connection failure never triggers automatic protocol fallback. This catalog defines implementation and qualification scope, not a promise that every version supports every operation.

Profile	Interface	Constraint
bms.siemens.desigo-wsi	Versioned WSI/REST	Build-specific sessions/heartbeat; verified northbound OPC is DA, not assumed UA.
bms.schneider.ebo-ews	SOAP EWS / qualified SmartConnector	Version-qualified provider; event ACK is distinct from alarm state.
bms.jci.metasys-rest	Pinned REST / SSE	Separate API entitlement; inspect each batch result.
bms.honeywell.ebi-api	Qualified vendor SDK/web services	Obtain vendor contract; old specifications do not establish current capabilities.
bms.honeywell.webs-niagara	OEM Niagara station	Separate from EBI; qualify OEM modules and licenses.
bms.tridium.niagara-local	Qualified local oBIX/module	No assumption of universal REST or ACK.
bms.tridium.nds-cloud	NDS cloud API	Separate entitlement; not a critical local-control fallback when WAN fails.
bms.delta.enteliweb	Qualified API/BACnet export	Confirm API add-on and method-level contract.
bms.alc.webctrl-events	Outbound event HTTP/MQTT	Event export does not establish point write or native ACK.
bms.alc.webctrl-soap-sdk	SOAP / Java SDK	Case-specific Java SDK license; qualify SOAP methods separately.
bms.distech.ecnet-niagara	OEM Niagara station	ECLYPSE controller REST is not the EC-Net supervisor API.

Evidence and version qualifications are in the BMS research and its 45 linked primary sources. BMS/connectors profiles are disabled designs. Connector-Manifest.schema.json describes a separate deployment projection that only the admission service may sign after acceptance.

14.4 Sources of truth, tenancy and time

Data	Authority	Rule
Native alarm occurrence/ACK	BMS/controller	SmartAlarm mirrors it; native ACK rechecks occurrence/branch.
Point raw value/quality	Qualified BMS interface	Preserve source_time and received_at separately; do not fabricate a missing source timestamp.
Asset/space/zone map	Approved SmartAlarm map revision	Native renames do not change point_id; rebinding needs a revision/diff.
Rescue incident/task/human status	SmartAlarm incident/rescue domain	BMS normal, occupancy and AI do not establish human safety.
Physical actuation	Local controller + Physical Command aggregate	Local override wins; SmartAlarm does not replace local cause-and-effect.

Storage keys carry tenant_id, region_id and site_id; resource lookup follows scope checking. Vendor BMS cloud does not change health/location/care residency policy. No health data is exported to BMS by default: only approved incident summaries, zones and task status enter the designated namespace. Event time is UTC; UI uses site timezone. Clock skew beyond policy thresholds marks evidence uncertain and raises an alert.

14.5 HA, capacity and operations

Observation SaaS workers may be active-active, with partitions by tenant/site/connection and ordering per native occurrence where a source sequence exists. Edge active/standby may both read only within upstream budgets, but only one writer may egress. A database lease is insufficient: fencing must be enforced at the command gateway, credential or an endpoint that verifies the token. Standby never replays old commands as compensation. Where effective fencing cannot be established, control failover is a manual procedure while read-only HA remains possible.

Size actual requests: R_poll = Σ ceil(points_group/batch_size)/poll_period + renewal + reconciliation + retries. Design example, not a benchmark:10,000 points, batch 100 and 30 seconds require 3.34 steady poll requests/s; without batch the rate is 333.34 requests/s and admission must reject it if the source budget is insufficient. COV/SSE reduces steady load but initial snapshots, alarm bursts and reconnects still need budgets. Max inflight, records/page, jitter and per-connection token buckets are mandatory admission settings.

Minimum journal capacity = peak events/s × bytes/event × offline seconds × safety factor, plus index/WAL. Example 50 events/s × 1 KB × 24 h × 1.5 ≈ 6.48 GB before indexes; measure actual payloads before sizing disks. A full queue must not silently drop alarms or command receipts: mark degraded, stop low-priority discovery/history, backpressure telemetry, alert capacity and mark any unrecoverable data interval. Do not promise RPO0 without durable upstream receipt and sufficient disk/replication.

Mandatory separate measures are source→edge, edge→regional, backlog age, last successful read, stale ratio, subscription renew failures, reconciliation gaps, unknown commands and certificate/license expiry. A healthy edge heartbeat does not prove fresh BMS data. Site-specific latency/SLO targets require SAT approval for the actual protocol and load; a universal default is not a vendor commitment.

14.6 Architecture decisions and handover

ADR	Decision	Consequence
BMS-01	One canonical domain, separate vendor/version profiles	Adding vendors leaves incident/rescue core stable without losing native semantics.
BMS-02	Separate observation and mutation authorities	A read account cannot ACK/write through a side path.
BMS-03	BMS commands reuse the physical executor	One physical-effect ledger, no competing state machine.
BMS-04	Qualification binds digests and revisions	Build/map/license/cause-effect changes require reassessment.
BMS-05	Snapshots never replay side effects	Recovering an alarm mirror does not retransmit PA or commands.

Handover includes a signed deployment manifest, checksummed point/alarm dictionary, network diagram/allowlist, credential ownership, vendor entitlement, build-specific FAT/SAT, measured load, offline/unknown/local-override runbooks, backup/restore and named site ownership.

Required appendices within the same baseline

Rescue speakers and voice navigation adds WF-21–22. Positioning, optimization, AI, and care profiles adds WF-23–27. Both documents are components of the full design, not optional features for a later phase. Screen specifications provides FRs for each screen ID.

SmartAlarm — Detail Design / Detailed Technical Design



Version 3.0 · production architecture design baseline, not deployed · 24/10/2026. This document is used together with API-Contract.json, Event-Schemas.json, Screen-Specifications.md, and Acceptance-Matrix.csv. The attached API contract is a machine-readable core baseline; vendor integrations require their own binding specifications once actual devices are available.

1. Domain data model

IDs use UUIDv7 or an equivalent UUID with uniqueness; external IDs exist only within their source namespace. All tenant-data tables include tenant_id, region_id, created_at in UTC, updated_at, and version bigint. Internal foreign keys use composite tenant_id+id within the same DB; cross-service references use IDs and snapshot versions, without cross-database foreign keys.

Entity	Main fields and constraints	Proposed retention / owner
Tenant	id, home_region, status, policy_version, default_locale; region immutable through normal APIs	Contract end + policy / Tenant
Membership	tenant,user,role,site_scope,valid_from,valid_until,revoked_at; unique tenant,user,role,scope	Audit retains history / IAM
Location	id,parent_id,type,name_i18n,geometry,timezone,version; parent in same tenant; cycle forbidden	Versions retained long term / Location
PlanRevision	plan_id,revision,asset_id,scale,exit_graph,reviewer,published_at; unique plan,revision	Based on incident references / Location
Device	id,external_id,connector_id,location_id,cert_thumbprint,capabilities,health; unique tenant,connector,external_id	Device lifetime + audit / Registry
Signal	id,source,event_id,type,observed_at,received_at,schema_version,payload_ref,location_snapshot,mode	30 days hot + archive policy / Ingestion
Incident	id,hazard,severity,state,home_region,writer_epoch,owner,location_ref,version,opened_at,closed_at	Proposed default of 365 days / Incident

Entity	Main fields and constraints	Proposed retention / owner
IncidentSignal	incident_id,signal_id,relation,actor,reason; unique incident,signal	Based on incident / Incident
SOS	id,client_request_id,person_ref,incident_ref,category,state,location_report,sharing_policy,version	365 days, precise location separate / Assistance
PersonObservation	id,person_ref,incident_ref,status,reporter_type,observed_at,received_at,location_ref	Append-only; precise location 30 days / Assistance
Task	id,incident_ref,unit_ref,state,offer_deadline,version,assignment_epoch,completed_at	365 days / Dispatch
WarningVersion	warning_id,version,content_i18n,area,audience_snapshot,expiry,status,approvals	365 days / Warning
DeliveryAttempt	id,warning_ref,recipient_ref,channel,attempt_no,provider_id,state,receipt_at; effect_key unique	90 days metadata / Delivery
Command	id,target,capability,params_digest,nonce,expires_at,approval,state,device_receipt,writer_epoch	365 days / Command
Mission	id,drone_ref,incident_ref,pilot_ref,plan_version,clearance_ref,state,flight_log_ref	365 days by policy / Command
Evidence	id,tenant,object_key,sha256,mime,size,scan_state,classification,hold_id	90 days, legal hold override / Evidence
Consent	subject,purpose,recipient_scope,fields,policy_version,granted_at,revoked_at	Version history / Privacy
AuditEntry	id,actor,action,object_ref,object_version,reason,trace_id,at,digest,previous_digest	365+ according to policy / Audit
Outbox / Inbox	outbox: event_id,payload,attempt,next_attempt; inbox: consumer,event_id,processed_at	Replay horizon + margin / each service
PhysicalBindingRevision	binding_id,revision,controller_id,regional_model,firmware,gateway_model,gateway_firmware,license_ref,point_map_digest,zone_revision,cause_effect_revision,status,acceptance_ref; unique tenant,binding_id,revision	Referenced configuration history; approved policy / Physical integration
PhysicalPoint	id,binding_id,binding_revision,controller_id,zone_id,raw_address,semantic,access_mode,unit,scale,offset,byte_order,word_order,enum_map,stale_after,feedback_for; unique tenant,binding_id,binding_revision,raw_address	Binding lifetime + incident/test references / Physical integration
PhysicalObservation	id,binding_id,point_id,binding_revision,source_event_id,observed_at,received_at,value,quality,controller_mode,maintenance_state,raw_evidence_ref; append-only	Proposed 30 days hot; archive by incident/test policy / Physical integration
PhysicalCapability	id,binding_id,revision,operation,parameter_schema,allowed_targets,authorization_policy,local_interlocks,command_ttl,retry_class,feedback_point_ids,completion_criterion,unknown_outcome_policy,approval_evidence_ref	Versioned approval history / Command
PhysicalCommandDetail	command_id,binding_revision,capability_revision,zone_revision,cause_effect_revision,expected_point_versions,feedback_deadline,physical_state,evidence_refs; one-to-one Command	Inherits Command retention / Command
Impairment	id,scope_refs,functions_affected,reason,owner,planned_until,notifications,compensatory_plan_ref,status,restoration_evidence_refs,version	Site inspection policy; legal hold applies / Physical readiness
PhysicalTestRun	id,binding_revision,test_plan_ref,mode,authorization_ref,impairment_ref,operator,witness,started_at,ended_at,result,evidence_refs	Site inspection/acceptance policy / Physical readiness
CompatibilityAssessment	id,binding_revision,hardware_serials,firmware_versions,license_refs,certificate_refs,fat_run_refs,sat_run_refs,open_defects,signoffs,status	Configuration lifecycle and site policy / Physical readiness
RescueEquipmentMission	id,incident_ref,dispatch_task_ref,equipment_ref,operator_ref,capability_revision,mission_scope,local_console_required,flight_mission_ref,state,limitations,result_evidence_refs	Proposed 365 days; sensitive evidence separately / Rescue equipment operations

Retention values are proposed design defaults, not legal conclusions. Legal holds prevent purging related objects. Deletion retains reference tombstones to preserve audit integrity, without retaining PII beyond its purpose. Search/indexes/backups require deletion propagation and an expiration strategy; deleting from OLTP alone is insufficient.

Indexes and concurrency

Signal unique(tenant_id,source,event_id); index(tenant_id,observed_at DESC), partition by received_at month. Incident index(tenant_id,state,severity,opened_at DESC), spatial index on geometry; partial index for active records. SOS unique(tenant_id,client_request_id), index(tenant_id,incident_ref,state). Task updates use `WHERE version=:expected AND assignment_epoch=:epoch`; affected_rows=0 returns 409. Delivery unique(effect_key); provider callback unique(provider,event_id). Outbox index(status,next_attempt_at) and worker leasing prevent concurrent processing of the same row.

RLS uses tenant context bound from the verified principal within a transaction, not directly trusted headers. Connection pools must reset context. A tenant ID in the URL is insufficient authorization. Region routing checks the membership directory and signed region mapping; wrong-region requests receive safe redirect metadata or 409 REGION_MISMATCH.

Physical storage and revision invariants

Every physical entity also carries the common tenant_id, region_id, created_at, updated_at and version fields above. Use composite references within a service; cross-service references retain the accepted snapshot digest. PhysicalObservation deduplication is unique(tenant_id,binding_id,source_event_id,point_id) when a stable source event ID exists; otherwise the adapter generates a stable receipt ID and documents that it cannot prove source-level deduplication. Index observations by tenant_id,point_id,observed_at and partition by received_at. A point's latest view uses source sequence/time and quality, not arrival order alone.

Binding revisions are immutable after acceptance. Activation atomically changes the active revision only after compatibility acceptance and readiness checks; outstanding command approvals pin their original revisions and are invalidated for dispatch when these no longer match the active binding. physical_state cannot become COMPLETED_VERIFIED without the declared evidence criterion. Impairment restoration and capability re-enable are different transactions; clearing a dashboard alarm cannot restore a device or authorize writes. Snapshot reconciliation compares software maintenance_state with controller mode and reports conflict.

2. API conventions

Base `/v1`; JSON UTF-8, RFC3339 timestamps, cursor pagination. Bearer authorization + verified tenant context; optional X-Tenant-ID is only a selector, checked against token membership by the server. `Idempotency-Key` is mandatory for POST requests creating SOS, sending warnings, offering dispatch tasks, and issuing commands; scope is principal+tenant+route+key, with a request-body hash. Same key with different payload → 409 IDEMPOTENCY_MISMATCH; same key with same payload → previous response. Receipt only after durable commit. Proposed key retention is 7 days for human mutations; source-event deduplication follows an archive replay horizon proposed at a minimum of 90 days, with older replay through a controlled path. Command expiry is independent of deduplication retention.

`If-Match` version/ETag for updates; 412 VERSION_MISMATCH or domain 409 according to API schema. Error envelope: code, message_key, params, trace_id, retryable, field_errors; no exposed stack. 400 malformed, 401 authentication, 403 permission, 404 inaccessible/not found, 409 domain conflict, 412 precondition, 413 payload, 422 validation, 429 quota, 503 unavailable. GET retries are bounded; mutations retry only with an idempotency key.

API	Permission	Request/response and side effect
POST /events	event.ingest + source binding	CloudEvent; 202 durable receipt, not incident success
GET /incidents	incident.read scope	Filters severity/state/site/from; cursor result
POST /incidents	incident.create	hazard,severity,location,signal_refs; 201 aggregate + outbox
POST /incidents/{id}/transitions	incident.transition	expected version,to,reason; validate state guard
POST /sos	sos.create	client_request_id,category,location,note; 201 server receipt
POST /sos/{id}/updates	sos.self_update or sos.manage	append note/location/cancellation request; no deletion

API	Permission	Request/response and side effect
POST /safety-reports	safety.self_report	incident_ref,status,sharing preference; save observation + fan-out event
POST /tasks	task.assign	unit,incident,scope,deadline; offer + workflow timer
POST /tasks/{id}/accept	task.accept + assigned unit	expected version; atomic owner confirmation
POST /warnings	warning.author	localized content,audience,expires; create draft version
POST /warnings/{id}/approvals	warning.approve	version,digest,decision,reason; separate actor guard
POST /warnings/{id}/send	warning.send	approved version; effect key, immutable audience snapshot
GET /deliveries	delivery.read	warning,channel,state; no private address returned without scope
POST /commands	command.request	capability,target,parameters,expiry; 202 requested, not yet actuated
POST /commands/{id}/approve	command.approve	digest,version; publish intent after interlock checks
POST /connectors/{id}/tests	connector.test	fixture_set,manifest_version; sandbox results
POST /connectors/{id}/replay	connector.replay	event IDs,reason; forbid command replay
POST /evidence/upload-sessions	evidence.write	mime,size,classification; short-lived scoped URL
POST /privacy-requests	privacy.self_request	type,scope; identity verification workflow
GET /stream	authenticated subscriptions	event cursor + projection changes; no mutation through socket
POST /physical-bindings	physical.binding.author	controller_id,regional_model,firmware,gateway,point_map,zone_revision,cause_effect_revision; create immutable candidate revision
POST /physical-bindings/{id}/validate	physical.binding.test	revision,fixture_set; schema/point/capability validation, no live actuation
POST /physical-bindings/{id}/activate	physical.binding.activate	expected_version,revision,acceptance_ref; reject absent signoff or incompatible readiness
GET /physical-points	physical.point.read	controller,zone,binding_revision,quality; scoped point projection
GET /physical-points/{id}/observations	physical.point.read	from,to,cursor; values, units, source/receive time, quality and revisions
POST /physical-commands	physical.command.request	capability_id,target,binding_revision,zone_revision,cause_effect_revision,parameters,expires_at; create Command + PhysicalCommandDetail, not actuation
POST /physical-commands/{id}/approvals	physical.command.approve	expected_version,digest,decision; separate actor where policy requires, local guards still apply
POST /physical-commands/{id}/reconciliations	physical.command.reconcile	expected_version,observations,evidence_refs,reason; no automatic resend; preserve UNKNOWN history
POST /impairments	physical.impairment.manage	scope,functions_affected,reason,owner,planned_until,compensatory_plan_ref; create tracked impairment
POST /impairments/{id}/restoration-checks	physical.impairment.restore	expected_version,checklist,evidence_refs,witness; verify before closure; no implicit command
POST /physical-test-runs	physical.test.execute	binding_revision,test_plan_ref,mode,authorization_ref,impairment_ref; approved test lifecycle
POST /physical-test-runs/{id}/evidence	physical.test.record	expected_version,step_id,expected,observed,evidence_refs; append evidence
POST /compatibility-assessments	physical.compatibility.author	binding_revision,hardware,firmware,licenses,certificates,fat_run_refs,sat_run_refs; create review record

API	Permission	Request/response and side effect
POST /compatibility-assessments/{id}/acceptance	physical.compatibility.accept	expected_version,decision,signoffs,open_defects; reject unresolved critical defects
POST /rescue-equipment-missions	rescue.equipment.assign	incident_ref,dispatch_task_ref,equipment_ref,operator_ref,mission_scope,capability_revision; qualified operator workflow, not motion

All physical POST mutations require Idempotency-Key; applicable updates also require expected_version/If-Match. Existing `/commands` routes remain the common command interface; `/physical-commands` is a specialized facade over the same Command aggregate, not a second actuator queue. Unknown/unapproved capabilities return 422 CAPABILITY_NOT_APPROVED; revision drift returns 409 BINDING_REVISION_MISMATCH; active impairment/local inhibit returns 409 PHYSICAL_INTERLOCK or IMPAIRMENT_ACTIVE; stale prerequisites return 409 PHYSICAL_STATE_STALE. Reconciliation cannot claim a result without supporting evidence. Permissions are site/controller/capability scoped, not tenant-wide by default.

3. Canonical event envelope and semantics

```
{
  "specversion": "1.0",
  "id": "019a0000-0000-7000-8000-000000000041",
  "source": "urn:smartalarm:connector:gw-12",
  "type": "com.smartalarm.signal.fire.observed.v1",
  "subject": "devices/cam-a05-03",
  "time": "2026-10-23T08:40:12Z",
  "datacontenttype": "application/json",
  "tenantid": "tenant-northstar",
  "regionid": "eu-cell-1",
  "mode": "actual",
  "data": {
    "schema_version": "1.0.0",
    "observed_at": "2026-10-23T08:40:11Z",
    "hazard": "fire.possible_flame",
    "location": {"id":"zone-a05-03","method":"registry","version":12},
    "confidence": 0.82,
    "model_version": "flame-v3",
    "evidence_refs": ["evidence-31"]
  }
}
```

The gateway verifies tenantid against credentials; the payload is not the source of truth. source is a registered identity namespace. Event schemas have a version-specific additional-property policy; invalid fields are quarantined with a reason. Detector confidence does not map directly to severity; severity follows policy and impact. observed_at may be delayed or affected by clock errors; server received_at is always present; a skew flag is stored separately. Location geometry uses GeoJSON longitude,latitude; inbound CAP explicitly converts from lat,lon and has polygon tests.

Proposed topics: `sa.<region>.<mode>.signals.v1`, `incidents.v1`, `assistance.v1`, `dispatch.v1`, `warnings.v1`, `delivery.v1`, `commands.v1`, `audit.v1`, `deadletters.v1`. Do not create a topic per user. Broker ACLs are service-specific, with actual/exercise mode isolation. Partition keys use aggregate ID with tenant; telemetry keys use device ID. DLQs contain no secrets; they include an encrypted payload reference, reason, and replay policy.

Physical event contract

Use the same CloudEvents envelope, verified source binding and actual/exercise isolation. Proposed types are `com.smartalarm.physical.point.observed.v1`, `com.smartalarm.physical.command.state_changed.v1`, `com.smartalarm.physical.impairment.changed.v1`, `com.smartalarm.physical.test.completed.v1`, `com.smartalarm.physical.binding.accepted.v1`, and `com.smartalarm.rescue.equipment_mission.updated.v1`. Event data includes schema_version, aggregate_id, aggregate_version, binding_revision where relevant, source provenance and evidence references. The canonical point payload example is a design fixture, not a live observation:

```
{
  "schema_version": "1.0.0",
  "aggregate_id": "point-example",
  "aggregate_version": 1,
  "tenant_id": "tenant-example",
  "site_id": "site-example",
  "controller_id": "controller-example",
  "binding_id": "binding-example",
  "binding_revision": 1,
  "point_id": "point-example",
  "zone_id": "zone-example",
  "semantic": "pump_running",
  "value": true,
  "unit": "boolean",
  "observed_at": "2026-10-23T08:40:11Z",
  "received_at": "2026-10-23T08:40:12Z",
  "quality": "GOOD",
  "controller_mode": "AUTO",
  "maintenance_state": "IN_SERVICE",
  "cause_effect_revision": 1,
  "source_event_id": "source-example-1",
  "raw_evidence_ref": "evidence-example"
}
```

`pump_running` is not `waterflow_confirmed` and neither means fire extinguished. Units/scaling are defined by the accepted point contract; missing sensors remain UNKNOWN. Command events carry command_id, capability_revision, previous_state, state, controller_receipt_ref, feedback_observation_ids, unknown_reason and source/receive times as applicable. Impairment events carry scope/functions and restoration status; test events carry authorization and evidence references; mission events carry operator and supported-interface mode. Do not publish raw medical notes in equipment events.

4. Sequence: signal → incident → warning

```
sequenceDiagram
  participant E as Edge
  participant I as Ingestion
  participant B as Broker
  participant C as Correlation
  participant S as Incident
  participant W as Workflow
  participant D as Delivery
  E->>I: event + source credential + stable ID
  I->>I: verify / durable store + outbox
  I-->>E: received (not incident confirmed)
  I->>B: signal.accepted (retryable publish)
  B->>C: at-least-once delivery
  C->>S: create candidate (idempotency key)
  S->>S: commit aggregate + outbox
  S->>B: incident.created
  B->>W: start versioned playbook
  W->>W: approved template or human approval
  W->>D: enqueue warning version + audience
  D->>D: channel attempts + feedback tracking
```

If the Incident API times out after commit, the caller retries with the same key and receives the existing incident. If the broker is down, the outbox persists; ingestion still has bounded durable capacity and a degraded state, and does not ACK before commit. If correlation fails, queue-lag alerts and raw-signal triage remain available to operators; manual SOS is not blocked.

For physical alarms, the source is the approved local controller/gateway and preserves alarm/supervisory/trouble distinctions. Local automatic suppression operates on its own approved inputs and cause-and-effect; this cloud sequence only creates the incident and response workflow. Closing the incident does not silence/reset the panel, stop a fire pump, close a valve, or cancel local protection. Such actions require distinct permitted capabilities and site authority.

5. Sequence: offline SOS and reconciliation

The client creates a UUID request, saves it in an encrypted local outbox, and displays “not sent”. When online, it POSTs with the same request ID; the server commits before returning a receipt. If the response is lost, the client retries and the server returns the existing SOS ID. Only then does the client display received; human ACK arrives as a separate event. Offline location updates always include observed_at; the server retains history without overwriting a newer location simply because an update arrived late. Offline cancellation is a queued request, not a promise to stop dispatch. For a lost device or revoked account, the app purges secure cache according to policy when possible; it does not promise remote wipe while a device remains offline.

6. Workflows and deadlines

Workflow ID = tenant:aggregate:playbook_version. Timers run server-side; client cron polling is not a safety mechanism. Each step has an input schema, activity idempotency key, deadline, retry policy, compensation policy, and manual fallback. Retries use exponential backoff+jitter with maximum attempts/time; business rejections are not retried. An escalation step key of workflow+step+recipient prevents unbounded duplicate sends. Warning expiry cancels unsent jobs; sent jobs require a cancellation/update message and cannot be recalled by deleting DB records.

Situation	Proposed baseline timer	On expiry
P1 incident without operator ACK	60 s	Shift supervisor, then regional backup at 120 s
Unaccepted task offer	60 s	Expire offer; dispatcher selects another unit
Possible fall check-in	Configurable 30 s	Create care request; do not make a diagnosis
Physical command	TTL 30 s or capability-specific	Expire before dispatch; reconcile unknown after dispatch
Webhook delivery	Backoff for up to 24 h or shorter event expiry	DLQ + owner alert

These product defaults require site validation; policy includes min/max and an approver, preventing administrators from arbitrarily changing timeouts during an incident without recording a version.

Physical workflows in baseline 1.0 are WF-28 point/zone onboarding; WF-29 physical supervision; WF-30 authorized actuation, feedback and UNKNOWN reconciliation; WF-31 impairment/test/restoration; WF-32 FAT/SAT compatibility; WF-33 rescue-equipment mission. Their human deadlines and command TTL are capability/site configuration with approvers. No universal software timeout authorizes release, reset, silence or automatic pump stopping. The 30 s command default above is only a proposal and must be overridden or rejected when inconsistent with the approved equipment design.

7. Correlation algorithm and hazard expansion

Exact deduplication precedes fuzzy correlation. Candidate queries use tenant, mode, hazard compatibility, location graph, and event window; scoring uses time distance, same zone, source diversity, and semantic match. Scores are suggestions unless an automation rule is approved. Rule inputs include source confidence/quality; two cameras using the same model are not counted as two wholly independent pieces of evidence. When several incident candidates exist, the operator chooses; source corrections append correction events instead of editing original events.

The hazard registry contains key, localized labels, severity policy, required fields, playbook binding, sensitivity, default audience, detector schema, and retention class. Add “gas leak”, “industrial spill”, “lone worker”, “missing person”, or “infrastructure outage” through registry/schema/plugins after domain review, without changing hard-coded enums in every service. A new category does not automatically acquire physical-command permissions.

8. Command safety protocol

Requests contain command_id, device_id, capability, canonical params digest, tenant, writer_epoch, nonce, issued_at, expires_at, requester, and approval evidence. Approval signs the digest, so parameter changes invalidate it. The gateway checks credentials, epoch, TTL, capability, zone, local interlocks, and used nonces. Non-idempotent capabilities return retry_class=manual_reconcile; they are not automatically resent when ACK is lost. Local controller results record source time and observed physical state when available. Success requires the capability-specific completion criterion; accepted/executing/unknown must appear separately in the UI.

HA command workers use distributed fencing; failover does not reissue expired commands. Local safety interlocks may reject commands even after cloud approval. Emergency stop appears only when the device supports the correct semantics; a stop-request receipt does not prove the device has stopped. The drone gateway includes flight-stack capability negotiation, altitude frame, link health, pilot override, and lost-link policy; video streaming is separate from the command channel. MAVLink command cancel is not treated as a universal guarantee when unsupported by the flight stack.

Physical command execution sequence

Cloud policy/approval → OT DMZ authenticated conduit → site edge durable command ledger/fencing → approved vendor gateway → local controller → physical device → independent feedback where available. The local protection sequence and manual controls do not depend on this cloud path. The adapter's access profile is observe-only unless exact model/firmware/license and site approval establish a specific write capability; a BACnet WriteProperty advertisement alone is insufficient. Use the restrictions in Physical Device Catalog and Rescue/IoT Device Catalog.

Physical states are DRAFT, PENDING_APPROVAL, AUTHORIZED, SENT, ACCEPTED, EXECUTING, COMPLETED_VERIFIED, REJECTED, EXPIRED, FAILED, UNKNOWN and RECONCILED. Map legacy Command `succeeded` only when its physical completion criterion is verified; retain the richer physical_state in PhysicalCommandDetail. Bind approval to target, capability revision, point/zone/cause-and-effect revisions, parameter digest, TTL and writer epoch. Edge checks freshness, local mode, impairment, allowed operation and feedback availability before sending; the local controller retains final interlocks. After send, a timeout is UNKNOWN unless trustworthy evidence proves failure. Reconciliation appends evidence and a reason; it never silently replays the command.

Wet-pipe thermal sprinkler heads are not generic addressable actuators. Controlled-release assemblies keep their approved local releasing logic; no generic relay bypass, AI-confidence trigger or LLM-generated cause-and-effect is allowed. Approved supervised I/O follows exact electrical, supervision and feedback design. `release_output_active`, valve position, pump running, pressure and waterflow are different observations. Manual-console-only rescue equipment receives a mission assignment, not invented drive/aim/discharge API calls. An unverified remote emergency-stop interface cannot be presented as guaranteed physical stop.

9. Localization vi / en / ja

UI keys use forms such as `incident.status.active`; versioned resource bundles and ICU MessageFormat handle plural/select without concatenated fragments. User locale priority: explicit choice → profile → supported OS locale → tenant default → reviewed English. Language selection does not change region or authorization. Font stack: system UI + Noto Sans when deploying a font bundle; Japanese uses Noto Sans JP/system CJK; no forced uppercase for Japanese; minimum body line height 1.5, with 40% text expansion for testing. Dates/numbers/units follow locale, with timezone shown alongside commands/incidents spanning countries.

Content templates are entities with `template_id`, `version`, `locale`, `text`, `variables_schema`, `reviewer`, `approved_at`, and `voice_asset_ref`. Warning approval references exact locale versions. If an approved translation is missing, clearly show the fallback language and require the publisher to confirm policy; do not generate and automatically publish an AI translation. Do not block all P1 communication when an emergency fallback template is preapproved. Original messages/user chat retain their language; assistive translations are labeled machine translated and do not replace original evidence. TTS voices follow locale and always have transcripts; text remains readable if TTS fails. Critical translations require native-speaker/domain-reviewer checks before release.

Physical terminology must consistently distinguish “request authorized”, “controller accepted”, “physical outcome verified”, “outcome unknown”, “test”, “impaired” and “restored” in vi/en/ja. Preserve manufacturer point labels alongside reviewed localized semantics. Translations never change register semantics, control polarity, parameter units or authority. Speaker TTS/language support is model-specific; use reviewed recordings where the researched payload lacks a verified locale or API.

10. Evidence, chat, and privacy

Baseline upload policy: 20 MB/file for images/documents/short clips; MIME sniffing + extension allowlist + antivirus + decompression limits; quarantine before scanning. Signed upload URLs are scoped to object keys with short expiry; download URLs check policy and TTL, with no public bucket. Video session tokens are incident-scoped, with audited access and region-resident relays. Webhook payloads do not send raw medical notes by default. Chat retention follows incident/social class, with abuse reporting/restricted threads and a shared attachment policy. Safety fan-out snapshots include group membership versions, but revocation is rechecked at send time.

Precise location has field-level protection; generalized site status is the family default. Retention purges include DB, object store, search index, caches, and export links; backups expire and restores reapply tombstones. Conflicts between legal holds and data-subject deletion go to privacy review; requests are not silently ignored. Audit hash chains/tamper-evident storage support modification detection, without claiming absolute immutability if all authority is compromised.

FAT/SAT records include exact serial/firmware/license/configuration, expected and observed point states, timestamps, witnesses, approved test authorization, deviations and signed restoration. Store raw captures separately with hashes and access controls; redact secrets without discarding traceable source references. A product brochure or component certificate cannot set field_accepted. Acceptance references the specific installed binding and approved scope; firmware, point-map or cause-and-effect changes reopen affected review.

11. Frontend runtime and state design

V2 presentation contract and transition tests

`getContent()` selects a renderer by group; `SAEnterprise.render(s,c)` selects composition by screen ID; `SAVisual.scene()` draws schematics using locale and state. A generic `type` property must not determine every layout. `data-go` means navigation and `data-demo` means illustrative interaction; neither is API or actuator success. Existing FRs and `Screen-Registry.json` remain the source contract.

Production applies the following rules before rendering and speech. These are implementation rules, not new API schemas:

Input / transition	Required result and assertion
Heading unknown, floor known	Retain valid plan/landmarks; remove relative arrows and “turn left/right.” A QR position fix does not establish heading.
Floor unknown / route withdrawn / evidence stale	Cancel active and queued speech; remove old route overlay and instructions in the same update; retain warning, reported-location entry and assistance. Back/history cannot restore the old route.
Late snapshot older than a withdrawn revision	Reject the obsolete update by revision; HTTP response time does not establish route freshness.
SOS local pending → durable receipt → human ACK	Advance only on evidence for the correct request ID; clicks or hash navigation never create receipts.
Command accepted, readback absent	F11/F12 remain UNKNOWN; no physical retry action appears. Identify missing evidence and reconciliation.
Enter C27 / change language / leave guidance	Cancel old audio; C27 remains silent after return, reload, state changes or new notifications.

Component tests assert text and accessibility properties alongside SVG, rather than image snapshots alone. Integration tests change revisions during speech, interrupt connectivity between send and receipt, delay ACK and assert local override. E2E tests measure focus, targets, reflow, keyboard access and equivalent map text in VI/EN/JA. The current prototype has fixed sample

speech, local state and illustrative data; it does not contain production mechanisms proving these assertions.

Server-state caches are keyed by tenant+region+user scope+resource; workspace switching purges or fully partitions them. Client stores hold only drafts/UI preferences and are not the source of incident truth. A server-rendered shell or static app + BFF is selected per deployment; strict CSP, no tokens in localStorage, secure cookies for web/session BFF. Native secure keystores hold tokens, with an encrypted offline database. Websocket sequence gap → snapshot refresh; stale banners show age. Command buttons are disabled when capability/location/policy staleness exceeds thresholds, with a reason and manual channel.

Form pattern: labels always visible; required/error indicators per field; pending submission state; receipt after server ACK; conflicts show server version and draft without automatic overwrite. List pattern: shareable filters without PII in URLs; accessible tables, keyboard focus, selectable rows with context. Maps always have list alternatives; floor controls support keyboards; safety information is never conveyed solely by color. SOS does not depend solely on a hold gesture; an explicit button + confirmation provides a fast path according to accessibility/site policy.

F01–F20 extend the coordinated screen set to 233 screens. Physical views include point/zone inventory, quality and freshness, accepted capability evidence, target review, approval, separate receipt/feedback timeline, UNKNOWN reconciliation, impairment/test/restoration and FAT/SAT acceptance, plus rescue-equipment assignment. FR-PHY-01–FR-PHY-16 governs these behaviors. Do not enable an action from a stale projection; server/edge recheck. A local-only or unsupported action shows its reason and authorized manual workflow. No AI response can submit an actuator command or alter controller configuration.

12. Test architecture

Unit tests for state guards/rules/unit conversion. Contract tests for OpenAPI/events/adapters. Integration tests for outbox/inbox, workflow restart, DB tenant isolation, provider callbacks, and replay. E2E tests by actor and vi/en/ja, offline/reconnect, accessibility, and role changes. Performance tests for steady/burst/fan-out, soak, noisy neighbors, and media overload. Chaos tests for broker failure, DB failover, workflow-worker termination, provider timeouts, region partition, and clock skew. Security tests for BOLA, source spoofing, SSRF URL validation, upload malware fixtures, replay nonces, key revocation, and support break-glass. Separate hardware-in-loop tests for each actuator/drone vendor in an authorized environment.

Synthetic test data, deterministic clocks for deadlines, and mock providers represent accepted/delivered/unknown separately. Passing prototype navigation is not equivalent to passing production E2E. The acceptance matrix includes actor, trigger, expected evidence, and architecture owner so independent QA can design tests rather than merely mirror implementation.

Physical acceptance adds exact-configuration FAT and installed SAT. Test every enabled point/zone mapping and capability, negative cross-tenant/site/zone authorization, wrong revision, expired approval, duplicate event, missing feedback, and accepted-without-completion. Disconnect cloud, power off edge, restart gateway, isolate a partial point group, skew clocks and restore connectivity; verify local automatic protection continues and old intents do not actuate. Test hardware under an approved impairment plan with qualified local staff; simulator success is not authority for live discharge. FMEA assigns site owners and residual-risk decisions for wrong-zone actuation, lost feedback, supply failure, relay fault, test/actual confusion and compromised gateway. The detailed test/governance gates are in Standards and Deployment.

13. Key runbooks

Failure	Detection	Containment / recovery	Verify
Ingestion lag	oldest event age + p95	Reserve SOS lane; scale consumers; quarantine faulty source	No lost receipts, replay preserved IDs
Provider outage	failed/unknown attempts + callback age	Circuit break; alternate channel; operator visibility	Human ACK tracked separately; no duplicate flood
DB primary loss	health + write errors	Fence; promote verified replica; refresh service connections	Committed SOS present; no dual writer
Region unavailable	multi-vantage synthetic	Approved DR routing; epoch increase; outbox reconcile	Residency unchanged or explicitly approved; RPO/RTO measured

Failure	Detection	Containment / recovery	Verify
False detector storm	incident rate + operator feedback	Disable specific rule/model deployment; retain raw signals	Manual SOS unaffected; rollback version audited
Lost drone link	telemetry age	Expose unknown; local pilot/lost-link procedure	Receipt and physical state reconciled; no blind command replay
Compromised device	abnormal source behavior	Revoke cert; quarantine; rotate binding; coverage warning	Denied traffic; replacement device validated
Translation error	report/locale review	Withdraw affected template version; reviewed correction	Version lineage and recipients of correction visible
Cloud or edge loss at site	heartbeat/quality age, local panel observation	Mark remote integration unavailable; preserve local protection; buffer/reconcile supported observations	Independent automatic sequence verified; no stale write replay
Wrong zone or binding revision	point walkdown, digest mismatch	Suspend affected remote capability; retain local operation; correct/reaccept mapping	Point-to-point and negative cross-zone tests signed
Physical outcome unknown	accepted/sent without required feedback	UNKNOWN; qualified local verification; alternative response; no blind retry	Evidence-linked reconciliation with unchanged command history
Impairment or test overrun	planned_until, controller mode mismatch	Notify site responsible person; follow approved compensatory plan and restoration procedure	Local restored state, witness and signoff before closure
Rescue equipment link/console unavailable	operator report, supported telemetry age	Operator follows vendor lost-link/stop procedure; mission shows unknown/unavailable	Local physical condition and operator handback recorded

14. Delivery decomposition for the full build

Concurrent workstreams belong to the same complete scope: platform/IAM/regions; events/connectors/devices; incident/workflow/dispatch; Companion/Field; Command/admin; content/i18n; media/privacy/audit; observability/DR/security; hardware integration. Dependency order: contracts/domain invariants → platform foundations → vertical flows → complete role surfaces → integrations + resilience → independent acceptance. Administrative UI and error states are not omitted merely because they rarely appear in demos.

Definition of Ready: approved actors/permissions, state transitions, API/event schemas, error cases, translated copy, test fixtures. Definition of Done: code + migrations + contract tests + UI in three locales + accessible errors + dashboards/runbook + rollback + security/tenant tests + reviewer signoff. Production go-live requires domain/site authority validation; this document is a detailed review baseline, not acceptance signoff.

Full-build workstreams now explicitly include physical point/zone integration, OT DMZ/site edge, approved vendor gateways, capability/feedback contracts, impairment/test governance, FAT/SAT, and rescue-equipment operations. Definition of Ready for a physical write includes documented manufacturer permission, site cause-and-effect and observable completion; Definition of Done includes exact-configuration acceptance and a local restoration runbook. Unsupported actions remain visibly unavailable while the full designed scope is retained. The proposed operations/events above must be kept aligned with the separate JSON contracts during implementation.

15. BMS implementation detail design

This is the implementation contract for WF-34–37 and FR-BMS-01–40. Canonical endpoints are /bms/connections/... in API-Contract.json 1.1. Draft research names such as bms-systems/bms-release-intents are conceptual and do not create public aliases. No public API accepts arbitrary vendor methods or raw endpoints. The JSON contract adds23 operations and 7 BMS event types while preserving all 44 existing operations.

15.1 Storage schema and transactions

The following tables extend regional PostgreSQL. Large raw payloads and evidence live in same-region object storage with hashes/references in tables. Business foreign keys include tenant/site in composite keys to prevent cross-tenant joins. Use UTC timestamptz and a transactionally monotonic bigint revision. RLS supplements service authorization rather than replacing scope checks.

Table / key	Columns and implementation constraints	Index / proposed retention
bms_connection PK(tenant_id,site_id,connection_id)	region_id, profile_id/version, vendor_build, adapter_digest, endpoint_ref, credential_ref, state, revision; UNIQUE(tenant_id,site_id,connection_id); endpoint refs are registry FK	(tenant_id,site_id,state); metadata through lifecycle plus approved audit retention
bms_map_revision PK(scope,map_revision)	connection_id, mapping_digest, base_revision, state, approved_by, approved_at; immutable after approval; one active via partial UNIQUE(scope) WHERE state=active	(scope,state,created_at); retain every revision referenced by an incident/command
bms_point_binding PK(scope,map_revision,point_id)	native_address_hash, asset_id, space_id, zone_id, type, units, scale, offset, quality_policy, stale_after; UNIQUE(scope,map_revision,native_address_hash); scale finite/nonzero when conversion needs inversion	(scope,asset_id); (scope,zone_id); no hard delete of referenced bindings
bms_qualification PK(scope,qualification_id)	adapter_digest, vendor_build, map_revision, license_ref, capabilities, expires_at, signature_ref, status; admission verifies all evidence references	(scope,status,expires_at); expiry never deletes history
bms_observation PK(scope,observation_id)	point_id, source_time nullable, received_at, value_json, unit, quality, map_revision, raw_ref/hash, source_seq nullable; append-only	partition received_at; (scope,point_id,received_at DESC); hot 30 days is an approval-required assumption; incident hold overrides
bms_alarm_occurrence PK(scope,alarm_id)	native_id, occurrence_id, branch_id nullable, active_state, native_ack_state, reviewed_at, source_time, quality, revision, incident_id; UNIQUE(scope,native_id,occurrence_id,normalized_branch)	(scope,active_state,severity); never purge active/unknown alarms; policy-based archive
bms_alarm_receipt PK(scope,receipt_id)	source_event_id/cursor, raw_hash, received_at, alarm_id, mapping_revision, replay_mode; duplicate receipts retain audit link	UNIQUE(scope,source_stream,source_event_id) if source ID stable; otherwise deterministic fingerprint plus collision evidence
bms_snapshot_generation PK(scope,generation_id)	kind, started_at, ended_at, complete, gap_detected, cursor_ref, expected_count nullable, actual_count; only complete scoped generation can reconcile absence	(scope,kind,started_at DESC); retain incident-relevant gaps
bms_command_link PK(scope,operation_id)	operation_kind, physical_command_id nullable, alarm_id nullable, preview_id, approval_digest, idempotency_key, request_hash, state, expires_at, revision; physical_command_id FK existing command	UNIQUE(scope,operation_kind,idempotency_key); no deletion while outcome unknown
bms_writer_lease PK(scope)	epoch bigint, owner_instance, lease_until, egress_fencing_ref; epoch advances atomically; egress validates current epoch	UNIQUE(scope); lease/epoch history to audit
bms_inbox / bms_outbox	inbox UNIQUE(tenant_id,event_source,event_id); outbox aggregate_id/revision/type/body/created_at/published_at; projection+inbox and state+outbox each atomic	outbox unpublished index; receipt retention >= upstream retry/replay horizon; proposed 90 days requires approval
bms_lifecycle_job PK(scope,job_id)	kind,state,from_revision,to_digest,steps_json,approval_id,rollback_ref,started_at,finished_at; resumable steps have idempotency markers	(scope,state,submitted_at); lifecycle audit survives restart

Never use a time-window dedupe key as an alarm occurrence identity. Without a stable source occurrence ID/sequence, an adapter derives occurrences from observed active→inactive→active cycles and records generation/uncertainty across gaps; it does not invent missed cycles. Direct panel and BMS feeds create two receipts/provenance records. Correlation may link one incident but must not merge native ACK identities.

15.2 Adapter SDK and scheduler

```
AdapterContext = {scope, workload_identity, approved_endpoint, secret_provider,
  profile_revision, map_revision, capability_snapshot, request_budget, journal, clock}
probe(context) -> ProductIdentity + SupportedInterfaces + Diagnostics
discover(context, allowlist, cursor) -> Page<NativePointDescriptor>
```

```
readPoints(context, native_ids, cursor) -> Page<NativeObservation>
subscribePoints(context, approved_filter, resume_token) -> SubscriptionHandle
readActiveAlarms(context, cursor, generation) -> Page<NativeAlarm>
subscribeAlarms(context, approved_filter, resume_token) -> SubscriptionHandle
readHistory(context, native_id, interval, cursor) -> Page<HistoricalObservation>
acknowledge(context, validated_native_ack, egress_epoch) -> NativeReceipt
execute(context, validated_physical_command, egress_epoch) -> NativeReceipt
readback(context, accepted_target, predicate) -> Evidence
relinquish(context, validated_release, egress_epoch) -> NativeReceipt
drain(context) -> OutstandingWork + RetainedAuthority
close(context) -> CloseEvidence
```

Mutation methods are never callable directly from UI/API controllers. Runtime supplies ValidatedNativeAck/ValidatedPhysicalCommand objects minted by the policy executor, not arbitrary JSON. Adapter support is distinct from admission; missing or unverified methods return unsupported errors, never fabricated success. Plugins pin immutable image digests. Windows/Java vendor SDKs run in OS-specific sidecars rather than in the shared API process.

Priority queues separate command receipts/local overrides/critical alarms, active snapshots, point refresh, and discovery/history. Unknown alarm severity is not silently downgraded. COV leases renew with jitter and token budgets. Polling fallback is an approved strategy within the same contract, never an automatic new protocol/account. Retry is limited to proven idempotent/read operations within budget. Reconnect uses capped exponential backoff with jitter; deadlines include queue time. An open circuit marks stale/unknown by evidence age; it never substitutes zero for missing data.

15.3 Ingestion and alarm reconciliation

1. Authenticate source and journal raw receipt/hash/cursor before transport ACK where supported. Validate scope, mapping revision, type/unit/status. Decode/scale failures are quarantined and excluded from routing; retain a privacy-limited raw reference.
2. Regional ingest uses an inbox unique key and atomic observation/alarm-mirror plus outbox transaction. Same event ID and digest returns the previous receipt; reused ID with changed digest is an auditable conflict.
3. Active state and native_ack_state are independent axes. SmartAlarm reviewed is a third axis and does not send ACK. Native ACK binds the current occurrence_id, branch_id and native_transition_token, valid expiry, authority and approval.
4. Lost subscriptions mark the generation incomplete and retain active/unknown alarms. Reconnect establishes the stream and obtains an active snapshot/ConditionRefresh according to the vendor contract, buffering/reconciling interleaved events and recording gaps where ordering is unproven.
5. Only a demonstrably complete snapshot within its exact scope may reconcile absence. Sources without completeness guarantees cannot auto-clear missing alarms; readback or review is required. Historical backfill carries replay mode and never re-triggers PA, dispatch or commands.
6. Return-to-normal updates native inactive state; incident closure still follows rescue workflow. Source suppression/disable/maintenance has an owner and expiry and never deletes evidence.

15.4 Commands, approvals and fencing

Connection states are draft→validating→qualified→active_read; active_control requires control qualification and approved handback. Degraded/suspended/draining deny new commands; retired is terminal. Resume_read never automatically restores control. Adapter upgrades, map publication or entitlement changes invalidate old qualification.

The BMS operation receipt is a transport projection: persisted→dispatch_pending→accepted_by_gateway→accepted_by_controller→feedback_confirmed, with rejected/expired/unknown/superseded branches. The existing Physical Command aggregate remains authoritative; the projection does not independently decide physical state. effect_confirmed=true requires feedback_confirmed and valid feedback_ref. For native_ack, feedback is readback of the correct occurrence acknowledgment, not a plant effect. physical_command_id is required for physical_command and null for native_ack.

```
BEGIN
  lock connection revision; verify If-Match == expected_revision
  verify token scope + current qualification + immutable preview/approval digest
  insert operation using UNIQUE(scope,operation_kind,idempotency_key)
  on duplicate: same request_hash => return prior receipt; else409
  insert transactional outbox(operation_id, deadline, approved envelope)
```

```
COMMIT  -- only now 202 persisted
edge consumes -> journal + check current epoch at egress
recheck expiry/map/adapter/interlocks/local mode -> execute once if admissible
transport timeout after possible send -> UNKNOWN -> readback/reconcile; no blind replay
feedback predicate + stable period + evidence identity -> Physical Command transition
regional projection follows authoritative command event
```

Approval digests bind tenant/site, connection/point/native address, unit/value, priority, adapter digest, map/qualification/cause-effect revision, feedback predicate and expiry. Approvers cannot edit the payload; edits require a new preview. Target/capability allowlists and approval levels are site policy rather than generic protocol enums. Requester/approver separation applies when required; an LLM cannot approve or establish qualification.

BACnet priority arrays provide no owner ID or application TTL. Do not relinquish by writing NULL into a slot that might be owned by another actor. Check effective ownership, reserved-priority policy, local mode and handback. Where ownership is unproven, stop for local reconciliation. Cloud TTL prevents new dispatch but does not guarantee release after edge power loss. Automatic fail-safe requires a qualified local controller/watchdog. Otherwise admission must reject an auto-relinquish promise and require the manual runbook.

15.5 Canonical API, errors and pagination

Method / path	Request → response	Permission
GET /bms/connections	— → BMSConnectionPage	bms.connection.read
POST /bms/connections	BMSConnectionCreate → BMSConnectionView	bms.connection.create
GET /bms/connections/{connection_id}	— → BMSConnectionView	bms.connection.read
POST /bms/connections/{connection_id}/discoveries	BMSDiscoveryRequest → BMSJob	bms.discovery.run
POST /bms/connections/{connection_id}/map-revisions	BMSMapDraftCreate → BMSMapRevision	bms.mapping.edit
POST /bms/connections/{connection_id}/validations	BMSValidationRequest → BMSJob	bms.mapping.validate
POST /bms/connections/{connection_id}/qualifications	BMSQualificationCreate → BMSJob	bms.qualification.submit
POST /bms/connections/{connection_id}/activations	BMSActivationRequest → BMSJob	bms.connection.activate
GET /bms/connections/{connection_id}/points	— → BMSPointPage	bms.point.read
GET /bms/connections/{connection_id}/alarms	— → BMSAlarmPage	bms.alarm.read
POST /bms/connections/{connection_id}/alarms/{alarm_id}/native-acknowledgements	BMSNativeAckRequest → BMSMutationReceipt	bms.alarm.ack_native
POST /bms/connections/{connection_id}/command-previews	BMSCommandPreview → BMSCommandPreviewResult	bms.command.preview
POST /bms/connections/{connection_id}/commands	BMSCommandSubmit → BMSMutationReceipt	bms.command.submit
GET /bms/connections/{connection_id}/commands/{command_id}	— → BMSMutationReceipt	bms.command.read
POST /bms/connections/{connection_id}/commands/{command_id}/reconciliations	BMSCommandReconcile → BMSMutationReceipt	bms.command.reconcile
POST /bms/connections/{connection_id}/reconciliations	BMSReconciliationRequest → BMSJob	bms.reconciliation.run
POST /bms/connections/{connection_id}/lifecycle-transitions	BMSLifecycleRequest → BMSJob	bms.lifecycle.change
GET /bms/connections/{connection_id}/health	— → BMSHealth	bms.health.read
GET /bms/connections/{connection_id}/jobs/{job_id}	— → BMSJob	bms.job.read
GET /bms/connections/{connection_id}/policies/request-budget	— → BMSRequestBudgetView	bms.policy.budget.read
PATCH /bms/connections/{connection_id}/policies/request-budget	BMSRequestBudgetUpdate → BMSRequestBudgetView	bms.policy.budget.change

Method / path	Request → response	Permission
GET /bms/connections/{connection_id}/policies/incident-export	— → BMSIncidentExportView	bms.policy.export.read
PATCH /bms/connections/{connection_id}/policies/incident-export	BMSIncidentExportUpdate → BMSIncidentExportView	bms.policy.export.change

POST requires Idempotency-Key; existing-resource mutations add If-Match and expected_revision. Collections use opaque tenant/site/filter-bound snapshot-stable cursors with limit1–500. A changed filter invalidates the cursor with 400. Never return an unbounded point inventory. Tenant/region bind from token; body cannot select another tenant. Credential_ref and endpoint_ref must be scoped registry resources, not client URLs.

401/403 distinguish authentication/authority;404 hides out-of-scope resources;409 signals state/idempotency conflict;412 indicates ETag/evidence drift;422 indicates semantic/unit/domain failure;429 supplies a budget-based Retry-After;503 does not imply acceptance. If the ledger committed before response loss, the same key returns the old receipt. Vendor HTTP 200 batches still require per-item inspection; missing/partial outcomes are recorded in job results rather than marked universally successful. Completed jobs expose result_ref and structured problem. UI polling backs off; cancel stops undispatched work only and never undoes a physical command.

15.6 Events and version compatibility

CloudEvent type	Payload
com.smartalarm.bms.point_observed.v1	BMSPointView
com.smartalarm.bms.alarm_changed.v1	BMSAlarmView
com.smartalarm.bms.command_changed.v1	BMSMutationReceipt
com.smartalarm.bms.connection_changed.v1	BMSConnectionView
com.smartalarm.bms.health_changed.v1	BMSHealth
com.smartalarm.bms.map_published.v1	BMSMapRevision
com.smartalarm.bms.reconciliation_completed.v1	BMSReconciliationCompleted

Keep the existing CloudEvents envelope, require time for the BMS namespace and validate strict payloads. Envelope tenantid/regionid must match payload tenant_id/region_id and workload certificate scope; this is a service invariant, not something JSON Schema can prove. id is stable after durable ingest; source identifies adapter deployment without secrets. Consumers atomically store inbox and projection. Side effects originate only from the policy engine with a separate dedupe key, not from every alarm_changed event. Optional v1 additions follow compatibility policy; meaning/type/enum changes require a v2 event and consumer migration. Profile version, adapter digest, map revision and qualification version evolve independently.

15.7 Migration, upgrade and recovery

M1 expands the database with new tables/indexes without changing old keys; online-index migration follows a lock-time runbook. M2 deploys BMS-aware consumers with ingestion disabled. M3 deploys a read-only lab adapter with pinned digest/schema and validates mapping, units and alarm gaps. M4 publishes a signed map and runs shadow correlation without dispatch, comparing samples to the BMS console. M5 enables site alarm/incident integration after FAT/SAT; ACK/write remain disabled. M6 enables capabilities individually after authority, interlock, feedback and handback checks. These are release gates within the full design, not scope cuts.

Upgrade/rollback: drain rejects new mutations; wait for journaled outbox work and record pending/unknown commands. If the controller retains authority, complete local handback before stopping the writer. Preserve cursor, generation, mapping digest and prior image. A new adapter undergoes read qualification; semantic/native-ID changes need map migration and rebinding tests. Rolling back an image does not undo physical effects or native ACK, and does not automatically reactivate an old qualification. Database/edge restore always starts suspended/read-reconcile, advances epoch, reconciles snapshots and command readback, then requests renewed control admission.

15.8 Mandatory implementation tests

Contract tests cover23 request/response schemas,7 event payloads, reference resolution, preservation of44 existing APIs, tenant isolation, read-only ACK/write denial, expiry/revision and effect_confirmed invariants. Simulator tests cover subscription loss, duplicate/late/out-of-order delivery, units/endianness, active alarms at startup, incomplete generations, wrong-occurrence ACK, post-send timeout and split-brain fencing. Real-source FAT/SAT verifies entitlements, native methods, ACK semantics, timestamps/quality, actual batch/rate limits, local override/interlocks, feedback criteria, retained BACnet priority after power loss and handback.

Evidence binds adapter digest, vendor build, mapping and qualification; simulation is insufficient for field_accepted. Executable JSON checks are in work/test_bms_technical_contracts.py with a separately packaged QA report. Schema tests do not establish authentication, current expiry, signer authority or physical effects; service, integration and hardware tests remain distinct gates.

BMS06 budget policy applies through GET/PATCH request-budget with CAS and approval. history_backfill_paused pauses only historical backfill, not active alarm reconciliation. Upstream tested quotas and priority floors remain mandatory. BMS08 incident-export policy pins the approved namespace, target binding and field allowlist. Quarantine sets enabled=false, keeps raw/provenance evidence and stops new exports without changing native alarm state. Re-enable requires review of causation/echo and current export_incident qualification. No health/person fields are accepted by the contract.

Required appendices within the same baseline

Rescue speakers and voice navigation adds WF-21–22. Positioning, optimization, AI, and care profiles adds WF-23–27. Both documents are components of the full design, not optional features for a later phase. Screen specifications provides FRs for each screen ID.

SmartAlarm — Rescue loudspeakers, voice navigation and hazard avoidance



Production baseline 3.0 · additional requirements dated 23/10/2026. This document is a mandatory extension of the BRD/FRD/Basic/Detail Design, not an idea for a later phase.

Current design baseline: 3.0 · BMS integration: 24 October 2026. Original research/requirement dates above are preserved.

1. Business requirement BR-09

Support people without smartphones through loudspeakers, and support smartphone users with step-by-step voice navigation. The same incident must produce consistent content across the app, area loudspeakers, displays and response teams. The objective is to deliver approved guidance to the right people and zones while making data limitations visible. Never describe a route as “absolutely safe”; devices and data cannot observe every hazard on the ground.

2. WF-21 — Public address / rescue loudspeakers

Actors: PA author, approver, live announcer, device admin and local controller. Devices: PA amplifier/controller, IP speaker, SIP audio endpoint where a supporting adapter exists, site gateway loudspeakers, and drone and vehicle loudspeaker payloads. Each device declares its zone coverage polygon/indoor zone, priority capabilities, audio formats, playback feedback, mic streaming support, local override, battery/power/network and health.

Standard flow: select incident → select zones → preview coverage gaps → select approved vi/en/ja recording/TTS → priority/repetition/expiry → approve exact message+zone digest → dispatch → controller ACK → per-speaker playback state → reconcile failures. Selecting “entire area” must display the speaker count and zones before approval. Preapproved emergency templates may be used for rapid response under local policy, while retaining actor/audit and scope.

Playback model: queued → accepted_by_controller → buffering → playing → completed, or rejected/failed/unknown/preempted/expired. “Completed” means the device reported playing the entire content; it does not mean everyone heard or understood it. Acoustic feedback, if equipped, must be separate evidence and cannot be inferred from a network heartbeat. A failed speaker creates a coverage gap and a fallback task for patrol/mobile loudspeaker/app/SMS according to policy.

Priority arbitration prioritizes the local alarm path approved by the site authority; the cloud must not disable or override mandatory fire alarm systems. Routine audio may be preempted; equivalent emergency content requires an arbiter/zone lease to avoid overlapping messages. A stop command stops only the specified broadcast instance if the capability supports it; it does not automatically disable the entire fire alarm. Reconnection does not replay expired announcements. Recording/microphone/live speech have separate permissions, push-to-talk or explicit start/stop; opening the screen alone does not activate the microphone.

Live rescue: request microphone permission after an explicit action; approve session zones; short-lived media token; local + remote indicators; maximum duration; lease heartbeat; timeout automatically stops the live channel under controller policy. Do not assume every PA supports two-way audio. The proposed audio transport is WebRTC between the announcer and regional media gateway; the gateway converts to SIP/RTP/vendor protocol according to capability, with TLS/SRTP where supported and an isolated OT network. Recorded/TTS assets are cached at the edge with checksum/version so an approved local procedure remains playable during WAN loss.

3. WF-22 — Smartphone voice navigation

Actors: citizen/responder; inputs include location report, floor anchor, heading quality, route graph version, hazard restrictions, accessibility profile, destination capacity and instruction template locale. Outputs: ordered steps with landmark, direction when reliable, distance estimate, expected node, voice text, expiry, route version and reason for replan. The app provides play/pause/repeat, text transcript, speech rate, haptic cues and offline fallback. Voice must not automatically activate and reveal a user's location in discreet mode.

Flow: receive alert → confirm floor/location (outdoor GPS; indoor QR/BLE/UWB/manual depending on deployment) → select mobility needs → request route → check freshness/authorization → display route and selection rationale → play each step → update location → advance/replan → reach assembly point → user self-reports safety. Entering a geofence does not automatically confirm safety. Phone orientation does not equal travel direction; when heading quality is low, use landmarks such as “go to the WEST STAIR sign”, not left/right. If the floor is unknown or map registration is incorrect, do not provide turn-by-turn guidance; instruct the user to find a location code or use SOS.

The app avoids hazards using a constrained route graph, rather than simply choosing a vector away from a polygon's center. A route farther from a fire may still enter a dead end, flooded area or unsuitable floor. Outdoor routes check network walkability, closures, polygon/hazard evolution and destination status; indoor routes use site-reviewed doors, stairs, refuge areas, exits and directional constraints. Elevators are not available by default, only when an approved evacuation plan explicitly permits the specific capability. An accessibility profile excludes stairs if the user cannot use them; if no suitable path exists, propose assistance/a refuge area under the site plan rather than inventing a route.

4. Additional services and ownership

Service	Owner data	API/events
Hazard & route guidance	Hazard polygons/restrictions, route request, route version, graph validation	POST /guidance/routes; guidance.updated/withdrawn
PA orchestration	Broadcast instance, zone lease, language sequence, repetition, device receipts	POST /broadcasts; broadcast.playback.changed
Speech content	Reviewed audio/TTS text, voice config, pronunciation lexicon, locale versions	POST /speech/previews; speech.asset.approved
Real-time media gateway	Short-lived live session/transport state, no business authority	POST /broadcasts/{id}/live-session; media.session.closed

Mapping owns the graph/plan, Incident owns restriction approval references, and Command service executes approved PA intent through adapters. PA cannot bypass command policy; Guidance cannot independently change incident severity. These service additions use production cell deployment, HA and observability like the core services.

5. Data design and route algorithm

RouteNode(id,tenant,plan_revision,level,geometry,type,landmark_i18n);
RouteEdge(id,from,to,direction,width_m,stairs,accessible,door_state,approval_ref).
HazardRestriction(id,incident_ref,geometry/zone,level_range,source,certainty,severity,observed_at,valid_until,approved_by,version).
RoutePlan(id,person_session,graph_revision,hazard_version,profile,destination,steps,issued_at,expires_at,state).
Broadcast(id,incident,scope_digest,content_version,locale_order,priority,repeat_count,expires_at,state).
PlaybackReceipt(broadcast,device,sequence,state,device_at,received_at,evidence_type).

Algorithm baseline: validate location/floor confidence → snapshot graph+restrictions+profile → remove forbidden/blocked/unknown-critical edges and expired/closed destinations → apply accessibility constraints → find feasible paths using Dijkstra/A* with distance, congestion estimates and exposure penalties → choose a path only if hard safety constraints pass → produce landmark steps with localized templates → sign/cache route version with expiry. An exposure penalty cannot allow a hard exclusion to be crossed because a route is “shorter”. Unknown edges are handled conservatively under policy; unknown is not clear. Dynamic predictive hazard models are advisory for operators unless separately validated; do not automatically treat fire-spread forecasts as certain evidence.

Replan triggers: hazard version changes, new closure, destination closes, deviation exceeds threshold, location/heading quality degrades, route TTL expires. Debounce and hysteresis prevent continual direction changes; a new hazard cutting the route triggers immediate withdrawal without waiting for debounce. The client must cancel queued TTS from the old route before playing new guidance. Until a valid new route exists, announce that the route cannot be verified and display SOS/approved local instructions; do not replay an old turn because the app is offline.

Thresholds such as location age ≤ 10 s for step guidance, hazard recheck ≤ 30 s and route TTL ≤ 60 s are design assumptions requiring site trials; they do not apply uniformly to every hazard/indoor technology. Hazard-specific rules define approved maximum age, source reliability and fallback.

6. I18n, audio and intelligibility

The three locales vi-VN/en/ja-JP have separate text and voice versions with identical semantics and variables. The app uses the individual's language; loudspeakers use a language order based on audience/site policy. Avoid cycles so long that they delay the main instruction: design an immediate primary-language message and bounded subsequent versions, with operator preview of total duration. Critical TTS phrases undergo pronunciation review; place names have a lexicon and approved recordings where TTS is unstable. Sentence breaks, units and floor pronunciation must suit each language. Maximum volume is not the default solution: coverage/intelligibility require acoustic surveys and site acceptance.

Offline: cached graph/audio/instructions carry revision and expiry. When hazard data is too old, use only an approved offline site procedure with a stale-data label; do not continue claiming a dynamic hazard-avoiding route. Bluetooth/headphones, silent mode, permission and OS interruptions make audio delivery uncertain; the UI provides visual/haptic alternatives. Voice starts according to interaction/permission and consented policy; OS critical-alert entitlement is a separate integration task, and the app cannot be assumed always able to activate the speaker.

7. API details

POST /v1/guidance/routes: location{method,accuracy_m,observed_at,floor_ref,heading,heading_accuracy}, incident_id, mobility_profile, locale; response route_id/version, graph_revision,hazard_revision,expires_at,quality,steps[]. Each step has landmark_i18n_key, optional direction enum, optional distance_m, speech_template_ref and prohibited-actions. Insufficient data returns 422 LOCATION_UNCERTAIN or 409 NO_VERIFIED_ROUTE with a safe fallback object, not HTTP success with an empty route.

POST /v1/broadcasts: incident_id,zone_refs[],content_versions[],priority,repeat_count,expires_at,idempotency_key; 201 draft. POST /{id}/approve: exact digest + version; POST /{id}/start: command intent after policy checks, 202 does not yet mean playback. GET /{id}/receipts: per-device accepted/playing/completed/unknown. POST /{id}/stop: instance-scoped, permission, controller capability. POST /{id}/live-session: session TTL, approved zones, media endpoint token; the UI does not display credentials. Device telemetry ingestion is source-bound and monotonically sequenced so late receipts cannot roll state back.

8. Mandatory acceptance scenarios

V2 addition: visuals, speech and validity form one decision

C35/C36 put a verified landmark and one next action before technical fields. C03 is a floor plan, not automatically a usable route. C37 shows withdrawal; C38 shows insufficient position/floor/heading evidence; C39 must switch to the correct outdoor context. Unknown heading removes turn arrows. Recognizing a landmark must not be expressed as a promise that a route segment is safe.

Test scenario	Required result
Withdraw a route while C35 is speaking	Stop audio and clear queued speech for the old revision; remove the overlay and directions in the same update. C37 opens a replacement only with a newly approved revision; otherwise offer assistance.
QR confirms floor but provides no heading	Show position source, landmark and unknown heading; do not speak “turn left/right.” C38 offers manual entry or assistance.
Stop / change locale / navigate away	Old audio stops; replay uses only the currently valid instruction. Equivalent text is available without relying on TTS or a speaker icon.
Discreet mode C27	No automatic sound/vibration; quick exit never triggers speech; returning does not resume previous audio.
P04/P09 receive gateway ACK without playback	Separately show command receipt, playback state, unknown zones and local audibility verification; never report that everyone heard it.

The prototype provides user-triggered Web Speech play/stop controls and sample wording, not an approved live emergency message. Playback on a reviewer's device does not establish the correct voice, place-name pronunciation, field intelligibility or speaker coverage. Production acceptance checks revision-bound text/audio, template variables, VI/EN/JA, local-panel priority and disconnection. Public-address alarm policy and discreet phone audio policy are separate; one toggle must not control both.

PA-01 broadcasts to exactly the 8 selected speakers, never another site. PA-02 a network-online speaker with an amplifier fault still reports a coverage gap. PA-03 controller ACK does not display “everyone has heard”. PA-04 reconnection after expiry does not play old messages. PA-05 cloud cannot override local fire-controller priority. PA-06 two announcers competing for a zone have only one valid lease; the other sees a conflict. PA-07 the mic opens only after user action and closes when the session expires. PA-08 a drone payload without feedback displays unknown, never a fabricated complete state.

NAV-01 an unknown floor produces no left/right turn instruction. NAV-02 a hazard cutting an active route withdraws the route and cancels old TTS. NAV-03 wheelchair users are not routed through stairs. NAV-04 no feasible path switches to assistance/a refuge area under the approved plan. NAV-05 network loss and expired hazard TTL do not retain a current-route label. NAV-06 vi/en/ja pronounce landmarks, units and directions correctly; text/audio parity. NAV-07 the app route and PA use the same guidance version or display a mismatch requiring operator resolution. NAV-08 arrival does not automatically mark safe. NAV-09 emergency navigation requires no lengthy data-entry task. NAV-10 unreliable heading uses landmarks, not relative directions.

9. Mockup coverage

C35–C40: step-by-step navigation, uncertainty, reroute, outdoor, voice settings. P01–P09: zone/coverage, author/approve, playback, health, live mic, mobile speaker, content and priority. S15–S16: graph editor/test/approval. O35–O36: hazard restriction and consistency/receipt. Offline/error/accessibility states share X01–X10 and the state selector. All are mandatory baseline screens of the full product.

10. Baseline 1.0 — coordinating voice with physical firefighting and rescue

WF-28–WF-33 add physical equipment to the same incident, but PA, suppression release and vehicle control remain separate capability domains. PA receives approved hazard/guidance versions and site status; flow-switch, pump-run or valve-open signals are not proof that a fire is extinguished and cannot automatically trigger an all-clear message. Controller cause-and-effect changes require engineering review/FAT/SAT, not generation by TTS or AI.

The certified voice alarm and local fire-alarm path retain priority; SmartAlarm adapters must respect the certification scope of the installed configuration. Unknown/failed PA receipts create coverage gaps and field-team tasks. Speaker test/impairment exposes scope, compensating measures and expiry; fire watch does not end merely because the cloud reconnects. Local takeover revokes the live lease, retains recording/audit under policy and never automatically reopens the microphone.

Drone audio uses verified equipment combinations: Matrice 4T+AS1 remains an operator workflow while its exact API is unverified; Matrice 400+V1 has a vendor-documented SDK path, still requiring operation-by-operation verification on the selected firmware. English/Japanese support does not imply native Vietnamese TTS; use reviewed recordings when necessary. F19 records pilot, mission authority, broadcast scope, expiry, local abort and audio state; speaker playback does not grant autonomous flight permission. Colossus supports suppression/transport missions but its public API is unverified: dispatch only to a qualified vendor-console operator, never convert a message into movement/discharge.

Additional acceptance: PA-PHY-01 cloud live audio cannot preempt local fire priority; PA-PHY-02 unverified suppression telemetry does not produce all-clear content; PA-PHY-03 local override ends remote authority and reconnect does not replay; PA-PHY-04 test zones do not broadcast to actual audiences; PA-PHY-05 expired drone messages do not play after recovery; PA-PHY-06 absent SDK playback feedback remains unknown and requires operator evidence. Tests include witnessed on-site intelligibility; a prototype does not replace acceptance. See device catalog and F09–F19 in screen specifications.

11. BMS to warnings and voice — execution contract

WF-35 connects FR-BMS-11/13/19/21 with WF-21/22. Source alarms pass normalization, occurrence deduplication and approved zone mapping before creating WarningIntent. The anti-repeat key includes tenant/site, incident occurrence, approved response-rule revision, message revision and actual/exercise mode; received text/timestamps alone are insufficient. Native ACK, historical replay, state refresh and writeback echoes cannot create new broadcasts. Semantically changed alarms produce policy-approved updates/cancellations rather than blindly repeating old content.

Step	Required conditions and behavior	Output / failure
Resolve scope	Map BMS point → equipment → fire compartment/affected zone → audience/PA group; keep zone types separate	Missing/quarantined mapping creates an operator task, never a guessed site-wide broadcast
Select content	Use site-approved VI/EN/JA templates, priority and emergency authority; LLM only analyzes evidence	Missing approved translation uses a reviewed fallback and records the fallback locale; no automatic unreviewed translation broadcast
Select route	Door/fan/lift states are evidence for the route engine; check quality/freshness/uncertainty/accessibility	Stale/conflicting evidence withdraws the route or requires human assistance; no relative turns with unknown heading
Deliver	Bind message revision, audience snapshot and mode before fan-out; local PA controller retains priority	Separate queued/sent/device accepted/playback reported/human acknowledgement; never claim everyone heard it
Update / withdraw	Atomically invalidate old map/voice/haptics within the guidance session; online clients acknowledge the new revision	Offline clients show cached information and age; reconcile before new guidance after reconnect
End	BMS normal/ACK cannot automatically end guidance or the incident	Incident commanders verify rescue and closure criteria separately

Transaction boundary. Incident service decides response intents; warning service owns message/audience revisions and idempotency; PA adapters execute only qualified capabilities; apps receive guidance session/route revision/expiry/source quality. Commit an outbox before external delivery; transport retries follow channel contracts without creating new WarningIntents. BMS link-loss escalation and occupant messages are separate flows to avoid alarm floods.

Integrated acceptance. Inject the same alarm through direct panel and BMS: only one approved response is created. Replay a day of history without broadcasting. Native BMS ACK must not stop still-needed voice guidance. Door-constraint changes withdraw map and voice together. PA local override blocks remote delivery while retaining the operator task. LLM failure must not delay predetermined warnings. Site intelligibility is measured independently of an API playback report.

SmartAlarm — Positioning, evacuation optimization, real-time AI and proactive rescue



Mandatory part of full-product baseline 3.0 · 24/10/2026. Adds BR-10 (positioning/optimization), BR-11 (AI decision support), BR-12 (proactive assistance). Uses the same authorization, event, region, audit, localization and command boundaries as Basic/Detail Design.

1. WF-23 — Fused positioning and presence determination

The Positioning service accepts outdoor GNSS; QR/NFC checkpoints; BLE beacon/RTLS; UWB anchors; Wi-Fi positioning where the platform permits; wearable/vehicle location; manual reports and responder observations. Do not assume every phone supports UWB or that operating systems allow continuous Wi-Fi scanning. Each technology requires a site survey, calibration, device capability and permission. Site onboarding records the support matrix, measured expected accuracy, coverage holes and fallback.

LocationObservation: subject_ref, source, source_device, observed_at, received_at, lat/lon or local x/y/z, coordinate_frame, floor_ref, uncertainty_m, floor_confidence, heading_deg, heading_uncertainty, method, calibration_version, permission_basis. An indoor coordinate frame has a transform to geodetic coordinates and a plan revision, validated using reference points. Fusion must not average coordinates from two floors: establish a floor hypothesis first, then use outlier rejection, motion constraints and a quality-weighted estimator; Kalman/particle filtering is an engineering choice that must be benchmarked per site. Last known location always carries a timestamp; interpolation through unobserved areas is not presented as fact.

Presence differs from location: registered occupant, expected attendance, observed present, last-seen, reported absent and unknown. An access-control entry is only evidence of presence, not proof that the person remains there. Outdoor GPS cannot establish a floor. People without devices can still have roster/check-in/visitor records and manual accountability. Data provenance is not lost when an operator changes a location; create a new observation.

Acceptance: conflicting-floor sources are flagged; losing a beacon does not make a person absent/safe; stale location becomes unknown; users who deny tracking can still use manual SOS; tenants cannot query history outside purpose/scope; calibration updates do not silently alter coordinates in old incidents.

2. WF-24 — Evacuation option optimization

“Most effective” means the best among feasible options given available data, with hard safety constraints first and time/distribution optimization second. Absolute optimality is not promised when field data is incomplete. Proposed objective: minimize weighted travel time + queue delay + uncertainty cost + exposure cost; hard constraints prohibit hazard exclusions, closed/unknown critical doors, unsuitable stairs, unavailable destinations and capacity violations. People with limited mobility have a separate feasible graph; those needing assisted evacuation receive a rendezvous/refuge area + responder task, not a route that assumes independent travel.

The planner produces 1 primary + up to 2 alternatives, displaying reason, assumptions, validity and route provenance. Crowd optimization uses aggregate occupancy estimates, measured corridor/stair throughput and uncertainty intervals; it does not send everyone to one exit merely because it is the shortest path. Flow assignment is by cohort/zone, with rate-limited route changes and operator approval for changes to zone strategy. Hysteresis prevents oscillation; emergency hazard withdrawal is immediate. Do not publish ETAs precise to the second without calibration; display intervals and confidence.

Inputs: validated graph; current hazard snapshot; presence quality; exit/shelter availability; assistance profile; actual restrictions; verified weather/road data for outdoor use. Predictions from fire/flood simulation are a separate scenario layer and do not automatically overwrite observed hazards. The simulation dashboard must display assumptions and a “not real incident” environment boundary.

Algorithm contract

1. Filter the snapshot by incident, tenant, floor, time and freshness.
2. Build the available graph using hard exclusions; reject insufficient floor/position quality.
3. Estimate edge time from distance/speed profile and congestion; do not arbitrarily infer speed from diagnoses, and prefer self-reported ability/assessed profiles.
4. Find feasible routes, compare robust cost under uncertainty, and retain alternatives that do not share a bottleneck where possible.
5. Apply destination capacity and cohort balancing; publish route decisions with graph/hazard/policy versions.
6. Monitor deviation/new hazard/capacity; replan; invalidate queued old audio; provide the operator with the reason and affected cohort.

Acceptance: routes never cross prohibited polygons; stair-excluded profiles have no stairs; unknown capacity is not treated as unlimited; new hazards withdraw routes for the correct recipients; no feasible path creates assistance; route failure does not block manual SOS; compare results against site-approved scenarios and physical walkthroughs before go-live.

3. WF-25 — Real-time AI / LLM analysis

The AI Copilot service subscribes to read-only incident projections and an authorized retrieval corpus. Tasks: consolidate timelines, detect contradictory/missing data, suggest verification questions, cross-check playbooks, summarize sensor/camera observations, compare options computed by the route engine, and draft warnings/translations and after-action reports. The model does not independently create location nodes, approve warnings, issue commands, diagnose or close incidents.

Pipeline: event updates → coalesce by incident → build scoped context → retrieve approved SOP versions → model inference → structured result validation → evidence/permission/freshness guard → operator review. Answers include assertion, evidence_refs, observed_at, uncertainty, conflicts, proposed_action, required_approval and expires_at. Any important factual claim without evidence must be marked unsupported and cannot be used as an instruction. The LLM cannot modify the route graph; it may only call the read-only tool `compare_routes(route_ids)` through an allowlisted service, with route IDs supplied by the deterministic planner.

Proposed real-time budgets: incident projection refresh p95 ≤ 2 s; copilot summary target p95 ≤ 10 s when the model is available at the tested context size. The UI displays generated_at and data_cutoff; after TTL, content becomes stale and cannot appear live. Critical alarms, SOS, routing rules and dispatch continue during AI timeout/outage. Do not wait for an LLM to send alarms under approved policy. Streaming token text must not be played over loudspeakers before validation and approval finish.

Security: retrieved chats, external metadata and documents are all untrusted content. Prompt content cannot control system policy/tool scopes. Tool execution goes through a policy proxy, with no arbitrary network/code, result schema validation, prompt-injection fixtures and source citations. Regional model endpoints follow residency requirements; sensitive health fields are redacted or reduced to minimal functional needs. Configuration/contracts prohibiting model-vendor training on tenant data must be confirmed; self-hosted deployment may be selected where needed. Context logging is redacted, retention follows data class, and chain-of-thought is not stored; retain model/version, prompt template version, evidence IDs, structured answer and user decision.

Evaluation: scenario sets by hazard/locale, factual grounding, omitted-critical-fact, unsafe-action suggestion, prompt injection, cross-tenant leakage, hallucinated exit, stale evidence, translation meaning, calibration/abstention. Promotion requires thresholds agreed with domain reviewers, not merely generic BLEU/accuracy. Human acceptance does not automatically become a training label; feedback has a review workflow. An AI score does not independently decide who deserves rescue first.

4. WF-26 — Rescue care profiles and user management

The identity profile (name, contact, age/DOB as needed) is separate from the sensitive Assistance Profile. The profile includes self-reported functional needs: mobility, stairs capability, wheelchair/walking aid, hearing/vision, communication language, cognitive assistance, caregiver dependency, evacuation equipment, medically relevant emergency notes and optional condition history. Collect only what is needed for rescue; the UI does not require a comprehensive disease list to use the app. An age group may suffice instead of exact DOB. Do not automatically infer impairment from age or a condition name.

Profile sources: self, authorized guardian/caregiver, authorized clinician/occupational-health role under a service agreement, or responder observation. Record source, verification status, effective_at, last_reviewed_at, consent/legal basis, visibility scope and expiration review. A self-reported profile cannot be labeled “medically verified”. Users/caregivers can edit; conflicts between sources require review without losing history. Guardian relationships must be verified, with field-scoped rights and revocation; minors/dependent adults need jurisdiction-specific processes, without assuming a simple global consent model.

Authorization: citizens view/edit their own profiles; caregivers act within authorization; dispatchers see functional needs and reasons for assistance, not detailed medical histories by default; assigned responders see the emergency subset required for their task; only an authorized medical role sees deeper notes; platform admins cannot view them. Health-record access is audited, and emergency break-glass has scope/time/reason. Social groups see only consented status, not impairments or medical notes.

5. WF-27 — Proactive prioritization before SOS

Triggered when an incident affects a site/zone: build a candidate list from **presence evidence + impact zone + assistance profile + no recent confirmation**. Absence of SOS is not evidence that help is unnecessary. Groups: known exposed and needs assistance; possibly exposed/stale presence; registered but presence unknown; confirmed away. Known exposed people receive proactive contact/tasks under policy; uncertain presence first creates a welfare check and an uncertainty warning, rather than presenting the person as definitely in room X.

Prioritization is decision support: hazard imminence/exposure, inability to self-evacuate, isolation/no caregiver, route infeasibility, last contact and source confidence. Do not use wealth/nationality/social importance; age alone is not a criterion determining a person's value. Urgency categories P1/P2/P3 have explicit reasons, confidence and evidence freshness, not an absolute black-box “rescue score”. Operator adjustments require reason/audit; force override is limited to appropriate roles. Human clinical triage, when needed, follows expert-approved protocols rather than LLM diagnosis.

Flow: incident active → intersect impacted locations/presence → evaluate approved assistance policy → create ProactiveCase with idempotency key incident+person+policyVersion → notify/check-in where appropriate → operator review → dispatch assistance task → responder welfare observation/handover → close case. A subsequent SOS from the same person must be linked to the case, rather than dispatching two teams because there are two sources. No response escalates the case; it does not automatically mean “confirmed injured“. If a person reports being outside the affected area, retain correction evidence and re-evaluate; do not erase the audit trail.

Acceptance: people needing assistance enter the queue without SOS; registered people with stale locations are shown as uncertain; missing profiles do not default to “healthy”; every reason has evidence; simultaneous triggers do not duplicate tasks; a new SOS links to the existing case; revoked caregivers cannot read profiles; age changes do not automatically raise/lower priority without context; actors without health scope see only minimal assistance needs.

6. Entities / APIs / events

AssistanceProfile(id,person,version,functional_needs,emergency_notes_ref,source,verification,review_due,sharing_policy).
CareRelationship(subject,caregiver,authority_ref,scopes,expires,revoked).
PresenceEvidence(subject,location,kind,observed_at,confidence,source).
ProactiveCase(id,incident,person,policy_version,priority,reasons,evidence_refs,presence_quality,state,task_ref).
RouteOptimizationRun(id,input_snapshot,algorithm_version,constraints,alternatives,selected,approver).
AIAnalysis(id,incident,model_version,context_cutoff,evidence_refs,result,guard_status,expires,user_decision).

APIs: `/assistance-profiles` field-level scopes and version; `/care-relationships` verified grants; `/presence-observations` source-bound; `/incidents/{id}/proactive-cases` authorized list; `/proactive-cases/{id}/decisions` reason + expected version; `/guidance/optimizations` computed alternatives; `/incidents/{id}/ai-analyses` asynchronous job with status; `/ai-analyses/{id}/reviews` accept/reject draft, never command endpoint. Events: presence.updated, assistance_profile.updated, proactive_case.created, guidance.optimized, ai.analysis.ready; health payloads are minimal, with details fetched only within authorized scope.

Additional services: Position & Presence, Assistance Profile & Care Relationships, Proactive Assistance Policy, Route Optimization, AI Copilot/Gateway/Evaluation. Deploy within the regional cell, encrypt sensitive fields, and keep exact health data out of generic event broker payloads, logs/search/global control plane.

7. Human factors and notifications

V2 visual decisions for positioning, AI and care

Q01–Q03 place source, observation age, floor and quality beside the plan. Heading is independent: QR may confirm a position without orientation; stale points must not become a certain movement trail. C38 retains reported-location entry and assistance when no route is verified. Q04/Q05 show excluded alternatives and reasons beside the recommendation; green color or shorter distance does not replace feasibility evidence.

Q06/Q07 use claim/evidence/unknown composition, exposing sources, cutoff and unknowns before approval. Q08/Q09 retain model governance, region and deterministic fallback. AI must not infer safety from silence, invent exit nodes outside the approved graph, grant equipment authority or invent rescue ETAs. C08 shows arrival time only with a confirmed source; assignment does not prove acceptance or approach.

H08/H09 and O11/O12/O34 prioritize people by functional need, exposure and presence evidence rather than an aggregate count without provenance. No SOS is not the same as no assistance need. Medical records require their own scope; assistance cards do not expose diagnoses to every role. A self-reported safe state carries a subject and timestamp and does not apply to family members or the whole site.

V2 acceptance covers unknown heading with known floor, conflicting BLE/QR sources, withdrawal while AI drafts, stale presence without SOS, no capable team, delayed ACK, keyboard/screen-reader users and discreet mode. Results retain uncertainty, suppress invalid guidance, offer an actionable assistance step and invent neither success nor ETA. Prototype plans, AI suggestions and people lists are illustrative; they do not establish positioning performance or field rescue capability.

The prioritization screen displays “Needs mobility assistance · observed in the affected area · updated 2 minutes ago”, rather than only a condition name/score. People without profiles are labeled “Assistance information unavailable”, not “Normal”. Users actively choose how to be contacted; text/voice/haptics follow preferences. Japanese/Vietnamese/English use reviewed terminology and avoid stigmatizing labels. Dependents have an emergency profile subset and caregiver workflows; being added as a contact does not give a caretaker access to all data.

8. Verification gates

Site survey and indoor-positioning ground truth; route walkthrough and blocked-exit drill; crowd model calibration; clinical/domain review of assistance policies; privacy review for each jurisdiction; AI grounding/injection tests in all three locales; hardware-in-loop and live responder exercises. These are release conditions for the relevant components and cannot be replaced by a dashboard prototype or a generic “real-time AI” claim.

9. Baseline 1.0 — physical feedback, routing and AI boundaries

WF-29 supplies alarm/supervisory/trouble/flow/pressure/pump/valve observations to hazard context with source, observed_at, quality and approved point/zone mapping revision. Dependent decisions use only sufficiently fresh data mapped to the correct floor/zone. Pump-running or valve-open does not prove the hazard ended; the route engine cannot reopen excluded edges without site verification. Impairment/test/maintenance and unavailable fire-water may create restrictions/review tasks under approved policy, not inferred fire-spread predictions.

Loss of controller connection, mapping changes, revoked local authority or a new hazard intersecting a route triggers reassessment of route freshness/constraints; withdraw routes and cancel queued old speech when required. Robot/drone location is a separate source with uncertainty; equipment pose is not a casualty's location. Stretcher/assisted evacuation missions link existing ProactiveCase/task records and retain owner/handover to avoid duplicate teams. Equipment completion does not automatically confirm personal safety; use a correctly typed responder observation or self-report.

AI may explain evidence, contradictions and gaps and draft options. It cannot write cause-and-effect logic, grant connector capabilities, approve release, clear impairment, control robots/drones or edit certificates. Telemetry/vendor documents are untrusted input; capability allowlists and authorization remain outside the model. Routing/SOS/dispatch services continue during AI outages.

Additional acceptance: PHY-AI-01 pump ACK does not remove a hazard restriction; PHY-AI-02 stale points or wrong-floor mappings cannot produce a current route; PHY-AI-03 prompt injection in vendor metadata cannot issue commands; PHY-AI-04 robot tasks link existing assistance cases; PHY-AI-05 device completion does not automatically mark safe; PHY-AI-06 test telemetry cannot update actual incidents. See F01–F20 screen specifications and device catalog.

10. BMS evidence for positioning, routing, AI and proactive rescue

Mapping and confidence for the route engine

FR-BMS-06/08/19 add RouteConstraintEvidence: tenant/site, point/equipment, topology revision, affected route edges, source path, raw/mapped value, quality, nullable observedAt, receivedAt, source clock quality and policy expiry. Gateway reception does not make source observations newer. Never replace missing with zero or door-unlocked with door-open. Rooms, HVAC zones, fire compartments and navigation graphs are distinct relationships approved by a site engineer.

For each update, resolve affected edges, verify mapping revision, quality and freshness, apply approved constraints, then calculate candidates consistent with mobility needs. Stale/bad/conflicting evidence must not optimistically reduce hazard cost. If evidence cannot sustain a route, withdraw it within the same guidance session and offer assistance. Lift availability is a BMS state, not permission to use lifts for evacuation without an approved site plan. Fan/damper command receipts do not establish smoke-control effectiveness.

Presence and proactive rescue

FR-BMS-20 uses counts/PIR/access events as additional area-presence evidence. Occupancy counts do not identify people; badge access does not establish current location. Combine sources with timestamps, accuracy, consent and conflict rules; preserve lineage so BMS mirrors do not double confidence. Health/functional profiles remain restricted to assistance roles; do not export clinical records or detailed age/health data through BMS incident writeback. People without SOS can enter a check queue when exposure and assistance evidence justify it, but priority must be explainable and reviewed by authorized personnel.

AI contract and failure behavior

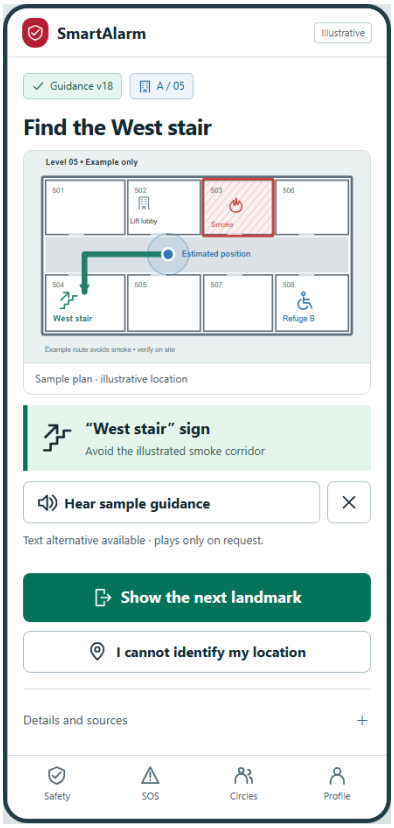
FR-BMS-38 supplies an authorized LLM projection containing evidence references/revisions/quality/age, not credentials or raw-control handles. BMS alarm text, descriptions and labels are untrusted data, not model instructions. Prompt injection in source content cannot enable tools or change policy. AI returns advisory claims mapped to evidence and uncertainty; only deterministic services with required approvals create action intents. Unavailable, expired or unsupported model advice is discarded while alarm intake, routing safety constraints and PA rules remain operational.

Specific acceptance

1. Door-open with an old source timestamp cannot promote a route to verified.
2. BMS occupancy=0 cannot erase an SOS located in the zone or lower its priority by itself.
3. Direct panel and mirrored BMS signals cannot double confidence.
4. Smoke/temperature type or unit drift quarantines mapping and invalidates related routing evidence.
5. Alarm text instructing an LLM to control equipment cannot invoke write/ACK; retain security evidence.
6. BMS network loss with cloud still available makes C37/C38 expose missing evidence and H08 retain unconfirmed people, never all-clear.

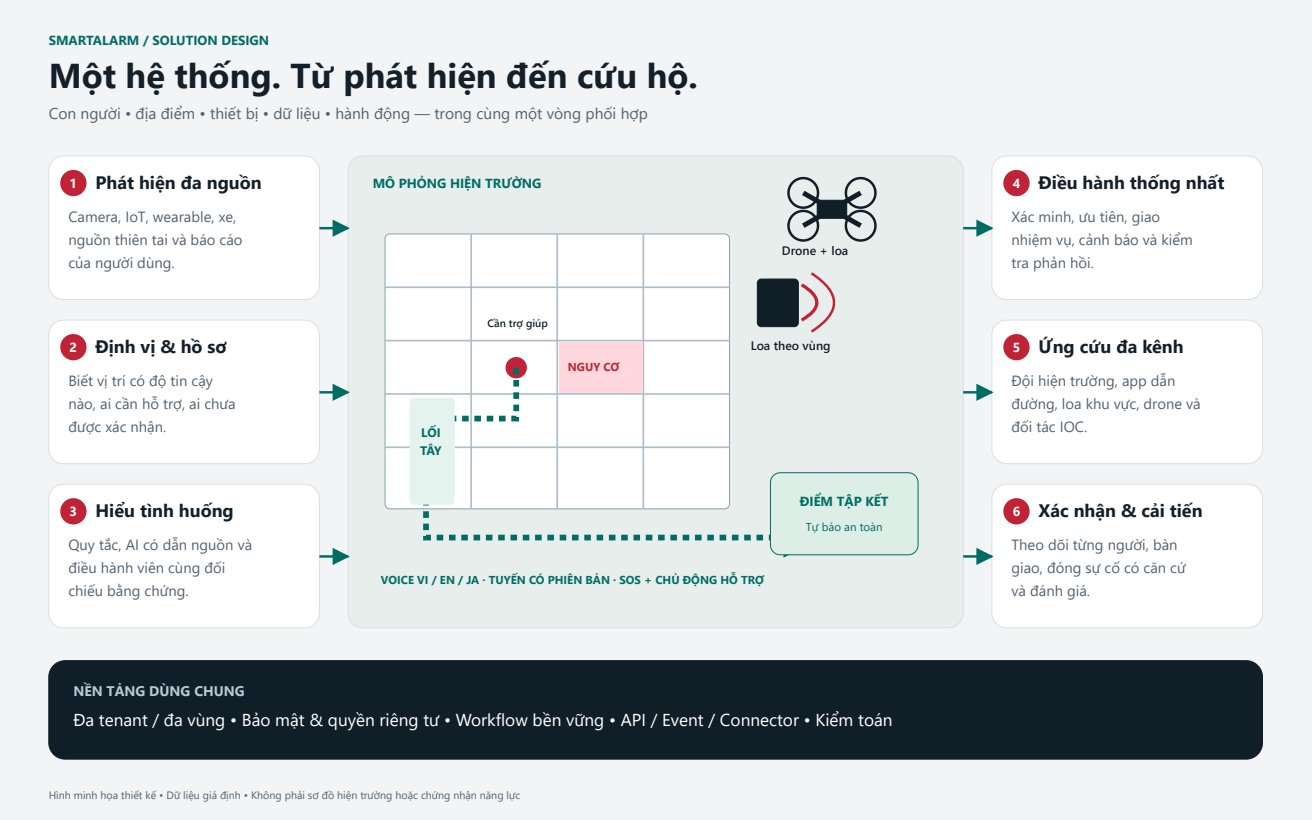
Fields/events are shared with Basic/Detail Design; no second location source of truth or Command aggregate is introduced. BMS-T16/17 and FR-BMS-19/20/38 trace to Q04–Q09, H08–H11 and C35–C38.

SmartAlarm — UI/UX & Design System



Evacuation guidance — illustrative scenario.

Revision 2.0 · Full product · Vietnamese / English / Japanese. See Prototype, screen catalog, individual screen specifications, and tokens.



1. V2 visual-first design: what, where, next action

Decision order: hazard → affected place → next action → uncertainty. Citizen emergency screens are guidance surfaces, not miniature administration dashboards. A labeled hazard pictogram, task-specific scene and concise next action occupy the first decision layer. Full field values and operational rules remain available in labeled disclosures and the design inspector. Internal implementation rules do not become the product subtitle.

Composition follows the actor and task before the legacy layout type. The citizen sees the relevant landmark and assistance action; the responder sees destination, hazard and assignment state; the commander sees incident scope, open needs, map and unit capacity; the administrator sees the object, version and change impact. Do not draw a map for every screen: permissions use a matrix, translations use paired text, execution uses an evidence timeline, and physical systems use a control schematic.

Use purpose-drawn SVG scenes and icon-plus-short-label controls. Generic glyphs, color-only status, decorative photos and cinematic emergency backgrounds are excluded from the critical path. Camera evidence is source material with permission, capture time and caption. All prototype geometry is illustrative, not a surveyed plan.

Unknown/stale/pending must be as legible as confirmed. Display observed time and source, not a fixed “updated 8 seconds ago” claim. A receipt is distinct from human acknowledgment and from physical effect. Self-reported safe is labeled as such; absence of a distress signal never means safe.

This revision is a design contract and prototype specification. It does not claim completed usability testing or proven three-second recognition. See the UX audit and the 233-screen matrix.

2. Information architecture by user

Surface	Priority tasks	Navigation and principles
Companion	Receive warnings, evacuate, SOS, support family	Home / SOS / Groups / Profile; prioritize guidance and receipts during incidents
Field	Accept work, arrive, record observations, hand over	Shift / Assignments / Incident / Outbox; show owner and pending sync

Surface	Priority tasks	Navigation and principles
Command	Triage, dispatch, accountability, warning, PA, AI	Site/region scope fixed in the header; incident workspace connects all actions
Site operations	Location, graph, device, detector, calibration	Edit → test → review → publish; explicit version states
Connect / Developer	Adapter, schema, sandbox, health, DLQ	Capability before enabling; secrets never displayed in tables/logs
Care	Profile, consent, relationship, proactive cases	Separate functional needs from medical notes; minimal role-based views
Platform	Cell, release, DR, security, support access	Metadata by default; platform-operator status alone does not unlock tenant PII
Localization	VI/EN/JA, terms, templates, voice	Source version → translation → independent review → publication

Screen IDs remain stable across name/locale changes; shared screens are reused through context without duplicating logic per industry. Taxonomy/playbook/site profiles determine content differences. Maps have floor/location context, a list alternative, and source timestamps.

3. V2 tokens and measured accessibility targets

Design-Tokens.json is synchronized to prototype-v2.css. The CSS currently declares these values directly; synchronization is not a claim that the prototype imports the JSON. Production must generate both from one versioned source. The JSON retains primitive → semantic → component layers and records CSS variable bindings, responsive breakpoints, reference contrast calculations and acceptance targets.

Element	V2 value / behavior
Canvas / ink / muted / divider	#F3F5F4 / #142B35 / #52666F / #D3DFDF
Urgent / confirmed / information / uncertain	#B51E32 / #08745E / #155FA0 / #8B5B00
Navigation / review stage	#192F39 / #E8EEEF
Verified guidance / uncertain guidance	#0A5F50 with white; #FFF1CD with #5F3E00
Body	16 px, line-height 1.45 in current CSS; user text enlargement must remain usable
Headings	h1 30 px; mobile 26 px; urgent 27 px; h2 21 px; h3 17 px
Guidance instruction	25 px; mobile 23 px
Font	Segoe UI, Arial, Yu Gothic, Meiryo, sans-serif
Controls	Main buttons/inputs 48 px minimum; mobile primary 56 px; review chrome/disclosures 44 px; phone navigation 52 px
SOS choices	132 px minimum; "other" row 72 px; pictogram 44 px with visible label
Focus	3 px blue outline, 4 px offset; verify against actual adjacent surfaces
Radius	Button/card 8 px; scene 10 px; frame 12 px; desktop mobile-preview frame 28 px
Motion	No defined transition durations; reduced motion disables animation/transitions and smooth scrolling
Responsive	Compact layout at 1120 px; mobile at 720 px; mobile-preview maximum 444 px, focus preview 460 px

A subtle divider is not automatically an accessible control boundary. Test boundaries, text and focus in the rendered context. Target normal-text contrast ≥ 4.5:1 and applicable large-text/non-text contrast ≥ 3:1. WCAG 2.2 minimum target size is 24 CSS px with defined exceptions; project touch targets are deliberately larger. Reference contrast values in JSON apply only to flat color pairs, not the complete screen. Use label, shape and pattern in addition to color; unknown never inherits confirmed green. Test 320 CSS-pixel reflow, 200% text enlargement, keyboard focus, screen readers, one-handed interaction, reduced motion and long VI/EN/JA content.

4. Component specifications

Component	Data and behavior	States
Incident badge	severity + state + source/freshness; click opens incident	new/active/contained/closed/unknown
Receipt card	ID, server time, process state; server receipt differs from human ACK	local pending/received/acknowledged/failed
SOS action	Clear CTA, with an Unknown option; no mandatory hold gesture for users unable to perform it	ready/confirming/sending/pending/received
Voice step	landmark, heading quality, step index, transcript, repeat/pause	playing/paused/replanning/stale/no verified route
Map/floor	version, floor, overlays, legend, accessible text list	loading/partial/offline/stale/no location
Priority case	person, exposure, functional need, presence quality, reasons	proposed/reviewed/assigned/contacted/resolved
Approval panel	exact digest, author, approver, target, effect, TTL	draft/pending/approved/expired/rejected/conflict
Device command	Separate requested/accepted/completed; capability-specific feedback	pending/accepted/executing/succeeded/unknown
Speaker receipt	device + broadcast version + playback + fault	buffering/playing/completed/preempted/unknown
AI answer	cutoff, evidence links, uncertain claims, human decision	generating/validated/unsupported/stale/unavailable
Data grid	sortable/searchable, stable row key, scoped export	loading/empty/filter-empty/denied/error
Health record	purpose, access reason, minimal fields, audit	limited view/authorized/break-glass/expired

Approval actions use explicit verbs, not just “OK”. Delete/deactivate/revoke include an impact explanation; destructive actions do not share the color/position of safe primary actions. Undo appears only when the backend actually supports reversing the effect; do not promise undo for an executed physical command.

5. Clickable review flows

- Fire response: O03 → O05 → O06 → O04 → O08 → O09 → O10 → O11 → O12; citizen C02 → C03 → C05 → C06 → C07 → C08 → C11 → C12.
- Proactive rescue: H10 → H08 → H09 → O09 → R03 → R04 → R05 → R06 → R07 → R08. Later SOS requests link to the case according to the backend design.
- Voice route: C35 → C36; secondary links C37/C38 cover rerouting and uncertainty; C40 configures voice. S15 → S16 is graph validation; O35 → O36 distributes versions.
- PA: P01 → P02 → P03 → P04 → P05; P06 live session review; P07 connects a drone mission; P09 arbitration policy.
- AI: Q06 → Q07 → Q05 → O36; evidence links to H09/O05, with no automatic command-execution button.
- Integration: I01 → I02 → I03 → I04 → I05 → I06; D05 → D06 is adapter conformance.
- Localization: L01 → L02 → L03 → L01; L04 voice, L05 fallback, L06 terms.

Other screens have individual deep links in the catalog. The prototype implements navigation, search/filter, basic form validation, choice selection, local sample chat, and speech samples if supported by the OS. It has no backend or real control; production business states are described in the FRD/Detail Design. Some sample input values remain in English as source content; labels/titles/navigation switch between VI/EN/JA. English sample data is not a complete translation of production warning templates.

6. Internationalization & accessibility

UI locale and content locale are separate: Japanese users may read an original message entered by a Vietnamese user, alongside a labeled assistive translation. Critical templates have approved vi/en/ja versions, text/audio parity, and locale-specific reviewers. Do not concatenate words into sentences; ICU placeholders use typed variables and guard against lost/swapped variables across translations. Field names, validation, empty/error states, and date/time must all be localized. Test Japanese line breaks, full-width punctuation, and IME; test Vietnamese diacritics with fonts and truncation. Avoid excessive uppercase/letter spacing for Japanese.

All actions support keyboards; modals trap focus correctly and return focus to the trigger on closing; live regions announce receipts/errors sufficiently without continuously reading telemetry. Screen readers have map descriptions and alternative lists. Support 200% user zoom, mobile text scaling, high contrast, and reduced motion; do not unnecessarily lock orientation. App voice has transcripts; people who cannot hear use text/haptics; people with limited vision have screen readers and CTA labels. Directions are more than arrows; always include text/landmarks.

7. Critical state matrix

Situation	Required display	Alternative action
SOS has not reached server	Local pending, timestamp, no success icon	Retry with same request ID; configured local calling option
Location/floor unknown	Method/age/uncertainty; no falsely precise pin	Enter location, scan anchor, SOS
Hazard updated	Route withdrawn, stop old speech, new version	Reviewed guidance / assistance
No route	Clearly state that a route cannot be verified	Approved refuge/assisted evacuation, no fabricated route
No response to welfare check	Unknown/no contact	Escalation according to policy
Missing health profile	No assistance information, not “healthy”	Proactively ask about needs when making contact
PA ACK but speaker fault	Per-device unknown/fault, coverage gap	Team/portable speaker/other channel
AI stale/unsupported	Cutoff, sources, warning; action disabled	Manual review / deterministic services
Conflicting update	Server version vs draft	Review/merge, no silent overwrite
Region degradation	Impacted scope and freshness	Approved failover, cached approved content

8. V2 responsive and composition architecture

The current CSS uses 1120 px and 720 px breakpoints. Desktop shows a role-specific scene/workspace with an adjacent decision rail. At 1120 px the command rail and split views move below the scene; at 720 px the review shell collapses and product content uses a single task column. Mobile actions stack or use an explicit two-action arrangement; they must not disappear. Main mobile actions are 56 px high. The current implementation does not guarantee a sticky action bar: acceptability depends on verifying the action and its necessary alternative remain discoverable without covering the scene or content.

The review catalog and specification inspector are separate from product content. Focus mode removes them. Source fields can be progressively disclosed, but critical constraints and recovery controls remain visible. Screen-specific composition is defined in each of the 233 screen specifications, not inferred from a generic legacy layout label. Wallboards remain read-only and omit restricted medical details.

Scenes provide meaningful titles/descriptions and text equivalents. Drawn arrows require verified heading, route version and hazard evidence. If those fail, remove the implied direction and expose manual location, verified refuge or assistance as appropriate. Receipt ladders distinguish local, server, human and physical evidence; they do not advance merely because navigation occurs.

9. V2 usability validation plan — not yet conducted

The three-second hazard/place/next-action recognition goal is a hypothesis, not a result. Run counterbalanced baseline-versus-V2 tasks with citizens, responders, dispatchers and physical-system operators; include older, low-vision, cognitive/access-needs, one-handed and multilingual participants. Use safe simulations, consent and a stop option. Measure correct recognition, time-to-first-correct-action, completion, mistaken pending/unknown interpretations, accidental actions and recovery after loss of connection, heading or route approval. Report errors and role-specific distributions, not only averages.

Test small unfamiliar icons alone and with labels, long VI/EN/JA copy, reduced motion, color-vision variation and assistive technology. Agree release criteria before testing; fast but wrong is failure. Screenshot checks and structural QA are not user studies or accessibility certification. The audit contains the study protocol and domain failure cases.

10. References

WCAG 2.2 is the target web standard. The design references modern component systems and native conventions while retaining emergency-response domain semantics. No certification or actual usability measurement is claimed yet.

11. Baseline 1.0 — physical firefighting and rescue interfaces

The full product now contains **233 screens and 33 workflows**. PhysicalSystems F01–F20 add controller/point/zone/cause-and-effect onboarding (WF-28), physical supervision (WF-29), conditional control (WF-30), impairment/maintenance (WF-31), FAT/SAT/certification (WF-32), and rescue equipment missions (WF-33). Existing screen IDs remain stable. Do not provide a generic “fight fire” button for devices with different physical consequences.

F01/F07/F08 place source, observed_at, quality and test/impairment/local-override state beside each signal. Pump-running, pressure, valve-open and flow remain separate; controller ACK never has the color or label “fire extinguished”. Unknown, fault, stale and power loss cannot share a green indicator. Dashboards do not aggregate impaired zones into all-clear. Physical maps identify floor, source point, mapping revision and zone boundary.

F09→F10→F11 form review → authorize → feedback, not three clicks that imply execution. Review exposes exact target/operation/parameters/zone/version/expiry, consequences and local authority. Unverified capability/certificate/interlock blocks the production execution CTA with reasons and a path to F17; the prototype only navigates review screens. Requester and approver are independent; any payload edit invalidates approval. Default Enter, hidden-zone bulk selection and ambiguous gestures cannot initiate release. Two-person confirmation does not replace controller interlocks.

F11 separates acceptance, controller execution reports and independent physical feedback. F12 reconciles unknown outcomes through supported readback/qualified observation; it provides no release-retry action to discover the outcome. F13 prominently shows local override and revokes the remote lease; return to remote needs explicit authorized action, never reconnection alone.

F14/F18 expose permit, scope, compensating measure, owner, expiry, overdue escalation and witnessed restoration. Permit expiry does not make impaired equipment healthy. F15–F17 pin BOM/firmware/mapping/cause-and-effect/test plan; expected and observed fields are separate, with no automatic pass. Missing certificate-scope evidence blocks release. F19 shows robot/drone mission, qualified operator, water supply, access/load limits, local stop and casualty procedure; an unknown public API permits task dispatch only, not a fabricated joystick/discharge control.

Additional usability tasks: distinguish flow from valve contact; identify incomplete ACK evidence; detect test zones; handle local control; dispatch a robot despite link loss; locate missing SAT evidence. Keyboard/screen-reader output includes blocking reasons and source/freshness, not color alone. State transitions retain a path to the local response team during cloud outages. See screen specifications and device catalog.

BMS integration revision 2.1

The full catalog contains **241 screens and37workflows**: the original233IDs remain, with BMS01–BMS08 added to WF-34..37. FR-BMS01..08 are screen contracts; FR-BMS-01..40 are domain requirements and are not duplicate IDs. The8new views are available in the same VI/EN/JA prototype and gallery.

The BMS workspace supports engineering and incident operations. It prioritizes topology, state comparison, source freshness and authority before writable forms. Raw protocol identifiers and units remain intact beside localized labels. Color is paired with labels, hatching and distinct layout. No visually green transport receipt may substitute for physical feedback.

Scre en	Primary visual	Main decision	Workf low
BMS 01	BMS → edge → SmartAlarm chain; capability matrix beside the identity dossier.	Is the exact product/build/site identified, and which operations have sufficient evidence?	WF-34
BMS 02	Native point → equipment → served zone → route edge topology with a separate mapping/evidence table.	Is this a service, room-containment or fire-compartment relation, and what evidence establishes it?	WF-34
BMS 03	Native condition and incident comparison with separate receive/link/native-ACK/response-closure timeline states.	Which occurrence is this, who acknowledged it in which system, and what remains unknown?	WF-35
BMS 04	Priority ladder and winning authority, with separate demand/controller-report/physical-feedback blocks and prominent local override.	Who currently controls the equipment, and is this requested effect admissible?	WF-36
BMS 05	Side-by-side before/after diff highlights unit/type changes, with dependent rules and routes below.	How did this same-ID point change meaning, and which rules/routes are affected?	WF-37
BMS 06	Zone-quality heatmap with stale hatching, alarm/telemetry/history queues and request-budget calculation.	Which sources are stale, and which workload can be reduced to protect alarms?	WF-37
BMS 07	Old edge → journal/cursor → standby edge recovery lane, with separate evidence checklist and writer lock.	Is the old writer actually fenced, and which alarms/commands remain unreconciled?	WF-37
BMS 08	Panel → BMS → SmartAlarm → writeback causation path; the echo leg has its own shape/label while raw receipts are retained.	Is this event new evidence, a duplicate, or our own echo?	WF-35

State, interaction and access contract

Use server-projected connection states draft/validating/qualified/active_read/active_control/degraded/draining/suspended/retired without inferring them from a connected socket. Mutations follow their own receipt projection and link to PhysicalCommand where relevant. Preserve unknown and superseded states. On changes to target, occurrence, profile, unit, revision, approval or source epoch, discard the previous preview and request fresh server evidence.

All privileged controls require both user authority and admitted connector capability. Disabled controls expose a visible reason; they do not rely on a tooltip. A read-only user may inspect the allowed evidence but cannot approve their own request. Keyboard focus moves to preview results; changing inputs invalidates previous results. Status updates use aria-live without forcing unrelated focus changes. At390px, panels stack, tables wrap and topology becomes vertical; diagrams are descriptive regions, not the only text alternative.

Prototype scope: local scenario selection, validation previews, queue simulation, handover checklist and quarantine preview are implemented. The prototype sends no API call, native ACK or equipment command. Production persistence, authorization and actual vendor behavior are specified by the API contract, FRD and Detail Design and still require implementation and FAT/SAT. The supplemental review does not claim human usability validation.

Screen contracts · API contract · BMS UI QA · UX acceptance matrix

SmartAlarm — Physical device compatibility catalog



Concept illustration — proposed integration, not an installed system.

Research date: **2026-10-23**. Status: **document-verified candidates; no hardware integration, procurement, or field acceptance completed**. This catalog supplements WF-03, WF-14, WF-15, and WF-17. The Vietnamese and English editions contain the same scope and conclusions.

Current design baseline: 3.0 · BMS integration: 24 October 2026. Original research/requirement dates above are preserved.

1. Deployment decision

Use manufacturer-supported interfaces to observe existing fire panels, pump controllers, and releasing panels. Keep their local automatic protection functions operational independently of SmartAlarm. A protocol connection, a writable BACnet property, a pump-running signal, and confirmed water discharge are four different capabilities. None alone demonstrates that SmartAlarm may safely initiate suppression or that water reaches the protected hazard.

The six candidates below are alternatives or complementary components, not a single certified assembly. Availability, exact firmware, regional approval, optional parts, and permitted actions require project-specific vendor confirmation. A manufacturer's listing of its own component does not certify the complete SmartAlarm installation.

2. Six concrete candidates

ID / product binding	Documented interface and feedback	Command boundary and unresolved items	Proposed SmartAlarm use
PHY-01 — Siemens Cerberus PRO FC922 / FC924; FCA2034-A1 S2 or FCA2035-A1 S3	FC922/FC924 are identified in the panel datasheet A6V10349710. The US accessory datasheet A6V10349679 describes S2 BACnet as monitoring-only; S3 retains monitoring-only BACnet while adding separate remote-viewer control.	Do not equate remote-viewer control with BACnet actuation. Obtain regional SKU, firmware, license scope, PICS/object export, and supported event states. No suppression-release capability is established here.	Read event/state data through a site edge adapter; preserve panel/point identifiers and communication quality.

ID / product binding	Documented interface and feedback	Command boundary and unresolved items	Proposed SmartAlarm use
PHY-02 — Honeywell NOTIFIER BACNET-GW-3 with NFS-320 / NFS2-640 / NFS2-3030 and appropriate NFN interface	DN-6877:C, 2014-08-04: BACnet/IP event objects; up to 14 NFN nodes and 15,000 objects; NCM/HS-NCM connection and browser configuration.	The listing C6, 2019-02-21 restricts use to supplementary functions and excludes triggering mass-notification announcements. Confirm current supply/support, firmware, chassis, power, and exact panel compatibility.	Supplementary monitoring; never treat this gateway as a replacement for the primary alarm or as a proven suppression command path.
PHY-03 — Simplex BACpac 4100-6111 with 4100ES / 4100U	S4100-0051 Rev.6, 10/2020, pp.1–4: 15,000-point variant; separate 4100-6038 RS-232 module; BACnet IP/MS/TP or Modbus TCP/RTU. Supplementary status interface.	Its PICS includes WriteProperty support; this does not establish an authorized panel-release command. Obtain the actual point mapping and permitted write semantics. Confirm software and separately ordered hardware.	Read-only status adapter for a specifically bound panel configuration.
PHY-04 — Tornatech GPX electric fire-pump controller, V2 communication profile	The GPx-V2 Modbus bulletin maps motor-run, voltage/current, pressure, and fault feedback. It also names remote/manual and automatic-start entries.	That short map does not fully specify write functions, enable conditions, persistence, or retry behavior. Do not automatically apply its V2 map to V2.1/current firmware. Exact controller suffix, options, and current register contract remain required.	Pump health and demand monitoring; command capability stays disabled until vendor-supported, project-approved validation.
PHY-05 — Tornatech GPD diesel fire-pump controller, V2 communication profile	The separate GPD-V2 Modbus bulletin maps engine-run, batteries, pressure, and diesel faults, plus named start entries.	GPX and GPD maps are not interchangeable. Confirm firmware, engine interface, installed sensors, writable registers, and automatic/manual mode behavior. Missing sensors must remain unknown.	Diesel readiness and pump-state monitoring with a dedicated adapter profile.
PHY-06 — Potter PFC-4410G3, stock 3006735	5401600 Rev.B, 4/23: pre-action/deluge releasing panel; alarm, trouble, supervisory, and waterflow/discharge-confirmation relays; Ethernet programming; proprietary RS-485 P-Link.	Ethernet/P-Link does not prove a public release API. Confirm output ratings, chosen program, compatible solenoid, feedback wiring, and optional relay modules. Do not assume releasing output equals waterflow confirmation.	Read approved relay indications through a supervised interface; local releasing panel owns suppression sequencing.

No additional interface license is established for PHY-02–06 by this review; this is not a claim that all configurations or engineering tools are free or included. Obtain a complete bill of materials and license statement.

3. Version and lifecycle findings that change implementation

Siemens regional distinction. The 2018 Siemens FC72x catalog describes FCA2034-A1 with basic and FCA2035-A1 with extended third-party BACnet commands. This differs from the US accessory wording above. Therefore the adapter key must include regional panel family and firmware, not just “Cerberus PRO” or the license label. The exact commands available at a particular site remain unverified.

NOTIFIER network constraints. Manual LS10014-000NF-E:D2, 2019-04-30, p.9 specifies a static IP and no NAT support. Proposed topology: keep the BACnet client in the permitted site network and export normalized events outward; do not assume a cloud VPN/NAT arrangement is supported.

Real pump start interfaces exist. Tornatech's current GPX manual V2026.09 r0, p.16 documents remote manual-start contact closure and remote automatic/deluge-start contact opening. These are physical controller inputs, not evidence of a cloud command API. Its GPX product page also identifies remote alarm contacts. Obtain the job-specific wiring diagram, contact ratings, supervision, and fail-state behavior before designing an I/O interface.

The GPD manual V2026.09 r0, pp.13–15 separately documents contact-based remote starts. Automatic stopping is not enabled by default and requires authority-having-jurisdiction authorization; commissioning requires an authorized provider. SmartAlarm must not introduce an unvalidated remote-stop function.

Select the current releasing panel deliberately. Potter's official releasing-panel catalog marks PFC-4410RC obsolete. The G3 is the candidate for new procurement, subject to vendor confirmation. Its product page explicitly identifies email monitoring as a non-UL-listed feature; it is not the safety event transport. Use the G3 manual 5403751 Rev.G, 11/25 and the manufacturer's compatible releasing-device list to request confirmation of the exact panel/solenoid/valve assembly. No valve pairing is approved by this catalog.

4. Proposed integration contract — SmartAlarm design, not manufacturer claims

For each installation, bind `manufacturer + regional_model + controller_serial + firmware + gateway_model + gateway_firmware + license + point_map_revision`. Store every observed point with source address, semantic type, engineering unit, source timestamp where available, receive timestamp, quality, and communication status. A point-map revision must be reviewable and rollback-capable.

Proposed semantic	Required evidence	Must not be inferred
`fire_alarm_active`	Valid panel point/state mapping	Camera confidence alone is not a panel alarm
`pump_start_demand`	Documented demand contact/register	Motor actually running
`pump_running`	Valid controller run feedback	Adequate pressure, flow, or extinguishment
`system_pressure`	Installed transducer, unit/scaling, calibration	Flow at a specific protected zone
`release_output_active`	Documented releasing circuit/output feedback	Valve movement or water discharge
`waterflow_confirmed`	Commissioned waterflow/pressure feedback and zone binding	Fire extinguished or every nozzle operating
`communication_lost`	Link supervision and timeout	Normal, safe, pump stopped, or failed actuation

The default adapter capability is `observe`. A potential `request_start` or other command requires a distinct signed capability record describing authority, exact operation, conditions, local interlocks, TTL, duplicate handling, feedback, and recovery. An uncertain response stays `unknown`; reconnection must not replay expired physical requests. Do not add reset/silence/disable/stop actions merely because a protocol supports writing.

5. Vendor RFI before procurement

RFI	Required written answer/evidence
RFI-01 — Identity and lifecycle	Regional SKU, orderable status, end-of-support date, firmware matrix, substitutions, and spare-part availability
RFI-02 — Complete package	Gateway/modules, licenses, engineering tools, power/backup, enclosure, isolation, protection, and authorized commissioning scope
RFI-03 — Read contract	Current PICS/register map, supported event/state types, polling/subscription limits, address conventions, scaling, sequence/time behavior, and communication-loss semantics
RFI-04 — Command contract	Explicit allowed/prohibited operations; write function/property; enabling configuration; physical input semantics; local priority; TTL/replay behavior; confirmation and rollback limits
RFI-05 — Physical assembly	Approved valve/solenoid/controller combinations, electrical loads, hydraulic design responsibility, pressure/flow sensor placement, cause-and-effect matrix, and supervision
RFI-06 — Regional acceptance	Applicable adopted codes and editions, component listings and limitations, system designer approval, AHJ/site acceptance requirements; no blanket international-compliance assertion
RFI-07 — Cybersecurity/support	Network segmentation, credentials, supported updates, vulnerability process, maintenance access, logging, outage behavior, and recovery procedures

6. Commissioning evidence and release gate

The following are proposed SmartAlarm acceptance requirements, not a replacement for manufacturer or local mandatory tests.

1. Record serial numbers, firmware, license proof, approved drawings, point map, and baseline controller configuration. Obtain approval of the complete cause-and-effect matrix.

2. In a controlled test arrangement, verify each mapped alarm, supervisory, fault, pump demand/run, pressure, releasing-output, and waterflow indication against the physical system. Record expected versus observed states and timestamps.
3. Test broken links, adapter restart, stale data, duplicate events, power/backup changes, and reconnect. Verify that local automatic protection continues and no stale event creates an unintended command.
4. If any command is authorized, test its approved preconditions, denial paths, local override, feedback loss, and recovery with the qualified site team. Do not perform live discharge merely to validate software connectivity.
5. Obtain signed factory/site acceptance records, calibration records, operator training, maintenance ownership, and an approved impairment/restoration procedure. Mark the adapter `field_accepted` only for that exact configuration.

Procurement recommendation: first bind to the installed panel/controller family, then buy the supported interface and required options. For a greenfield water-suppression installation, have the fire-protection designer specify the physical hydraulic and releasing system before selecting the SmartAlarm adapter. No purchase or vendor contact has been made.

7. Evidence register and access limits

All sources below were reviewed on **2026-10-23**. Document dates are publisher revision dates, not retrieval dates. Online presence does not establish current stock or support. Only manufacturer sources support product claims; no reseller claim is used.

Evidence	Publisher document / date	Access and scope
E01	Siemens A6V10349710; A6V10349679	Official indexed title/accessory passages; direct PDF retrieval was blocked. Revision must be obtained with the quotation.
E02	Siemens FC72x catalog, ©2018	Official indexed license passage; full download exceeded browser size limit. Regional distinction only; not a current command contract.
E03	NOTIFIER DN-6877:C, 2014-08-04	Official indexed datasheet passages.
E04	NOTIFIER listing C6, 2019-02-21; manual D2, 2019-04-30	Official indexed limitation/configuration passages; direct retrieval unavailable. Request controlled copies and current revisions.
E05	Simplex S4100-0051 Rev.6, 10/2020	Official six-page PDF read; selection and PICS tables examined.
E06	Tornatech GPx-V2; GPD-V2	Official two-page maps read; V2 identity verified, no printed revision date established. URL upload path is not a revision guarantee.
E07	Tornatech GPX; GPD, both V2026.09 r0	Official manuals read, particularly starting/stopping and commissioning; current manuals do not by themselves validate the older V2 register maps.
E08	Potter 5401600 Rev.B, 4/23; 5403751 Rev.G, 11/25	Official datasheet and manual retrieved; interfaces and revision verified.
E09	Potter G3 product page; releasing-panel catalog; device compatibility list	Live manufacturer pages reviewed; compatibility-list link retrieved but subsequent fetch timed out, so no specific valve/solenoid pairing is asserted.

Open evidence gaps remain procurement gates, not proof of incompatibility. This research demonstrates feasible integration paths and specific restrictions; it does not report completed interoperability or certify a deployment.

UX2-PHYSICAL · Visual supervision and physical feedback

The v2 console uses a pipe/valve/pump mimic, zone coverage and explicit device state. A requested command, gateway acknowledgement and observed physical effect are separate states. Unknown or stale feedback must use a labeled pattern and timestamp, never a green success state. Review-only actions in the prototype do not operate equipment. Local fire protection authority, interlocks and commissioning requirements in this document remain binding.

Verify F08–F20 with communication loss, conflicting local state and delayed feedback. Operator access to the schematic must not bypass command authorization. PA playback feedback is not proof of audibility to every occupant. Drone/robot task acceptance is not proof of arrival or rescue.

See UI/UX design, screen specifications and redesign audit.

8. Physical devices through BMS — implementation and qualification

A BMS is an observation/coordination path to physical systems, not a replacement for controller qualification. Register both native device identity and the BMS mirror path; a panel read directly and through BMS is still one physical source. F01/F07/F08 show source path, time, quality and mapping revision; BMS03 retains lineage across alarm paths.

Vendor profiles incorporated into the existing catalogue

Profile	Product and interface	Evidence and qualification
`bms.siemens.desigo-wsi`	Desigo CC REST / WSI subscriptions	V8 specification describes REST read/write, events and history. WSI authentication has version-specific session/heartbeat behavior. Verified northbound OPC is DA, not UA. Qualify project/extension, HTTPS and PICS for a BACnet binding. Siemens WSI
`bms.schneider.ebo-ews`	EcoStruxure Building Operation SOAP EWS / SmartConnector REST	The SmartConnector2.6/EWS1.2 guide was tested against EBO7.1.2. EBO3+ uses SoapEwsRestProvider. Do not assume root discovery or bulk support. Acknowledgement targets an alarm event, not every alarm-state transition. EBO EWS
`bms.jci.metasys-rest`	Metasys Version-pinned REST / SSE	API v6 documents objects, command schemas, alarms, histories and SSE. Inspect each batch result rather than HTTP200 alone. Monitoring and Commanding needs separate licensing; the Metasys name alone does not establish compatibility. Metasys API
`bms.honeywell.ebi-api`	Enterprise Buildings Integrator Vendor SDK / web services	EBI2025 advertises enterprise web services, but the public page does not provide an implementation contract. Older R600 guide specifications discuss OPC DA/A&E/HDA and APIs; they do not establish equivalent REST or UA capability in EBI2025. EBI
`bms.honeywell.webs-niagara`	WEBs N4 Niagara OEM station interface	WEBs is Niagara-based and needs an OEM profile. Modules, station exports, permissions and licensing determine capabilities. Do not reuse EBI APIs. WEBs N4
`bms.tridium.niagara-local`	Niagara local station / Supervisor Qualified oBIX / licensed module	Supervisor4.6 documentation and the 2025 driver catalog identify oBIX. A marketplace REST module is a separate capability, not universal built-in REST. Alarm acknowledgement requires a qualified contract. Niagara local
`bms.tridium.nds-cloud`	Niagara Data Service Cloud API	NDS is a separate cloud service for histories/tags, live read/write and alarm queries. Prerequisites include SMA and a reseller. Alarm acknowledgement is unverified; WAN cloud is not a critical local-control fallback. Niagara Data Service
`bms.delta.enteliweb`	enteliWEB Vendor API / BACnet Web Services	The 4.23 catalog, updated November2022, lists an -API add-on. Confirm current entitlement; a brochure is insufficient for authentication, endpoints and lifecycle implementation. enteliWEB
`bms.alc.webctrl-events`	WebCTRL event/telemetry integration Outbound GET/POST / MQTT add-on	The 2025 sheet mentions alarm REST and outbound alarm-triggered GET/POST; MQTT is an add-on. Outbound events do not prove REST point commands or native acknowledgement. Initial scope is event/telemetry intake. WebCTRL
`bms.alc.webctrl-soap-sdk`	WebCTRL SOAP / SDK SOAP / licensed Java SDK	SOAP needs a version-specific method contract; JavaSDK licensing is case-by-case. Qualify read, acknowledgement, commands and history by method and account scope. WebCTRL SDK

Profile	Product and interface	Evidence and qualification
`bms.distech.ecnet-niagara`	EC-Net Facilities / EC-Net4 Niagara OEM station interface	EC-Net Facilities is the renamed Niagara-based EC-Net4. ECLYPSE REST belongs to the controller family, not automatically the EC-Net Supervisor. Qualify OEM modules and station access. EC-Net

Building-services boundaries

Subsystem	Required mapped data	Control and acceptance boundary
Fire panel / releasing	Alarm/trouble/supervisory, occurrence, zone, inhibit, local state	A BMS mirror does not grant reset/silence/release authority. Only qualified interfaces/models and approved cause-and-effect may be considered for commands.
Pumps/valves/ tanks/flow	Run contact, pressure, valve limits, tank level, waterflow, quality/source time	Run/valve-open does not prove adequate flow. Use independent observations and expected effects; unknown remains unknown.
HVAC / smoke control	Fan command versus run, damper command versus limit, smoke zone, override	Comfort HVAC and smoke-control modes have different authority. Do not operate fans from an alarm rule without site-engineer approval.
Doors / access / lifts	Door contact, lock state, accessibility, lift availability/recall	Door-open does not prove a safe route; lift availability does not establish fire evacuation suitability. Never bypass controller interlocks.
PA / exit signage	Zone/group, emergency priority, device report, playback/fault	BMS signals are qualified rule triggers, not substitutes for PA supervision or intelligibility acceptance.

Required commissioning bundle for each BMS path

Record physical-controller manufacturer/model/firmware, BMS product/build, interface/provider/license, endpoint/principal scope, raw address and applicable byte order, units/enums/status mappings, topology, update semantics, latency/freshness budgets, read-only fixtures, any command allowlist, local priority/override, expected feedback and test permits. F17 stores bundle hashes/revisions and sign-off. Research JSON profiles are design-only; runtime admission requires evidence for this exact combination. Brochures or BTL listings do not establish per-method command permission.

SmartAlarm — Rescue, firefighting and IoT device catalog



Concept illustration — proposed integration, not an installed system.

Research snapshot: **2026-10-23**. Six deployment candidates researched against official manufacturer documentation. Published capability is not proof of integration with SmartAlarm; all adapters remain proposed until bench and site acceptance. This catalog adds physical intervention to sensing, warning, positioning and dispatch. No equipment has been purchased or commissioned.

Current design baseline: 3.0 · BMS integration: 24 October 2026. Original research/requirement dates above are preserved.

1. Evidence and deployment decision

ID	Exact candidate	Physical role	Integration evidence	Proposed pilot disposition
DEV-01	AXIS C1310-E Mk II Network Horn Speaker	Fixed-site audible instructions	Published VAPIX and SIP interfaces	Bench adapter, then acoustic site test
DEV-02	Bosch PRAESENSA with PRA-SCL system controller	Supervised public address / emergency voice system	Published Open Interface	Accredited integrator review before any emergency control
DEV-03	DJI Matrice 4T + DJI AS1 Speaker	Mobile aerial instructions and search support	Operator features verified; exact AS1 command API unverified	Operator-assisted workflow initially
DEV-04	DJI Matrice 400 + Zenmuse V1	Aerial voice delivery with developer integration path	Official PSDK V1 widget control and MSDK product statement	SDK proof of concept, then flight exercise
DEV-05	Shark Robotics Colossus, water/foam monitor and casualty-evacuation configuration	Physical suppression, reconnaissance and casualty transport	Vendor remote control; public integration API unknown	Task dispatch with trained vendor-console operator
DEV-06	FLIR A700 with Smart Sensor configuration	Thermal observation and early-warning evidence	Published MQTT, Modbus TCP and REST	Read-only sensor gateway pilot

2. DEV-01 — Fixed network horn

Verified: The C1310-E Mk II datasheet lists VAPIX, SIP, SRTP/SIPS, speaker-test results and audio-clip-playing events. These are device capabilities, not evidence that people heard the message. Official datasheet.

Proposed integration: Edge adapter maps approved message/zone intent to the pinned VAPIX API and routes live audio through a tested SIP media gateway. Maintain a speaker-to-zone inventory; distinguish command receipt, playback evidence and human acknowledgment. Never infer coverage from network-online status alone.

RFI / limits: Request exact firmware and API versions, audio upload/play/stop methods, authentication, clip capacity, feedback semantics, management licensing and lifecycle/support dates. Confirm whether the purchased package needs separate audio-manager or third-party integration licenses; no free entitlement is assumed. Do not specify this network horn as a certified replacement for the site's mandatory fire voice alarm without an applicable certificate and approved design.

SAT-01: Play an approved recording only in the selected zone; test wrong-zone rejection, amplifier/acoustic fault, certificate failure, power loss, WAN loss and expired-command reconnection. Measure intelligibility under actual noise and occupancy, including Vietnamese/English/Japanese recordings.

3. DEV-02 — Supervised emergency PA

Verified: PRA-SCL exposes an Open Interface and supervises stored messages; its official product record identifies the controller. PRA-SCL product. The vendor publishes native and C#/.NET Open Interface documentation. Programming manual. Configuration documentation includes TLS and an option to disable emergency control through the interface. Configuration manual.

Certification boundary: The vendor describes PRAESENSA's EN 54-16 capability; applicability depends on the installed system. Official system page. Earlier official configuration instructions explicitly warn that Open Interface use can compromise evacuation-standard compliance. Integration restrictions.

Proposed integration: Regional SmartAlarm command policy → local approved adapter → controller; start with health/events and authorized general announcements. The local certified alarm path retains priority. Emergency initiation, acknowledgment or reset requires the integrator's documented permitted design; a generic API acknowledgment is not acoustic completion.

RFI / limits: Obtain the exact current firmware/manual/certificate set, permitted interface commands, zone/priority arbitration, failure supervision, redundancy, power backup and approved fire-panel interface. Confirm required controller, amplifier, PSU, call station, loudspeaker and optional subsystem/software licenses in a complete bill of materials. A controller alone is not a complete certified installation. Older Bosch download links may redirect; use the current vendor documentation portal and pin downloaded revisions at procurement.

SAT-02: Test local fire priority during SmartAlarm announcements, loss of adapter/network/controller, supervised circuit faults, message expiry, redundant failover, and prohibited emergency-reset requests. The integrator and site authority must accept the complete installed configuration.

4. DEV-03 — Compact drone and AS1

Verified: DJI identifies AS1 compatibility with Matrice 4 Series and supports recording, imported media, TTS and live broadcasting. Its FAQ currently lists Mandarin and English for AS1 TTS and lists aircraft SDK support. These statements do not establish a supported SmartAlarm API for each AS1 action. Official Matrice 4 support.

Proposed integration: SmartAlarm prepares an approved mission and audio asset; the pilot uses DJI's supported operator workflow, then records the operational result. Treat SDK-based speaker start/stop/upload/status and live-audio injection as **UNKNOWN pending exact model/firmware documentation and demonstration**. Do not reuse a generic PSDK or Dock speaker command solely because its name resembles the required operation.

RFI / limits: Obtain an AS1-specific SDK capability matrix, SDK licensing/access conditions, supported firmware, stream/codec/file limits, cancellation behavior and feedback states. For Vietnamese/Japanese, validate approved imported recordings; do not claim native TTS support. Confirm payload-dependent flight endurance, legal operating conditions, maintenance and operator training.

SAT-03: Demonstrate correct-language playback, intelligibility under rotor noise, operator abort, lost radio link, depleted battery, duplicated mission requests and stale queued content. Verify no automatic takeoff follows an alert. Demonstrate manual fallback if integration is unavailable.

5. DEV-04 — Matrice 400 and Zenmuse V1

Verified: DJI documents V1 compatibility with Matrice 400, live broadcasting through Pilot 2 and an MSDK integration route. Official V1 product. Its PSDK documentation explicitly describes controlling the V1 on Matrice 400 through widget values. PSDK custom-widget interface. V1 support lists Japanese among Matrice 400 TTS languages, but not Vietnamese. Official V1 support.

Proposed integration: A version-pinned mobile or onboard adapter translates approved voice commands into only the documented supported SDK operations. Separate mission/flight authorization from payload playback authorization. SDK availability does not establish parity with all Pilot 2 or FlightHub 2 functions, and does not prove unrestricted cloud-to-speaker live streaming.

RFI / limits: Require exact SDK versions, keys/activation, compatible host hardware, widget definitions, command/status schema, media transport, permissions and callback semantics. Confirm any FlightHub 2 subscriptions separately if that route is selected. Test approved Vietnamese recordings; confirm Japanese pronunciation and firmware-specific TTS. Request a complete airframe/payload/mount/controller/battery bill of materials.

SAT-04: On the purchased combination, exercise upload/play/stop/repeat/volume and reported state only where officially exposed. Test concurrent operators, cancellation, link loss and pilot takeover. A callback success must not be shown as “people heard”. Conduct a permitted outdoor exercise before operational use.

6. DEV-05 — Physical firefighting and casualty rescue robot

Verified: Colossus is remotely controlled and offers water/foam, thermal/video and stretcher configurations. The current page advertises flow up to 3,000 L/min; actual suppression requires the appropriate water supply and equipment. The page does not establish a public third-party control API. Official Colossus.

Proposed integration: SmartAlarm dispatches a robot mission to a trained team, tracks operator acceptance and evidence, and imports video/telemetry only through a vendor-supported interface after verification. Keep driving, monitor aiming, water discharge and casualty movement under the qualified operator's control. **API, ROS, MQTT, ONVIF, remote E-stop and autonomous navigation compatibility are UNKNOWN unless the supplier publishes or supplies applicable documentation.** No protocol is inferred from a camera or radio being present.

RFI / limits: Request exact robot generation, module SKUs, supplied console, API/SDK availability and commercial terms, video export, telemetry, loss-link behavior, independent emergency stop, safety interlocks, heat-exposure limits, hose pressure/flow requirements, foam compatibility, transport and maintenance. Confirm payload/stretcher constraints and patient-handling procedure. Reconcile differing module counts and slope figures on vendor materials before contracting; this catalog makes no operational commitment from those figures.

SAT-05: With the fire/rescue authority, test access width, floor loading, ramps/debris, hose drag, water-supply failure, nozzle reaction, loss link, stop distance, thermal limits and battery reserve. Perform casualty loading/transport using a training manikin and approved crew procedure. Confirm that SmartAlarm duplicates or outages cannot initiate movement or discharge.

7. DEV-06 — Thermal IoT sensor

Verified: The official Axxx table lists MQTT, Modbus TCP and REST alarm/readout paths for the A700 Smart Sensor family. A500 rows differ, so A500 must not be substituted based on the family name. Official product/configuration table. FLIR lists optional Advanced Smart Sensor configuration and separately purchased Wi-Fi/visual-camera options. Official configuration support.

Proposed integration: Start read-only: edge gateway reads ROI temperatures/alarms and emits source-bound observations with device time, receipt time, calibration/configuration version and quality. A threshold event is evidence requiring the relevant alarm policy, not automatic proof of a confirmed fire or authority to release suppression.

RFI / limits: Require exact A700 SKU, Smart Sensor edition, lens, firmware and the relevant MQTT payload schema, Modbus register map and REST manual. Confirm authentication/TLS availability per selected interface; do not assume Modbus is encrypted. Confirm ROI/analysis limits, calibration, environmental enclosure and optional licenses. Source values are not site accuracy guarantees.

SAT-06: Use a controlled reference target; test emissivity/reflection effects, occlusion, dirty optics, temperature threshold/hysteresis, timestamp drift, network interruption, duplicate/replayed observations and faulty-sensor states. Verify degraded sensor data cannot silently become a clear/no-hazard assertion.

8. Cross-device procurement and release gate

For each purchased configuration, record model/serial, firmware, licenses, interface revision, credential owner, network zone, allowed commands, approval role, local override, expiry, receipt semantics, maintenance owner and support end date. Attach vendor evidence and bench/SAT results. No candidate is “production integrated” until these records and negative tests pass.

Physical action follows **approved incident task → qualified operator/authority → device interlocks → execution evidence → reconciliation**. SmartAlarm coordinates the response; the certified fire system, pilot and rescue operator retain their respective local control responsibilities. An API success, a hot pixel, a location estimate or an AI recommendation must never independently establish that a person is safe or that a rescue action succeeded.

UX2-PHYSICAL · Visual supervision and physical feedback

The v2 console uses a pipe/valve/pump mimic, zone coverage and explicit device state. A requested command, gateway acknowledgement and observed physical effect are separate states. Unknown or stale feedback must use a labeled pattern and timestamp, never a green success state. Review-only actions in the prototype do not operate equipment. Local fire protection authority, interlocks and commissioning requirements in this document remain binding.

Verify F08–F20 with communication loss, conflicting local state and delayed feedback. Operator access to the schematic must not bypass command authorization. PA playback feedback is not proof of audibility to every occupant. Drone/robot task acceptance is not proof of arrival or rescue.

See UI/UX design, screen specifications and redesign audit.

9. Rescue equipment through BMS — information routing and ownership

Equipment may be visible through a direct PA API, BMS mirror, drone gateway or vendor console. EquipmentRegistry retains one equipment identity with multiple source endpoints, each with separate capability evidence. A BMS mirror of speakers, robots or water monitors cannot inherit direct-connector command authority. Actual/exercise mode propagates end-to-end; exercise data cannot create real equipment tasks.

Equipment	BMS data role	Required integration	Acceptance
Emergency PA	Ready/fault, zone, emergency priority and available playback reports	Map site groups; the direct PA adapter owns sessions; BMS provides qualified triggers/evidence	Echo/replay cannot rebroadcast; local override wins; playback does not prove audibility
Ordinary IP speaker	Model-dependent connectivity/playback	Classify separately from life-safety PA; do not replace supervised power/wiring design	Power loss/offline is unknown/unavailable, never evidence of PA certification
Drone	Alarm/site coordinates/zone and task context	BMS does not fly drones; route tasks through the drone gateway and pilot authority with geofence/no-fly/permits	Accepted is not arrived; BMS loss does not change drone lost-link behavior
Robot / water monitor	Access constraints, water-supply observations and target zone	Commands use qualified interfaces or operator tasks at vendor consoles; separate readback	BMS pump-run does not establish water at the robot; telemetry cannot trigger arbitrary suppression
IoT / wearable	Point observations and aggregate presence	Normalize source time/quality/provenance; avoid direct-plus-mirror double counts	Occupancy/wearable silence does not establish personal safety

Coordination runbook. BMS alarms first enter incident/correlation processing, then capability-based task/team selection. Task context includes location uncertainty, related equipment/cause-and-effect and source age. Responders confirm local access and observed state before effects. Handover preserves task owner/time/evidence and cannot complete automatically on BMS alarm clear. Test with FR-BMS-13/19/20/21/25/27 and WF-33/35/36; signed site permits remain deployment prerequisites.

SmartAlarm — External Integration Standards and Deployment



Concept illustration — proposed integration, not an installed system.

Research/design baseline 3.0 · 24 October 2026 · English. Scope: physical fire protection integration, deployment engineering, and acceptance. This is a proposed integration design, not a certificate, installation approval, or substitute for the site fire-protection design. Manufacturer/model research is documented separately. The architecture below is a SmartAlarm proposal; cited standards summaries describe verified public scope rather than unverified clauses.

1. Decision and boundary

SmartAlarm shall support real physical fire-protection systems through approved interfaces while preserving autonomous local protection. It may ingest fire/supervisory/trouble states, coordinate responders, publish approved evacuation guidance, and request explicitly authorized actions. It must not require the cloud, Internet, a mobile app, or an LLM to complete the local automatic protective sequence.

Do not implement a universal “turn on sprinklers” API. Conventional heat-operated sprinkler heads act locally and individually; an interface to a fire panel does not imply that those heads are electrically addressable. Deluge, preaction, gas suppression, pumps, smoke control, and robotic monitors have different controllers, prerequisites, and consequences. Their permitted actions must be identified by system type and exact manufacturer configuration. The distinction between individual heat operation and software control is supported by the Pennsylvania Office of the State Fire Commissioner.

Three approvals remain separate: the applicable local code/edition and authority having jurisdiction (AHJ); the equipment listing/certification and manufacturer compatibility constraints; and acceptance of the complete installed integration. A protocol, a compliant component, or a cloud test does not establish all three. Record local adoption explicitly: an example official 2026 notice from Poudre Fire Authority cites earlier editions for specific inspection work, illustrating why the newest publication is not automatically the project edition.

2. Verified standards and interface matrix

Publication versions below were checked on 23 October 2026. The project must separately freeze adopted editions, amendments, equipment approvals, and contractual requirements. Exact paywalled clauses, installation dimensions, hydraulic calculations, and test intervals have not been inferred from public summaries.

Reference / verified version	Verified scope	Proposed SmartAlarm use and limit
NFPA 72, 2025	Fire alarm, signaling, emergency communication, and mass notification	Map alarm/supervisory/trouble and approved communication interfaces; no claim that SmartAlarm is a listed fire alarm panel. NFPA
NFPA 13, 2025	Design and installation of automatic sprinkler systems	Preserve the engineered sprinkler installation; collect waterflow/valve supervision through approved interfaces. Software does not replace sprinkler design. NFPA
NFPA 20, 2025	Selection and installation of stationary fire-protection pumps	Integrate approved controller status; any permitted remote request remains subordinate to the approved controller design. NFPA
NFPA 25, 2026	Inspection, testing, and maintenance of water-based fire protection	Track deficiencies, impairment, restoration, and evidence; schedules must be set from the adopted text and equipment instructions. Edition corroborated by NFPA's 2026 handbook and ANSI's NFPA 25 record.
EN 54-2 / -4 / -13	Control/indicating equipment; power supplies; component compatibility	Require exact certificates, configurations, and applicable national editions; no single “EN 54 certified system” shortcut. Public scope: BSI fire standards guide. Edition/amendment freeze remains a procurement task for these parts.
EN 54-16:2008 / EN 54-24:2008	Voice alarm control/indicating equipment; voice alarm loudspeakers	Use an approved voice alarm chain with zone mapping, priority, power/fault monitoring, and commissioned performance; an ordinary IP speaker is not automatically equivalent. BSI certification guide
ISO 7240-5:2018	Point-type heat-detector requirements and testing	Record detector class and approved application; not a universal environmental suitability claim. ISO
ISO 7240-16:2007 / -19:2007, both confirmed in 2025	Emergency sound control equipment; emergency sound-system design, installation, commissioning, and service	Distinguish component evidence from installed audibility/intelligibility and operating procedures. ISO part 16, ISO part 19
BACnet / ISO 16484-5:2026, edition 8, published August 2026	Building-system data communication and object-oriented representation	Require supported objects/services, revision, writable-property allowlist, priorities, and quality semantics. Interoperability does not authorize a life-safety action. ISO, BACnet committee
Modbus Application Protocol V1.1b3; Modbus Security	Transactions/function codes; TLS-based authentication and integrity for the security profile	Require exact register map, scaling, byte/word order, access mode, and supported security profile. Do not assume ordinary Modbus/TCP has Modbus Security. Modbus Organization
OPC UA / IEC 62541; verified part 2:2026	OPC UA security model	Require server namespace, node IDs, profiles, certificates, security policy, user rights, and status/timestamp semantics. Ordinary OPC UA use is not proof of a safety-certified control path. IEC
IEC 62443-3-2:2020; ISA/IEC 62443 series	IACS risk assessment, zones/conduits, target security levels; lifecycle security	Produce site risk assessment and zone/conduit controls. Do not assign a security level or claim certification merely because the gateway uses TLS. IEC, ISA
MQTT 5.0, OASIS Standard 7 March 2019	Publish/subscribe messaging	Edge telemetry transport with authentication, ACLs, expiry, deduplication, and bounded queues; QoS is not physical completion. OASIS
CloudEvents; deployed schema version must be pinned	Common event description/envelope	Carry source, type, event ID, time, and versioned domain payload; does not certify detection or actuation. CloudEvents
CAP 1.2, OASIS Standard 1 July 2010	Common alert exchange	Public advisories with source/update/cancel/area/expiry handling; never a direct suppression-release protocol. OASIS
MAVLink Command Protocol; vendor/flight-stack version pinned per aircraft	Command transport and command acknowledgments	Capability-specific gateway and pilot authority; ACK and physical outcome remain distinct. Protocol support grants no flight authorization. MAVLink
Aviation rules of the operating jurisdiction	Operator, aircraft, airspace, and mission authorization	Record jurisdiction, permission, pilot, flight constraints, and emergency procedure. The FAA public-safety guide is a United States example, not a worldwide permit.

Reference / verified version	Verified scope	Proposed SmartAlarm use and limit
WCAG 2.2	Web content accessibility	AA design target, with keyboard, focus, contrast, target, status, and text alternatives verified independently; does not establish acoustic coverage. W3C

Fire suppression media, special hazards, pressure equipment, electrical installation, smoke-control systems, and local fire/building law can add other standards. The site designer must identify these from the actual system; this matrix is not an exhaustive code list.

3. Proposed deployment architecture

```
flowchart TB
  CLOUD[SmartAlarm regional cloud: incident / policy / approval / audit]
  DMZ[OT DMZ: authenticated broker or reverse connection / firewall / monitoring]
  EDGE[Site edge: durable events / signed policy / command ledger / fencing]
  VG[Approved vendor gateway: supported API or supervised interface]
  LOCAL[Local approved fire / releasing / pump / voice controllers]
  DEV[Detectors / valves / pumps / suppression devices / speakers]
  MAN[Authorized local controls / manual stations / pilot]
  CLOUD <-->|events and bounded approved intents| DMZ
  DMZ <--> EDGE
  EDGE <--> VG
  VG <--> LOCAL
  LOCAL <--> DEV
  MAN --> LOCAL
  DEV -->|local initiating inputs| LOCAL
```

The local controller/device path runs independently of the cloud, DMZ, edge, identity service, and LLM. Existing approved hardwired/manual paths remain authoritative. SmartAlarm's edge is an integration component unless separately assessed and approved for another role; “edge” does not itself mean a safety controller.

Place cloud-facing services in an OT DMZ with explicit conduits to site systems. Prefer an authenticated site-initiated connection where supported; prohibit public Internet access to panel buses and general-purpose remote desktop into fire controllers. Separate telemetry ingestion from command authorization. Broker ACLs, firewalls, service identities, signed configurations, certificate rotation, and bounded maintenance access enforce that separation. These are proposed controls derived from the zone/conduit approach, not a claimed IEC 62443 certification.

Use a manufacturer-approved gateway or supervised I/O only where the exact design permits it. Specify input supervision, fault detection, isolation, wiring, electrical ratings, feedback, and commissioning evidence in the approved installation documents. No generic relay board bypasses a releasing panel, pump controller, interlock, or evacuation priority controller. Do not attach undocumented writes to a maintenance port. If only a common alarm contact exists, report common alarm—not a fabricated detector address or zone.

4. Integration permission tiers

Tier	Permitted behavior	Evidence required before enabling
T0 Inventory	Register assets/documents; no live connection	Site owner, exact models, location and ownership
T1 Observe	Read alarm, supervisory, trouble, flow, pressure, health, and status where available	Approved read path, point list, semantic mapping, FAT/SAT
T2 Operational coordination	Create incident/tasks; publish approved communications through the permitted voice interface	Approved message/zone/priority policy and verified delivery feedback
T3 Restricted control request	Only explicitly approved target/capability combinations	Manufacturer authorization, site engineering, AHJ/other acceptance as applicable, signed cause-and-effect revision, local interlocks, completion evidence

Tier	Permitted behavior	Evidence required before enabling
T4 Local automatic protection	Executes in the local approved protection system	Existing/local system design and acceptance; not granted to the cloud by a software entitlement

Default new connectors to T1 or lower. Alarm acknowledgment, silence, reset, disable/isolate, pump requests, and suppression release are distinct capabilities. Do not make a single generic “control” permission enable them all. Some remain local-only. An incident commander may coordinate the incident without possessing authority to reset or inhibit protection.

5. Asset, point, zone, and cause-and-effect contract

Each binding is versioned and reviewable before activation. Minimum contract fields:

Object	Required information
Asset	tenant_id, site_id, asset_id, controller_id, manufacturer, exact model, firmware, serial, controller/gateway relationship, protocol revision, certificate/listing reference and scope, maintenance owner, maintenance_state, impairment_ref
Point	point_id, controller_id, zone_id, raw address/object/node/register, type, read/write permissions, unit, scale, byte/word order if applicable, enumeration, normal/alarm/trouble/supervisory mapping, validity, observed_at, received_at, timestamp source, quality, stale_after
Zone	zone_id, physical boundary/floor, fire compartment reference, detection/suppression/voice zone types, point membership, exclusions, revision and approver
Capability	capability_id, allowed target types, parameter schema/ranges, local preconditions/interlocks, manual/automatic authority, authorization_policy, command_ttl, expiry, idempotency_key, retry class, feedback_point_ids, completion criterion, unknown_outcome_policy
Cause and effect	revision, initiating condition, required local confirmations, effect, affected zone, priority, delay if approved, interlocks/inhibits, reset authority, failure response, test case and signoff
Evidence	source time, receive time, source quality, raw payload reference/hash, connector/firmware/config versions, operator and approval, command/result correlation

Never equate a location zone with a suppression zone automatically. A floor may have several control zones and a control zone may cross UI boundaries. Changes to device addressing or zoning invalidate the affected mapping and require review/test before write capabilities resume. Quality states are GOOD, BAD, STALE, UNKNOWN, and TEST; absence of data is not NORMAL.

A cause-and-effect record is the approved engineer/vendor/site document reflected in software, not a rule invented by an LLM. For example, a locally approved releasing sequence may report its initiation, pending state, interlock condition, and discharge feedback to SmartAlarm. SmartAlarm records and coordinates these states; it must not infer authorization to release from camera confidence or silence alarms as a side effect of closing an incident.

6. Command lifecycle and physical evidence

Proposed states: DRAFT → PENDING_APPROVAL → AUTHORIZED → SENT → ACCEPTED → EXECUTING → COMPLETED_VERIFIED; alternatives REJECTED, EXPIRED, FAILED, UNKNOWN, and RECONCILED. A controller may omit EXECUTING; the adapter must declare which states are observable. UNKNOWN is mandatory when the command may have arrived but the physical outcome is unverified.

Bind approval to command_id, target, capability, parameter digest, zone revision, cause-and-effect revision, requester, approver, writer_epoch, nonce, issued_at, and expires_at. Edge and vendor gateway enforce allowlists, freshness, authorization, and local controller acceptance. Local protection/interlocks take precedence. Mutating parameters or zone revisions invalidates approval. Failover fences the old writer; it never blindly repeats an uncertain physical action.

Record transport receipt, controller acceptance, and physical feedback separately. A pump-request ACK is not pressure/flow evidence; a voice-buffer ACK is not confirmed playback or intelligibility; a valve command is not confirmed position. Completion criteria are capability-specific and may require more than one instrument. Network timeout after transmission produces

UNKNOWN, blocks automatic non-idempotent retry, and opens reconciliation with local staff/controller status. A software cancellation cannot promise to reverse released agent or a completed physical action. Reset and restoration follow a separate authorized process.

7. Failure-mode analysis and degraded operation

This is the initial qualitative FMEA; the site team assigns severity, likelihood, detectability, residual risk, and owners using site evidence. No generic numerical risk score is asserted.

Failure / consequence	Detection	Proposed containment and acceptance evidence
Cloud/Internet loss / remote visibility lost	Heartbeat age and connection state	Local automatic protection continues; edge buffers events; UI shows stale; prove with disconnection test
Edge loss / integration unavailable	Edge heartbeat and independent panel observation	Local sequence remains autonomous; cloud shows unavailable; restore from durable ledger without replaying uncertain writes; power-off edge test
Gateway restart or split brain / duplicate effect	Epoch/lease, durable command ledger	One fenced writer; uncertain writes reconciled, not replayed; reboot/failover tests
Wrong point or zone map / wrong action location	Independent point walkdown and revision digest	Disable affected write binding; verify every enabled target and negative cross-zone case
Stuck relay or failed feedback / false completion	Supervision and independent controller feedback where approved	No inferred success; local fault response and physical inspection under approved procedure
Lost power or depleted backup / incomplete coverage	Controller/power trouble and maintenance evidence	Preserve approved backup design; verify site-specific fail states and power restoration
Alarm storm / delayed critical event	Queue age, source rate, priority metrics	Reserve critical processing, bound queues, flag loss; local sequence never waits for broker
Compromised gateway / unauthorized request	Identity, allowlist, audit, anomaly monitoring	Revoke conduit/credential and isolate integration without disabling local protection
Maintenance mode confused with actual / unintended effect or ignored alarm	Explicit mode and local panel state	Separate test identities/routes, physical procedure, authorized test window, restoration checks
Clock skew or stale observation / unsafe premise	Source/receive times and quality flag	Block stale remote request; retain local protection; no timestamp rewrite hiding uncertainty
Controller ACCEPTED but no outcome / operator assumes success	Feedback deadline and missing independent evidence	UNKNOWN, local verification, alternative response; no blind retry
Voice zone unavailable / occupants not warned	Per-zone/controller fault and commissioned checks	Display coverage gap; initiate approved alternative communication procedure

8. Impairment, testing, and change governance

An impairment record contains affected assets/zones/functions, reason, start/end plan, responsible person, required notifications, compensatory measures determined by the site authority, access controls, and restoration proof. SmartAlarm does not decide that a degraded fire system is safe merely because alerts are acknowledged. Unplanned impairment escalates to the named site responsible person.

Test mode must have an explicit boundary. Software exercise labels alone cannot prevent a physical output from operating. Use only approved isolation/test arrangements, a responsible local technician, a reviewed test script, site notifications, and the required authorization. Simulator results do not authorize live agent discharge, pump operation, or emergency broadcasts. Live tests follow the adopted standard, vendor method, and local authority/site plan.

Configuration and firmware changes require impact assessment, signed version, rollback/recovery plan, and regression of affected cause-and-effect paths. No remote software rollback is assumed to restore a physical system state. Close impairment only after local restoration, normal supervisory state, target verification, outstanding-defect review, and signed handback. Test frequency and compensatory measures remain site-specific; this document invents neither.

9. RFI, survey, FAT, SAT, and handover

Gate	Required deliverable	Exit evidence
RFI / procurement	Exact panel/controller/device models, firmware, regional variants, supported APIs/registers, read/write permissions, gateway licenses, listing/certificates, compatible accessories, lifecycle and maintenance support	Vendor written confirmation of each proposed capability; unresolved items stay disabled
Site survey	As-built drawings, fire strategy, hydraulic/system boundaries, cause-and-effect, zone/point schedule, communications/power, AHJ/adopted editions, operational roles and impairment process	Designer/site owner reviews mapping and identifies missing documentation
Detailed binding design	Network zones/conduits, point contracts, command state/feedback, local-only controls, approved interface/wiring, migration and rollback plans	Engineering/vendor/site approval and authority acceptance where applicable
FAT	Representative exact hardware/firmware or documented scope-limited simulator; positive/negative mappings, access control, bad quality, clock skew, duplicates, lost ACK, timeout, expiry, restart, failover, no-Internet operation	Signed results, raw evidence, defects resolved; simulator limitations explicit
SAT	Installed point-to-point verification, physical zone confirmation, approved cause-and-effect tests, local autonomy on loss of cloud/edge, supervised faults, voice coverage/performance per approved plan, restoration	Witnessed site evidence; every enabled command has verified result semantics; no unresolved critical defect
Handover / operation	As-built configuration, credential ownership, backup/restore, training, runbooks, maintenance schedule, impairment contacts, support boundaries	Site owner acceptance; inspection/authority records where required; ongoing change control

Acceptance means more than a green dashboard. For each enabled physical capability, demonstrate that a denied request cannot actuate, stale/expired requests cannot actuate, an uncertain outcome is shown as UNKNOWN, local controls retain priority, and cloud failure cannot stop the independent automatic sequence. Record response times measured in the approved test; do not claim a universal latency target without the site/controller acceptance criterion.

10. Evidence register and remaining engineering decisions

All links in section 2 and this register were consulted on **2026-10-23**. Evidence type is public official scope/catalog/technical guidance; none is a full installation conformity assessment.

Evidence group	Official sources	Confirmed / remaining limitation
E01 Fire alarm and suppression	NFPA 72, NFPA 13, NFPA 20, NFPA 25 handbook, ANSI NFPA 25	Titles/editions/scope verified; adopted clauses/amendments must be checked for the project. NFPA 13 scope was available in official indexed content; direct retrieval was intermittent.
E02 Detection/voice	BSI components, BSI voice, ISO 7240-5, ISO 7240-16, ISO 7240-19	Public part scopes; national EN editions/amendments and exact product certificates require procurement verification.
E03 Building/OT data	ISO BACnet, BACnet committee, Modbus, IEC OPC UA	Protocol scope/version; not vendor compatibility or safety approval.
E04 OT security	IEC 62443-3-2, ISA series	Risk/lifecycle framework; site target levels and evidence remain to be engineered.
E05 Events and warning	MQTT, CloudEvents, CAP	Messaging scope; no guarantee of physical completion.

Evidence group	Official sources	Confirmed / remaining limitation
E06 Drone/accessibility /local context	MAVLink, FAA, WCAG, Pennsylvania sprinkler explanation, Poudre adopted-edition example	Interface, jurisdiction example, accessibility target, and local physical context; not global compliance.

Open decisions before implementation: deployment countries and authorities; installed system types and exact variants; permission to read versus write each point; vendor gateway and certificate scope; approved cause-and-effect; local-only actions; physical completion feedback; cybersecurity risk assessment; site-specific acceptance criteria and responsible signatories. These are required engineering inputs, not reasons to replace the full physical integration scope with a demo.

UX2-PHYSICAL · Visual supervision and physical feedback

The v2 console uses a pipe/valve/pump mimic, zone coverage and explicit device state. A requested command, gateway acknowledgement and observed physical effect are separate states. Unknown or stale feedback must use a labeled pattern and timestamp, never a green success state. Review-only actions in the prototype do not operate equipment. Local fire protection authority, interlocks and commissioning requirements in this document remain binding.

Verify F08–F20 with communication loss, conflicting local state and delayed feedback. Operator access to the schematic must not bypass command authorization. PA playback feedback is not proof of audibility to every occupant. Drone/robot task acceptance is not proof of arrival or rescue.

See UI/UX design, screen specifications and redesign audit.

11. BMS integration deployment contract — site implementation

Engineering bundle and commissioning sequence

1. **Survey:** inventory supervisors, servers/engines/controllers, product/build/firmware, API/module/provider/license, protocol/PICS, point/alarm dictionaries, network routes/ports, principal scopes, local emergency authority and maintenance windows. Record unknowns explicitly; unknown capabilities remain disabled.
2. **Read-only qualification:** configure endpoint allowlists, secret references/certificates, clocks and measured budgets. Discovery is bounded to approved roots/scope, cancellable and logged. Verify authentication, pagination, subscriptions/polling, native timestamps, quality, reconnect and active-alarm reconciliation before production admission.
3. **Mapping qualification:** stage and run golden fixtures for raw addresses, types, endianness, scale/offset, units, enums, severity ordering, native occurrences and ACK tokens. Map equipment/room/HVAC/fire-compartment/route/PA groups with typed relationships; verify direct/mirror lineage. Publish revisions with impact graphs and independent review.
4. **Runtime admission:** registry records adapter binary/hash, profile/mapping versions, qualified build/license and site acceptance. Servers calculate effective capabilities from current evidence. Exercises use separate namespaces and connector sessions; UI switches cannot enable writes.
5. **Site SAT:** measure actual load/bursts and clock quality; execute BMS-T01–20/BMS-P-01–28 per capability. Local protection/override remains cloud-independent. Record deviations, rollback/handover, owners and signed acceptance; reads/history/ACK/commands/release receive separate sign-off.
6. **Handover:** deliver data-freshness dashboards, alarm-gap runbooks, certificate/license expiry, support scopes, backup/restore, upgrade/cutover, priority-ownership cleanup and retirement. Update inventory and evidence when builds or equipment change.

Network and security

Site edges terminate BMS/OT communication in approved segments; northbound communication crosses authenticated OT-DMZ conduits with tenant/site binding and restricted egress. Do not expose BACnet/Modbus/OPC DA directly to the Internet. BACnet/SC, OPC UA security or Modbus Security requires correctly supported endpoint profiles; legacy TCP is not implicitly

encrypted. OPC DA adapters require a site-approved compatible Windows/DCOM bridge/runtime and cannot be relabeled OPC UA. Secrets reference a vault; audits/exports remove tokens and clinical data. Apply NIST OT segmentation and site risk assessment without treating a standard as whole-system certification. NIST SP800-82r3.

Protocol-specific contract

Interface	Semantics to verify	Prohibited inference
BACnet IP/MS-TP/SC	Objects/properties, PICS, COV lifetime, notification/ACK, status flags, priority arrays and relinquish default	WriteProperty does not grant life-safety authority; application TTL cannot clear retained slots automatically; cloud leases do not prove slot ownership
OPC UA	Namespace URI, stable NodeIds, status/timestamps, security policy, A&C occurrence/branch/EventId, ConditionRefresh and history capability	Subscription alone does not load all active alarms; refresh is not replay; DA/A&E/HDA are not UA
Modbus	Register maps, function codes, zero/one-based addresses, byte/word order, scale, quality provenance and polling budgets	Register changes do not create native alarm lifecycle/ACK; plain TCP is not Modbus Security
Native REST/SOAP /SSE	Authentication/session expiry, pagination/tokens, object IDs, event semantics, per-method scopes/licenses, partial batch failures and restart cursors	HTTP200 cannot confirm every batch item; event ACK need not change whole-alarm state; outbound webhooks do not establish inbound command APIs

Recovery, failover and prohibited shortcuts

Lost source cursors or restarts move connections to reconciling. Commit active-snapshot completion markers; partial snapshots cannot clear missing alarms. Replay/history carries mode and cannot create broadcasts/commands. Duplicates retain receipts/canonical lineage; incident-writeback echoes are isolated before intake rules. Edge journals are durable before acceptance; disk pressure generates backpressure/gap alarms, not silent discard. Write fencing must operate at real egress; if unproven, standby remains observe-only with operator recovery. Local override revokes remote eligibility; reconnect never restores old setpoints automatically. BACnet slots retained after power loss need site-approved local watchdogs or cleanup; cloud TTL alone cannot guarantee release.

Acceptance evidence and operating contract

Each case records test revision, deployment fingerprint, input fixtures/expected observations, raw native statuses/timestamps, policy/approval results, redacted transport evidence, any physical witnesses, result and exceptions. Measure source→edge, edge commit→cloud, cloud→operator, request→native receipt and receipt→independent feedback separately. Freshness thresholds differ from latency SLAs and depend on hazard/site; historian backfills cannot starve alarm queues. Certificate/license rotation measures interruptions and reconciles data rather than suppressing errors to report healthy. Invalidate admission evidence for the affected scope while retaining unrelated capabilities only where evidence still holds.

Sources and implementation artifacts

The existing Physical Device Catalog incorporates 11 source-backed vendor profiles. FRD defines FR-BMS-01–40/WF-34–37. Basic Design and Detail Design are the primary implementation specifications; the BMS source ledger retains 45 sources and research limitations. Engineering JSON profiles are not runtime authorization; activation requires site-specific evidence and the corresponding runtime policy/schema.

SmartAlarm — UX redesign audit and screen contract

Decision baseline: 23 October 2026 · 233 screens · English companion

Executive decision

SmartAlarm needs different visual workspaces for a person escaping danger, a responder completing a task, a commander coordinating an incident, and an administrator changing configuration. Repeated card grids and explanatory paragraphs do not communicate those jobs. The redesign places hazard, location, current evidence and next action before supporting details. Every registered ID, role, field, action and destination remains preserved.

This is a design audit and acceptance specification, not a user-testing report, safe-evacuation proof, WCAG conformance claim or evidence of live operation. Three-second recognition is a **hypothesis to test**, not an achieved result. Prototype data and simulated receipts remain explicitly identified. The JSON companion contains one bilingual record per screen and its original contract. Context classification describes the predominant job; it never changes authorization.

1. Evidence and concrete defects

The review inspected Screen-Registry.json, prototype.js, UIUX-Design.md and Screen-Specifications.md. The written baseline already describes role-based surfaces and uncertainty, but the previous renderer did not realize those distinctions consistently.

Implementation evidence	Consequence	Required correction
content(s) selects shared composition primarily from s.type.	Residents, dispatchers and maintainers decode similar cards despite different decisions.	Select role/task composition first; reuse smaller components inside it.
floor(s) draws one static floor and route; map(s) draws one campus.	A schematic can imply situation knowledge it does not possess.	Bind geometry to context and evidence; mark illustrations and provide list equivalents.
render() places localized SMARTALARM.rules in the product subtitle.	Internal implementation rules compete with urgent guidance.	Move rules to the inspector; product copy states what, where and next action.
ui.fresh contains constant “Updated 8s ago”.	Data appears current without new observations.	Show explicit simulation time or age calculated from real observed time; distinguish fetched time.
Generic glyphs and repeated dot legends represent different entities.	Users must learn arbitrary symbols under pressure.	Labeled pictograms distinguish people, blocked paths, refuges, speakers and controllers.
Mobile navigation is reused across citizen and responder surfaces.	Tasks, decline, assistance, transcript or outbox alternatives become hard to discover.	Use role-specific navigation and verify critical alternatives at narrow widths.
One generic non-ready message replaces task content.	Offline SOS, stale navigation and denied medical access look equivalent.	Show the unavailable capability, retained evidence and valid next step.
Broad states omit route withdrawal, uncertain heading and physical-result reconciliation.	Happy-path polish conceals consequential failure decisions.	Add domain-specific transitions and test them, not only isolated error screenshots.
Primary clicks show a transition toast then navigate.	Navigation can be mistaken for delivery or execution.	Separate local intent, server receipt, human acknowledgment and physical evidence.

These are source observations and design risks, not measured incident outcomes. Source review cannot establish comprehension, touch usability or screen-reader behavior.

2. Hierarchy by role

Surface	First reading	Main visual	Action and disclosure
Citizen in danger	Hazard, affected place, one instruction	Relevant landmark or floor segment with uncertainty	Help/SOS or validated next step; voice, transcript and manual location reachable
Citizen everyday	Local warning and own/family status	Compact area summary and labeled people states	Active warning first; social and privacy features outside urgent guidance
Responder	Assigned task, destination, hazards	Approach/floor view with assistance targets	Accept/decline, arrival, observation or handover by phase; visible outbox
Command	Priority, unanswered needs, ownership	Situation map, actionable queue and unit lanes	Incident-scoped dispatch/warning; full history in details
Public address	Zones, approved message, priority owner	Coverage map and playback-evidence chain	Preview/approve/stop within authority; local priority visible
Physical systems	Exact equipment, authority, evidence	Control schematic and cause/effect graph	Review gates and reconciliation, no shortcut around interlocks
Care	Functional need, exposure, last-known evidence	Need/capability matching and presence matrix	Assign assistance; separate medical permission boundary
Site/integration	Object/version and change impact	Graph, mapping, diff or dependencies	Draft, test, review, publish; disclose advanced fields
Platform/developer/localization	Scope and validity	Compatibility grid, trace, region topology or paired translation	Sandbox/release/publication gates; protected secrets and tenant boundaries

Visual-first does not mean maps everywhere. T04 needs a permission matrix; D02 a request/response split view; L03 a version comparison.

3. Emergency composition and visual policy

Order: **what happened → where it matters → what to do now → what is uncertain**. The first viewport contains one dominant action and its necessary alternative. A large help button must not push affected floor or route validity below the fold. Icons always have short visible labels; meaning never depends on color alone.

Use diagrams for spatial facts and device relationships. Decorative photography, dramatic flames and cinematic backgrounds do not belong in the critical path. Camera evidence has permission, capture time, source and readable caption. Prototype drawings are illustrative, not surveyed plans.

Position includes source, time and accuracy. Unknown heading removes directional arrows: use a named verified landmark, orientation-free plan, manual correction or assistance. Unknown floor, blocked segment, expired hazard evidence and withdrawn guidance are different states. Withdraw obsolete route lines and queued speech together; a replacement is not approved until its exact version is approved.

SOS permits unknown location and manual entry. Saved locally, received by server, accepted by team and resolved are different. Self-reported safety applies only to the person, audience and timestamp; it does not prove household or incident safety. Local controller authority, interlocks and handover rules survive cloud/network failure.

4. Findings by domain

- **A Access:** make region, organization and effective role explicit. Device revocation shows the selected target and consequence. Accessibility preview begins before emergencies.
- **C Citizen:** separate calm home, warnings, SOS, guidance and social circles. C35–C39 require actual guidance-state variations; C25 needs explicit fall-response choices; C27 honors discreet audio/haptics.

- **R Responder:** acceptance deadlines and ownership precede navigation. Observation, outbox and receiver handover are separate; synchronization must preserve notes.
- **O Command:** show open SOS and unconfirmed people beside unit capacity. Connect warning audience, approved version and delivery evidence. O35/O36 expose route restriction, withdrawal and unreachable people.
- **S Sites:** published geometry differs from drafts. Compare plan revisions, accessible edges, calibration, test evidence and coverage gaps before publication.
- **I Integrations:** visualize capability and trust boundaries. Test tenant spoof, duplicate events and expired certificates. Drone control stays with authorized pilot and local safety boundaries.
- **T Administration:** use scope matrices, escalation ladders, retention timelines and capacity charts. Make emergency reserved capacity explicit.
- **X System states:** each is a task-specific recovery. X01 retains pending SOS identity; X02 offers manual location; X09 marks stale evidence and unavailable commands.
- **G Platform:** show regional writer authority without exposing tenant records. Recovery evidence, support expiry and rollback are separate gates.
- **D Developer:** request/response, traces and versions dominate. Sandbox is unmistakable; copied examples never expose secrets.
- **L Localization:** align source, translation, variables and transcript. Missing critical content and stale versions block publication.
- **P Public address:** coverage, exact message, priority and stop matter more than a generic audio player. Receipt is not audibility; preserve local priority and lost-link behavior.
- **H Care:** match functional needs to assistance, restricting medical detail separately. Expected attendance, observed presence and unknown are different.
- **Q Intelligence:** evidence and uncertainty remain beside recommendations. Compare excluded routes as well as selected options; AI has sources, cutoff and human authority.
- **F Physical systems:** equipment topology, exact command and physical feedback replace generic cards. Unknown outcomes trigger reconciliation; capability evidence is not universal remote-control permission.

5. Accessibility and localization acceptance

WCAG 2.2 is the accessibility reference. WAI cognitive patterns inform design rather than adding a conformance level. Clear visible labels support comprehension. WCAG minimum target size is 24 × 24 CSS pixels with defined exceptions; this design targets 48–56 CSS pixels for emergency/field controls and 44 CSS pixels for routine controls, verified on actual layouts. WCAG 2.2, Target Size (Minimum), Clear Visible Labels.

Test keyboard access, visible unobscured focus, programmatic names matching visible labels, status announcements, text enlargement and 320 CSS-pixel reflow. Measure normal-text contrast of at least 4.5:1 and applicable large-text/non-text contrast of at least 3:1. Map alternatives expose equivalent named entities and decisions without vision or dragging. Declared tokens alone do not establish compliance.

Add shape, pattern and text to color; test monochrome and common color-vision deficiencies. Reduced motion removes nonessential motion; continuous flashing is not an urgency mechanism. Voice includes transcript, replay and stop. Test one-handed use on touch hardware, reachability and accidental taps.

Fluent readers review VI/EN; preserve and regression-test existing JA rendering. Test long translations, Vietnamese diacritics and mixed-script locations. Test small/unfamiliar pictograms without labels to reveal ambiguity, then with labels in task context. Never claim universal icon comprehension or standards certification without evidence.

6. Controlled usability study — proposed, not conducted

Recruit citizens, responders, dispatchers and site/physical-system operators, including older people, low vision, cognitive/access needs, one-handed use and second-language readers. Use consent, non-distressing simulation and a stop option; never create real danger or send live alerts.

Counterbalance baseline/new design and scenario order while holding hazard facts and action choices constant. Measure time and accuracy identifying hazard, place and next action; successful task completion; unknown/stale/pending misread as safe/received; accidental consequential actions; recovery after route withdrawal, uncertain heading, offline SOS and unknown actuation. Report medians and upper-tail times, errors, sample size and confidence intervals where supported; do not hide role differences in one average.

Three seconds is a product target for short controlled exposure, followed by the participant explaining the interpretation. Fast but wrong fails. Set release thresholds before the study with the safety owner, informed by risk and pilot data. Any observed unsafe interpretation triggers investigation; no errors in a small sample does not prove safety.

Compare icon-only/icon-plus-label, small-screen/desktop, reduced-motion/standard and first/second-language conditions. Add assistive-technology walkthroughs and physical one-hand trials. Screenshot review cannot replace these sessions.

7. Verification gates and sources

1. Preserve exactly 233 unique IDs and original fields, role, action, destination, rules and workflow.
2. Execute each screen-specific validation; record fixture, viewport, locale, input method, expected state and observed result.
3. Exercise route withdrawal C03/C35–C39, SOS pending C07/X01, authority and unknown actuation O18/O19/F09–F13, acoustic uncertainty P04 and unknown presence H10.
4. Render all VI/EN screens and inspect narrow/enlarged-text layouts, focus, names, contrast and target geometry; archive evidence and defects.
5. Conduct the proposed study before making recognition-time claims.
6. Separately validate production authorization, local interlocks, integrations, timestamps and device feedback. A prototype transition proves none of these.

HSE emphasizes relevant alarms with defined responses and respect for human limitations. This informs action priority, ownership and alarm-flood control. The process-industry guidance does not certify SmartAlarm or replace site-specific assessment. HSE alarm management. WAI also advocates making objects and usage clear; role composition is this project's interpretation. WAI understandable content objective.

Sources were checked for this review as of **23 October 2026**. No external source establishes this prototype's usability.

8. Complete screen-by-screen redesign matrix

Every entry is a design obligation. JSON retains the full original fields, rules, workflow and transition. Folded details remain discoverable and keyboard-accessible; critical constraints stay in the decision layer.

A01 — Welcome & region

Context: configuration · **Role:** Citizen · **FR:** FR-A01

Primary visual: Region selector with language samples and readable text-size preview

Primary question: For welcome & region, what do Region, Language establish, and is “Continue” available to Citizen in this scope?

Primary action: Continue → A02

Secondary disclosure: Keep Region, Language in the decision layer. Put the supporting details for Accessibility, full source/version history and FR-A01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a welcome & region fixture for Citizen; change Region and make Accessibility unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Continue” reaches A02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A02 — Sign in

Context: configuration · **Role:** All roles · **FR:** FR-A02

Primary visual: Workspace identity beside passkey and SSO entry

Primary question: For sign in, what do Email, Workspace establish, and is “Sign in” available to All roles in this scope?

Primary action: Sign in → A03

Secondary disclosure: Keep Email, Workspace in the decision layer. Put the supporting details for Method, full source/version history and FR-A02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a sign in fixture for All roles; change Email and make Method unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Sign in” reaches A03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A03 — Verify identity

Context: configuration · **Role:** All roles · **FR:** FR-A03

Primary visual: Verification method and code slots with remaining attempt feedback

Primary question: For verify identity, what do Method, Verification code establish, and is “Verify” available to All roles in this scope?

Primary action: Verify → A04

Secondary disclosure: Keep Method, Verification code in the decision layer. Put the supporting details for Trusted device, full source/version history and FR-A03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a verify identity fixture for All roles; change Method and make Trusted device unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Verify” reaches A04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A04 — Workspace switcher

Context: configuration · **Role:** All roles · **FR:** FR-A04

Primary visual: Organization tiles showing region and effective role before entry

Primary question: For workspace switcher, what do Northstar, Harbor Campus establish, and is “Enter workspace” available to All roles in this scope?

Primary action: Enter workspace → O01

Secondary disclosure: Keep Northstar, Harbor Campus in the decision layer. Put the supporting details for Personal, full source/version history and FR-A04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a workspace switcher fixture for All roles; change Northstar and make Personal unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Enter workspace” reaches O01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A05 — Recover access

Context: configuration · **Role:** All roles · **FR:** FR-A05

Primary visual: Recovery method selector and masked destination receipt

Primary question: For recover access, what do Email, Recovery method establish, and is “Send recovery link” available to All roles in this scope?

Primary action: Send recovery link → A02

Secondary disclosure: Keep Email, Recovery method in the decision layer. Put the supporting details for Request, full source/version history and FR-A05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a recover access fixture for All roles; change Email and make Request unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send recovery link” reaches A02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A06 — Accept invitation

Context: configuration · **Role:** All roles · **FR:** FR-A06

Primary visual: Invitation identity, role scope and expiry comparison

Primary question: For accept invitation, what do Organization, Role establish, and is “Accept invitation” available to All roles in this scope?

Primary action: Accept invitation → A07

Secondary disclosure: Keep Organization, Role in the decision layer. Put the supporting details for Region, Invitation, full source/version history and FR-A06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a accept invitation fixture for All roles; change Organization and make Invitation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Accept invitation” reaches A07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A07 — Permissions & consent

Context: configuration · **Role:** Citizen · **FR:** FR-A07

Primary visual: Separate permission switches with purpose pictograms

Primary question: For permissions & consent, what do Notifications, Location establish, and is “Save choices” available to Citizen in this scope?

Primary action: Save choices → C01

Secondary disclosure: Keep Notifications, Location in the decision layer. Put the supporting details for Emergency sharing, Camera, full source/version history and FR-A07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a permissions & consent fixture for Citizen; change Notifications and make Camera unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save choices” reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A08 — Session & device security

Context: configuration · **Role:** All roles · **FR:** FR-A08

Primary visual: Device silhouettes with current-session badge and last activity

Primary question: For session & device security, what do This device, Work tablet establish, and is “Review session” available to All roles in this scope?

Primary action: Review session → A09

Secondary disclosure: Keep This device, Work tablet in the decision layer. Put the supporting details for Old phone, full source/version history and FR-A08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a session & device security fixture for All roles; change This device and make Old phone unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review session” reaches A09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A09 — Revoke session

Context: configuration · **Role:** All roles · **FR:** FR-A09

Primary visual: Selected device and revocation impact confirmation

Primary question: For revoke session, what do Device, Effect establish, and is “Revoke selected session” available to All roles in this scope?

Primary action: Revoke selected session → A08

Secondary disclosure: Keep Device, Effect in the decision layer. Put the supporting details for Reason, full source/version history and FR-A09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a revoke session fixture for All roles; change Device and make Reason unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Revoke selected session” reaches A08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A10 — Profile & accessibility

Context: configuration · **Role:** All roles · **FR:** FR-A10

Primary visual: Live accessibility preview beside language and motion settings

Primary question: For profile & accessibility, what do Name, Language establish, and is “Save profile” available to All roles in this scope?

Primary action: Save profile → C01

Secondary disclosure: Keep Name, Language in the decision layer. Put the supporting details for Text size, Reduce motion, Timezone, full source/version history and FR-A10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

Validation: Use a profile & accessibility fixture for All roles; change Name and make Timezone unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save profile” reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C01 — Safety home

Context: configuration · **Role:** Citizen · **FR:** FR-C01

Primary visual: Local hazard summary above family check-in and connection state

Primary question: For safety home, what do Your area, Active alert establish, and is “View active alert” available to Citizen in this scope?

Primary action: View active alert → C02

Secondary disclosure: Keep Your area, Active alert in the decision layer. Put the supporting details for Family, Connection, full source/version history and FR-C01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a safety home fixture for Citizen; change Your area and make Connection unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View active alert” reaches C02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C02 — Alert detail

Context: emergency · **Role:** Citizen · **FR:** FR-C02

Primary visual: Hazard pictogram, affected place and one authoritative instruction

Primary question: For alert detail, what do Severity, Source establish, and is “View guidance” available to Citizen in this scope?

Primary action: View guidance → C03

Secondary disclosure: Keep Severity, Source in the decision layer. Put the supporting details for Updated, Instruction, full source/version history and FR-C02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a alert detail fixture for Citizen; change Severity and make Instruction unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View guidance” reaches C03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C03 — Evacuation guidance

Context: emergency · **Role:** Citizen · **FR:** FR-C03

Primary visual: Floor schematic with current position uncertainty, blocked corridor and accessible refuge

Primary question: For evacuation guidance, what do Location, Plan establish, and is “I need assistance” available to Citizen in this scope?

Primary action: I need assistance → C05

Secondary disclosure: Keep Location, Plan in the decision layer. Put the supporting details for Route status, Accessible route, full source/version history and FR-C03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: If route approval, floor or hazard freshness becomes invalid, withdraw the route line and show assistance/refuge options validated for the site.

Validation: Use a evacuation guidance fixture for Citizen; change Location and make Accessible route unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “I need assistance” reaches C05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C04 — Voice & accessible guidance

Context: emergency · **Role:** Citizen · **FR:** FR-C04

Primary visual: Large audio controls paired with synchronized readable transcript

Primary question: For voice & accessible guidance, what do Language, Audio establish, and is “Return to plan” available to Citizen in this scope?

Primary action: Return to plan → C03

Secondary disclosure: Keep Language, Audio in the decision layer. Put the supporting details for Transcript, Haptics, full source/version history and FR-C04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a voice & accessible guidance fixture for Citizen; change Language and make Haptics unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Return to plan” reaches C03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C05 — SOS type

Context: emergency · **Role:** Citizen · **FR:** FR-C05

Primary visual: Distinct fire, medical, collision and threat pictograms with short labels

Primary question: For sos type, what do Fire or trapped, Medical or fall establish, and is “Continue with selected need” available to Citizen in this scope?

Primary action: Continue with selected need → C06

Secondary disclosure: Keep Fire or trapped, Medical or fall in the decision layer. Put the supporting details for Road collision, Personal threat, Other need, full source/version history and FR-C05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a sos type fixture for Citizen; change Fire or trapped and make Other need unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Continue with selected need” reaches C06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C06 — SOS location & details

Context: emergency · **Role:** Citizen · **FR:** FR-C06

Primary visual: Location pin with accuracy boundary and manual correction

Primary question: For sos location & details, what do Location, Accuracy establish, and is “Review SOS” available to Citizen in this scope?

Primary action: Review SOS → C07

Secondary disclosure: Keep Location, Accuracy in the decision layer. Put the supporting details for People needing help, Note, full source/version history and FR-C06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a sos location & details fixture for Citizen; change Location and make Note unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review SOS” reaches C07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C07 — Confirm SOS

Context: emergency · **Role:** Citizen · **FR:** FR-C07

Primary visual: SOS review showing need, location and recipients before sending

Primary question: For confirm sos, what do Category, Location establish, and is “Send SOS — demo” available to Citizen in this scope?

Primary action: Send SOS — demo → C08

Secondary disclosure: Keep Category, Location in the decision layer. Put the supporting details for Recipients, Sharing, full source/version history and FR-C07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Sending may remain local pending; display server receipt separately and keep a safe alternate contact path, without repeated duplicate SOS.

Validation: Use a confirm sos fixture for Citizen; change Category and make Sharing unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Send SOS — demo” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C08 — SOS tracking

Context: emergency · **Role:** Citizen · **FR:** FR-C08

Primary visual: Receipt ladder distinguishing queued, server-received and human-accepted

Primary question: For sos tracking, what do 10:42:12, 10:42:18 establish, and is “Contact assigned team” available to Citizen in this scope?

Primary action: Contact assigned team → C09

Secondary disclosure: Keep 10:42:12, 10:42:18 in the decision layer. Put the supporting details for 10:43:02, Current, full source/version history and FR-C08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a sos tracking fixture for Citizen; change 10:42:12 and make Current unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Contact assigned team” reaches C09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C09 — Incident conversation

Context: emergency · **Role:** Citizen · **FR:** FR-C09

Primary visual: Incident conversation showing sender role and delivery state per message

Primary question: For incident conversation, what do Operator, You establish, and is “Send demo message” available to Citizen in this scope?

Primary action: Send demo message → C09

Secondary disclosure: Keep Operator, You in the decision layer. Put the supporting details for System, full source/version history and FR-C09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a incident conversation fixture for Citizen; change Operator and make System unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send demo message” reaches C09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C10 — Update or cancel SOS

Context: emergency · **Role:** Citizen · **FR:** FR-C10

Primary visual: Existing SOS beside proposed update and cancellation effect

Primary question: For update or cancel sos, what do Request, Action establish, and is “Submit update” available to Citizen in this scope?

Primary action: Submit update → C08

Secondary disclosure: Keep Request, Action in the decision layer. Put the supporting details for Reason, Notice, full source/version history and FR-C10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a update or cancel sos fixture for Citizen; change Request and make Notice unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Submit update” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C11 — Confirm personal safety

Context: emergency · **Role:** Citizen · **FR:** FR-C11

Primary visual: Three explicit personal-status choices with sharing precision

Primary question: For confirm personal safety, what do Status, Location establish, and is “Confirm my safety” available to Citizen in this scope?

Primary action: Confirm my safety → C12

Secondary disclosure: Keep Status, Location in the decision layer. Put the supporting details for Share with, Precision, full source/version history and FR-C11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Personal confirmation is self-reported, audience-scoped and timestamped; it must not mark an entire household or incident safe.

Validation: Use a confirm personal safety fixture for Citizen; change Status and make Precision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Confirm my safety” reaches C12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C12 — Safety receipt

Context: emergency · **Role:** Citizen · **FR:** FR-C12

Primary visual: Personal report and separate delivery receipts for each audience

Primary question: For safety receipt, what do Personal status, Groups establish, and is “View delivery details” available to Citizen in this scope?

Primary action: View delivery details → C13

Secondary disclosure: Keep Personal status, Groups in the decision layer. Put the supporting details for Contacts, Incident, full source/version history and FR-C12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a safety receipt fixture for Citizen; change Personal status and make Incident unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View delivery details” reaches C13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C13 — Notification inbox

Context: configuration · **Role:** Citizen · **FR:** FR-C13

Primary visual: Alert inbox ordered by urgency with source and expiration

Primary question: For notification inbox, what do Fire alert, Safety confirmation establish, and is “Open alert” available to Citizen in this scope?

Primary action: Open alert → C02

Secondary disclosure: Keep Fire alert, Safety confirmation in the decision layer. Put the supporting details for Family update, full source/version history and FR-C13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a notification inbox fixture for Citizen; change Fire alert and make Family update unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open alert” reaches C02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C14 — Nearby hazards

Context: emergency · **Role:** Citizen · **FR:** FR-C14

Primary visual: Nearby hazard map with list equivalent and measured distance uncertainty

Primary question: For nearby hazards, what do Flood watch, Road closure establish, and is “View hazard” available to Citizen in this scope?

Primary action: View hazard → C15

Secondary disclosure: Keep Flood watch, Road closure in the decision layer. Put the supporting details for Fire, full source/version history and FR-C14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a nearby hazards fixture for Citizen; change Flood watch and make Fire unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “View hazard” reaches C15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C15 — Disaster advisory

Context: emergency · **Role:** Citizen · **FR:** FR-C15

Primary visual: Advisory boundary and validity band beside source certainty

Primary question: For disaster advisory, what do Source, Certainty establish, and is “View shelter information” available to Citizen in this scope?

Primary action: View shelter information → C16

Secondary disclosure: Keep Source, Certainty in the decision layer. Put the supporting details for Valid until, Area, full source/version history and FR-C15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a disaster advisory fixture for Citizen; change Source and make Area unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View shelter information” reaches C16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C16 — Shelters & assembly points

Context: emergency · **Role:** Citizen · **FR:** FR-C16

Primary visual: Shelter cards comparing accessibility, current availability and distance

Primary question: For shelters & assembly points, what do North shelter, West center establish, and is “Open shelter details” available to Citizen in this scope?

Primary action: Open shelter details → C17

Secondary disclosure: Keep North shelter, West center in the decision layer. Put the supporting details for Campus North, full source/version history and FR-C16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a shelters & assembly points fixture for Citizen; change North shelter and make Campus North unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open shelter details” reaches C17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C17 — Shelter details

Context: emergency · **Role:** Citizen · **FR:** FR-C17

Primary visual: Shelter entrance diagram and verified access/capacity status

Primary question: For shelter details, what do Status, Accessibility establish, and is “Save shelter offline” available to Citizen in this scope?

Primary action: Save shelter offline → C18

Secondary disclosure: Keep Status, Accessibility in the decision layer. Put the supporting details for Capacity, Address, full source/version history and FR-C17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a shelter details fixture for Citizen; change Status and make Address unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save shelter offline” reaches C18 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C18 — Offline safety pack

Context: configuration · **Role:** Citizen · **FR:** FR-C18

Primary visual: Offline pack contents with version and last synchronization age

Primary question: For offline safety pack, what do Campus plan, Contacts establish, and is “Review cached plan” available to Citizen in this scope?

Primary action: Review cached plan → C03

Secondary disclosure: Keep Campus plan, Contacts in the decision layer. Put the supporting details for Guidance, Last sync, full source/version history and FR-C18 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a offline safety pack fixture for Citizen; change Campus plan and make Last sync unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review cached plan” reaches C03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C19 — My circles

Context: configuration · **Role:** Citizen · **FR:** FR-C19

Primary visual: Circle membership summary with outstanding check-in count

Primary question: For my circles, what do Family, Campus establish, and is “Open family” available to Citizen in this scope?

Primary action: Open family → C20

Secondary disclosure: Keep Family, Campus in the decision layer. Put the supporting details for Travel team, full source/version history and FR-C19 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a my circles fixture for Citizen; change Family and make Travel team unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open family” reaches C20 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C20 — Circle conversation

Context: configuration · **Role:** Citizen · **FR:** FR-C20

Primary visual: Group conversation with member check-in context

Primary question: For circle conversation, what do System, Jamie establish, and is “View member status” available to Citizen in this scope?

Primary action: View member status → C22

Secondary disclosure: Keep System, Jamie in the decision layer. Put the supporting details for Morgan, full source/version history and FR-C20 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a circle conversation fixture for Citizen; change System and make Morgan unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “View member status” reaches C22 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C21 — Create or edit circle

Context: configuration · **Role:** Citizen · **FR:** FR-C21

Primary visual: Circle membership and sharing scope preview

Primary question: For create or edit circle, what do Name, Membership establish, and is “Save circle” available to Citizen in this scope?

Primary action: Save circle → C19

Secondary disclosure: Keep Name, Membership in the decision layer. Put the supporting details for Safety updates, Location sharing, full source/version history and FR-C21 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a create or edit circle fixture for Citizen; change Name and make Location sharing unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save circle” reaches C19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C22 — Circle member status

Context: configuration · **Role:** Citizen · **FR:** FR-C22

Primary visual: Member status board separating reported safe from unknown

Primary question: For circle member status, what do Alex, Jamie establish, and is “Send check-in request” available to Citizen in this scope?

Primary action: Send check-in request → C20

Secondary disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, full source/version history and FR-C22 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a circle member status fixture for Citizen; change Alex and make Morgan unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send check-in request” reaches C20 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C23 — Emergency contacts

Context: configuration · **Role:** Citizen · **FR:** FR-C23

Primary visual: Contact order with channel pictograms and disclosure preview

Primary question: For emergency contacts, what do Primary, Secondary establish, and is “Save contacts” available to Citizen in this scope?

Primary action: Save contacts → C01

Secondary disclosure: Keep Primary, Secondary in the decision layer. Put the supporting details for Channel, Share, full source/version history and FR-C23 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a emergency contacts fixture for Citizen; change Primary and make Share unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save contacts” reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C24 — Care & fall detection

Context: configuration · **Role:** Citizen · **FR:** FR-C24

Primary visual: Wearable connection and fall-check-in escalation sequence

Primary question: For care & fall detection, what do Wearable, Detection establish, and is “Review test alert” available to Citizen in this scope?

Primary action: Review test alert → C25

Secondary disclosure: Keep Wearable, Detection in the decision layer. Put the supporting details for Check-in window, Escalation, full source/version history and FR-C24 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a care & fall detection fixture for Citizen; change Wearable and make Escalation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review test alert” reaches C25 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C25 — Possible fall check-in

Context: emergency · **Role:** Citizen · **FR:** FR-C25

Primary visual: Fall check-in with large help and dismiss controls and time remaining

Primary question: For possible fall check-in, what do Evidence, Check-in window establish, and is “I need help now” available to Citizen in this scope?

Primary action: I need help now → C08

Secondary disclosure: Keep Evidence, Check-in window in the decision layer. Put the supporting details for No response, Alternative, full source/version history and FR-C25 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: No response remains unconfirmed; a stopped timer is not a help receipt, and accessibility must not depend on a press-and-hold gesture.

Validation: Use a possible fall check-in fixture for Citizen; change Evidence and make Alternative unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “I need help now” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C26 — Road incident report

Context: emergency · **Role:** Citizen · **FR:** FR-C26

Primary visual: Road orientation sketch with manual direction and injury entry

Primary question: For road incident report, what do Position, Direction establish, and is “Review SOS” available to Citizen in this scope?

Primary action: Review SOS → C07

Secondary disclosure: Keep Position, Direction in the decision layer. Put the supporting details for Vehicles, Injuries, Attachment, full source/version history and FR-C26 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a road incident report fixture for Citizen; change Position and make Attachment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review SOS" reaches C07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C27 — Discreet assistance

Context: emergency · **Role:** Citizen · **FR:** FR-C27

Primary visual: Quiet high-contrast assistance controls showing sound and vibration state

Primary question: For discreet assistance, what do Sound, Haptics establish, and is "Send discreet request — demo" available to Citizen in this scope?

Primary action: Send discreet request — demo → C08

Secondary disclosure: Keep Sound, Haptics in the decision layer. Put the supporting details for Sharing, Location, full source/version history and FR-C27 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Never autoplay speech or vibration against the discreet selection; reveal pending versus received without a conspicuous success animation.

Validation: Use a discreet assistance fixture for Citizen; change Sound and make Location unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Send discreet request — demo" reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C28 — My reports & history

Context: configuration · **Role:** Citizen · **FR:** FR-C28

Primary visual: Personal request history showing open, resolved and pending receipts

Primary question: For my reports & history, what do SOS-041, SOS-028 establish, and is "View request" available to Citizen in this scope?

Primary action: View request → C08

Secondary disclosure: Keep SOS-041, SOS-028 in the decision layer. Put the supporting details for Check-in, full source/version history and FR-C28 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a my reports & history fixture for Citizen; change SOS-041 and make Check-in unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and "View request" reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C29 — Privacy & data requests

Context: configuration · **Role:** Citizen · **FR:** FR-C29

Primary visual: Privacy scope preview showing live location and export/deletion choices

Primary question: For privacy & data requests, what do Live location, Circle visibility establish, and is “Review data request” available to Citizen in this scope?

Primary action: Review data request → C30

Secondary disclosure: Keep Live location, Circle visibility in the decision layer. Put the supporting details for Export, Deletion, full source/version history and FR-C29 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a privacy & data requests fixture for Citizen; change Live location and make Deletion unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review data request” reaches C30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C30 — Data request receipt

Context: configuration · **Role:** Citizen · **FR:** FR-C30

Primary visual: Data request progress and verified delivery destination

Primary question: For data request receipt, what do Type, Scope establish, and is “Back to privacy” available to Citizen in this scope?

Primary action: Back to privacy → C29

Secondary disclosure: Keep Type, Scope in the decision layer. Put the supporting details for Status, Delivery, full source/version history and FR-C30 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a data request receipt fixture for Citizen; change Type and make Delivery unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Back to privacy” reaches C29 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R01 — Shift & readiness

Context: field · **Role:** Responder · **FR:** FR-R01

Primary visual: Unit readiness checklist with capability and shift status

Primary question: For shift & readiness, what do Unit, Shift establish, and is “View assignments” available to Responder in this scope?

Primary action: View assignments → R02

Secondary disclosure: Keep Unit, Shift in the decision layer. Put the supporting details for Capabilities, Readiness, full source/version history and FR-R01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a shift & readiness fixture for Responder; change Unit and make Readiness unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View assignments” reaches R02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R02 — Assignment inbox

Context: field · **Role:** Responder · **FR:** FR-R02

Primary visual: Assignment queue ranked by urgency and acceptance deadline

Primary question: For assignment inbox, what do INC-2041, INC-2037 establish, and is “Open assignment” available to Responder in this scope?

Primary action: Open assignment → R03

Secondary disclosure: Keep INC-2041, INC-2037 in the decision layer. Put the supporting details for Coverage, full source/version history and FR-R02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a assignment inbox fixture for Responder; change INC-2041 and make Coverage unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open assignment” reaches R03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R03 — Accept or decline task

Context: field · **Role:** Responder · **FR:** FR-R03

Primary visual: Task identity, person location and hazard summary above accept/decline

Primary question: For accept or decline task, what do Incident, Person establish, and is “Accept assignment” available to Responder in this scope?

Primary action: Accept assignment → R04

Secondary disclosure: Keep Incident, Person in the decision layer. Put the supporting details for Location, Hazards, full source/version history and FR-R03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a accept or decline task fixture for Responder; change Incident and make Hazards unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Accept assignment” reaches R04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R04 — Navigation & arrival

Context: field · **Role:** Responder · **FR:** FR-R04

Primary visual: Destination and approach map showing route validity and arrival action

Primary question: For navigation & arrival, what do Assignment, Destination establish, and is “Report arrival” available to Responder in this scope?

Primary action: Report arrival → R05

Secondary disclosure: Keep Assignment, Destination in the decision layer. Put the supporting details for Location freshness, Route, full source/version history and FR-R04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a navigation & arrival fixture for Responder; change Assignment and make Route unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Report arrival” reaches R05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R05 — Field situation

Context: field · **Role:** Responder · **FR:** FR-R05

Primary visual: Field floor map with blocked zones, open SOS and teammate locations

Primary question: For field situation, what do Open SOS, Blocked area establish, and is “Open assistance record” available to Responder in this scope?

Primary action: Open assistance record → R06

Secondary disclosure: Keep Open SOS, Blocked area in the decision layer. Put the supporting details for Plan, Team, full source/version history and FR-R05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a field situation fixture for Responder; change Open SOS and make Team unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open assistance record” reaches R06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R06 — Assistance & triage record

Context: field · **Role:** Responder · **FR:** FR-R06

Primary visual: Person-centered observation form with condition and assistance actions

Primary question: For assistance & triage record, what do Person, Contact establish, and is “Save observation” available to Responder in this scope?

Primary action: Save observation → R07

Secondary disclosure: Keep Person, Contact in the decision layer. Put the supporting details for Condition, Action, full source/version history and FR-R06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a assistance & triage record fixture for Responder; change Person and make Action unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save observation” reaches R07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R07 — Task status & handover

Context: field · **Role:** Responder · **FR:** FR-R07

Primary visual: Task handover timeline showing receiver acceptance separately

Primary question: For task status & handover, what do Task, State establish, and is “Submit handover” available to Responder in this scope?

Primary action: Submit handover → R08

Secondary disclosure: Keep Task, State in the decision layer. Put the supporting details for Receiver, Evidence, full source/version history and FR-R07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a task status & handover fixture for Responder; change Task and make Evidence unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Submit handover” reaches R08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R08 — Task completion

Context: field · **Role:** Responder · **FR:** FR-R08

Primary visual: Completion evidence and receiving team acknowledgment

Primary question: For task completion, what do Person, Handover establish, and is “Back to shift” available to Responder in this scope?

Primary action: Back to shift → R01

Secondary disclosure: Keep Person, Handover in the decision layer. Put the supporting details for Incident, Next, full source/version history and FR-R08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a task completion fixture for Responder; change Person and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Back to shift” reaches R01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R09 — Team communications

Context: field · **Role:** Responder · **FR:** FR-R09

Primary visual: Team channel with command identity and pending message markers

Primary question: For team communications, what do Command, Rescue 02 establish, and is “Send demo message” available to Responder in this scope?

Primary action: Send demo message → R09

Secondary disclosure: Keep Command, Rescue 02 in the decision layer. Put the supporting details for System, full source/version history and FR-R09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

Validation: Use a team communications fixture for Responder; change Command and make System unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send demo message” reaches R09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R10 — Evidence capture

Context: field · **Role:** Responder · **FR:** FR-R10

Primary visual: Evidence preview with incident binding and privacy redaction

Primary question: For evidence capture, what do Type, Incident establish, and is “Queue attachment” available to Responder in this scope?

Primary action: Queue attachment → R11

Secondary disclosure: Keep Type, Incident in the decision layer. Put the supporting details for Caption, Privacy, full source/version history and FR-R10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor's receipt.

Validation: Use a evidence capture fixture for Responder; change Type and make Privacy unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Queue attachment" reaches R11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R11 — Offline outbox

Context: field · **Role:** Responder · **FR:** FR-R11

Primary visual: Outbox grouped by queued, retryable and conflicting items

Primary question: For offline outbox, what do Observation, Photo establish, and is "Resolve sync conflict" available to Responder in this scope?

Primary action: Resolve sync conflict → R12

Secondary disclosure: Keep Observation, Photo in the decision layer. Put the supporting details for Status update, full source/version history and FR-R11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor's receipt.

Validation: Use a offline outbox fixture for Responder; change Observation and make Status update unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and "Resolve sync conflict" reaches R12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R12 — Resolve field conflict

Context: field · **Role:** Responder · **FR:** FR-R12

Primary visual: Local observation beside server change with preserve-note option

Primary question: For resolve field conflict, what do Local, Server establish, and is "Keep note and refresh" available to Responder in this scope?

Primary action: Keep note and refresh → R02

Secondary disclosure: Keep Local, Server in the decision layer. Put the supporting details for Your note, Action, full source/version history and FR-R12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor's receipt.

Validation: Use a resolve field conflict fixture for Responder; change Local and make Action unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Keep note and refresh" reaches R02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O01 — Command overview

Context: command · **Role:** Operator · **FR:** FR-O01

Primary visual: Priority incident strip, capacity board and site situation map

Primary question: For command overview, what do Active incidents, Unacknowledged P1 establish, and is "Open priority incident" available to Operator in this scope?

Primary action: Open priority incident → O04

Secondary disclosure: Keep Active incidents, Unacknowledged P1 in the decision layer. Put the supporting details for Open SOS, Available units, full source/version history and FR-O01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a command overview fixture for Operator; change Active incidents and make Available units unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open priority incident” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O02 — Live situation map

Context: command · **Role:** Operator · **FR:** FR-O02

Primary visual: Multi-incident geographic map with event freshness by layer

Primary question: For live situation map, what do Fire, Flood establish, and is “Open incident” available to Operator in this scope?

Primary action: Open incident → O04

Secondary disclosure: Keep Fire, Flood in the decision layer. Put the supporting details for Road collision, Map freshness, full source/version history and FR-O02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a live situation map fixture for Operator; change Fire and make Map freshness unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open incident” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O03 — Signal triage queue

Context: command · **Role:** Operator · **FR:** FR-O03

Primary visual: Signal evidence queue showing duplicate suspicion and source quality

Primary question: For signal triage queue, what do EV-3101, EV-3102 establish, and is “Inspect signal” available to Operator in this scope?

Primary action: Inspect signal → O05

Secondary disclosure: Keep EV-3101, EV-3102 in the decision layer. Put the supporting details for EV-3103, full source/version history and FR-O03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a signal triage queue fixture for Operator; change EV-3101 and make EV-3103 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Inspect signal” reaches O05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O04 — Incident workspace

Context: command · **Role:** Operator · **FR:** FR-O04

Primary visual: Incident map beside open SOS, unconfirmed people and action ownership

Primary question: For incident workspace, what do INC-2041, Open SOS establish, and is “Open dispatch” available to Operator in this scope?

Primary action: Open dispatch → O08

Secondary disclosure: Keep INC-2041, Open SOS in the decision layer. Put the supporting details for Self-reported safe, Unconfirmed people, full source/version history and FR-O04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a incident workspace fixture for Operator; change INC-2041 and make Unconfirmed people unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open dispatch” reaches O08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O05 — Signal evidence review

Context: command · **Role:** Operator · **FR:** FR-O05

Primary visual: Evidence frame with observed time, confidence and mapped location

Primary question: For signal evidence review, what do Source, Observed establish, and is “Create incident from signal” available to Operator in this scope?

Primary action: Create incident from signal → O06

Secondary disclosure: Keep Source, Observed in the decision layer. Put the supporting details for Confidence, Location, full source/version history and FR-O05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a signal evidence review fixture for Operator; change Source and make Location unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Create incident from signal” reaches O06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O06 — Create incident

Context: command · **Role:** Operator · **FR:** FR-O06

Primary visual: Incident creation preview linking category, severity and owner

Primary question: For create incident, what do Category, Severity establish, and is “Create demo incident” available to Operator in this scope?

Primary action: Create demo incident → O04

Secondary disclosure: Keep Category, Severity in the decision layer. Put the supporting details for Location, Evidence, Owner, full source/version history and FR-O06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a create incident fixture for Operator; change Category and make Owner unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Create demo incident” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O07 — Correlate or split incidents

Context: command · **Role:** Supervisor · **FR:** FR-O07

Primary visual: Side-by-side incident lineage with merge/split impact

Primary question: For correlate or split incidents, what do Primary, Related establish, and is “Save correlation” available to Supervisor in this scope?

Primary action: Save correlation → O04

Secondary disclosure: Keep Primary, Related in the decision layer. Put the supporting details for Evidence, Decision, full source/version history and FR-O07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a correlate or split incidents fixture for Supervisor; change Primary and make Decision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save correlation” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O08 — Dispatch board

Context: command · **Role:** Dispatcher · **FR:** FR-O08

Primary visual: Unit lanes showing capabilities, tasks and acceptance state

Primary question: For dispatch board, what do Rescue 02, Medical 01 establish, and is “Review assignment” available to Dispatcher in this scope?

Primary action: Review assignment → O09

Secondary disclosure: Keep Rescue 02, Medical 01 in the decision layer. Put the supporting details for Security 03, Task, full source/version history and FR-O08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a dispatch board fixture for Dispatcher; change Rescue 02 and make Task unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review assignment” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O09 — Assign responder

Context: command · **Role:** Dispatcher · **FR:** FR-O09

Primary visual: Assignment preview binding exact unit, task and response deadline

Primary question: For assign responder, what do Incident, Unit establish, and is “Assign team — demo” available to Dispatcher in this scope?

Primary action: Assign team — demo → O10

Secondary disclosure: Keep Incident, Unit in the decision layer. Put the supporting details for Task, Acceptance deadline, full source/version history and FR-O09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a assign responder fixture for Dispatcher; change Incident and make Acceptance deadline unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Assign team — demo” reaches O10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O10 — Response timeline

Context: command · **Role:** Operator · **FR:** FR-O10

Primary visual: Chronological response milestones with source-specific receipts

Primary question: For response timeline, what do 10:40:12, 10:40:19 establish, and is “Review people status” available to Operator in this scope?

Primary action: Review people status → O11

Secondary disclosure: Keep 10:40:12, 10:40:19 in the decision layer. Put the supporting details for 10:42:18, 10:43:02, full source/version history and FR-O10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a response timeline fixture for Operator; change 10:40:12 and make 10:43:02 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review people status” reaches O11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O11 — People accountability

Context: command · **Role:** Operator · **FR:** FR-O11

Primary visual: People board separating unconfirmed, assisted and self-reported safe

Primary question: For people accountability, what do Alex, Jamie establish, and is “Open person” available to Operator in this scope?

Primary action: Open person → O12

Secondary disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Scope, full source/version history and FR-O11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a people accountability fixture for Operator; change Alex and make Scope unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open person” reaches O12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O12 — Person assistance detail

Context: command · **Role:** Operator · **FR:** FR-O12

Primary visual: Assistance need, consent scope and last-known location

Primary question: For person assistance detail, what do Location, Consent establish, and is “Assign assistance” available to Operator in this scope?

Primary action: Assign assistance → O09

Secondary disclosure: Keep Location, Consent in the decision layer. Put the supporting details for Status, Freshness, full source/version history and FR-O12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a person assistance detail fixture for Operator; change Location and make Freshness unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Assign assistance” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O13 — Compose warning

Context: command · **Role:** Warning author · **FR:** FR-O13

Primary visual: Warning composer beside affected-area and instruction preview

Primary question: For compose warning, what do Category, Area establish, and is “Preview audience” available to Warning author in this scope?

Primary action: Preview audience → O14

Secondary disclosure: Keep Category, Area in the decision layer. Put the supporting details for Language, Instruction, Expiry, full source/version history and FR-O13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a compose warning fixture for Warning author; change Category and make Expiry unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Preview audience” reaches O14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O14 — Warning preview & audience

Context: command · **Role:** Warning author · **FR:** FR-O14

Primary visual: Audience footprint and channel/language coverage comparison

Primary question: For warning preview & audience, what do Audience, Channels establish, and is “Submit for approval” available to Warning author in this scope?

Primary action: Submit for approval → O15

Secondary disclosure: Keep Audience, Channels in the decision layer. Put the supporting details for Translations, Exclusions, full source/version history and FR-O14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a warning preview & audience fixture for Warning author; change Audience and make Exclusions unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Submit for approval” reaches O15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O15 — Approve warning

Context: command · **Role:** Warning approver · **FR:** FR-O15

Primary visual: Exact warning version and author/approver separation

Primary question: For approve warning, what do Author, Approver establish, and is “Approve demo warning” available to Warning approver in this scope?

Primary action: Approve demo warning → O16

Secondary disclosure: Keep Author, Approver in the decision layer. Put the supporting details for Version, Reason, full source/version history and FR-O15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a approve warning fixture for Warning approver; change Author and make Reason unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve demo warning” reaches O16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O16 — Delivery & escalation

Context: command · **Role:** Operator · **FR:** FR-O16

Primary visual: Delivery funnel separating provider receipt and human acknowledgment

Primary question: For delivery & escalation, what do Push, SMS establish, and is “Review failed deliveries” available to Operator in this scope?

Primary action: Review failed deliveries → O17

Secondary disclosure: Keep Push, SMS in the decision layer. Put the supporting details for Human acknowledgments, Next escalation, full source/version history and FR-O16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a delivery & escalation fixture for Operator; change Push and make Next escalation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review failed deliveries” reaches O17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O17 — Delivery failure review

Context: command · **Role:** Operator · **FR:** FR-O17

Primary visual: Failed-recipient cohort beside fallback-channel proposal

Primary question: For delivery failure review, what do Provider, Failure establish, and is “Queue fallback — demo” available to Operator in this scope?

Primary action: Queue fallback — demo → O16

Secondary disclosure: Keep Provider, Failure in the decision layer. Put the supporting details for Affected, Fallback, full source/version history and FR-O17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a delivery failure review fixture for Operator; change Provider and make Fallback unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Queue fallback — demo” reaches O16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O18 — Physical command approval

Context: command · **Role:** Authorized controller · **FR:** FR-O18

Primary visual: Equipment target schematic with interlocks and expiring command digest

Primary question: For physical command approval, what do Target, Command establish, and is “Approve simulated command” available to Authorized controller in this scope?

Primary action: Approve simulated command → O19

Secondary disclosure: Keep Target, Command in the decision layer. Put the supporting details for Interlocks, Expires, full source/version history and FR-O18 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Expired command, failed interlock or changed target invalidates approval; require review of the exact command version.

Validation: Use a physical command approval fixture for Authorized controller; change Target and make Expires unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve simulated command” reaches O19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O19 — Physical command status

Context: command · **Role:** Authorized controller · **FR:** FR-O19

Primary visual: Command timeline separating transport receipt and physical effect

Primary question: For physical command status, what do 10:44:00, 10:44:03 establish, and is “Refresh command evidence” available to Authorized controller in this scope?

Primary action: Refresh command evidence → O19

Secondary disclosure: Keep 10:44:00, 10:44:03 in the decision layer. Put the supporting details for 10:44:04, Current, full source/version history and FR-O19 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Controller acknowledgment without physical feedback remains unknown; show reconciliation rather than repeat-command success.

Validation: Use a physical command status fixture for Authorized controller; change 10:44:00 and make Current unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Refresh command evidence” reaches O19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O20 — Incident closure review

Context: command · **Role:** Supervisor · **FR:** FR-O20

Primary visual: Closure blockers highlighting open SOS and unresolved people

Primary question: For incident closure review, what do Open SOS, Unconfirmed people establish, and is “Close demo incident” available to Supervisor in this scope?

Primary action: Close demo incident → O21

Secondary disclosure: Keep Open SOS, Unconfirmed people in the decision layer. Put the supporting details for Units, Reason, Approver, full source/version history and FR-O20 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Open SOS and unresolved people remain visible blockers or require explicit governed disposition; closing the view is not incident closure.

Validation: Use a incident closure review fixture for Supervisor; change Open SOS and make Approver unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Close demo incident” reaches O21 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O21 — After-action review

Context: command · **Role:** Supervisor · **FR:** FR-O21

Primary visual: Response interval chart with delivery gaps and improvement owners

Primary question: For after-action review, what do First acknowledgment, Assignment acceptance establish, and is “Create improvement action” available to Supervisor in this scope?

Primary action: Create improvement action → O22

Secondary disclosure: Keep First acknowledgment, Assignment acceptance in the decision layer. Put the supporting details for Delivery gaps, Improvement, full source/version history and FR-O21 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a after-action review fixture for Supervisor; change First acknowledgment and make Improvement unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Create improvement action” reaches O22 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O22 — Improvement action

Context: command · **Role:** Supervisor · **FR:** FR-O22

Primary visual: Improvement task with accountable owner and due-date progress

Primary question: For improvement action, what do Action, Owner establish, and is “Save action” available to Supervisor in this scope?

Primary action: Save action → O21

Secondary disclosure: Keep Action, Owner in the decision layer. Put the supporting details for Due date, Linked incident, full source/version history and FR-O22 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a improvement action fixture for Supervisor; change Action and make Linked incident unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save action” reaches O21 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O23 — Drill scheduler

Context: command · **Role:** Supervisor · **FR:** FR-O23

Primary visual: Drill scenario schedule clearly separated from live environment

Primary question: For drill scheduler, what do Scenario, Environment establish, and is “Open drill control” available to Supervisor in this scope?

Primary action: Open drill control → O24

Secondary disclosure: Keep Scenario, Environment in the decision layer. Put the supporting details for Participants, Start, full source/version history and FR-O23 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a drill scheduler fixture for Supervisor; change Scenario and make Start unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open drill control” reaches O24 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O24 — Drill control room

Context: command · **Role:** Supervisor · **FR:** FR-O24

Primary visual: Drill progress board with explicit actuator isolation state

Primary question: For drill control room, what do Mode, Live actuators establish, and is “Review drill report” available to Supervisor in this scope?

Primary action: Review drill report → O21

Secondary disclosure: Keep Mode, Live actuators in the decision layer. Put the supporting details for Test recipients, Completed steps, full source/version history and FR-O24 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a drill control room fixture for Supervisor; change Mode and make Completed steps unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review drill report” reaches O21 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O25 — Shift handover

Context: command · **Role:** Supervisor · **FR:** FR-O25

Primary visual: Outgoing/incoming responsibility transfer with unresolved actions

Primary question: For shift handover, what do Outgoing, Incoming establish, and is “Record handover” available to Supervisor in this scope?

Primary action: Record handover → O01

Secondary disclosure: Keep Outgoing, Incoming in the decision layer. Put the supporting details for Open incidents, Outstanding actions, Acknowledgment, full source/version history and FR-O25 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a shift handover fixture for Supervisor; change Outgoing and make Acknowledgment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Record handover” reaches O01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O26 — Shelter operations

Context: command · **Role:** Operator · **FR:** FR-O26

Primary visual: Shelter capacity board including unknown and expired reports

Primary question: For shelter operations, what do North shelter, West center establish, and is “Edit shelter” available to Operator in this scope?

Primary action: Edit shelter → O27

Secondary disclosure: Keep North shelter, West center in the decision layer. Put the supporting details for South shelter, Last refresh, full source/version history and FR-O26 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a shelter operations fixture for Operator; change North shelter and make Last refresh unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit shelter” reaches O27 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O27 — Edit shelter status

Context: command · **Role:** Operator · **FR:** FR-O27

Primary visual: Shelter status editor beside capacity and validity preview

Primary question: For edit shelter status, what do Shelter, Status establish, and is “Publish demo update” available to Operator in this scope?

Primary action: Publish demo update → O26

Secondary disclosure: Keep Shelter, Status in the decision layer. Put the supporting details for Available capacity, Accessibility, Valid until, full source/version history and FR-O27 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a edit shelter status fixture for Operator; change Shelter and make Valid until unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish demo update” reaches O26 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O28 — Regional disaster feed

Context: command · **Role:** Operator · **FR:** FR-O28

Primary visual: Regional advisory queue with source identity and supersession

Primary question: For regional disaster feed, what do CAP-041, CAP-039 establish, and is “Inspect advisory” available to Operator in this scope?

Primary action: Inspect advisory → O29

Secondary disclosure: Keep CAP-041, CAP-039 in the decision layer. Put the supporting details for CAP-038, full source/version history and FR-O28 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a regional disaster feed fixture for Operator; change CAP-041 and make CAP-038 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Inspect advisory” reaches O29 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O29 — CAP advisory review

Context: command · **Role:** Operator · **FR:** FR-O29

Primary visual: CAP message lineage with update/cancel references and expiry

Primary question: For cap advisory review, what do Sender, Identifier establish, and is “Create local response” available to Operator in this scope?

Primary action: Create local response → O06

Secondary disclosure: Keep Sender, Identifier in the decision layer. Put the supporting details for Message type, References, Expires, full source/version history and FR-O29 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a cap advisory review fixture for Operator; change Sender and make Expires unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Create local response” reaches O06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O30 — Search & audit timeline

Context: command · **Role:** Auditor · **FR:** FR-O30

Primary visual: Audit timeline with actor, scope and immutable event details

Primary question: For search & audit timeline, what do Query, Create establish, and is “Inspect audit entry” available to Auditor in this scope?

Primary action: Inspect audit entry → T11

Secondary disclosure: Keep Query, Create in the decision layer. Put the supporting details for Dispatch, Export, full source/version history and FR-O30 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a search & audit timeline fixture for Auditor; change Query and make Export unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect audit entry” reaches T11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S01 — Site directory

Context: configuration · **Role:** Site admin · **FR:** FR-S01

Primary visual: Site map and directory with region boundary

Primary question: For site directory, what do Harbor Campus, Riverside establish, and is “Open site” available to Site admin in this scope?

Primary action: Open site → S02

Secondary disclosure: Keep Harbor Campus, Riverside in the decision layer. Put the supporting details for North highway, full source/version history and FR-S01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a site directory fixture for Site admin; change Harbor Campus and make North highway unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open site” reaches S02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S02 — Site structure

Context: configuration · **Role:** Site admin · **FR:** FR-S02

Primary visual: Building-floor-zone tree with selected spatial context

Primary question: For site structure, what do Building A, Floor 05 establish, and is “Edit location” available to Site admin in this scope?

Primary action: Edit location → S03

Secondary disclosure: Keep Building A, Floor 05 in the decision layer. Put the supporting details for Zone A05-03, Assembly, full source/version history and FR-S02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a site structure fixture for Site admin; change Building A and make Assembly unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit location” reaches S03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S03 — Location editor

Context: configuration · **Role:** Site admin · **FR:** FR-S03

Primary visual: Parent-location path beside coordinate and floor preview

Primary question: For location editor, what do Name, Parent establish, and is “Save location” available to Site admin in this scope?

Primary action: Save location → S02

Secondary disclosure: Keep Name, Parent in the decision layer. Put the supporting details for Coordinates, Floor reference, Timezone, full source/version history and FR-S03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a location editor fixture for Site admin; change Name and make Timezone unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save location” reaches S02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S04 — Floor plan editor

Context: configuration · **Role:** Site admin · **FR:** FR-S04

Primary visual: Plan canvas showing exits, refuges and revision differences

Primary question: For floor plan editor, what do Plan, Revision establish, and is “Review plan publication” available to Site admin in this scope?

Primary action: Review plan publication → S05

Secondary disclosure: Keep Plan, Revision in the decision layer. Put the supporting details for Exits, Accessible refuge, Hazard overlay, full source/version history and FR-S04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a floor plan editor fixture for Site admin; change Plan and make Hazard overlay unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review plan publication” reaches S05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S05 — Plan approval & versions

Context: configuration · **Role:** Supervisor · **FR:** FR-S05

Primary visual: Plan revision comparison with published/draft state

Primary question: For plan approval & versions, what do Revision 13, Revision 12 establish, and is “Publish reviewed draft — demo” available to Supervisor in this scope?

Primary action: Publish reviewed draft — demo → S04

Secondary disclosure: Keep Revision 13, Revision 12 in the decision layer. Put the supporting details for Revision 11, full source/version history and FR-S05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a plan approval & versions fixture for Supervisor; change Revision 13 and make Revision 11 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Publish reviewed draft — demo” reaches S04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S06 — Device inventory

Context: configuration · **Role:** Device admin · **FR:** FR-S06

Primary visual: Device inventory grouped by physical function and health

Primary question: For device inventory, what do CAM-A05-03, SMK-A05-03 establish, and is “Open device” available to Device admin in this scope?

Primary action: Open device → S07

Secondary disclosure: Keep CAM-A05-03, SMK-A05-03 in the decision layer. Put the supporting details for GW-012, WR-401, full source/version history and FR-S06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a device inventory fixture for Device admin; change CAM-A05-03 and make WR-401 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open device” reaches S07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S07 — Device detail & health

Context: configuration · **Role:** Device admin · **FR:** FR-S07

Primary visual: Device diagram with heartbeat, certificate and firmware evidence

Primary question: For device detail & health, what do Certificate, Last heartbeat establish, and is “Review device configuration” available to Device admin in this scope?

Primary action: Review device configuration → S08

Secondary disclosure: Keep Certificate, Last heartbeat in the decision layer. Put the supporting details for Firmware, Location, full source/version history and FR-S07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a device detail & health fixture for Device admin; change Certificate and make Location unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review device configuration” reaches S08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S08 — Provision device

Context: configuration · **Role:** Device admin · **FR:** FR-S08

Primary visual: Enrollment sequence tying serial, site and capability claim

Primary question: For provision device, what do Serial, Enrollment token establish, and is “Enroll simulated device” available to Device admin in this scope?

Primary action: Enroll simulated device → S07

Secondary disclosure: Keep Serial, Enrollment token in the decision layer. Put the supporting details for Location, Capabilities, full source/version history and FR-S08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a provision device fixture for Device admin; change Serial and make Capabilities unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Enroll simulated device" reaches S07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S09 — Camera wall

Context: configuration · **Role:** Operator · **FR:** FR-S09

Primary visual: Permission-gated camera tiles with timestamp and missing-feed state

Primary question: For camera wall, what do CAM-A05-01, CAM-A05-02 establish, and is "Open detection evidence" available to Operator in this scope?

Primary action: Open detection evidence → O05

Secondary disclosure: Keep CAM-A05-01, CAM-A05-02 in the decision layer. Put the supporting details for CAM-A05-03, Access, full source/version history and FR-S09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a camera wall fixture for Operator; change CAM-A05-01 and make Access unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Open detection evidence" reaches O05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S10 — Maintenance & calibration

Context: configuration · **Role:** Device admin · **FR:** FR-S10

Primary visual: Maintenance window alongside backup-coverage map

Primary question: For maintenance & calibration, what do Device, Window establish, and is "Schedule maintenance" available to Device admin in this scope?

Primary action: Schedule maintenance → S06

Secondary disclosure: Keep Device, Window in the decision layer. Put the supporting details for Backup coverage, Approver, full source/version history and FR-S10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a maintenance & calibration fixture for Device admin; change Device and make Approver unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Schedule maintenance" reaches S06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S11 — Rule library

Context: configuration · **Role:** Rule author · **FR:** FR-S11

Primary visual: Versioned rule list showing hazard family and enabled scope

Primary question: For rule library, what do Fire rule v4, Fall rule v2 establish, and is “Edit rule” available to Rule author in this scope?

Primary action: Edit rule → S12

Secondary disclosure: Keep Fire rule v4, Fall rule v2 in the decision layer. Put the supporting details for Road collision v1, Flood threshold v3, full source/version history and FR-S11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a rule library fixture for Rule author; change Fire rule v4 and make Flood threshold v3 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit rule” reaches S12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S12 — Rule editor & simulation

Context: configuration · **Role:** Rule author · **FR:** FR-S12

Primary visual: Condition-action flow beside simulation-only indicator

Primary question: For rule editor & simulation, what do Rule, Condition establish, and is “View simulation results” available to Rule author in this scope?

Primary action: View simulation results → S13

Secondary disclosure: Keep Rule, Condition in the decision layer. Put the supporting details for Window, Action, Mode, full source/version history and FR-S12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a rule editor & simulation fixture for Rule author; change Rule and make Mode unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “View simulation results” reaches S13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S13 — Rule test & approval

Context: configuration · **Role:** Rule approver · **FR:** FR-S13

Primary visual: Test-case outcome grid with false-merge evidence

Primary question: For rule test & approval, what do Fixture cases, False merges establish, and is “Return rule for changes” available to Rule approver in this scope?

Primary action: Return rule for changes → S12

Secondary disclosure: Keep Fixture cases, False merges in the decision layer. Put the supporting details for Version, Reviewer, full source/version history and FR-S13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a rule test & approval fixture for Rule approver; change Fixture cases and make Reviewer unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Return rule for changes” reaches S12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S14 — AI model monitoring

Context: configuration · **Role:** Model operator · **FR:** FR-S14

Primary visual: Model quality trend with sample provenance and drift

Primary question: For ai model monitoring, what do Model, Labeled sample establish, and is “Open review queue” available to Model operator in this scope?

Primary action: Open review queue → O03

Secondary disclosure: Keep Model, Labeled sample in the decision layer. Put the supporting details for False-positive review, Drift, full source/version history and FR-S14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a ai model monitoring fixture for Model operator; change Model and make Drift unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open review queue” reaches O03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I01 — Connector catalog

Context: configuration · **Role:** Integration admin · **FR:** FR-I01

Primary visual: Connector catalog grouped by event, advisory and control capability

Primary question: For connector catalog, what do MQTT sensors, CAP authority establish, and is “Configure connector” available to Integration admin in this scope?

Primary action: Configure connector → I02

Secondary disclosure: Keep MQTT sensors, CAP authority in the decision layer. Put the supporting details for MAVLink gateway, IOC bridge, full source/version history and FR-I01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a connector catalog fixture for Integration admin; change MQTT sensors and make IOC bridge unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Configure connector” reaches I02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I02 — Connector setup

Context: configuration · **Role:** Integration admin · **FR:** FR-I02

Primary visual: Connector setup flow with endpoint-region-capability preview

Primary question: For connector setup, what do Type, Endpoint establish, and is “Test connection” available to Integration admin in this scope?

Primary action: Test connection → I03

Secondary disclosure: Keep Type, Endpoint in the decision layer. Put the supporting details for Region, Auth, Capabilities, full source/version history and FR-I02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a connector setup fixture for Integration admin; change Type and make Capabilities unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Test connection” reaches I03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I03 — Schema mapping

Context: configuration · **Role:** Integration admin · **FR:** FR-I03

Primary visual: Source-to-target field mapping with units and unknown-field lane

Primary question: For schema mapping, what do Source field, Target establish, and is “Run contract tests” available to Integration admin in this scope?

Primary action: Run contract tests → I04

Secondary disclosure: Keep Source field, Target in the decision layer. Put the supporting details for Unit, Location, Unknown fields, full source/version history and FR-I03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a schema mapping fixture for Integration admin; change Source field and make Unknown fields unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Run contract tests” reaches I04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I04 — Connector test console

Context: configuration · **Role:** Integration admin · **FR:** FR-I04

Primary visual: Contract-test matrix separating accepted and rejected fixtures

Primary question: For connector test console, what do Valid event, Duplicate event establish, and is “Enable demo connector” available to Integration admin in this scope?

Primary action: Enable demo connector → I05

Secondary disclosure: Keep Valid event, Duplicate event in the decision layer. Put the supporting details for Wrong tenant, Expired certificate, full source/version history and FR-I04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a connector test console fixture for Integration admin; change Valid event and make Expired certificate unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Enable demo connector” reaches I05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I05 — Connector health

Context: configuration · **Role:** Integration admin · **FR:** FR-I05

Primary visual: Connector health lanes showing lag, failure and data age

Primary question: For connector health, what do MQTT bridge, IOC bridge establish, and is “Inspect dead letters” available to Integration admin in this scope?

Primary action: Inspect dead letters → I06

Secondary disclosure: Keep MQTT bridge, IOC bridge in the decision layer. Put the supporting details for CAP feed, Drone bridge, full source/version history and FR-I05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a connector health fixture for Integration admin; change MQTT bridge and make Drone bridge unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect dead letters” reaches I06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I06 — Dead-letter & replay

Context: configuration · **Role:** Integration admin · **FR:** FR-I06

Primary visual: Dead-letter payload diff and replay eligibility

Primary question: For dead-letter & replay, what do Event, Error establish, and is “Replay simulated event” available to Integration admin in this scope?

Primary action: Replay simulated event → I05

Secondary disclosure: Keep Event, Error in the decision layer. Put the supporting details for Correction, Replay, full source/version history and FR-I06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a dead-letter & replay fixture for Integration admin; change Event and make Replay unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Replay simulated event” reaches I05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I07 — API clients & scopes

Context: configuration · **Role:** Tenant admin · **FR:** FR-I07

Primary visual: API scope grid with masked credentials and expiry

Primary question: For api clients & scopes, what do Client, Scopes establish, and is “Save client” available to Tenant admin in this scope?

Primary action: Save client → I08

Secondary disclosure: Keep Client, Scopes in the decision layer. Put the supporting details for Region, Secret, Expiry, full source/version history and FR-I07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a api clients & scopes fixture for Tenant admin; change Client and make Expiry unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save client” reaches I08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I08 — Webhook subscriptions

Context: configuration · **Role:** Integration admin · **FR:** FR-I08

Primary visual: Webhook route diagram showing signature and retry policy

Primary question: For webhook subscriptions, what do URL, Events establish, and is “Send test payload” available to Integration admin in this scope?

Primary action: Send test payload → I04

Secondary disclosure: Keep URL, Events in the decision layer. Put the supporting details for Signature, Retries, full source/version history and FR-I08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a webhook subscriptions fixture for Integration admin; change URL and make Retries unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Send test payload” reaches I04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I09 — Partner exchange agreements

Context: configuration · **Role:** Tenant admin · **FR:** FR-I09

Primary visual: Partner disclosure boundary with retention and approval state

Primary question: For partner exchange agreements, what do Partner, Purpose establish, and is “Save agreement draft” available to Tenant admin in this scope?

Primary action: Save agreement draft → I10

Secondary disclosure: Keep Partner, Purpose in the decision layer. Put the supporting details for Fields, Retention, Approval, full source/version history and FR-I09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a partner exchange agreements fixture for Tenant admin; change Partner and make Approval unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save agreement draft” reaches I10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I10 — IOC exchange history

Context: configuration · **Role:** Operator · **FR:** FR-I10

Primary visual: Exchange timeline with payload scope and receipt identity

Primary question: For ioc exchange history, what do INC-2041, SOS-041 establish, and is “Inspect payload” available to Operator in this scope?

Primary action: Inspect payload → I04

Secondary disclosure: Keep INC-2041, SOS-041 in the decision layer. Put the supporting details for Update 4, full source/version history and FR-I10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a ioc exchange history fixture for Operator; change INC-2041 and make Update 4 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Inspect payload” reaches I04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I11 — Drone fleet

Context: configuration · **Role:** Drone operator · **FR:** FR-I11

Primary visual: Fleet readiness cards showing battery, availability and pilot control

Primary question: For drone fleet, what do DR-01, DR-02 establish, and is “Plan mission” available to Drone operator in this scope?

Primary action: Plan mission → I12

Secondary disclosure: Keep DR-01, DR-02 in the decision layer. Put the supporting details for DR-03, Control, full source/version history and FR-I11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a drone fleet fixture for Drone operator; change DR-01 and make Control unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Plan mission” reaches I12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I12 — Drone mission planning

Context: configuration · **Role:** Drone operator · **FR:** FR-I12

Primary visual: Mission geofence with pilot, weather and authority checklist

Primary question: For drone mission planning, what do Purpose, Geofence establish, and is “Review mission approval” available to Drone operator in this scope?

Primary action: Review mission approval → I13

Secondary disclosure: Keep Purpose, Geofence in the decision layer. Put the supporting details for Pilot, Weather, Authority, full source/version history and FR-I12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a drone mission planning fixture for Drone operator; change Purpose and make Authority unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review mission approval” reaches I13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I13 — Drone preflight & approval

Context: configuration · **Role:** Drone supervisor · **FR:** FR-I13

Primary visual: Preflight gate showing lost-link plan and explicit pilot approval

Primary question: For drone preflight & approval, what do Pilot, Battery establish, and is “Approve simulated mission” available to Drone supervisor in this scope?

Primary action: Approve simulated mission → I14

Secondary disclosure: Keep Pilot, Battery in the decision layer. Put the supporting details for Geofence, Clearance, Lost-link plan, full source/version history and FR-I13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a drone preflight & approval fixture for Drone supervisor; change Pilot and make Lost-link plan unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve simulated mission” reaches I14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I14 — Drone mission telemetry

Context: command · **Role:** Drone operator · **FR:** FR-I14

Primary visual: Telemetry strip with stale-data warning and local-control ownership

Primary question: For drone mission telemetry, what do Flight state, Altitude establish, and is “Review mission record” available to Drone operator in this scope?

Primary action: Review mission record → I15

Secondary disclosure: Keep Flight state, Altitude in the decision layer. Put the supporting details for Battery, Telemetry age, Command, full source/version history and FR-I14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a drone mission telemetry fixture for Drone operator; change Flight state and make Command unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review mission record” reaches I15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I15 — Drone mission record

Context: configuration · **Role:** Drone operator · **FR:** FR-I15

Primary visual: Mission evidence timeline with flight log and accountable pilot

Primary question: For drone mission record, what do Mission, Incident establish, and is “Return to fleet” available to Drone operator in this scope?

Primary action: Return to fleet → I11

Secondary disclosure: Keep Mission, Incident in the decision layer. Put the supporting details for Flight log, Outcome, Pilot, full source/version history and FR-I15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

Validation: Use a drone mission record fixture for Drone operator; change Mission and make Pilot unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Return to fleet” reaches I11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T01 — Tenant overview

Context: configuration · **Role:** Tenant admin · **FR:** FR-T01

Primary visual: Tenant health summary with region and capacity allocation

Primary question: For tenant overview, what do Region, Sites establish, and is “Manage organization” available to Tenant admin in this scope?

Primary action: Manage organization → T02

Secondary disclosure: Keep Region, Sites in the decision layer. Put the supporting details for Licensed devices, Operational health, full source/version history and FR-T01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a tenant overview fixture for Tenant admin; change Region and make Operational health unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Manage organization” reaches T02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T02 — Organization & regions

Context: configuration · **Role:** Tenant admin · **FR:** FR-T02

Primary visual: Organization region boundary beside locale and timezone controls

Primary question: For organization & regions, what do Organization, Home region establish, and is “Save settings” available to Tenant admin in this scope?

Primary action: Save settings → T01

Secondary disclosure: Keep Organization, Home region in the decision layer. Put the supporting details for Locale, Timezone, Cross-region transfer, full source/version history and FR-T02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a organization & regions fixture for Tenant admin; change Organization and make Cross-region transfer unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save settings” reaches T01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T03 — Users & invitations

Context: configuration · **Role:** Tenant admin · **FR:** FR-T03

Primary visual: Membership roster with invitation lifecycle and role badges

Primary question: For users & invitations, what do Alex, Jamie establish, and is “Edit membership” available to Tenant admin in this scope?

Primary action: Edit membership → T04

Secondary disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Service client, full source/version history and FR-T03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a users & invitations fixture for Tenant admin; change Alex and make Service client unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit membership” reaches T04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T04 — Role & scope editor

Context: configuration · **Role:** Tenant admin · **FR:** FR-T04

Primary visual: Permission-by-site matrix with explicit denied capabilities

Primary question: For role & scope editor, what do Role, Sites establish, and is “Save role” available to Tenant admin in this scope?

Primary action: Save role → T03

Secondary disclosure: Keep Role, Sites in the decision layer. Put the supporting details for Permission, Restriction, full source/version history and FR-T04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a role & scope editor fixture for Tenant admin; change Role and make Restriction unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save role” reaches T03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T05 — On-call & escalation

Context: configuration · **Role:** Supervisor · **FR:** FR-T05

Primary visual: Escalation ladder with timed owner changes and calendar coverage

Primary question: For on-call & escalation, what do Primary, After 60 seconds establish, and is “Save escalation” available to Supervisor in this scope?

Primary action: Save escalation → T01

Secondary disclosure: Keep Primary, After 60 seconds in the decision layer. Put the supporting details for After 120 seconds, Calendar, full source/version history and FR-T05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a on-call & escalation fixture for Supervisor; change Primary and make Calendar unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save escalation” reaches T01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T06 — Message templates & languages

Context: configuration · **Role:** Content editor · **FR:** FR-T06

Primary visual: Multilingual message preview with variable and fallback status

Primary question: For message templates & languages, what do Template, English establish, and is “Preview template” available to Content editor in this scope?

Primary action: Preview template → O14

Secondary disclosure: Keep Template, English in the decision layer. Put the supporting details for Japanese, Fallback, Variables, full source/version history and FR-T06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a message templates & languages fixture for Content editor; change Template and make Variables unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Preview template” reaches O14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T07 — Hazard taxonomy & playbooks

Context: configuration · **Role:** Tenant admin · **FR:** FR-T07

Primary visual: Hazard taxonomy tree connected to draft playbook steps

Primary question: For hazard taxonomy & playbooks, what do Category, Severity default establish, and is “Save draft playbook” available to Tenant admin in this scope?

Primary action: Save draft playbook → T08

Secondary disclosure: Keep Category, Severity default in the decision layer. Put the supporting details for Playbook, Sensitivity, full source/version history and FR-T07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a hazard taxonomy & playbooks fixture for Tenant admin; change Category and make Sensitivity unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save draft playbook” reaches T08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T08 — Playbook approval

Context: configuration · **Role:** Supervisor · **FR:** FR-T08

Primary visual: Versioned playbook step sequence and timeout responsibilities

Primary question: For playbook approval, what do Version, Trigger establish, and is “Publish demo playbook” available to Supervisor in this scope?

Primary action: Publish demo playbook → T07

Secondary disclosure: Keep Version, Trigger in the decision layer. Put the supporting details for Steps, Timeout, Owner, full source/version history and FR-T08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a playbook approval fixture for Supervisor; change Version and make Owner unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish demo playbook” reaches T07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T09 — Retention & legal holds

Context: configuration · **Role:** Privacy officer · **FR:** FR-T09

Primary visual: Retention timeline with legal-hold overrides

Primary question: For retention & legal holds, what do Precise location, Incident record establish, and is “Save policy draft” available to Privacy officer in this scope?

Primary action: Save policy draft → T10

Secondary disclosure: Keep Precise location, Incident record in the decision layer. Put the supporting details for Evidence, Legal hold, full source/version history and FR-T09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a retention & legal holds fixture for Privacy officer; change Precise location and make Legal hold unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save policy draft” reaches T10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T10 — Privacy request review

Context: configuration · **Role:** Privacy officer · **FR:** FR-T10

Primary visual: Data export preview showing identity proof and evidence exclusions

Primary question: For privacy request review, what do Identity, Data classes establish, and is “Approve demo export” available to Privacy officer in this scope?

Primary action: Approve demo export → C30

Secondary disclosure: Keep Identity, Data classes in the decision layer. Put the supporting details for Incident evidence, Decision, full source/version history and FR-T10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a privacy request review fixture for Privacy officer; change Identity and make Decision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve demo export” reaches C30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T11 — Audit record

Context: configuration · **Role:** Auditor · **FR:** FR-T11

Primary visual: Immutable audit event linking actor, object and trace

Primary question: For audit record, what do Actor, Action establish, and is “Return to audit search” available to Auditor in this scope?

Primary action: Return to audit search → O30

Secondary disclosure: Keep Actor, Action in the decision layer. Put the supporting details for Object, Time, Trace, full source/version history and FR-T11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a audit record fixture for Auditor; change Actor and make Trace unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Return to audit search” reaches O30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T12 — Service health & recovery

Context: configuration · **Role:** Platform operator · **FR:** FR-T12

Primary visual: Dependency-health topology with recovery readiness

Primary question: For service health & recovery, what do EU ingestion, Notification provider establish, and is “Open recovery runbook” available to Platform operator in this scope?

Primary action: Open recovery runbook → T13

Secondary disclosure: Keep EU ingestion, Notification provider in the decision layer. Put the supporting details for Database replica, Last restore drill, full source/version history and FR-T12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a service health & recovery fixture for Platform operator; change EU ingestion and make Last restore drill unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open recovery runbook” reaches T13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T13 — Recovery runbook

Context: configuration · **Role:** Platform operator · **FR:** FR-T13

Primary visual: Recovery checklist with verified completion evidence per step

Primary question: For recovery runbook, what do Step 1, Step 2 establish, and is “Record drill outcome” available to Platform operator in this scope?

Primary action: Record drill outcome → T12

Secondary disclosure: Keep Step 1, Step 2 in the decision layer. Put the supporting details for Step 3, Step 4, full source/version history and FR-T13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a recovery runbook fixture for Platform operator; change Step 1 and make Step 4 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Record drill outcome” reaches T12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T14 — Usage & subscription

Context: configuration · **Role:** Billing admin · **FR:** FR-T14

Primary visual: Capacity consumption chart with emergency reservation

Primary question: For usage & subscription, what do Plan, Sites establish, and is “Review capacity policy” available to Billing admin in this scope?

Primary action: Review capacity policy → T15

Secondary disclosure: Keep Plan, Sites in the decision layer. Put the supporting details for Devices, SMS budget, Emergency behavior, full source/version history and FR-T14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a usage & subscription fixture for Billing admin; change Plan and make Emergency behavior unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review capacity policy” reaches T15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T15 — Capacity & quota policy

Context: configuration · **Role:** Platform operator · **FR:** FR-T15

Primary visual: Queue policy comparison showing P1 reserves and hard-limit effects

Primary question: For capacity & quota policy, what do P1 reserved queue, Telemetry sampling establish, and is “Save capacity draft” available to Platform operator in this scope?

Primary action: Save capacity draft → T14

Secondary disclosure: Keep P1 reserved queue, Telemetry sampling in the decision layer. Put the supporting details for SOS ingestion, Warning threshold, Hard limit behavior, full source/version history and FR-T15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a capacity & quota policy fixture for Platform operator; change P1 reserved queue and make Hard limit behavior unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save capacity draft” reaches T14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T16 — Analytics & export

Context: configuration · **Role:** Analyst · **FR:** FR-T16

Primary visual: Distribution chart separating median, P95 and missing receipts

Primary question: For analytics & export, what do Period, Median acknowledgment establish, and is “Prepare demo report” available to Analyst in this scope?

Primary action: Prepare demo report → T17

Secondary disclosure: Keep Period, Median acknowledgment in the decision layer. Put the supporting details for P95 acknowledgment, Unconfirmed delivery, Export, full source/version history and FR-T16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a analytics & export fixture for Analyst; change Period and make Export unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Prepare demo report” reaches T17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T17 — Report schedule

Context: configuration · **Role:** Analyst · **FR:** FR-T17

Primary visual: Report schedule preview with timezone and recipient disclosure

Primary question: For report schedule, what do Report, Audience establish, and is “Save schedule draft” available to Analyst in this scope?

Primary action: Save schedule draft → T16

Secondary disclosure: Keep Report, Audience in the decision layer. Put the supporting details for Timezone, Schedule, PII, full source/version history and FR-T17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

Validation: Use a report schedule fixture for Analyst; change Report and make PII unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save schedule draft” reaches T16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X01 — No connection / SOS pending

Context: emergency · **Role:** Citizen · **FR:** FR-X01

Primary visual: Locally saved SOS receipt with server-receipt absence prominent

Primary question: For no connection / sos pending, what do Local record, Server receipt establish, and is “View offline outbox” available to Citizen in this scope?

Primary action: View offline outbox → R11

Secondary disclosure: Keep Local record, Server receipt in the decision layer. Put the supporting details for Retry, Alternative, full source/version history and FR-X01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a no connection / sos pending fixture for Citizen; change Local record and make Alternative unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View offline outbox” reaches R11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X02 — Permission unavailable

Context: emergency · **Role:** Citizen · **FR:** FR-X02

Primary visual: Manual location entry beside unavailable-permission explanation

Primary question: For permission unavailable, what do Automatic location, Alternative establish, and is “Enter location manually” available to Citizen in this scope?

Primary action: Enter location manually → C06

Secondary disclosure: Keep Automatic location, Alternative in the decision layer. Put the supporting details for Privacy, SOS, full source/version history and FR-X02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a permission unavailable fixture for Citizen; change Automatic location and make SOS unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Enter location manually” reaches C06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X03 — No active incidents

Context: configuration · **Role:** Operator · **FR:** FR-X03

Primary visual: Empty incident view with coverage and unknown-device evidence

Primary question: For no active incidents, what do Filters, Coverage establish, and is “Review device health” available to Operator in this scope?

Primary action: Review device health → S06

Secondary disclosure: Keep Filters, Coverage in the decision layer. Put the supporting details for Last sync, Unknown devices, full source/version history and FR-X03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a no active incidents fixture for Operator; change Filters and make Unknown devices unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review device health” reaches S06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X04 — Access denied

Context: configuration · **Role:** All roles · **FR:** FR-X04

Primary visual: Workspace selection recovery without disclosing denied object

Primary question: For access denied, what do Request, Data establish, and is “Choose workspace” available to All roles in this scope?

Primary action: Choose workspace → A04

Secondary disclosure: Keep Request, Data in the decision layer. Put the supporting details for Next step, Audit, full source/version history and FR-X04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a access denied fixture for All roles; change Request and make Audit unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Choose workspace” reaches A04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X05 — Service unavailable

Context: emergency · **Role:** All roles · **FR:** FR-X05

Primary visual: Service outage panel with cached-content age and alternate path

Primary question: For service unavailable, what do Status, Cached content establish, and is “Open offline pack” available to All roles in this scope?

Primary action: Open offline pack → C18

Secondary disclosure: Keep Status, Cached content in the decision layer. Put the supporting details for Submission, Recovery, full source/version history and FR-X05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a service unavailable fixture for All roles; change Status and make Recovery unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open offline pack” reaches C18 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X06 — Unsaved changes & conflict

Context: configuration · **Role:** Site admin · **FR:** FR-X06

Primary visual: Local/server version comparison preserving unsaved edits

Primary question: For unsaved changes & conflict, what do Your version, Current version establish, and is “Review latest version” available to Site admin in this scope?

Primary action: Review latest version → S05

Secondary disclosure: Keep Your version, Current version in the decision layer. Put the supporting details for Your changes, Resolution, full source/version history and FR-X06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a unsaved changes & conflict fixture for Site admin; change Your version and make Resolution unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review latest version” reaches S05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X07 — Empty search & filter reset

Context: configuration · **Role:** Operator · **FR:** FR-X07

Primary visual: Search query and active filters beside reset control

Primary question: For empty search & filter reset, what do Query, Scope establish, and is “Reset search” available to Operator in this scope?

Primary action: Reset search → O30

Secondary disclosure: Keep Query, Scope in the decision layer. Put the supporting details for Period, Suggestion, full source/version history and FR-X07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a empty search & filter reset fixture for Operator; change Query and make Suggestion unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Reset search” reaches O30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X08 — Expired session

Context: configuration · **Role:** All roles · **FR:** FR-X08

Primary visual: Session-expired panel explaining protected draft recovery

Primary question: For expired session, what do Draft, Session establish, and is “Sign in” available to All roles in this scope?

Primary action: Sign in → A02

Secondary disclosure: Keep Draft, Session in the decision layer. Put the supporting details for Sensitive data, Next, full source/version history and FR-X08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a expired session fixture for All roles; change Draft and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Sign in” reaches A02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X09 — Loading & stale data

Context: emergency · **Role:** Operator · **FR:** FR-X09

Primary visual: Stale-data band showing source age and unavailable commands

Primary question: For loading & stale data, what do Current view, Connection establish, and is “Return to command” available to Operator in this scope?

Primary action: Return to command → O01

Secondary disclosure: Keep Current view, Connection in the decision layer. Put the supporting details for Commands, Next, full source/version history and FR-X09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a loading & stale data fixture for Operator; change Current view and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Return to command” reaches O01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X10 — Attachment rejected

Context: configuration · **Role:** Responder · **FR:** FR-X10

Primary visual: Rejected attachment preview with reason and allowed replacement

Primary question: For attachment rejected, what do File, Reason establish, and is “Choose another attachment” available to Responder in this scope?

Primary action: Choose another attachment → R10

Secondary disclosure: Keep File, Reason in the decision layer. Put the supporting details for Allowed, Action, full source/version history and FR-X10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

Validation: Use a attachment rejected fixture for Responder; change File and make Action unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Choose another attachment” reaches R10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G01 — Regional operations

Context: configuration · **Role:** Platform operator · **FR:** FR-G01

Primary visual: Regional cell topology with residency boundaries

Primary question: For regional operations, what do EU cell, APAC cell establish, and is “Inspect regional cell” available to Platform operator in this scope?

Primary action: Inspect regional cell → G02

Secondary disclosure: Keep EU cell, APAC cell in the decision layer. Put the supporting details for Global directory, Residency, full source/version history and FR-G01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a regional operations fixture for Platform operator; change EU cell and make Residency unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect regional cell” reaches G02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G02 — Regional cell detail

Context: configuration · **Role:** Platform operator · **FR:** FR-G02

Primary visual: Cell replication and writer-epoch status

Primary question: For regional cell detail, what do Writer epoch, Broker lag establish, and is “Review recovery readiness” available to Platform operator in this scope?

Primary action: Review recovery readiness → G03

Secondary disclosure: Keep Writer epoch, Broker lag in the decision layer. Put the supporting details for Database, DR target, full source/version history and FR-G02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a regional cell detail fixture for Platform operator; change Writer epoch and make DR target unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review recovery readiness” reaches G03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G03 — Disaster recovery drill

Context: configuration · **Role:** Platform operator · **FR:** FR-G03

Primary visual: Recovery drill timeline with target RPO/RTO and evidence

Primary question: For disaster recovery drill, what do Cell, Mode establish, and is “Record drill evidence” available to Platform operator in this scope?

Primary action: Record drill evidence → G04

Secondary disclosure: Keep Cell, Mode in the decision layer. Put the supporting details for Backup, Target RPO, Target RTO, full source/version history and FR-G03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a disaster recovery drill fixture for Platform operator; change Cell and make Target RTO unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Record drill evidence” reaches G04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G04 — Recovery verification

Context: configuration · **Role:** Platform operator · **FR:** FR-G04

Primary visual: Recovery proof grid separating fencing and record durability

Primary question: For recovery verification, what do Writer fencing, SOS records establish, and is “Review regional operations” available to Platform operator in this scope?

Primary action: Review regional operations → G01

Secondary disclosure: Keep Writer fencing, SOS records in the decision layer. Put the supporting details for Outbox, Residency, full source/version history and FR-G04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a recovery verification fixture for Platform operator; change Writer fencing and make Residency unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review regional operations” reaches G01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G05 — Tenant provisioning

Context: configuration · **Role:** Platform admin · **FR:** FR-G05

Primary visual: Tenant provisioning dependency plan and isolation boundary

Primary question: For tenant provisioning, what do Tenant, Home region establish, and is “Review provisioning plan” available to Platform admin in this scope?

Primary action: Review provisioning plan → G06

Secondary disclosure: Keep Tenant, Home region in the decision layer. Put the supporting details for Isolation, Plan, Policy set, full source/version history and FR-G05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a tenant provisioning fixture for Platform admin; change Tenant and make Policy set unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review provisioning plan” reaches G06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G06 — Tenant lifecycle

Context: configuration · **Role:** Platform admin · **FR:** FR-G06

Primary visual: Tenant lifecycle state diagram with migration/deletion gates

Primary question: For tenant lifecycle, what do Status, Region migration establish, and is “Request support access” available to Platform admin in this scope?

Primary action: Request support access → G07

Secondary disclosure: Keep Status, Region migration in the decision layer. Put the supporting details for Deletion, Support access, full source/version history and FR-G06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a tenant lifecycle fixture for Platform admin; change Status and make Support access unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Request support access” reaches G07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G07 — Time-bound support access

Context: configuration · **Role:** Security operator · **FR:** FR-G07

Primary visual: Support request showing minimal scope and access expiry

Primary question: For time-bound support access, what do Tenant, Scope establish, and is “Submit access request” available to Security operator in this scope?

Primary action: Submit access request → G08

Secondary disclosure: Keep Tenant, Scope in the decision layer. Put the supporting details for Reason, Duration, PII access, full source/version history and FR-G07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a time-bound support access fixture for Security operator; change Tenant and make PII access unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Submit access request" reaches G08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G08 — Support access approval

Context: configuration · **Role:** Tenant admin · **FR:** FR-G08

Primary visual: Support authorization comparing requester and approved scope

Primary question: For support access approval, what do Requester, Scope establish, and is "Approve scoped access — sample" available to Tenant admin in this scope?

Primary action: Approve scoped access — sample → G06

Secondary disclosure: Keep Requester, Scope in the decision layer. Put the supporting details for Expiry, Audit, full source/version history and FR-G08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a support access approval fixture for Tenant admin; change Requester and make Audit unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Approve scoped access — sample" reaches G06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G09 — Release & schema compatibility

Context: configuration · **Role:** Release manager · **FR:** FR-G09

Primary visual: Release compatibility matrix across schema and mobile versions

Primary question: For release & schema compatibility, what do Incident service, Event schema establish, and is "Review release gate" available to Release manager in this scope?

Primary action: Review release gate → G10

Secondary disclosure: Keep Incident service, Event schema in the decision layer. Put the supporting details for Mobile API, Rollback, full source/version history and FR-G09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a release & schema compatibility fixture for Release manager; change Incident service and make Rollback unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Review release gate" reaches G10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G10 — Release gate & rollback

Context: configuration · **Role:** Release manager · **FR:** FR-G10

Primary visual: Release gate checklist with rollback viability and SLO trend

Primary question: For release gate & rollback, what do Contract tests, Tenant isolation establish, and is “Record release decision” available to Release manager in this scope?

Primary action: Record release decision → G09

Secondary disclosure: Keep Contract tests, Tenant isolation in the decision layer. Put the supporting details for SLO burn rate, Migration, full source/version history and FR-G10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a release gate & rollback fixture for Release manager; change Contract tests and make Migration unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Record release decision“ reaches G09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G11 — Security incidents

Context: configuration · **Role:** Security operator · **FR:** FR-G11

Primary visual: Security signal queue with affected trust boundary

Primary question: For security incidents, what do Device certificate, Webhook replay establish, and is “Open security case” available to Security operator in this scope?

Primary action: Open security case → G12

Secondary disclosure: Keep Device certificate, Webhook replay in the decision layer. Put the supporting details for Support session, Upload scan, full source/version history and FR-G11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a security incidents fixture for Security operator; change Device certificate and make Upload scan unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open security case“ reaches G12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G12 — Security case & containment

Context: configuration · **Role:** Security operator · **FR:** FR-G12

Primary visual: Containment timeline showing residual exposure and evidence

Primary question: For security case & containment, what do Case, Source establish, and is “Record containment” available to Security operator in this scope?

Primary action: Record containment → G11

Secondary disclosure: Keep Case, Source in the decision layer. Put the supporting details for Action, Coverage, Evidence, full source/version history and FR-G12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

Validation: Use a security case & containment fixture for Security operator; change Case and make Evidence unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Record containment” reaches G11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D01 — Developer workspace

Context: configuration · **Role:** Partner developer · **FR:** FR-D01

Primary visual: Developer resource map separating sandbox and production

Primary question: For developer workspace, what do API reference, Event schemas establish, and is “Open API explorer” available to Partner developer in this scope?

Primary action: Open API explorer → D02

Secondary disclosure: Keep API reference, Event schemas in the decision layer. Put the supporting details for SDK examples, Sandbox, full source/version history and FR-D01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a developer workspace fixture for Partner developer; change API reference and make Sandbox unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open API explorer” reaches D02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D02 — API explorer

Context: configuration · **Role:** Partner developer · **FR:** FR-D02

Primary visual: Request/response split pane with scope and idempotency preview

Primary question: For api explorer, what do Auth, Payload establish, and is “Run sandbox request” available to Partner developer in this scope?

Primary action: Run sandbox request → D03

Secondary disclosure: Keep Auth, Payload in the decision layer. Put the supporting details for Idempotency, Response, full source/version history and FR-D02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a api explorer fixture for Partner developer; change Auth and make Response unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Run sandbox request” reaches D03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D03 — Sandbox request results

Context: configuration · **Role:** Partner developer · **FR:** FR-D03

Primary visual: Sandbox response showing HTTP, durable receipt and trace separately

Primary question: For sandbox request results, what do HTTP status, Receipt establish, and is “Open schema registry” available to Partner developer in this scope?

Primary action: Open schema registry → D04

Secondary disclosure: Keep HTTP status, Receipt in the decision layer. Put the supporting details for Durability, Incident, Trace, full source/version history and FR-D03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a sandbox request results fixture for Partner developer; change HTTP status and make Trace unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open schema registry” reaches D04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D04 — Schema registry

Context: configuration · **Role:** Partner developer · **FR:** FR-D04

Primary visual: Schema version tree with compatibility and deprecation markers

Primary question: For schema registry, what do signal.fire.observed, incident.updated establish, and is “View schema contract” available to Partner developer in this scope?

Primary action: View schema contract → D05

Secondary disclosure: Keep signal.fire.observed, incident.updated in the decision layer. Put the supporting details for command.result, Deprecated versions, full source/version history and FR-D04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a schema registry fixture for Partner developer; change signal.fire.observed and make Deprecated versions unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View schema contract” reaches D05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D05 — Connector manifest editor

Context: configuration · **Role:** Partner developer · **FR:** FR-D05

Primary visual: Connector capability manifest with validation diff

Primary question: For connector manifest editor, what do Connector, Version establish, and is “Submit for conformance” available to Partner developer in this scope?

Primary action: Submit for conformance → D06

Secondary disclosure: Keep Connector, Version in the decision layer. Put the supporting details for Capabilities, Auth, Retry class, full source/version history and FR-D05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a connector manifest editor fixture for Partner developer; change Connector and make Retry class unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Submit for conformance” reaches D06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D06 — Conformance certification

Context: configuration · **Role:** Integration reviewer · **FR:** FR-D06

Primary visual: Conformance scenario matrix with security and backpressure results

Primary question: For conformance certification, what do Duplicate delivery, Tenant spoof establish, and is “Approve adapter version” available to Integration reviewer in this scope?

Primary action: Approve adapter version → I01

Secondary disclosure: Keep Duplicate delivery, Tenant spoof in the decision layer. Put the supporting details for Clock skew, Expired credentials, Backpressure, full source/version history and FR-D06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a conformance certification fixture for Integration reviewer; change Duplicate delivery and make Backpressure unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve adapter version” reaches I01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D07 — Webhook delivery trace

Context: configuration · **Role:** Partner developer · **FR:** FR-D07

Primary visual: Webhook retry timeline with exact delivery outcome

Primary question: For webhook delivery trace, what do 10:40:00, 10:40:01 establish, and is “Review subscription” available to Partner developer in this scope?

Primary action: Review subscription → I08

Secondary disclosure: Keep 10:40:00, 10:40:01 in the decision layer. Put the supporting details for 10:40:03, 10:40:07, full source/version history and FR-D07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a webhook delivery trace fixture for Partner developer; change 10:40:00 and make 10:40:07 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review subscription” reaches I08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D08 — Credential rotation

Context: configuration · **Role:** Partner developer · **FR:** FR-D08

Primary visual: Credential overlap timeline with secret-free rotation preview

Primary question: For credential rotation, what do Client, Current key establish, and is “Create rotation request” available to Partner developer in this scope?

Primary action: Create rotation request → I07

Secondary disclosure: Keep Client, Current key in the decision layer. Put the supporting details for Overlap window, New key, full source/version history and FR-D08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

Validation: Use a credential rotation fixture for Partner developer; change Client and make New key unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Create rotation request” reaches I07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L01 — Language coverage

Context: configuration · **Role:** Localization manager · **FR:** FR-L01

Primary visual: Language coverage grid highlighting critical missing content

Primary question: For language coverage, what do Vietnamese, English establish, and is “Open translation workspace” available to Localization manager in this scope?

Primary action: Open translation workspace → L02

Secondary disclosure: Keep Vietnamese, English in the decision layer. Put the supporting details for Japanese, Critical templates, full source/version history and FR-L01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

Validation: Use a language coverage fixture for Localization manager; change Vietnamese and make Critical templates unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open translation workspace” reaches L02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L02 — Translation editor

Context: configuration · **Role:** Translator · **FR:** FR-L02

Primary visual: Source/translation paired editor with protected variables

Primary question: For translation editor, what do Template, Source language establish, and is “Submit translation review” available to Translator in this scope?

Primary action: Submit translation review → L03

Secondary disclosure: Keep Template, Source language in the decision layer. Put the supporting details for Target language, Variables, Review, full source/version history and FR-L02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

Validation: Use a translation editor fixture for Translator; change Template and make Review unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Submit translation review” reaches L03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L03 — Translation review

Context: configuration · **Role:** Language reviewer · **FR:** FR-L03

Primary visual: Version-parity review showing text, variables and voice transcript

Primary question: For translation review, what do Source version, Target locale establish, and is “Approve translation version” available to Language reviewer in this scope?

Primary action: Approve translation version → L01

Secondary disclosure: Keep Source version, Target locale in the decision layer. Put the supporting details for Variable parity, Voice transcript, Reviewer, full source/version history and FR-L03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

Validation: Use a translation review fixture for Language reviewer; change Source version and make Reviewer unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve translation version” reaches L01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L04 — Voice & pronunciation

Context: configuration · **Role:** Content editor · **FR:** FR-L04

Primary visual: Pronunciation samples paired with exact transcript and fallback

Primary question: For voice & pronunciation, what do Locale, Voice establish, and is “Save voice configuration” available to Content editor in this scope?

Primary action: Save voice configuration → L01

Secondary disclosure: Keep Locale, Voice in the decision layer. Put the supporting details for Transcript, Site pronunciation, Fallback, full source/version history and FR-L04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

Validation: Use a voice & pronunciation fixture for Content editor; change Locale and make Fallback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save voice configuration” reaches L01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L05 — Locale fallback policy

Context: configuration · **Role:** Tenant admin · **FR:** FR-L05

Primary visual: Locale fallback tree showing missing-critical-content behavior

Primary question: For locale fallback policy, what do Supported, User preference establish, and is “Save language policy” available to Tenant admin in this scope?

Primary action: Save language policy → L01

Secondary disclosure: Keep Supported, User preference in the decision layer. Put the supporting details for Missing critical content, Fallback language, full source/version history and FR-L05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

Validation: Use a locale fallback policy fixture for Tenant admin; change Supported and make Fallback language unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save language policy” reaches L01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L06 — Emergency terminology

Context: configuration · **Role:** Language reviewer · **FR:** FR-L06

Primary visual: Emergency term comparison distinguishing safe, unknown and SOS

Primary question: For emergency terminology, what do SOS, Safe establish, and is “Review translation editor” available to Language reviewer in this scope?

Primary action: Review translation editor → L02

Secondary disclosure: Keep SOS, Safe in the decision layer. Put the supporting details for Unknown, full source/version history and FR-L06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

Validation: Use a emergency terminology fixture for Language reviewer; change SOS and make Unknown unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review translation editor” reaches L02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C31 — Circle membership & invitations

Context: configuration · **Role:** Citizen · **FR:** FR-C31

Primary visual: Circle roster with invitation state and sharing privileges

Primary question: For circle membership & invitations, what do Alex, Jamie establish, and is “Manage invitation” available to Citizen in this scope?

Primary action: Manage invitation → C32

Secondary disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Sharing, full source/version history and FR-C31 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a circle membership & invitations fixture for Citizen; change Alex and make Sharing unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Manage invitation” reaches C32 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C32 — Invite circle member

Context: configuration · **Role:** Citizen · **FR:** FR-C32

Primary visual: Invitation preview with recipient, role and location access

Primary question: For invite circle member, what do Circle, Contact establish, and is “Create invitation — sample” available to Citizen in this scope?

Primary action: Create invitation — sample → C31

Secondary disclosure: Keep Circle, Contact in the decision layer. Put the supporting details for Role, Invitation expiry, Location access, full source/version history and FR-C32 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a invite circle member fixture for Citizen; change Circle and make Location access unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Create invitation — sample” reaches C31 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C33 — Report or block conversation

Context: configuration · **Role:** Citizen · **FR:** FR-C33

Primary visual: Conversation report scope and block effect on safety warnings

Primary question: For report or block conversation, what do Conversation, Reason establish, and is “Submit report — sample” available to Citizen in this scope?

Primary action: Submit report — sample → C19

Secondary disclosure: Keep Conversation, Reason in the decision layer. Put the supporting details for Action, Site warnings, full source/version history and FR-C33 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a report or block conversation fixture for Citizen; change Conversation and make Site warnings unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Submit report — sample” reaches C19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C34 — Notification preferences

Context: configuration · **Role:** Citizen · **FR:** FR-C34

Primary visual: Separate social, safety and discreet notification preferences

Primary question: For notification preferences, what do Language, Social messages establish, and is “Save preferences” available to Citizen in this scope?

Primary action: Save preferences → C01

Secondary disclosure: Keep Language, Social messages in the decision layer. Put the supporting details for Safety alerts, Discreet requests, SMS, full source/version history and FR-C34 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a notification preferences fixture for Citizen; change Language and make SMS unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save preferences” reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O31 — Incident archive & reopen

Context: command · **Role:** Supervisor · **FR:** FR-O31

Primary visual: Closed-incident archive preserving original chronology

Primary question: For incident archive & reopen, what do INC-2032, INC-2028 establish, and is “Review reopening” available to Supervisor in this scope?

Primary action: Review reopening → O32

Secondary disclosure: Keep INC-2032, INC-2028 in the decision layer. Put the supporting details for INC-2010, full source/version history and FR-O31 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a incident archive & reopen fixture for Supervisor; change INC-2032 and make INC-2010 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review reopening” reaches O32 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O32 — Reopen incident

Context: command · **Role:** Supervisor · **FR:** FR-O32

Primary visual: Reopening preview showing new owner and retained evidence

Primary question: For reopen incident, what do Incident, Reason establish, and is “Reopen incident — sample” available to Supervisor in this scope?

Primary action: Reopen incident — sample → O04

Secondary disclosure: Keep Incident, Reason in the decision layer. Put the supporting details for Owner, Severity, History, full source/version history and FR-O32 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a reopen incident fixture for Supervisor; change Incident and make History unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Reopen incident — sample” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O33 — Federated incident sharing

Context: command · **Role:** Supervisor · **FR:** FR-O33

Primary visual: Partner exchange preview showing disclosed fields and authority

Primary question: For federated incident sharing, what do Partner, Fields establish, and is “Review partner exchange” available to Supervisor in this scope?

Primary action: Review partner exchange → I10

Secondary disclosure: Keep Partner, Fields in the decision layer. Put the supporting details for Authority, Agreement, full source/version history and FR-O33 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a federated incident sharing fixture for Supervisor; change Partner and make Agreement unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review partner exchange” reaches I10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O34 — Accessibility assistance queue

Context: command · **Role:** Dispatcher · **FR:** FR-O34

Primary visual: Accessibility queue matching functional needs to capable units

Primary question: For accessibility assistance queue, what do SOS-041, SOS-044 establish, and is “Assign capable unit” available to Dispatcher in this scope?

Primary action: Assign capable unit → O09

Secondary disclosure: Keep SOS-041, SOS-044 in the decision layer. Put the supporting details for SOS-046, Source, full source/version history and FR-O34 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

Validation: Use a accessibility assistance queue fixture for Dispatcher; change SOS-041 and make Source unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Assign capable unit” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C35 — Voice evacuation navigation

Context: emergency · **Role:** Citizen · **FR:** FR-C35

Primary visual: Next landmark instruction above floor route and position-quality band

Primary question: For voice evacuation navigation, what do Location, Next instruction establish, and is “Show next instruction” available to Citizen in this scope?

Primary action: Show next instruction → C36

Secondary disclosure: Keep Location, Next instruction in the decision layer. Put the supporting details for Position quality, Hazard data, full source/version history and FR-C35 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Stop obsolete speech when guidance version changes; stale position must not advance the next landmark.

Validation: Use a voice evacuation navigation fixture for Citizen; change Location and make Hazard data unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Show next instruction” reaches C36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C36 — Navigation step & orientation

Context: emergency · **Role:** Citizen · **FR:** FR-C36

Primary visual: Single navigation step with landmark picture and heading-confidence label

Primary question: For navigation step & orientation, what do Step, Instruction establish, and is “Repeat voice instruction” available to Citizen in this scope?

Primary action: Repeat voice instruction → C36

Secondary disclosure: Keep Step, Instruction in the decision layer. Put the supporting details for Heading, Accessible route, full source/version history and FR-C36 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Unknown heading removes directional arrows; use a verified landmark and manual orientation instead of fabricated left/right.

Validation: Use a navigation step & orientation fixture for Citizen; change Step and make Accessible route unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Repeat voice instruction” reaches C36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C37 — Route change notification

Context: emergency · **Role:** Citizen · **FR:** FR-C37

Primary visual: Withdrawn old route beside approved replacement and change reason

Primary question: For route change notification, what do Reason, Old route establish, and is “Review updated route” available to Citizen in this scope?

Primary action: Review updated route → C35

Secondary disclosure: Keep Reason, Old route in the decision layer. Put the supporting details for New instruction, Guidance version, full source/version history and FR-C37 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Withdraw old-route visuals and queued speech atomically; do not display replacement as approved before its version is accepted.

Validation: Use a route change notification fixture for Citizen; change Reason and make Guidance version unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review updated route” reaches C35 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C38 — Position uncertain / no route

Context: emergency · **Role:** Citizen · **FR:** FR-C38

Primary visual: Unknown floor/heading panel with manual location and assistance controls

Primary question: For position uncertain / no route, what do Floor, Heading establish, and is “Request assistance” available to Citizen in this scope?

Primary action: Request assistance → C05

Secondary disclosure: Keep Floor, Heading in the decision layer. Put the supporting details for Next step, Voice, full source/version history and FR-C38 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Do not invent a route or heading. Keep location correction and assistance reachable without permission recovery.

Validation: Use a position uncertain / no route fixture for Citizen; change Floor and make Voice unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Request assistance” reaches C05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C39 — Outdoor hazard avoidance

Context: emergency · **Role:** Citizen · **FR:** FR-C39

Primary visual: Outdoor hazard boundary, destination and unknown road segments

Primary question: For outdoor hazard avoidance, what do Area, Destination establish, and is “View next route step” available to Citizen in this scope?

Primary action: View next route step → C36

Secondary disclosure: Keep Area, Destination in the decision layer. Put the supporting details for Road condition, Unknown segments, full source/version history and FR-C39 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a outdoor hazard avoidance fixture for Citizen; change Area and make Unknown segments unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View next route step” reaches C36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C40 — Voice language & guidance settings

Context: configuration · **Role:** Citizen · **FR:** FR-C40

Primary visual: Voice sample, rate control and discreet-mode preview

Primary question: For voice language & guidance settings, what do Language, Voice establish, and is “Save guidance preferences” available to Citizen in this scope?

Primary action: Save guidance preferences → C35

Secondary disclosure: Keep Language, Voice in the decision layer. Put the supporting details for Speech rate, Haptics, Discreet mode, full source/version history and FR-C40 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

Validation: Use a voice language & guidance settings fixture for Citizen; change Language and make Discreet mode unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save guidance preferences” reaches C35 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P01 — Speaker zones & coverage

Context: command · **Role:** PA operator · **FR:** FR-P01

Primary visual: Speaker coverage map with uncovered zones and incident context

Primary question: For speaker zones & coverage, what do Building A, Outdoor campus establish, and is “Select broadcast zones” available to PA operator in this scope?

Primary action: Select broadcast zones → P02

Secondary disclosure: Keep Building A, Outdoor campus in the decision layer. Put the supporting details for Coverage gaps, Live incident, full source/version history and FR-P01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a speaker zones & coverage fixture for PA operator; change Building A and make Live incident unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Select broadcast zones” reaches P02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P02 — Compose rescue announcement

Context: command · **Role:** PA author · **FR:** FR-P02

Primary visual: Announcement text/audio preview tied to zones and expiry

Primary question: For compose rescue announcement, what do Zones, Priority establish, and is “Preview announcement” available to PA author in this scope?

Primary action: Preview announcement → P03

Secondary disclosure: Keep Zones, Priority in the decision layer. Put the supporting details for Languages, Source, Expires, full source/version history and FR-P02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a compose rescue announcement fixture for PA author; change Zones and make Expires unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Preview announcement” reaches P03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P03 — Preview & approve announcement

Context: command · **Role:** PA approver · **FR:** FR-P03

Primary visual: Exact message version with selected speakers and language sequence

Primary question: For preview & approve announcement, what do Message version, Selected speakers establish, and is “Approve sample playback” available to PA approver in this scope?

Primary action: Approve sample playback → P04

Secondary disclosure: Keep Message version, Selected speakers in the decision layer. Put the supporting details for Language sequence, Repetition, Approver, full source/version history and FR-P03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a preview & approve announcement fixture for PA approver; change Message version and make Approver unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve sample playback” reaches P04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P04 — Speaker playback monitoring

Context: command · **Role:** PA operator · **FR:** FR-P04

Primary visual: Playback progress distinguishing controller receipt from audibility

Primary question: For speaker playback monitoring, what do Controller, Speakers establish, and is “Inspect speaker health” available to PA operator in this scope?

Primary action: Inspect speaker health → P05

Secondary disclosure: Keep Controller, Speakers in the decision layer. Put the supporting details for Acoustic verification, Next, full source/version history and FR-P04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Speaker receipt cannot prove occupants heard the message; coverage gaps and unknown acoustic evidence remain prominent.

Validation: Use a speaker playback monitoring fixture for PA operator; change Controller and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect speaker health” reaches P05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P05 — Speaker device & self-test

Context: command · **Role:** Device admin · **FR:** FR-P05

Primary visual: Speaker power-network-amplifier chain with self-test age

Primary question: For speaker device & self-test, what do Power, Network establish, and is “Open coverage fallback” available to Device admin in this scope?

Primary action: Open coverage fallback → P01

Secondary disclosure: Keep Power, Network in the decision layer. Put the supporting details for Amplifier, Last self-test, Coverage fallback, full source/version history and FR-P05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a speaker device & self-test fixture for Device admin; change Power and make Coverage fallback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open coverage fallback” reaches P01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P06 — Live rescue microphone

Context: command · **Role:** Authorized announcer · **FR:** FR-P06

Primary visual: Live microphone session with selected zones and explicit stop control

Primary question: For live rescue microphone, what do Zones, Microphone establish, and is “Review live session — sample” available to Authorized announcer in this scope?

Primary action: Review live session — sample → P04

Secondary disclosure: Keep Zones, Microphone in the decision layer. Put the supporting details for Priority, Maximum session, Recording, full source/version history and FR-P06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a live rescue microphone fixture for Authorized announcer; change Zones and make Recording unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review live session — sample” reaches P04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P07 — Drone / vehicle speaker mission

Context: command · **Role:** Drone operator · **FR:** FR-P07

Primary visual: Mobile speaker mission footprint with pilot ownership

Primary question: For drone / vehicle speaker mission, what do Platform, Message establish, and is “Review mission” available to Drone operator in this scope?

Primary action: Review mission → I13

Secondary disclosure: Keep Platform, Message in the decision layer. Put the supporting details for Pilot, Broadcast area, Playback, full source/version history and FR-P07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a drone / vehicle speaker mission fixture for Drone operator; change Platform and make Playback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review mission” reaches I13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P08 — Audio library & TTS review

Context: configuration · **Role:** Content editor · **FR:** FR-P08

Primary visual: Language audio library with pronunciation review and transcript

Primary question: For audio library & tts review, what do PA-04 vi-VN, PA-04 en establish, and is “Edit voice content” available to Content editor in this scope?

Primary action: Edit voice content → L04

Secondary disclosure: Keep PA-04 vi-VN, PA-04 en in the decision layer. Put the supporting details for PA-04 ja-JP, Pronunciation, full source/version history and FR-P08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a audio library & tts review fixture for Content editor; change PA-04 vi-VN and make Pronunciation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit voice content” reaches L04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P09 — PA priority & arbitration

Context: configuration · **Role:** PA admin · **FR:** FR-P09

Primary visual: Priority arbitration stack showing local priority and lost-link behavior

Primary question: For pa priority & arbitration, what do Highest priority, Cloud rescue message establish, and is “Save arbitration policy” available to PA admin in this scope?

Primary action: Save arbitration policy → P01

Secondary disclosure: Keep Highest priority, Cloud rescue message in the decision layer. Put the supporting details for Routine audio, Lost link, Resume, full source/version history and FR-P09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

Validation: Use a pa priority & arbitration fixture for PA admin; change Highest priority and make Resume unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save arbitration policy” reaches P01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S15 — Evacuation graph editor

Context: configuration · **Role:** Route designer · **FR:** FR-S15

Primary visual: Evacuation graph with node/edge types and inaccessible links

Primary question: For evacuation graph editor, what do Nodes, Edges establish, and is “Review route validation” available to Route designer in this scope?

Primary action: Review route validation → S16

Secondary disclosure: Keep Nodes, Edges in the decision layer. Put the supporting details for Elevators, Map revision, full source/version history and FR-S15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a evacuation graph editor fixture for Route designer; change Nodes and make Map revision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review route validation” reaches S16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S16 — Route simulation & approval

Context: configuration · **Role:** Safety reviewer · **FR:** FR-S16

Primary visual: Scenario comparison showing blocked, unknown and wheelchair routes

Primary question: For route simulation & approval, what do Blocked corridor, Unknown floor establish, and is “Publish reviewed graph” available to Safety reviewer in this scope?

Primary action: Publish reviewed graph → S15

Secondary disclosure: Keep Blocked corridor, Unknown floor in the decision layer. Put the supporting details for Wheelchair profile, Expired hazard data, Reviewer, full source/version history and FR-S16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

Validation: Use a route simulation & approval fixture for Safety reviewer; change Blocked corridor and make Reviewer unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish reviewed graph” reaches S15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O35 — Hazard zones & route restrictions

Context: command · **Role:** Incident commander · **FR:** FR-O35

Primary visual: Hazard polygon editor with affected route segments and expiry

Primary question: For hazard zones & route restrictions, what do Hazard, Geometry establish, and is “Review guidance impact” available to Incident commander in this scope?

Primary action: Review guidance impact → O36

Secondary disclosure: Keep Hazard, Geometry in the decision layer. Put the supporting details for Certainty, Expires, Affected routes, full source/version history and FR-O35 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Expired geometry removes its authority; changed restrictions identify affected guidance versions before publication.

Validation: Use a hazard zones & route restrictions fixture for Incident commander; change Hazard and make Affected routes unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review guidance impact” reaches O36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O36 — Guidance distribution & receipts

Context: command · **Role:** Operator · **FR:** FR-O36

Primary visual: Guidance version distribution separating withdrawal and delivery receipts

Primary question: For guidance distribution & receipts, what do Guidance version, Smartphones establish, and is “Open speaker operations” available to Operator in this scope?

Primary action: Open speaker operations → P01

Secondary disclosure: Keep Guidance version, Smartphones in the decision layer. Put the supporting details for Speaker controller, Old route, Unreachable users, full source/version history and FR-O36 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Old-route withdrawal and new-version delivery have separate receipts; unreachable people remain visible and require alternate channels.

Validation: Use a guidance distribution & receipts fixture for Operator; change Guidance version and make Unreachable users unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open speaker operations” reaches P01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H01 — My rescue assistance profile

Context: configuration · **Role:** Citizen · **FR:** FR-H01

Primary visual: Functional assistance profile with mobility and communication pictograms

Primary question: For my rescue assistance profile, what do Age group, Mobility establish, and is “Review emergency sharing” available to Citizen in this scope?

Primary action: Review emergency sharing → H02

Secondary disclosure: Keep Age group, Mobility in the decision layer. Put the supporting details for Stairs, Communication, Source, full source/version history and FR-H01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a my rescue assistance profile fixture for Citizen; change Age group and make Source unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review emergency sharing” reaches H02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H02 — Health & assistance sharing

Context: configuration · **Role:** Citizen · **FR:** FR-H02

Primary visual: Separate functional-needs and medical-note disclosure previews

Primary question: For health & assistance sharing, what do Functional needs, Medical notes establish, and is “Save sharing preferences” available to Citizen in this scope?

Primary action: Save sharing preferences → H01

Secondary disclosure: Keep Functional needs, Medical notes in the decision layer. Put the supporting details for Family groups, Consent, Review date, full source/version history and FR-H02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a health & assistance sharing fixture for Citizen; change Functional needs and make Review date unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save sharing preferences” reaches H01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H03 — Caregiver relationships

Context: configuration · **Role:** Citizen · **FR:** FR-H03

Primary visual: Caregiver relationship graph with pending and expired authority

Primary question: For caregiver relationships, what do Jamie, Morgan establish, and is “Review caregiver authority” available to Citizen in this scope?

Primary action: Review caregiver authority → H04

Secondary disclosure: Keep Jamie, Morgan in the decision layer. Put the supporting details for Pending request, full source/version history and FR-H03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a caregiver relationships fixture for Citizen; change Jamie and make Pending request unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review caregiver authority” reaches H04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H04 — Caregiver authorization

Context: configuration · **Role:** Privacy officer · **FR:** FR-H04

Primary visual: Subject-caregiver scope comparison with authorization expiry

Primary question: For caregiver authorization, what do Subject, Caregiver establish, and is “Approve scoped relationship” available to Privacy officer in this scope?

Primary action: Approve scoped relationship → H03

Secondary disclosure: Keep Subject, Caregiver in the decision layer. Put the supporting details for Authority, Scope, Expires, full source/version history and FR-H04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a caregiver authorization fixture for Privacy officer; change Subject and make Expires unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve scoped relationship” reaches H03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H05 — Dependent person profile

Context: configuration · **Role:** Authorized caregiver · **FR:** FR-H05

Primary visual: Dependent profile with device link and review-age indicator

Primary question: For dependent person profile, what do Person, Mobility establish, and is “Request profile update” available to Authorized caregiver in this scope?

Primary action: Request profile update → H06

Secondary disclosure: Keep Person, Mobility in the decision layer. Put the supporting details for Device, Emergency note, Last reviewed, full source/version history and FR-H05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a dependent person profile fixture for Authorized caregiver; change Person and make Last reviewed unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Request profile update” reaches H06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H06 — Profile update & review

Context: configuration · **Role:** Authorized caregiver · **FR:** FR-H06

Primary visual: Profile version diff identifying source and review responsibility

Primary question: For profile update & review, what do Profile version, Change establish, and is “Submit profile update” available to Authorized caregiver in this scope?

Primary action: Submit profile update → H05

Secondary disclosure: Keep Profile version, Change in the decision layer. Put the supporting details for Source, Review status, Reason, full source/version history and FR-H06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a profile update & review fixture for Authorized caregiver; change Profile version and make Reason unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Submit profile update” reaches H05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H07 — People & assistance registry

Context: command · **Role:** Care coordinator · **FR:** FR-H07

Primary visual: Assistance registry with role-limited functional-needs columns

Primary question: For people & assistance registry, what do Alex, Jamie establish, and is “Open assistance case” available to Care coordinator in this scope?

Primary action: Open assistance case → H09

Secondary disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Visibility, full source/version history and FR-H07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a people & assistance registry fixture for Care coordinator; change Alex and make Visibility unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open assistance case” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H08 — Proactive rescue queue

Context: command · **Role:** Dispatcher · **FR:** FR-H08

Primary visual: Proactive rescue queue with explainable priority reasons

Primary question: For proactive rescue queue, what do Alex, Morgan establish, and is “Review priority reasons” available to Dispatcher in this scope?

Primary action: Review priority reasons → H09

Secondary disclosure: Keep Alex, Morgan in the decision layer. Put the supporting details for Jamie, Priority policy, full source/version history and FR-H08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a proactive rescue queue fixture for Dispatcher; change Alex and make Priority policy unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review priority reasons” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H09 — Priority evidence & decision

Context: command · **Role:** Dispatcher · **FR:** FR-H09

Primary visual: Exposure and self-evacuation evidence beside human priority decision

Primary question: For priority evidence & decision, what do Exposure, Self-evacuation establish, and is “Assign proactive assistance” available to Dispatcher in this scope?

Primary action: Assign proactive assistance → O09

Secondary disclosure: Keep Exposure, Self-evacuation in the decision layer. Put the supporting details for SOS, Reason, Priority, full source/version history and FR-H09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a priority evidence & decision fixture for Dispatcher; change Exposure and make Priority unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Assign proactive assistance” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H10 — Presence & accountability

Context: command · **Role:** Operator · **FR:** FR-H10

Primary visual: Presence evidence matrix separating observed, expected and unknown

Primary question: For presence & accountability, what do Observed present, Expected attendance establish, and is “Open proactive queue” available to Operator in this scope?

Primary action: Open proactive queue → H08

Secondary disclosure: Keep Observed present, Expected attendance in the decision layer. Put the supporting details for Presence unknown, Last-seen sources, Safe inference, full source/version history and FR-H10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Expected attendance and last seen are not live presence; never infer safety from absence of a distress signal.

Validation: Use a presence & accountability fixture for Operator; change Observed present and make Safe inference unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open proactive queue” reaches H08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H11 — Restricted medical record

Context: configuration · **Role:** Authorized medical role · **FR:** FR-H11

Primary visual: Restricted medical record with access purpose and disclosure log

Primary question: For restricted medical record, what do Person, Record establish, and is “Return to assistance” available to Authorized medical role in this scope?

Primary action: Return to assistance → H09

Secondary disclosure: Keep Person, Record in the decision layer. Put the supporting details for Source, Access reason, Disclosure, full source/version history and FR-H11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a restricted medical record fixture for Authorized medical role; change Person and make Disclosure unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Return to assistance” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H12 — Welfare check outcome

Context: field · **Role:** Responder · **FR:** FR-H12

Primary visual: Welfare observation with contact outcome and next responsible actor

Primary question: For welfare check outcome, what do Case, Person establish, and is “Save welfare observation” available to Responder in this scope?

Primary action: Save welfare observation → R06

Secondary disclosure: Keep Case, Person in the decision layer. Put the supporting details for Contact, Assistance, Location, Next, full source/version history and FR-H12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

Validation: Use a welfare check outcome fixture for Responder; change Case and make Next unavailable. Verify the visual and warning update, all 6 source fields remain reachable, and “Save welfare observation” reaches R06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q01 — Positioning source health

Context: command · **Role:** Positioning admin · **FR:** FR-Q01

Primary visual: Position-source coverage and health comparison by GNSS/BLE/UWB/QR

Primary question: For positioning source health, what do GNSS, BLE establish, and is “Inspect location quality” available to Positioning admin in this scope?

Primary action: Inspect location quality → Q02

Secondary disclosure: Keep GNSS, BLE in the decision layer. Put the supporting details for UWB, QR anchors, full source/version history and FR-Q01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a positioning source health fixture for Positioning admin; change GNSS and make QR anchors unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect location quality” reaches Q02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q02 — Position quality & history

Context: command · **Role:** Operator · **FR:** FR-Q02

Primary visual: Location history with floor confidence and conflicting-source markers

Primary question: For position quality & history, what do Current source, Floor confidence establish, and is “Open assistance context” available to Operator in this scope?

Primary action: Open assistance context → H09

Secondary disclosure: Keep Current source, Floor confidence in the decision layer. Put the supporting details for Observed, Heading, Conflicting source, full source/version history and FR-Q02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a position quality & history fixture for Operator; change Current source and make Conflicting source unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open assistance context” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q03 — Indoor calibration editor

Context: configuration · **Role:** Positioning admin · **FR:** FR-Q03

Primary visual: Calibration reference geometry with residual-error preview

Primary question: For indoor calibration editor, what do Coordinate frame, Reference points establish, and is “Review route graph” available to Positioning admin in this scope?

Primary action: Review route graph → S15

Secondary disclosure: Keep Coordinate frame, Reference points in the decision layer. Put the supporting details for Floor, Calibration error, Publish, full source/version history and FR-Q03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a indoor calibration editor fixture for Positioning admin; change Coordinate frame and make Publish unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review route graph” reaches S15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q04 — Evacuation optimization

Context: command · **Role:** Incident commander · **FR:** FR-Q04

Primary visual: Candidate-route comparison with excluded options and constraints

Primary question: For evacuation optimization, what do Primary, Alternative establish, and is “Review route comparison” available to Incident commander in this scope?

Primary action: Review route comparison → Q05

Secondary disclosure: Keep Primary, Alternative in the decision layer. Put the supporting details for Excluded, Objective, full source/version history and FR-Q04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a evacuation optimization fixture for Incident commander; change Primary and make Objective unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review route comparison” reaches Q05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q05 — Route comparison & decision

Context: command · **Role:** Safety reviewer · **FR:** FR-Q05

Primary visual: Route trade-off matrix showing uncertainty and human approval

Primary question: For route comparison & decision, what do Option A, Option B establish, and is “Publish guidance version” available to Safety reviewer in this scope?

Primary action: Publish guidance version → O36

Secondary disclosure: Keep Option A, Option B in the decision layer. Put the supporting details for Option C, Uncertainty, Approval, full source/version history and FR-Q05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a route comparison & decision fixture for Safety reviewer; change Option A and make Approval unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish guidance version” reaches O36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q06 — AI incident copilot

Context: command · **Role:** Operator · **FR:** FR-Q06

Primary visual: Evidence-linked AI summary with cutoff and authority boundary

Primary question: For ai incident copilot, what do Summary, Sources establish, and is “Review AI recommendation” available to Operator in this scope?

Primary action: Review AI recommendation → Q07

Secondary disclosure: Keep Summary, Sources in the decision layer. Put the supporting details for Data cutoff, Authority, full source/version history and FR-Q06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a ai incident copilot fixture for Operator; change Summary and make Authority unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review AI recommendation” reaches Q07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q07 — AI evidence & approval

Context: command · **Role:** Supervisor · **FR:** FR-Q07

Primary visual: Claim-evidence-unknown comparison before human decision

Primary question: For ai evidence & approval, what do Claim, Evidence establish, and is “Review route decision” available to Supervisor in this scope?

Primary action: Review route decision → Q05

Secondary disclosure: Keep Claim, Evidence in the decision layer. Put the supporting details for Unknown, Guard, Action, full source/version history and FR-Q07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a ai evidence & approval fixture for Supervisor; change Claim and make Action unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review route decision” reaches Q05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q08 — AI policies & model evaluation

Context: configuration · **Role:** AI governance lead · **FR:** FR-Q08

Primary visual: Evaluation results for grounding, injection and data exposure

Primary question: For ai policies & model evaluation, what do Model, Grounding test establish, and is “Open model governance” available to AI governance lead in this scope?

Primary action: Open model governance → Q09

Secondary disclosure: Keep Model, Grounding test in the decision layer. Put the supporting details for Prompt injection, Health data, Control tools, full source/version history and FR-Q08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a ai policies & model evaluation fixture for AI governance lead; change Model and make Control tools unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open model governance” reaches Q09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q09 — AI deployment governance

Context: configuration · **Role:** AI governance lead · **FR:** FR-Q09

Primary visual: Model/prompt version deployment preview with regional fallback

Primary question: For ai deployment governance, what do Region, Model version establish, and is “Save deployment draft” available to AI governance lead in this scope?

Primary action: Save deployment draft → Q08

Secondary disclosure: Keep Region, Model version in the decision layer. Put the supporting details for Prompt version, Data retention, Fallback, full source/version history and FR-Q09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a ai deployment governance fixture for AI governance lead; change Region and make Fallback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save deployment draft” reaches Q08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q10 — Simulation & crowd scenarios

Context: configuration · **Role:** Safety planner · **FR:** FR-Q10

Primary visual: Simulated crowd and blocked-exit scenario with mobility cohorts

Primary question: For simulation & crowd scenarios, what do Mode, Occupancy establish, and is “Review route validation” available to Safety planner in this scope?

Primary action: Review route validation → S16

Secondary disclosure: Keep Mode, Occupancy in the decision layer. Put the supporting details for Blocked exit, Mobility support, Result, full source/version history and FR-Q10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

Validation: Use a simulation & crowd scenarios fixture for Safety planner; change Mode and make Result unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review route validation” reaches S16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F01 — Physical systems overview

Context: command · **Role:** Physical supervisor · **FR:** FR-F01

Primary visual: Physical-system topology showing controller authority and evidence age

Primary question: For physical systems overview, what do Controller health, Zone status establish, and is “Review F07” available to Physical supervisor in this scope?

Primary action: Review F07 → F07

Secondary disclosure: Keep Controller health, Zone status in the decision layer. Put the supporting details for Evidence freshness, Local authority, Control permission, full source/version history and FR-F01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a physical systems overview fixture for Physical supervisor; change Controller health and make Control permission unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F07” reaches F07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F02 — Controller inventory

Context: configuration · **Role:** Integration engineer · **FR:** FR-F02

Primary visual: Controller inventory with model, protocol and verified capability

Primary question: For controller inventory, what do Controller model, Protocol binding establish, and is “Review F03” available to Integration engineer in this scope?

Primary action: Review F03 → F03

Secondary disclosure: Keep Controller model, Protocol binding in the decision layer. Put the supporting details for Operating mode, Site assignment, Capability evidence, full source/version history and FR-F02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a controller inventory fixture for Integration engineer; change Controller model and make Capability evidence unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F03” reaches F03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F03 — Controller capability detail

Context: configuration · **Role:** Integration engineer · **FR:** FR-F03

Primary visual: Allowed/forbidden operation matrix beside local override status

Primary question: For controller capability detail, what do Controller model, Allowed operations establish, and is “Review F04” available to Integration engineer in this scope?

Primary action: Review F04 → F04

Secondary disclosure: Keep Controller model, Allowed operations in the decision layer. Put the supporting details for Forbidden operations, Capability evidence, Local override, full source/version history and FR-F03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a controller capability detail fixture for Integration engineer; change Controller model and make Local override unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F04” reaches F04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F04 — Signal point mapping

Context: configuration · **Role:** Integration engineer · **FR:** FR-F04

Primary visual: Signal-address-to-meaning mapping with polarity and quality states

Primary question: For signal point mapping, what do Signal address, Signal meaning establish, and is “Review F05” available to Integration engineer in this scope?

Primary action: Review F05 → F05

Secondary disclosure: Keep Signal address, Signal meaning in the decision layer. Put the supporting details for Normal polarity, Quality mapping, Mapping revision, full source/version history and FR-F04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a signal point mapping fixture for Integration engineer; change Signal address and make Mapping revision unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F05” reaches F05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F05 — Physical zone mapping

Context: configuration · **Role:** Site engineer · **FR:** FR-F05

Primary visual: Physical/source zone overlay showing coordinate and coverage mismatch

Primary question: For physical zone mapping, what do Source zone, Target zone establish, and is “Review F06” available to Site engineer in this scope?

Primary action: Review F06 → F06

Secondary disclosure: Keep Source zone, Target zone in the decision layer. Put the supporting details for Coordinate frame, Coverage boundary, Mapping revision, full source/version history and FR-F05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a physical zone mapping fixture for Site engineer; change Source zone and make Mapping revision unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F06” reaches F06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F06 — Cause-and-effect matrix

Context: configuration · **Role:** Fire systems engineer · **FR:** FR-F06

Primary visual: Cause-to-effect graph with authority and inhibit gates

Primary question: For cause-and-effect matrix, what do Cause input, Approved effect establish, and is “Review F15” available to Fire systems engineer in this scope?

Primary action: Review F15 → F15

Secondary disclosure: Keep Cause input, Approved effect in the decision layer. Put the supporting details for Authority record, Inhibit condition, Change impact, full source/version history and FR-F06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a cause-and-effect matrix fixture for Fire systems engineer; change Cause input and make Change impact unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F15” reaches F15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F07 — Physical event detail

Context: command · **Role:** Physical supervisor · **FR:** FR-F07

Primary visual: Physical event provenance card linking observed evidence and incident

Primary question: For physical event detail, what do Source event, Signal meaning establish, and is “Review F08” available to Physical supervisor in this scope?

Primary action: Review F08 → F08

Secondary disclosure: Keep Source event, Signal meaning in the decision layer. Put the supporting details for Observed time, Evidence quality, Incident linkage, full source/version history and FR-F07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a physical event detail fixture for Physical supervisor; change Source event and make Incident linkage unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F08” reaches F08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F08 — Fire-water status

Context: command · **Role:** Physical supervisor · **FR:** FR-F08

Primary visual: Water-system schematic separating pump, pressure and valve feedback

Primary question: For fire-water status, what do Pump feedback, Pressure observation establish, and is “Review F09” available to Physical supervisor in this scope?

Primary action: Review F09 → F09

Secondary disclosure: Keep Pump feedback, Pressure observation in the decision layer. Put the supporting details for Valve feedback, Water availability, Equipment impairment, full source/version history and FR-F08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a fire-water status fixture for Physical supervisor; change Pump feedback and make Equipment impairment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F09” reaches F09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F09 — Controlled release review

Context: command · **Role:** Authorized fire officer · **FR:** FR-F09

Primary visual: Release-prerequisite gate against exact equipment and expiring request

Primary question: For controlled release review, what do Requested operation, Target equipment establish, and is “Review F10” available to Authorized fire officer in this scope?

Primary action: Review F10 → F10

Secondary disclosure: Keep Requested operation, Target equipment in the decision layer. Put the supporting details for Capability evidence, Release prerequisites, Command expiry, full source/version history and FR-F09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a controlled release review fixture for Authorized fire officer; change Requested operation and make Command expiry unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F10” reaches F10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F10 — Command authorization

Context: command · **Role:** Independent command approver · **FR:** FR-F10

Primary visual: Immutable command digest with requester/approver and execution boundary

Primary question: For command authorization, what do Exact command digest, Requester identity establish, and is “Review F11” available to Independent command approver in this scope?

Primary action: Review F11 → F11

Secondary disclosure: Keep Exact command digest, Requester identity in the decision layer. Put the supporting details for Approver identity, Approval state, Execution boundary, full source/version history and FR-F10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a command authorization fixture for Independent command approver; change Exact command digest and make Execution boundary unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F11” reaches F11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results

separately from this specification.

F11 — Execution feedback

Context: command · **Role:** Physical supervisor · **FR:** FR-F11

Primary visual: Receipt-versus-physical-feedback timeline with unknown outcome

Primary question: For execution feedback, what do Command receipt, Physical feedback establish, and is “Review F12” available to Physical supervisor in this scope?

Primary action: Review F12 → F12

Secondary disclosure: Keep Command receipt, Physical feedback in the decision layer. Put the supporting details for Outcome state, Execution correlation, Retry permission, full source/version history and FR-F11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Accepted command and physical actuation are separate states; unknown outcome disables blind retry and opens reconciliation F12.

Validation: Use a execution feedback fixture for Physical supervisor; change Command receipt and make Retry permission unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F12” reaches F12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F12 — Unknown-result reconciliation

Context: command · **Role:** Physical supervisor · **FR:** FR-F12

Primary visual: Unknown-result reconciliation comparing readback and field evidence

Primary question: For unknown-result reconciliation, what do Command reference, Readback evidence establish, and is “Review F13” available to Physical supervisor in this scope?

Primary action: Review F13 → F13

Secondary disclosure: Keep Command reference, Readback evidence in the decision layer. Put the supporting details for Field confirmation, Reconciliation decision, Decision reason, full source/version history and FR-F12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Keep outcome unknown until correlated readback or authorized field evidence resolves it; retain contradictory evidence and reason.

Validation: Use a unknown-result reconciliation fixture for Physical supervisor; change Command reference and make Decision reason unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F13” reaches F13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F13 — Local override & handover

Context: command · **Role:** Site incident commander · **FR:** FR-F13

Primary visual: Local-control handover showing remote lease and receiving operator

Primary question: For local override & handover, what do Local controller state, Remote lease establish, and is “Review F01” available to Site incident commander in this scope?

Primary action: Review F01 → F01

Secondary disclosure: Keep Local controller state, Remote lease in the decision layer. Put the supporting details for Receiving operator, Handover scope, Return-to-remote gate, full source/version history and FR-F13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Loss of remote lease cannot take control back from the local controller; returning remote requires the documented handover gate.

Validation: Use a local override & handover fixture for Site incident commander; change Local controller state and make Return-to-remote gate unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F01” reaches F01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F14 — Impairment & test mode

Context: command · **Role:** Maintenance coordinator · **FR:** FR-F14

Primary visual: Impairment map with compensating measures, permit and expiry

Primary question: For impairment & test mode, what do Impairment scope, Permit reference establish, and is “Review F18” available to Maintenance coordinator in this scope?

Primary action: Review F18 → F18

Secondary disclosure: Keep Impairment scope, Permit reference in the decision layer. Put the supporting details for Compensating measure, Test isolation, Expiry and escalation, full source/version history and FR-F14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a impairment & test mode fixture for Maintenance coordinator; change Impairment scope and make Expiry and escalation unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F18” reaches F18 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F15 — Commissioning plan

Context: configuration · **Role:** Commissioning lead · **FR:** FR-F15

Primary visual: Commissioning sequence with test isolation and witness checkpoints

Primary question: For commissioning plan, what do Test configuration, Test scope establish, and is “Review F16” available to Commissioning lead in this scope?

Primary action: Review F16 → F16

Secondary disclosure: Keep Test configuration, Test scope in the decision layer. Put the supporting details for Test prerequisites, Witnesses, Evidence requirement, full source/version history and FR-F15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a commissioning plan fixture for Commissioning lead; change Test configuration and make Evidence requirement unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F16” reaches F16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F16 — Test-run evidence

Context: configuration · **Role:** Commissioning witness · **FR:** FR-F16

Primary visual: Expected/observed physical-effect comparison with fault injections

Primary question: For test-run evidence, what do Test run, Expected effect establish, and is “Review F17” available to Commissioning witness in this scope?

Primary action: Review F17 → F17

Secondary disclosure: Keep Test run, Expected effect in the decision layer. Put the supporting details for Observed effect, Fault injection, Result status, full source/version history and FR-F16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a test-run evidence fixture for Commissioning witness; change Test run and make Result status unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F17” reaches F17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F17 — Compatibility & certificate dossier

Context: configuration · **Role:** Compliance reviewer · **FR:** FR-F17

Primary visual: Certificate scope mapped to installed bill of materials and limitations

Primary question: For compatibility & certificate dossier, what do Installed bill of materials, Certificate scope establish, and is “Review F20” available to Compliance reviewer in this scope?

Primary action: Review F20 → F20

Secondary disclosure: Keep Installed bill of materials, Certificate scope in the decision layer. Put the supporting details for Interface limitation, Evidence validity, Release decision, full source/version history and FR-F17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a compatibility & certificate dossier fixture for Compliance reviewer; change Installed bill of materials and make Release decision unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F20” reaches F20 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F18 — Maintenance & restoration

Context: configuration · **Role:** Maintenance coordinator · **FR:** FR-F18

Primary visual: Restoration checklist with residual impairment and approval evidence

Primary question: For maintenance & restoration, what do Maintenance scope, Repair evidence establish, and is “Review F01” available to Maintenance coordinator in this scope?

Primary action: Review F01 → F01

Secondary disclosure: Keep Maintenance scope, Repair evidence in the decision layer. Put the supporting details for Restoration test, Closure approval, Residual impairment, full source/version history and FR-F18 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a maintenance & restoration fixture for Maintenance coordinator; change Maintenance scope and make Residual impairment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F01” reaches F01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F19 — Rescue equipment mission

Context: command · **Role:** Rescue mission commander · **FR:** FR-F19

Primary visual: Rescue-equipment mission envelope with qualified operator and limits

Primary question: For rescue equipment mission, what do Equipment configuration, Control interface establish, and is “Review F11” available to Rescue mission commander in this scope?

Primary action: Review F11 → F11

Secondary disclosure: Keep Equipment configuration, Control interface in the decision layer. Put the supporting details for Operator qualification, Mission constraints, Mission evidence, full source/version history and FR-F19 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a rescue equipment mission fixture for Rescue mission commander; change Equipment configuration and make Mission evidence unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F11” reaches F11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F20 — Physical connector onboarding

Context: configuration · **Role:** Integration engineer · **FR:** FR-F20

Primary visual: Vendor connector onboarding gates for credentials, import and activation

Primary question: For physical connector onboarding, what do Vendor interface, Transport security establish, and is “Review F02” available to Integration engineer in this scope?

Primary action: Review F02 → F02

Secondary disclosure: Keep Vendor interface, Transport security in the decision layer. Put the supporting details for Credential custody, Import scope, Activation gate, full source/version history and FR-F20 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

Validation: Use a physical connector onboarding fixture for Integration engineer; change Vendor interface and make Activation gate unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F02” reaches F02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

BMS revision2.1 scope note

This audit records the historical233-screen redesign. Revision2.1 adds8BMSscreens, bringing the current catalog to241screens. The historical audit counts are intentionally unchanged. New screen specifications and browser evidence are maintained separately.

Current screen specifications · BMS UI evidence